

Before Beginning:

‘I see that there are many young people here; as an old man, a little advice: Life can set us a lot of snares, a lot of bumps, we can fail a thousand times, in life, in the social struggle . . . , but, if we search for it, we’ll have the strength to get up again and start over. The most beautiful thing about the day is that it dawns. There is always a dawn after the night has passed. Don’t forget it, kids. The only losers are the ones who stop fighting.’

These words are not mine, José Mujica, President of Uruguay (2010-2015). He declined to live in the presidential palace or to use its staff during his presidency and used a 1987 Volkswagen Beetle and his 60-year-old bicycle as means of transportation. he would commute from his home: a cluttered, three-room shack the size of a studio apartment, crammed with a wood stove, overstuffed bookcases

1. Please discuss with your teachers or classmates here, without fail, any problems you have. Do not hesitate.
2. You should relax with extra curricular activities: exercise, play, listen to music/songs etc. Remember, internet is so designed to unleash flood gates for distraction; and uncontrolled use is known to lead to reduced Brain functions.
3. Never underestimate your capabilities, work and work and work. Remember Grandmaster Divya Deshmukh. She did not even have one GM norm, but defeated several Grandmasters on her way to the top.
4. Do not be fooled by initial classes; as days proceed there will be more and more challenging concepts and ideas with whom you have to make friendship.
5. Unlike school studies, now you need to learn to ‘think, workout problems and learn to present solutions’. You need to understand concepts and be able to use in a new problem.
6. Critical and free thinking as well as flowering of diverse creative ideas is our ancient culture. But considering the society we live in, you must apply diligence.
Firstly, in the present society, questioning a majority view, leads to

being trolled, harassed, lynched; there is no scope for argument and discovering truth. This is NOT so in our class. Even if a majority in the class says something which you doubt, you should not hesitate to express and others should learn to argue their case.

Secondly, very many in authority, including most constitutional functionaries, do not respect the constitution; very unfortunate. This is not so in our class; **you as well as me** should all respect the rules of the Institute and rules of mathematics. We should carefully present our mathematical arguments.

Thirdly, If you are asked to prove something; saying easy/simple/trivial/ is it not so?/ etc do not constitute proof. Please remember You use the word simple, not because someone else used it but, because You have done it once and found it simple Yourself.

7. Remember education is not about literacy/job alone. As our ancient wisdom says and as Ghazala Wahab puts it neatly: **It's about exposure to diverse ideas, being challenged by new concepts, and learning to go beyond inherited wisdom.** Above all, education adds to confidence and encourages people to overcome insecurities. ❤️❤️❤️

From a physiological perspective, the brain must translate the external sources of energy – sight, sound etc – and encode into electrical patterns which the brain can understand. The more elaborately we encode information at the moment of learning, the stronger the memory. The more a learner focuses on the meaning of the information being presented, the more elaborately he/she will process the information. When you are trying to drive a piece of information into your brain's memory systems, make sure exactly what the information means. Do not try to memorize the information by rote and pray the meaning will somehow reveal itself.

John Medina

Beginning

Grading system

Quiz (10 pts): August last week. Midsem (30 pts) September last week.

Quiz (10 pts): October last week. Semestral (50 pts) November last week.

TAs

The following Research Scholars kindly agreed to help us.

Aisha Negi: aisha@cmi.ac.in

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Sessions at time convenient to them and you can be fixed.

You must let one of us know.

Books:

You can look at any book on Real Analysis in the library. For example

Rudin (Principles of Mathematical Analysis) or

Courant and John (Introduction to Calculus and Analysis Vol 1) or

T M Apostol (Calculus Vol 1) or

R G Bartle and D R Sherbert (Introduction to Real Analysis)

and so on (and on).

You are free to ask me any questions you may have.

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We are supposed to do Analysis. We do Real Analysis. In our course, it is study of real numbers; sequences; functions. Are there other analyses? yes, complex analysis, Harmonic Analysis, Functional Analysis etc.

First we start with real numbers. But then, do we not already know numbers? let us see.

What is a real number?

.....

Ok let us see what is one? 2? 3?

Symbol for some Measure.

What is measure and what is symbol? Is it only measure of that thing or other things also?

.....

They are natural numbers.

Excellent. But what is number, Why do you say 'natural'?

.....

We see one apple, two chairs, three tables. Do we see 1,2,3?

They are symbols.

Yes, excellent, you are right. In a sense 1 is 'one-ness' and 2 is 'two-ness' and so on. We must appreciate our elementary school teachers who made us friends with these abstract symbols.

What is colour of your shirt?

pink

Good. What is pink?

.....

Ok what is colour?

.....

You see, strictly speaking we know very few things; we can recognize many things. We can recognize colours and that is good enough for all practical purposes until we see a colour and start quarrelling: I say it is red and you say it is not red but a shade of pink. Of course some learned scholar would come along and say: Listen, you are seeing a colour because a certain frequency of light is reflected from the object and colour names are codes for certain frequencies. But it takes some time to understand frequencies, reflection etc. It is not really necessary.

is there a number x such that $x \times x = 3$?

$\sqrt{3}$.

What is it?

a number whose square is 3.

But is there such a number? if you show me that there is such a number, we can probably name it $\sqrt{3}$, right?

.....

Is there a number x such that $x \times x = -3$? This looks similar to previous question.

No, square of every number is non-negative.

Excellent. So there are some rules which allow you to show certain things and do not allow certain other things. What are those rules. Are there fixed number of rules? Whenever we are unable to show something are we going to make new rules? for example government can make a new rule today to put us all in jail, right? For real number system there are few rules and we have to deduce EVERYTHING only from those rules. We can not make new rules.

Real Number system

There are few rules that real numbers obey and they can be listed once and for all. Everything, that we have been doing so far in our school, follows from those rules. We do not make new rules. So our first objective is to learn those rules and convince ourselves that all the past few years whatever

you did follows from those rules. Thus there is no contradiction to anything you learnt so far. Of course, we are not going to go through ALL the high school syllabus, we shall select few things and show that they follow from the rules listed.

These rules are grouped so that we can remember easily. Consider a set R . [What is a set? A set is a collection of objects. Something which belongs to the set is called an element of the set, write $x \in R$, and some thing which is not in the set is not an element of the set, write $x \notin R$. It does not matter to us what the elements are. Just as our experience with colours, this definition is good enough for us.]

Axiom set I: Addition

There is an operation $+$ which associates with every ordered pair (x, y) of elements of R an element $x + y$ in R . This obeys the following three rules.

(1-i) $x + y = y + x$; (1-ii) $(x + y) + z = x + (y + z)$;
 (1-iii) there is an element $a \in R$ such that $x + a = a + x = x$ for all x and ; for each $x \in R$ there is an element y , depending on x , such that $x + y = y + x = a$. You can write the last rule as

$$\exists a[(\forall x, x + a = a + x = x) \ \& \ (\forall x \exists y, x + y = y + x = a)]$$

To be strict I should write $(\forall x \in R)$ and $(\exists a \in R)$ and so on. But when the context is understood, we shall not complicate life by writing too much.

This $+$ is called addition. The first rule says addition is ‘commutative’. The second rule says it is ‘associative’. The third says there is a ‘(additive identity)zero’ which when added to anything does not change that thing. Moreover everything has something which when added brings it to ‘zero’. Do not try to remember the technical names, because we will have many as we go along. It is better to remember the property.

Is there a unique a satisfying (iii)? Yes. In fact there is a unique a stisfying the first property itself. Suppose $x + a = a + x = x$ and also $x + b = b + x = x$ for all x . Then from first equation (taking $x = b$), you see $a + b = b$. From second equation (taking $x = a$) you see $a + b = a$. Thus $a = b$. Thus identity is unique, we shall denote it by 0.

Given x , how many y are there such that $x + y = 0$?. Exactly one! Because if there are two, say, y and y^* , then $x + y = 0$ and $x + y^* = 0$. So

$$y = y + 0 = y + (x + y^*) = (y + x) + y^* = 0 + y^* = y^*$$

Justify each equality above by using one of the rules listed. Thus given x , the unique y with $x + y = 0$ is denoted: $-x$. Thus $x + (-x) = (-x) + x = 0$.

We abbreviate $x + (-z)$ to simply $x - z$.

If $x + y = x + z$ then $y = z$. This is because

$$\begin{aligned} y &= y + 0 = y + (x + (-x)) = (y + x) + (-x) \\ &= (z + x) + (-x) = z + (x + (-x)) = z + 0 = z. \end{aligned}$$

Justify each equality using one of the appropriate rules. Sometimes this result is called ‘cancellation rule’ because we are cancelling x from both sides of the equation in the hypothesis.

$-(-x) = x$ because $(-x) + x = x + (-x) = 0$.

Axiom Set II: Multiplication

For every ordered pair (x, y) of elements from R , there is an element $x \cdot y$ in R . This operation satisfies again three rules.

(2-i) $x \cdot y = y \cdot x$; (2-ii) $(x \cdot y) \cdot z = x \cdot (y \cdot z)$;

(2-iii) there is an element $c \in R, c \neq 0$ such that $x \cdot c = c \cdot x = x$ for all x ; Moreover for each $x \in R, x \neq 0$ there is an element y , depending on x , such that $x \cdot y = y \cdot x = c$.

You can write the last rule as

$$\exists c[(\forall x, x \cdot c = c \cdot x = x) \ \& \ (\forall x \neq 0 \ \exists y, x \cdot y = y \cdot x = c)]$$

Imitating earlier argument we can show (Do it please) that the c is unique. We denote it 1. Thus $1 \neq 0$. Also given $x \neq 0$, the y asserted in third rule is unique and denoted as y^{-1} or as more familiar $1/y$. To avoid confusion, we say 0 is additive identity and 1 is multiplicative identity. $(-x)$ is additive inverse and $1/x$ as multiplicative inverse. We also write $x \cdot (1/z)$ as simply x/z . Sometimes we also omit the dot, thus xz means $x \cdot z$. Sometimes we use $\times$ instead of $\cdot$ (for visibility)

If $x \neq 0$, then $x \cdot y = x \cdot z$ implies $y = z$. This is because

$$\begin{aligned} y &= 1 \cdot y = ((1/x) \times x) \times y = (1/x) \times (x \times y) \\ &= (1/x)(x \times z) = ((1/x) \times x) \times z = 1 \times z = z \end{aligned}$$

Thus for the cancellation law to hold we needed $x \neq 0$.

$1/(1/x) = x$ because $(1/x) \times x = x \times (1/x) = 1$. You see that at this moment the two operations $+$ and $\cdot$ are separate operations, nothing to do with each other. If you wish you could have taken $\cdot$ also as the earlier $+$ itself!

Axiom set III: $+$ and $\cdot$ are friends

There is only one rule.

(3-i) $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$. for all x, y, z

This is called ‘distributivity’ but no need to remember the name. This is an extremely useful rule. For example if you want to calculate $(299 \times 59) +$

$(299 \times 41) = 299(59 + 41) = 29900$. You do not have to simplify the two factors and then add. Here are more useful consequences.

(1). $(0 \times y) + 0 = 0 \times y = (0 + 0) \times y = (0 \times y) + (0 \times y)$ Thus by cancellation rule $0 = (0 \times y)$ for whatever be y . Thus $0 \times y = 0 \times z$ does not allow you to conclude $y = z$.

(2) If $xy = 0$ then either $x = 0$ or $y = 0$. This is observed as follows. Suppose $x \neq 0$. Then,

$$y = 1 \times y = (1/x \times x) \times y = (1/x) \times (x \times y) = (1/x) \times 0 = 0.$$

(3) $(-x) \times y = x \times (-y) = -(x \times y)$. This is because

$$(-x) \times y + (x \times y) = (-x + x) \times y = 0 \times y = 0.$$

Thus $(-x) \times y$ added to $(x \times y)$ gives zero and hence it must be additive inverse of $(x \times y)$, namely, $-(x \times y)$.

In particular, taking $y = -x$, we see

$$(-x) \times (-x) = -[x \times (-x)] = -[-(x \times x)] = x \times x.$$

where we used $(-(-z)) = z$.

Axiom set IV: linear order

There is a binary relation $<$ for pairs of elements x, y satisfying the following two rules:

(4-i) for all x, y , exactly one of the three holds: $x < y$, $x = y$, $y < x$.

(4-ii) If $x < y$ and $y < z$ then $x < z$.

(binary relation is a relation between pairs of elements).

Such an order as above is called linear order. We read it as 'less than'. We abbreviate the sentence $x < y$ or $x = y$ to $x \leq y$. We also write $y > x$ to say $x < y$. Thus also $y \geq x$ to say either $y > x$ or $y = x$. At this moment this relation has nothing to do with addition and multiplication.

Axiom set V: $+, \times, <$ are friends.

There are two rules, one for $(+, <)$ and one for $(\times, <)$.

(5-i) If $y < z$ then $x + y < x + z$ for all x .

(5-ii) If $0 < x$ and $0 < y$ then $0 < xy$.

There are several interesting consequences.

(1) $x > 0$ iff $-x < 0$. This is immediate, add $-x$ or x as needed to both sides.

(2) If $x \neq 0$, then $x^2 > 0$, where $x^2 = x \cdot x$. Indeed, if $x > 0$ then axiom already tells $x \cdot x > 0$. If $x < 0$ then $(-x) > 0$ and so $(-x)(-x) > 0$ but this is same as $x \cdot x$.

In particular $1 = 1 \times 1 > 0$. Using the above now, we conclude $-1 < 0$.

(3) If $y < z$ then for every $x > 0$ we have $xy < xz$. this is because adding $-y$ to both sides of the hypothesis implies $z - y > 0$ and by axiom $x(z - y) > 0$,

that is $xz - xy > 0$. Add xy both sides.

On the other hand if $y < z$ then for every $x < 0$ we have $xy > xz$. Just use $(-x) > 0$ and proceed.

(4) If $x > 0$ then $(1/x) > 0$. Indeed if $(1/x) = 0$ then $1 = x(1/x) = 0$ which is not true. If $(1/x) < 0$ then $(-1/x) > 0$. So $-1 = -(1/x)(x) = (-1/x)(x) > 0$ which is again false. Thus we must have $(1/x) > 0$

(5) If $0 < x < y$ then $0 < (1/y) < (1/x)$. Indeed we already know $(1/x) > 0$ and $(1/y) > 0$. Hence $(1/xy) = (1/x)(1/y) > 0$. Now use $0 < x < y$ and multiply by $(1/xy)$ to get the result.

6. If $x > 1$ then $x < x^2 < x^3 < \dots$. More precisely, for all n , $x^n < x^{n+1}$. You can use $x^{n+1} - x^n = x^n(x - 1) > 0$.

If $0 < x < 1$ then $x > x^2 > x^3 > \dots$. More precisely, for all n , $x^n > x^{n+1}$. You can use $x^n - x^{n+1} = x^n(1 - x) > 0$.

Geometry of the number line.

How do you visualize R . Do you think of real numbers put in a bag like apples or oranges? or all numbers made into a big heap? No. We think of real numbers arranged in an orderly manner in a straight horizontal line following a particular rule. remember x -axis and y -axis you drew in high school. In spite of our so called (high sounding terms) axioms, rules etc you need not change your mental picture of real number system as x -axis. we shall now explain this geometric picture. We usually draw the number line as below.

We plot the number zero arbitrarily.

0

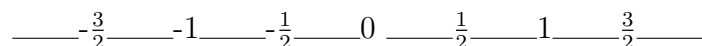
If $0 < x$ we say x is positive, If $x < 0$ we say x is negative. Remember x is positive if and only if $-x$ is negative. We make up our mind that numbers to the right of zero are positive and increase as you go away from zero. If you do not like, you can plot positive numbers to the left of zero. But we shall not.

So we plot 1 to the right of zero. Since $1 > 0$ we see $2 = 1 + 1 > 1 + 0 = 1$. Since $2 - 1 = 1 - 0$, we plot 2 at the same distance from 1 as 1 was from zero. Now argue in the same way and plot 3 to the right of 2 at the same distance as two was from one.

-2 -1 0 1 2

Since $0 - (-n) = n - 0$ we plot $-n$ to the left of zero at the same distance as n was from zero on the right.

Just to remind you, m/n means $m \cdot (1/n)$ and of course $1/n$ is the multiplicative inverse of n . Now we start plotting these numbers on the line. Since $(1/2) + (1/2) = 1$ we plot $1/2$ midway between zero and 1. This is because $1 - (1/2) = (1/2) - 0$. Now I leave the rest to your imagination.



The main point is that the mental picture about the arithmetic and geometry (whatever it may mean) of the number line we have, agrees with the picture given by the few rules we made above. And it is same as your high school picture.

There are two very important questions now.

Does every real number correspond to a point on the line? Yes.

Does every point of the line correspond to a number? Yes. We shall return to this matter later.

Now here is the final axiom.

Axiom set VI: Completeness axiom/ Least upper bound axiom/continuity axiom there is only one axiom. Suppose $S \subset R$ is a non-empty set which is bounded above, this means, there is a z such that $x \leq z$ for all $x \in S$. Such a z is called an upper bound for the set S .

(6-i) If S is nonempty and bounded above, then S has a least upper bound, this means, there is a u such that (a) $x \leq u$ for all $x \in S$ and (b) if $x \leq v$ for all $x \in S$ then $u \leq v$.

Note that with the above notation, (a) says that u is also an upper bound of S and (b) says that any other upper bound is at least as large as u , that is, u is the least upper bound. Thus the above says that there is a least upper bound, if the set is non-empty and bounded above. Can there be two such? No because if u_1 and u_2 are as stated above, then taking $v = u_2$ you get $u_1 \leq u_2$ and taking $v = u_1$ you get $u_2 \leq u_1$. Hence $u_1 = u_2$.

This completes the list of axioms. A set R with above operations and order satisfying the above axioms is called the set of real numbers. Elements of R are called real numbers. Three questions arise: (i) is there such a system? (ii) How many such systems are there? (iii) Would every thing we did all these years follow from these. Yes. There is such a system. There is only one such system, essentially. If there are two such systems, then you can set up a bijection which preserves all the operations. Thus it is only a renaming.

Finally, all you did so far follows from these rules.

First some notation. We denote $1 + 1$ as 2; $2 + 1$ as 3 and so on; all the natural numbers you are familiar with. The set of natural numbers is the set of all these. But if you do not want to use ‘so on’, here is precise definition of the set of natural numbers N . It is the smallest subset $S \subset R$ satisfying the two conditions: (i) $1 \in S$ and (ii) if $x \in S$ then $x + 1 \in S$. Is there such a set? Yes. Firstly the set R satisfies the two conditions. So consider all subsets of R satisfying these two conditions and take their intersections. Then this intersection clearly satisfies (i) and (ii) and is the smallest such set. But you need not bother, pretend that N is the set of strictly positive integers as you knew already.

[I am saying N is a subset of R such that blah blah. Ok, nothing wrong. But, strictly speaking, it is N that is constructed first; then all integers, then all fractions, then R . But since we are NOT constructing anything; just trying to understand the few basic rules of R that lead to everything else, we have taken this route.]

Here is an important theorem.

Archimedean Principle: The set N is not bounded above.

Proof is simple. Clearly N is non-empty. If it is bounded above then it must have a least upper bound, say, s . Since $s - 1 < s$, we conclude $s - 1$ is not an upper bound, so there is $m \in N$ such that $s - 1 < m$. But then $m + 1 \in N$ and $s < m + 1$ showing s is not an upper bound for N . This is called proof by contradiction.

Thus $(\forall x)(\exists n \in N)(x < n)$, because no matter what x you take it is not an upper bound for N . From this you see $(\forall x > 0)(\exists n \in N)(0 < 1/n < x)$ Indeed given x you apply previous observation with $(1/x)$ to get n such that $(1/x) < n$ and multiply both sides by (x/n) to get the stated conclusion.

This is Archimedean principle because Archimedes first observed this. For instance, he was calculating area under the curve $y = x^2$ between $x = 0$ and $x = 1$. Remember there was no concept of convergence or integration. To solve a problem, you first recall what you know about related matters and ask: how can I use my knowledge to solve this problem. (You must understand that people do not start with definitions and develop matters. People ask questions and find answers using their knowledge and then if the same idea appears again and again, you give a name to it, that

is, make a definition.)

We know areas of rectangles, can we use this knowledge to calculate the required area. Yes. Consider rectangles with sides $[0, 1/n], [1/n, 2/n], \dots, [(n-1)/n, n/n]$ and heights $(1/n)^2; (2/n)^2, \dots, (n/n)^2$ respectively. These rectangles cover the area under the curve. So area under the curve is at most

$$\sum_1^n \frac{1}{n} \frac{k^2}{n^2} = \frac{n(n+1)(2n+1)}{6n^3} = \frac{1}{6}(1 + \frac{1}{n})(2 + \frac{1}{n})$$

Of course this overestimates the required area. But inspection shows the extra area reduces as n becomes large and the extra area is shrinking to zero. Thus ‘ultimately’ we should get the area. On the other hand, using the Archimedean property, as n becomes large $1/n$ becomes smaller than any strictly positive number and the above expression ‘ultimately’ becomes

$$\frac{1}{6} \times 1 \times 2 = \frac{1}{3}.$$

Thus the area is $1/3$.

Let us return to the geometric picture. There is one main problem: Is every positive number somewhere on the right side of zero between the $1, 2, 3 \dots$ that we have plotted or are there numbers beyond all things we plotted? Luckily there is nothing beyond all these By Archimedean property. Stop and think.

Numbers in N are called natural numbers.

We define

$$Z = \{x : x = 0 \text{ or } x \in N; \text{ or } -x \in N\}$$

Or you can think as

$$Z = \{\dots\dots -2, -1, 0, 1, 2, \dots\dots\}$$

Numbers in Z are called integers. With this notation, elements of N are also called positive integers. Q stands for the set of all numbers of the form m/n where $m, n \in Z$ and $n \neq 0$. Of course $2/4$ is same as $13/26$ (show this). Numbers which are in Q are called rational numbers. For any x we define $x^2 = x \times x$, $x^3 = x^2 \times x$ and so on. More precisely, if we have defined x^n then $x^{n+1} = x^n \times x$. Of course, you think of x^n as x multiplied with itself n -times.

n -th roots for positive numbers:

We shall now prove a basic result.

Theorem: Given $x > 0$ and $n \geq 2$ in N , there is a unique $z > 0$ such that $z^n = x$

This z is denoted $\sqrt[n]{x}$ or $x^{1/n}$. Following highschool notation we shall denote $\sqrt[2]{x}$ as simply $\sqrt{x}$.

Proof: First observe that $0 < u < v$ implies $0 < u^n < v^n$. Either you can prove it by induction or use

$$v^n - u^n = (v - u)(v^{n-1} + v^{n-2}u + v^{n-3}u^2 + \cdots + u^{n-1}) > 0$$

This in particular implies that there can not be two different numbers $u > 0, v > 0$ such that $u^n = x$ and $v^n = x$.

We show two things

- (i) If $z > 0$ and $z^n < x$, then we can increase z a little bit so that its n -th power is still smaller than x . That is, there is $h > 0$ so that $(z + h)^n < x$.
- (ii) If $z > 0$ and $z^n > x$, then we can decrease z a little bit so that it is still positive and its n -th power is larger than x . That is, there is $h > 0$ so that $0 < z - h < z$ and $(z - h)^n > x$.

Let us see what happens if we have done this. Let $A = \{z > 0 : z^n < x\}$.

Is A non-empty? Yes, because if we take $z = x/(1 + x)$, then $0 < z < 1$ so that $z^n < z < x$ and hence $z \in A$.

Is this set bounded above? Yes, because if $u = 1 + x$, then $u > 1$ so that $x < u < u^n$ and hence $u \notin A$. But then of course $v \notin A$ for every $v > u$. In other words u is an upper bound of the set A . (Warning: After showing $u \notin A$ you can not conclude u is upper bound, who knows $u + 10$ may be in our set.)

Thus A has lub, name it s .

Can $s^n < x$? No, because then by (i) above we can get $(s + h)^n < x$, which means $s + h \in A$. But then $s < s + h \in A$, contradicting that s is an upper bound of A .

Can $s^n > x$? No, because then by (ii), there is $h > 0$ with $0 < s - h < s$ and $(s - h)^n > x$. This implies that $z < s - h$ for every $z \in A$. Thus $s - h$ is an upper bound of A contradicting that s is least upper bound.

There are only three possibilities: $s^n < x$, $s^n = x$, $s^n > x$. Since two of them are ruled out above, we must have $s^n = x$. completing the proof. Of course, the proof is complete provided we show the truth of the two statements we made at the beginning of the proof.

Let us prove (i). Let $z > 0$ and $z^n < x$. We need $h > 0$ so that $(z + h)^n < x$. Thought process: Let us make up our mind that we shall choose $0 < h < 1$.

Let us see what we want: want $(z + h)^n - z^n < x - z^n$

want $h[(z + h)^{n-1} + (z + h)^{n-2}z + \cdots + z^{n-1}] < x - z^n$.

This will be true if $hn(z + h)^{n-1} < x - z^n$.

This in turn will be true if we can choose $0 < h < 1$ so that $hn(z + 1)^{n-1} < x - z^n$.

We are done with rough calculations. We say now, choose

$$0 < h < 1; \quad \text{so that} \quad h < \frac{x - z^n}{n(z + 1)^{n-1}}.$$

Such a choice is possible, for example

$$h = \frac{1}{2} \min \left\{ 1, \frac{(x - z^n)}{n(z + 1)^{n-1}} \right\}.$$

You can convert the above rough calculation into a proof of the fact that, with this choice of h , we do have $(z + h)^n < x$.

Let us finally prove (ii). Let $z > 0$ and $z^n > x$. We need h , so that $0 < z - h < z$ and $(z - h)^n > x$. Again here is thought process.

We need $(z - h)^n - z^n > x - z^n$.

need, $z^n - (z - h)^n < z^n - x$.

need, $h[z^{n-1} + z^{n-2}(z - h) + \cdots + (z - h)^{n-1}] < z^n - x$.

This will be true if $0 < z - h < z$ and $hnz^{n-1} < z^n - x$.

Now rough calculation is over, we say choose

$$0 < h < z; \quad \text{and also} \quad h < \frac{z^n - x}{nz^{n-1}}.$$

As earlier, such a h can be chosen, for example, half of minimum of the two bounds we have above. The rough calculations above can be converted to showing that with this choice of h , (ii) is verified. ■

Many times we use mathematical induction. We should know what it is.

Mathematical induction:

Suppose for every $n \in N$ we have a mathematical statement P_n . Suppose we know the following: (i) P_1 is true. (ii) For every $k \geq 2$, If $P_1, P_2, \cdots, P_{k-1}$ are true then P_k is true.

Then P_n is true for every n .

You can replace (ii) by (ii) For every $k \geq 2$, If P_{k-1} is true then P_k is true.

Mathematical statement could be a formula, like

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

or a sentence like: ‘Every $n \geq 1$ is product of primes.’

Why should we believe this? Let A be the set of all n such that P_n is not true. If $A = \emptyset$, then we are done, each P_n is true. Suppose it is not empty. Then it must have a smallest element. Indeed, take any $m \in A$ and check one by one whether $1 \in A$, $2 \in A$ etc upto $m - 1 \in A$ (only finitely many to check). So let k be the smallest element of A . can $k = 1$? No because we know P_1 is true. So $k > 1$. Also k being smallest element of A , we know $P_1, \dots, P_{k-1}$ are true. But then by (ii) P_k must also be true. A contradiction to the assumption that $k \in A$.

Note that if you want to prove each P_n it is difficult unless you have one proof that works for every n . If you need to write proof of each P_n separately, then at least one sentence should be written to prove P_n , which means to prove all the P_n , you need to write infinitely many sentences; an impossible task.

We shall not spend too much time on cardinality, but do precisely understand a few things. We shall define number of elements $|A|$ for sets A .

we define $|\emptyset| = 0$, number of elements in emptyset is zero. We say that a set A has n elements if there is a bijection $f : A \rightarrow \{1, 2, 3, \dots, n\}$. What is a bijection?

if A and B are two sets we say $f : A \rightarrow B$ is a function if with each $a \in A$, f associates an element, to be denoted $f(a)$ from B . (It is possible to make precise the word ‘associates’ but now we shall not do.) The function f is said to be one-one or one to one if $a_1, a_2 \in A$ and $a_1 \neq a_2$ then $f(a_1) \neq f(a_2)$ in B . We say f is onto B if for every $b \in B$ there is an $a \in A$ so that $f(a) = b$. If function is both one-one and onto, then it is called a bijection. One-one functions are also called injections and onto functions are also called surjections. Thus bijections are both injective and surjective.

returning to number of elements in a set A , the definition makes precise what we do in practice: you pull $a \in A$ out and say one, then you pull another one out and say two and so on. That is how we count. A bijection

does precisely that.

Before proceeding further, we need to understand a tricky problem. Is the above definition good? Suppose I have a set A . Rhyin says he has a bijection of A with $\{1, 2, \dots, 15\}$ and Shreya says she has a bijection of A with $\{1, 2, \dots, 15, 16\}$. if both were correct then our definition is no good! We are sure this can not happen and one of them must be wrong. If they show us their functions, then we are sure to locate an error in one of them. But they refuse to show us their functions.
Question: How do you 'prove' one of them is surely wrong?

Cardinality/Number of elements:

Recall empty-set has zero number of elements, $|\emptyset| = 0$.

If A is a non-empty set, we say that A has n elements ($n \in \mathbb{N}$), in symbols $|A| = n$ if there is a bijection $f : \{1, 2, \dots, n\} \rightarrow A$.

We say that a set A is finite if $|A| = n$ for some $n = 0, 1, 2, \dots$. Otherwise, we say that the set A is infinite.

is 'number of elements' well defined?

If $|A| = 0$ then by definition A must be empty set. So no problems. We shall now show:

P_m : For any $n > m$ there is no bijection $\{1, 2, \dots, m\} \rightarrow \{1, 2, \dots, n\}$.

We prove it for $m = 1, 2, \dots$ by induction. We denote $A_m = \{1, 2, \dots, m\}$

$m = 1$. Take $n > 1$. Let f be any map of A_1 to A_n . If $f(1) \neq 1$, then 1 is not in the range of f . If $f(1) = 1$, then 2 is not in the range of f ; just remember $n \geq 2$ and $A_n = \{1, 2, \dots, n\}$. Thus f can not be onto.

done for $m = 1, 2, \dots, k-1$. Shall do for $m = k$. Let $n > k$ be fixed and if possible a function f on A_k onto A_n . We shall produce a function h on A_{k-1} onto A_{n-1} . But $n-1 > k-1$ and the statement P_{k-1} is true, leading to a contradiction.

Let $f(k) = a \in A_n$. We define h as follows. Take x with $1 \leq x \leq k-1$. If $f(x) < a$, then declare $h(x) = f(x)$; if $f(x) \geq a$ declare $h(x) = f(x) - 1$. Since value of f is at most n and $a \leq n$, It is clear that $h(x) \leq n-1$, so that h takes values in A_{n-1} . Is it onto A_{n-1} ? Yes, to see this, take y with $1 \leq y \leq n-1$. In case $y < a$, using the fact that f is onto A_n , we get x with $f(x) = y$. But then, $h(x) = y$ and of course $x \neq k$, so that $x \in A_{k-1}$. If $y \geq a$, then using the fact that $1 \leq y \leq n-1$ first conclude that $y+1 \leq n$ and hence there is a $x \in A_k$ with $f(x) = y+1$. Observe $x \neq k$ because $f(k) = a < y+1$. Thus $x \in A_{k-1}$ and $h(x) = y$. Thus whatever be $y \in A_{n-1}$ there is $x \in A_{k-1}$ with $h(x) = y$.

Finally, is h one-one? Suppose $h(x) = h(y) = j$. If $j < a$ then $h = f$ and f is one-one tells $x = y$. In case $j \geq a$, then $h = f - 1$ tells $x = y$.

This completes the proof. ■

We say that a set A is countably infinite if we can establish a bijection $f : \mathbb{N} \rightarrow A$. Thus you can count elements of A as $f(1), f(2), \dots$, but just that the

counting never ends, though sooner or later every element of A is counted.

We say A is countable if either it is finite or countably infinite. Otherwise, we say that A is uncountable.

Here are some facts: We shall not convert our Analysis course into a course on set theory. We just do enough for our use.

(i) Let $n \geq 1$ be an integer. Let $f : A \rightarrow B$ be a bijection, If $|A| = n$ then $|B| = n$. if $|B| = n$ then $|A| = n$.

This is because if $g : \{1, 2, \dots, n\} \rightarrow A$ is a bijection, then you can (please do) verify that $h(m) = f(g(m))$ is a bijection $h : \{1, 2, \dots, n\} \rightarrow B$, to show $|B| = n$. For the last sentence interchange the roles of A and B and observe that $f^{-1} : B \rightarrow A$ is well-defined and is bijection.

Many times in your later courses, you come across counting problems. Unfortunately, you cannot count directly! you setup a bijection with a set which you can count. So the simple fact above is indeed very useful.

(ii) Let $f : A \rightarrow B$ be a bijection,. If A is countably infinite, then B is also countably infinite. If B is countably infinite then so is A . Exactly the same proof as above works.

(iii) The set $N = \{1, 2, 3, \dots\}$ is countably infinite. Identity map would do, $f(m) = m$.

(iv) Any infinite subset of N is countably infinite. Let A be an infinite subset of N . Define, by induction, for each $n \in N$, an element $a_n \in A$ as follows. We use that every non-empty subset S of N has a minimum, that is, an element $s \in S$ such that $s \leq t$ for all $t \in S$. Note that such a minimum is indeed unique. define,

$$a_1 = \min A; \quad a_{n+1} = \min (A \cap \{a_1, \dots, a_n\}^c); \quad n > 1.$$

The set being infinite, it is non-empty and so a_1 is meaningful and also the above process does not stop at a finite stage. Let us observe

$$A = \{a_1, a_2, \dots\}$$

To see this, by definition each $a_i \in A$. Conversely, let $x \in A$. Then $B = \{y \in A : y \leq x\}$ is a finite subset of N because it is bounded above by x . It is not empty because $x \in B$. Hence, for some $m \geq 1$, there is a bijection

$g : \{1, 2, \dots, m\} \rightarrow B$. Clearly $x = a_m$.

Define

$$f : \{1, 2, \dots\} \rightarrow A; \quad f(n) = a_n.$$

This is one to one. Indeed, if $m < n$, then by definition of a_n we see that $a_n \neq a_m$. We have already observed above that if $x \in A$ then for some n , we have $x = a_n$. This shows that f is onto A .

(v) The set of integers, Z is countably infinite.

The set $B = \{2^n, n \geq 1\} \cup \{3^m, m \geq 1\} \cup \{5\}$ is countably infinite by above. Let $g : Z \rightarrow B$ be defined by $g(n) = 2^n$ if $n \geq 1$ and $g(m) = 3^{-m}$ if $m \leq -1$ and $g(0) = 5$. This gives required bijection. When the set is visibly infinite, we shall not specifically comment in the proof.

(vi) The set of pairs $S = \{(m, n) : m, n \in N\}$ is countable.

Put $f(m, n) = 2^m 3^n$ on S into N . This is bijection from S onto the range of f which is a sub set of N . Apply (iv).

In fact this shows that if A, B are countably infinite sets then the set of ordered pairs (x, y) with $x \in A$ and $y \in B$ is a countably infinite set.

(vii) The set of positive rational numbers is countably infinite.

Set of positive rational numbers is an infinite subset of the above by identifying a positive rational m/n where $m, n \geq 1$ and have no common factor with (m, n) .

(viii) The set of rational numbers is countable.

An argument used earlier: If f is a bijection $Q^+ \rightarrow N$, then we put on Q , $g(x) = 2^{f(x)}$ for $x \in Q^+$ and $g(x) = 3^{f(-x)}$ if $-x \in Q^+$ and $g(0) = 5$.

(ix) If for each $n = 1, 2, 3, \dots$ we have countable sets, then their union $\cup A_n$ is also countable set.

We can safely assume that $A_n \neq \emptyset$ for each n (why?). For each $n = 1, 2, \dots$, fix a function $f_n : A_n \rightarrow N$ a one-one function. We are not saying 'onto' because in case A_n is finite it would be impossible. Define a function $f : \cup A_n \rightarrow N$ as follows. Take $x \in \cup A_n$. Let i be the first integer such that $x \in A_i$. Put

$$f(x) = 2^i 3^{f_i(x)}.$$

This establishes a one-one function onto its range contained in N .

(x) The set of algebraic numbers is countable.

Recall algebraic number is a real number which is root of a polynomial with rational (or integer) coefficients. That is, a is algebraic number if there is a polynomial $P(z)$ with integer coefficients such that $P(a) = 0$. Since we are discussing real numbers we consider only real roots for now.

Fix an integer $n \geq 1$. If we show set A_n of algebraic numbers which are roots of n -th degree polynomials is a countable set then the set of algebraic numbers $\cup A_n$ would also be countable by above.

Let us fix $n \geq 1$. Consider one Polynomial $a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$ where each a_i is an integer. This polynomial has finitely many roots, in fact at most n . The set of such $(n+1)$ integer-tuples $(a_0, a_1, \dots, a_n)$ is countable set (why?) and so A_n is countable union of finite sets and hence countable.

(xi) The set of real numbers R is uncountable.

Clearly R is infinite. Let $f : N \rightarrow R$. We show that there is a number x which is not $f(n)$ for any $n \in N$. This will in particular show that there can not be any function on N with range all of R ; in particular, there can not be any bijection – showing that R is not countable. We start with an observation. We use the notation $[a, b] = \{x \in R : a \leq x \leq b\}$. In this proof whenever we use this interval then we ensure $a < b$ are real numbers.

Observation 1:

Given $x \in R$ and $[a, b]$ with $a < b$, there is $[c, d] \subset [a, b]$ with $c < d$ and $x \notin [c, d]$.

This is seen as follows. If $x \notin [a, b]$ then you can take $[c, d] = [a, b]$. If $x = a$, take $[\frac{a+b}{2}, b]$; if $x = b$ take $[a, \frac{a+b}{2}]$; finally if $a < x < b$ take $[\frac{x+b}{2}, b]$. Observe that $x < (x+b)/2$ so it is not in this interval.

observation 2:

If $[c, d] \subset [a, b]$ then $a \leq c \leq d \leq b$.

This is because $c \in [a, b]$ says $a \leq c$ and $d \in [a, b]$, says $d \leq b$. Of course by definition of interval $c \leq d$. Incidentally since in this proof we are considering only intervals with left point strictly smaller than right point, we actually have $c < d$. But in the general case $c \leq d$.

Returning to our problem, recall $f : N \rightarrow R$. Start with the interval $[0, 1] = [a, b]$. Apply claim to get $[a_1, b_1] \subset [a, b]$ such that $f(1) \notin [a_1, b_1]$ and $a_1 < b_1$. Then get $[a_2, b_2] \subset [a_1, b_1]$ such that $f(2) \notin [a_2, b_2]$ and $a_2 < b_2$. In general if you have obtained

$$[a, b] \supset [a_1, b_1] \supset [a_2, b_2] \supset \cdots \supset [a_n, b_n]$$

such that $f(k) \notin [a_k, b_k]$ for $k = 1, 2, \dots, n$, and $a_i < b_i$ for all $i \leq n$; then choose $[a_{n+1}, b_{n+1}] \subset [a_n, b_n]$ so that $f(n+1) \notin [a_{n+1}, b_{n+1}]$ and $a_{n+1} <$

b_{n+1} . Here we have repeatedly used observation 1. Now use observation 2 repeatedly to see

$$\begin{aligned} a &\leq a_1 \leq b_1 \leq b \\ a &\leq a_1 \leq a_2 \leq b_2 \leq b_1 \leq b \\ a &\leq a_1 \leq a_2 \leq a_3 \leq b_3 \leq b_2 \leq b_1 \leq b \end{aligned}$$

Thus

$$a \leq a_1 \leq a_2 \leq a_3 \leq \dots \leq \dots \leq b_3 \leq b_2 \leq b_1 \leq b$$

Let now $A = \{a_n : n \geq 1\}$, set consisting of all the a_n . This set is bounded above. In fact each b_k is an upper bound. Let $s = \sup A$. Then s being upper bound of A ; $a_n \leq s$ for all n ; further each b_n being an upper bound we also have $s \leq b_n$ for each n . Thus $s \in [a_n, b_n]$ for each n . In particular $s \neq f(n)$ for each $n = 1, 2, 3, \dots$.

This completes the proof. ■

In the proof above we see $s \in [a, b]$. We started with $[0, 1]$. But we could have started with any interval $[a, b]$ with $a < b$. Thus the above argument proves more.

(xi) Any interval $[a, b]$ with $a < b$ is uncountable.

(xii) The set of irrational numbers is uncountable. The set of irrational numbers in any interval $[a, b]$ with $a < b$, is uncountable.

This is because, Q is proved to be countable and if the above set is also countable then by an earlier fact their union, namely R (or $[a, b]$), will be countable; which is not the case.

(xiii) The set of transcendental numbers is uncountable. The set of transcendental numbers in any interval $[a, b]$ with $a < b$, is uncountable.

because the set of algebraic numbers is countable and above argument can be repeated.

Thus there are ‘more’ irrational numbers than rational numbers. Can we show one? yes. We can show many. For any prime p the number $\sqrt{p}$ is not rational. There should be no confusion because for us, at this stage, square root is the unique positive number whose square is the given number.

For example you know $\sqrt{2}$ is not a rational number. If you forgot here is a proof: If $\sqrt{2} = \frac{m}{n}$, where m, n have no common factors. Then $\frac{m^2}{n^2} = 2$ or $m^2 = 2n^2$. But then m^2 is even and so m must be even (show) and thus $m = 2k$, so $m^2 = 4k^2$. Hence $2k^2 = n^2$ showing that n must be even too. But

m, n have no common factor. This ‘proof by contradiction’ gives the result.

Did you see how we assumed several things. We assumed that a rational number x can be expressed as m/n where m, n have no common factors (earlier also we used this). This is because, by definition, $x = a/b$ where a, b are integers any common factor must be an integer k with $1 < k \leq \min\{a, b\}$, and there are only finitely many such integers in N . Take their maximum, say K . Let $m = a/K$ and $n = b/K$. Clearly $x = a/b = m/n$. If now, m, n have a common factor, say, $c \in N$ ($c > 1$), then $cK > K$ is a common factor of a, b contradicting that K is the maximum.

But $\sqrt{2}$ is algebraic; it is a root of $x^2 - 2 = 0$. We also showed there are more nonalgebraic (transcendental) numbers than algebraic numbers. Can we show one? Yes, but difficult to prove it is not algebraic. For example π (area of disc of radius one) and e (exponential, soon we define this).

Let me remind you that having square roots and cube roots (for positive numbers) is not our birth-right. Think of the set of rational numbers. It does not have a number whose square is 3. The set of real numbers R can provide a number whose square is 3, but is unable to provide a number whose square is (-3). Later, you will know that the set of complex numbers provides this also. Think about these matters *till* you are convinced that what we are doing in class **needs** to be done.

I would also like you to appreciate the subtle point in showing that **there is no one-to-one map on $\{1, 2, \dots, 11\}$ onto $\{1, 2, \dots, 13\}$** . You may start thinking: if I associate with 1, then with 2 etc, I run out of numbers in the domain but numbers remain in the range etc etc. This is fine and good thinking, but does not solve our problem. Your mind is already thinking that one set has 11 elements and another has 13 elements — but **this is precisely the point**: Does our definition of ‘number of elements’ exclude such a possibility whereby a set has 11 and also 13 elements?. It is important and necessary to feel what goes wrong, but you have to support the feeling with argument that no-one can refute. For example, someone says, I have such a function, but I will not show you. How do you convince him he is wrong? Even if he shows you his function and you point a mistake, that proves his function is wrong; tomorrow someone else may provide a bijection between these two sets. you want to rule out that.

I am calling your attention to tricky points in arguments so that you get a hang of **what is a proof** and how to write a proof.

You are likely to complain that I am being abstract. You have to keep two things separate: abstraction and clarity. We are not trying to be abstract. We want to be (i) clear about what we are saying and (ii) clear about why it is true. One important thing to keep in mind is that, in the middle of all this, we should

not lose track of the inherent beauty of ideas and arguments.

Exponentiation 1: $x \neq 0, n \in \mathbb{N}$

Definition: $x^1 = x$ and for $n \geq 1$, $x^{n+1} = x^n \cdot x = x \cdot x^n$

Here are some properties, known as laws of indices: In these statements it is understood that $m, n \in \mathbb{N}$ and $x \neq 0$.

1. $x^{m+n} = x^m x^n$
2. $(x^m)^n = x^{mn}$
3. $(xy)^n = x^n y^n$.
4. $x^n (1/x)^n = 1$ More generally for $m > n$, $x^m (1/x)^n = x^{m-n}$ and for $m < n$ $x^m (1/x)^n = (1/x)^{n-m}$.
5. If $0 < x < y$ then $x^n < y^n$.
6. Suppose $m < n$. Then $x^m > x^n$ if $0 < x < 1$ and $x^m < x^n$ if $x > 1$.

Here are the proofs:

(1) Fix any $m \geq 1$ We show by induction on n .

$$n = 1: x^{m+1} = x^m x \text{ (by definition)} = x^m x^1 \text{ (by definition of } x^1 \text{)}.$$

Done for $n = 1, 2, \dots, k$. Shall do for $n = k + 1$.

$$\begin{aligned} x^{m+(k+1)} &= x^{(m+k)+1} \text{ (property of addition)} = x^{m+k} x \text{ (definition)} \\ &= (x^m x^k) x \text{ (induction hypothesis)} = x^m (x^k x) \text{ (property of multiplication)} \\ &= x^m x^{k+1} \text{ (definition)}. \end{aligned}$$

(2) Again induction on n , fixing m .

$$n = 1: (x^m)^1 = x^m \text{ (definition)} = x^{m \times 1}.$$

Done $n \leq k$ Shall do for $n = k + 1$.

$$\begin{aligned} (x^m)^{k+1} &= (x^m)^k x^m \text{ Used } a^{k+1} = a^k a \text{ with } a = x^m. \\ &= x^{mk} x^m \text{ by induction hypothesis} = x^{mk+m} \text{ from part (i)} \\ &= x^{m(k+1)} \end{aligned}$$

(3) Induction. $n = 1$ follows because $(xy)^1 = xy$ by definition.

$$\begin{aligned} \text{Done for } n \leq k. \quad (xy)^{k+1} &= (xy)^k (xy) \text{ (by definition)} = (x^k y^k) xy \text{ induction hypothesis} \\ &= x^k x y^k y \text{ (multiplication commutative)} = x^{k+1} y^{k+1} \text{ by definition.} \end{aligned}$$

(4) First equality follows from above taking $y = 1/x$: $x^n (1/x)^n = (x \cdot 1/x)^n = 1^n = 1$. Of course, you might ask me why is $1^n = 1$. Prove by

induction on n again.

To prove the next part, suppose $m < n$.

$x^m(1/x)^n = x^m[(1/x)^m(1/x)^{n-m}]$ from part (1).

$= [x^m(1/x)^m](1/x)^{n-m}$ property of multiplication

$= 1(1/x)^{n-m}$ from what is seen just now $= (1/x)^{n-m}$.

You need to make sure, that you are talking of ‘positive’ exponents only because that is what we defined so far.

(5) and (6) are easy, partly observed while constructing n -th roots. ■

Exponentiation 2: $x \neq 0, n \in \mathbb{Z}$

Definition: $x^0 = 1$. If $n \geq 1$ then x^n is as above. If $n \leq -1$ define $x^n = (1/x)^{-n}$ where the right side is as defined above. Note that $n \leq -1$ implies $-n \geq 1$ and so $(1/x)^{-n}$ was defined above. In what follows $n \in \mathbb{Z}$ and $x \neq 0$

1. $x^n = (1/x)^{-n}$ and $x^n(1/x)^n = 1$.

More generally $x^m(1/x)^n = x^{m-n} = (1/x)^{n-m}$.

2. $x^{m+n} = x^m x^n$

3. $(x^m)^n = x^{mn}$

4. $(xy)^n = x^n y^n$.

5. If $0 < x < y$ then $x^n < y^n$ if $n \geq 1$ and $x^n > y^n$ if $n \leq -1$.

6. Suppose $m < n$. Then $x^m < x^n$ if $x > 1$; and $x^m > x^n$ if $0 < x < 1$.

Proofs: Throughout $x, y \neq 0$ and $n \in \mathbb{Z}$.

1. First equality: If $n = 0$ both sides are one. If $n < 0$, then definition of left side is right side. If $n > 0$, then definition of right side is left side if you observe $1/(1/x) = x$.

second equality: $n = 0$ both sides are one. $n \geq 1$, done in first exponentiation. For $n \leq -1$: $x^n(1/x)^n = (1/x)^{-n}x^{-n}$ by definition $= 1$ from the previous case applied to $1/x$.

Finally To prove $x^m(1/x)^n = x^{m-n} = (1/x)^{n-m}$. Observe that the second equality here is obvious because if $m - n < 0$ then this is definition of x^{m-n} , if $m - n > 0$ this is definition of $(1/x)^{n-m}$, while if $m - n = 0$ both sides are one. So it is enough to show that $x^m(1/x)^n$ equals one of the other two quantities.

If $m = 0$ or $n = 0$ or both equals zero, then the required equality is clear.

If both $m, n \geq 1$ then $x^m(1/x)^n$ equals x^{m-n} when $m \geq n$ and equals $(1/x)^{n-m}$ when $n \geq m$ (using exponentiation 1) and hence done.

If both $m, n \leq -1$ then by definition $x^m(1/x)^n$ equals $(1/x)^{-m}(x)^{-n}$ and since both $-m, -n$ are non-negative this last expression equals x^{-n+m} from

the case already done.

If $m \leq -1$ and $n \geq 1$ then by definition $x^m(1/x)^n$ equals $(1/x)^{-m}(1/x)^n = (1/x)^{n-m}$ from exponentiation 1; note both $n, -m$ are now at least one.

The case If $n \leq -1$ and $m \geq 1$ is similar.

(2) The case $m = 0$ and/or $n = 0$ is clear. If $m, n \geq 1$ this is exponentiation 1. If $m, n \leq -1$ then

$$x^m x^n = (1/x)^{-m} (1/x)^{-n} = (1/x)^{-n-m} = x^{m+n}$$

where first equality is definition of x^m, x^n ; second is exponentiation 1 because both $-m, -n$ are at least one; third equality is part (1) above. If $m \leq -1, n \geq 1$ then again from definition and part (1) above we see

$$x^m x^n = (1/x)^{-m} x^n = x^{n+m}$$

similar argument applies for $n \leq -1, m \geq 1$

Try your hands on the remaining parts to see if you have understood so far. ■

sequences and convergence:

All the numbers we know are rationals, of the form m/n from high school. We know others like $\sqrt{2}$ and π by geometrical constructions; triangles and circles. (We only pretend to understand these, if some one asks what is the 99th decimal place of $\sqrt{2}$ we can not really tell). Since real numbers are uncountable, what we know are very few. How do you discover new numbers? In real life, what you do not know, you would try to describe in terms of things you know. If you have a relative dropping by and you ask your parents who he/she is you are told: my brother's cousin's son's etc via a sequence of known relatives leading to the unknown. Or if you ask how to go to a particular place, someone may say: go to x and take a right, reach y then you take left, reach z etc, through a sequence of visible land marks leading to your destination. Of course in practice, this sequence must stop after a finite number of terms. We do exactly the same thing to describe unknown number, describe through a sequence of known numbers – the only difference is that this sequence is now unending; gets closer and closer to the number we have in mind.

Of course, there are other reasons too. you might want to add infinitely many numbers, which, of course, you can never do. So you add more and

more terms and see if you are getting close to something. Remember sum of infinite geometric series with common ratio $0 < r < 1$.

Or you might be trying to calculate area under a curve through a sequence of approximations via rectangles – because Area of rectangle is familiar from school. Remember Archimedes and area under the curve $y = x^2$ between $x = 0$ and $x = 1$

Or you might be trying to understand a phenomenon and our maths is unable to do exact calculation and give final answer. So you do some simulations on computer. You end up with different numbers in repeated simulations and you want to understand what exactly is the true answer? to which number are your simulations close to?

Or When you do advanced math and calculate some quantities, then the number you usually end up with is a complicated number. Unless you understand what that number is, you have not really achieved any real understanding of the phenomenon you are studying. And you understand that answer via known numbers, by producing a sequence of known numbers which are getting closer and closer to that complicated number..

definition: A sequence of real numbers is a function $f : N \rightarrow R$. The phrase sequence of real numbers is abbreviated to ‘sequence’ now.

Usually we think of a sequence by listing all values of the function in a line $f(1), f(2), f(3) \dots$, usually written $\{x_1, x_2, x_3, \dots\}$ or simply $\{x_n : n \geq 1\}$ or $\{x_n\}$. The number $f(n) = x_n$ is also called the n -th term of the sequence. Such a listing is sometimes easily understandable for the brain rather than a complicated formula giving f . Since N is infinite and we can not list infinitely many numbers we put dots; but you should NOT accept unless you know what is there in the dots; in other words whenever you are talking of a sequence and if the function is not presented before you, stop and ask, do I know what is the 101-st term? If you can not figure out stop and understand the sequence before proceeding.

Examples:

1. $3, 3, 3, 3, \dots$ Here $f(n) = 3 \quad \forall n$.
2. $5, -5, 5, -5, 5, -5, \dots$ Here $f(n) = 5$ for n odd; and -5 otherwise.
3. $1, 1/2, 1/3, 1/4, \dots$ Here $f(n) = 1/n \quad \forall n$
4. $1, -1/2, +1/3, -1/4, \dots$ Here $f(n) = 1/n$ for n odd, otherwise $-1/n$.
5. $f(n)$ is n -th prime. Thus the sequence is the list of primes. Here I should

not write $3, 5, 7, \dots$. There is a confusion. You may think it is sequence of odd numbers starting with 3. Also there is no formula here. But yet it is expressed as $\{p_1, p_2, p_3, \dots\}$ where p_n is the n -th prime. Thus it is not necessary to give a formula f , but the prescription should be clear as to what $f(n)$ is.

You can makeup your own sequences, plenty of them.

You can feel that the third sequence above is getting close to zero. When shall we say a sequence is getting closer and closer to some number a ? Intuitively speaking, no matter what small error you prescribe, after some stage all numbers of the sequence should be a ‘upto’ that error bound. This is made precise by the following.

Definition: We say that a sequence (x_n) converges to a number a if $(\forall \epsilon > 0) (\exists N) (\forall n > N) (a - \epsilon \leq x_n \leq a + \epsilon)$.

Thus no matter what $\epsilon > 0$ is given there must be an N so that for all $n > N$ we must have x_n in the interval $[a - \epsilon, a + \epsilon]$. The definition is ‘robust’. Can we replace $\leq$ above by $<$? Yes

$$(\forall \epsilon > 0) (\exists N) (\forall n > N) (a - \epsilon < x_n < a + \epsilon) \quad (*)$$

If the definition holds, and if $\epsilon > 0$ is given to you; use the definition with $\epsilon/2$. Thus after some stage we have $a - (\epsilon/2) \leq x_n \leq a + (\epsilon/2)$ which implies $(*)$. Conversely if $(*)$ holds then definition also holds.

Can we replace $n > N$ by $n \geq N$?

$$(\forall \epsilon > 0) (\exists N) (\forall n \geq N) (a - \epsilon \leq x_n \leq a + \epsilon) \quad (\dagger)$$

Yes. If you got N satisfying the definition then $N + 1$ satisfies $(\dagger)$. Of course if $(\dagger)$ holds then definition holds for the same N .

Observe $a - \epsilon < x < a + \epsilon$ is same as $-\epsilon < x - a < +\epsilon$ which, in turn, same as $|x - a| < \epsilon$. Thus we could also have said

$$(x_n \rightarrow a) \text{ means } (\forall \epsilon > 0) (\exists N) (\forall n > N) |x_n - a| < \epsilon$$

When (x_n) converges to a we write $x_n \rightarrow a$ instead of $\{x_n\} \rightarrow a$.

Warning: It is customary to use $\mathbb{N}$ for the set of natural numbers, but I just used N . In the above definition I used N for the existence of one natural number so that blah blah holds. You should not get confused. I do not want to change my notation now.

Example: $\{1/n\}$ converges to zero. That is $\frac{1}{n} \rightarrow 0$.

By Archimedean property, given $\epsilon > 0$, there is N such that $N > 1/\epsilon$, that

is, $1/N < \epsilon$. Then for any $n > N$ we have $1/n < 1/N < \epsilon$ and of course $1/n > 0 > -\epsilon$. Thus for all $n > N$ we have $0 - \epsilon < 1/n < 0 + \epsilon$ completing the proof. ■

Example:

$$1; 1 + 1; 1 + 1 + \frac{1}{2!}; 1 + 1 + \frac{1}{2!} + \frac{1}{3!}; \dots$$

$$x_1 = 1; \quad x_n = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{(n-1)!}; \quad n \geq 2$$

(x_n) converges. This is because firstly it is an increasing sequence, that is, $x_n \leq x_{n+1}$ for all n ; and secondly it is bounded above, that is, there is a number M such that $x_n \leq M$ for all n . Any such sequence converges by a theorem we show soon. The first property is clear from definition because x_{n+1} is obtained by adding one more positive term for x_n . Second property follows from

$$\begin{aligned} x_n &= 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{(n-1)!} \\ &\leq 1 + 1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^{n-2}} \leq 3 \end{aligned}$$

where we used

$$\frac{1}{k!} = \frac{1}{1 \cdot 2 \cdot 3 \dots k} \leq \frac{1}{2^{k-1}}$$

Is the limit of this sequence (x_n) a known number? No. We shall soon show that it is not rational. It is not even algebraic, but this fact we shall not show. The limit is denoted by e , for Euler. ■

Example:

$$x_n = \left(1 + \frac{1}{n}\right)^n$$

By binomial theorem

$$\begin{aligned} x_n &= 1 + \binom{n}{1} \frac{1}{n} + \binom{n}{2} \frac{1}{n^2} + \dots + \binom{n}{n} \frac{1}{n^n} \\ &= 1 + 1 + \frac{1}{2!} \left(1 - \frac{1}{n}\right) + \frac{1}{3!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) + \dots \\ &\quad + \frac{1}{n!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{n-1}{n}\right) \end{aligned}$$

Similar expansion shows that x_{n+1} is also such a sum each summand not less than the correspond summand of x_n ; and further it has one more extra positive term. Thus $x_n \leq x_{n+1}$. Observe from the above expansion,

$$x_n \leq 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!} \leq 3$$

where the last inequality is from previous example. So have we discovered another new number? No! We shall soon show that this limit is also e . ■

First let us observe some simple properties of limits.

Is convergence a good definition? Can $x_n \rightarrow a$ and $x_n \rightarrow b$ and $a \neq b$?

Theorem: $x_n \rightarrow a$ and $x_n \rightarrow b$ implies $a = b$.

Proof: If $a < b$ then let $\epsilon = (b - a)/2$. Using $x_n \rightarrow a$, we see after some stage, say, for $n > N_1$; $x_n < a + \epsilon = a + \frac{b-a}{2} = \frac{b+a}{2}$. Using $x_n \rightarrow b$, we see after some stage, say, for $n > N_2$; $x_n > b - \epsilon = b - \frac{b-a}{2} = \frac{b+a}{2}$. But this is a contradiction because if $k = N_1 + N_2$ then $x_k < (b + a)/2$ and also $x_k > (b + a)/2$.

If $b < a$, similar proof applies. ■

Theorem: Every increasing sequence bounded above converges to its supremum. Every decreasing sequence bounded below converges to its infimum.

Proof: Let (x_n) be increasing and bounded above. So the set consisting all numbers x_n is bounded above and hence has a supremum, say, s . To show $x_n \rightarrow s$, fix $\epsilon > 0$. Need to show N so that for $n > N$ the inequality $s - \epsilon < x_n < s + \epsilon$ holds. The rightside inequality anyway holds for all n because s is an upper bound. We show the leftside inequality. Because $s - \epsilon$ is not an upper bound (supremum is the smallest upper bound) for the sequence, we have an N such that $x_N > s - \epsilon$. But then the sequence being increasing, we conclude that for every $n > N$ also $x_n \geq x_N > s - \epsilon$. This completes proof of convergence.

The statement about decreasing sequences is similarly proved. ■

is convergence friendly with addition and additive inverse? Yes.

Theorem: $x_n \rightarrow a$ and $y_n \rightarrow b$ then $x_n + y_n \rightarrow a + b$.

Also if $x_n \rightarrow a$ **then** $-x_n \rightarrow -a$.

Note that here $(x_n + y_n)$ is the sequence whose n -th term is $x_n + y_n$. To prove the first statement, fix $\epsilon > 0$. Need to show N such that for $n > N$ we have $|(x_n + y_n) - (a + b)| < \epsilon$. Use $x_n \rightarrow a$ with $\epsilon/2$ to get N_1 so that for $n > N_1$, $|x_n - a| < \epsilon/2$. Use $y_n \rightarrow b$ with $\epsilon/2$ to get N_2 so that for $n > N_2$, $|y_n - b| < \epsilon/2$. Let $N = N_1 + N_2$. Then for $n > N$ we have

$$|(x_n + y_n) - (a + b)| = |(x_n - a) + (y_n - b)| \leq |x_n - a| + |y_n - b|$$

$$< (\epsilon/2) + (\epsilon/2) = \epsilon.$$

Second sentence is immediate because $|(-x_n) - (-a)| = |x_n - a|$. ■

Is convergence friendly with multiplication and multiplicative inverse? Yes. First let us observe a useful result.

Theorem: If $x_n \rightarrow a$ then (x_n) is bounded, that is, there is a number M such that $|x_n| \leq M$ for all n .

Proof: Take $\epsilon = 1$ and get N so that for $n > N$ we have $|x_n - a| < 1$. Let

$$M = \max\{|x_1|, |x_2|, \dots, |x_N|, |a| + 1\}$$

We claim this will do. Firstly, the set above being a finite set, it has a maximum. Also, by definition of M we see $|x_n| \leq M$ for $n \leq N$. If $n > N$ then we have $|x_n| = |a + x_n - a| \leq |a| + |x_n - a| \leq |a| + 1 \leq M$. ■

Theorem: $x_n \rightarrow a$ and $y_n \rightarrow b$ then $x_n y_n \rightarrow ab$.

Also if $x_n \rightarrow a$ and $a \neq 0$ then after some stage $x_n \neq 0$ and if you consider the sequence $(1/x_n)$ from then on then $(1/x_n) \rightarrow 1/a$.

To prove the first sentence, Let $|x_n| \leq M$ for all n , see previous theorem. Let $\epsilon > 0$ be given. First observe,

$$\begin{aligned} |x_n y_n - ab| &= |x_n y_n - x_n b + x_n b - ab| \leq |x_n y_n - x_n b| + |x_n b - ab| \\ &\leq |x_n| |y_n - b| + |b| |x_n - a| \leq M |y_n - b| + |b| |x_n - a| \quad (\spadesuit) \end{aligned}$$

Using $y_n \rightarrow b$ taking $\frac{\epsilon}{2(M+1)}$ get N_1 so that $|y_n - b| < \frac{\epsilon}{2(M+1)}$ for $n > N_1$. Similarly, using $x_n \rightarrow a$ and using $\frac{\epsilon}{2(|b|+1)}$ get N_2 so that $|x_n - a| < \frac{\epsilon}{2(|b|+1)}$ for $n > N_2$. Now for $n > N_1 + N_2$, we continue with $(\spadesuit)$ to see

$$|x_n y_n - ab| \leq M \frac{\epsilon}{2(M+1)} + |b| \frac{\epsilon}{2(|b|+1)} \leq \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

To prove the second sentence, taking $\epsilon = |a|/2$ see that for $n > N$, say,

$$a - \frac{|a|}{2} < x_n < a + \frac{|a|}{2}.$$

In case $a > 0$ this says $x_n > a - (a/2) = a/2$ so that $|x_n| > a/2$. In case $a < 0$ this says $x_n < a + (-a/2) = a/2$. That is, x_n is negative and $x_n < a/2$. Thus again $|x_n| > |a|/2$. Thus in any case

$$n > N \Rightarrow |x_n| > \frac{|a|}{2} > 0$$

Note that the above implies that for $n > N$

$$\frac{1}{|a||x_n|} < \frac{1}{|a|} \frac{2}{|a|} = M \text{ say}$$

Let us now consider the sequence after N onwards. That is $y_n = X_{n+N}$ for $n \geq 1$. But instead of using a new notation, we use x_n itself but keep in mind that $n > N$.

$$\left| \frac{1}{x_n} - \frac{1}{a} \right| = \frac{|x_n - a|}{|a| |x_n|} \leq M|x_n - a|$$

Given $\epsilon > 0$, let us choose $N_1 > N$ so that $|x_n - a| < \epsilon/M$ for $n > N_1$. Thus for $n > N_1$ we have

$$\left| \frac{1}{x_n} - \frac{1}{a} \right| < \epsilon. \quad \blacksquare$$

Is convergence friendly with order $<$ and $\leq$. Yes with $\leq$ and $\geq$ it is friendly and not friendly with $<$.

[I have deleted cartoons; apparently, they are copyrighted]

We write $x_n \rightarrow a$ as $\lim x_n = a$ or $\lim_n x_n = a$; read limit of x_n equals a .

Theorem: Convergence respects $\leq$; and also $\geq$; but does not respect $<$, and does not respect $>$.

That is, let $x_n \rightarrow a$ and $y_n \rightarrow b$. If $x_n \leq y_n$ for all n then $a \leq b$. However if $x_n < y_n$ for all n , it is not necessary that $a < b$.

In fact if $x_n \leq y_n$ for all $n \geq 1000000$ then you can say $a \leq b$.

For the second sentence consider $x_n = 0$ for all n and $y_n = 1/n$ then $x_n < y_n$ for all n but $\lim x_n < \lim y_n$ is False, both limits are zero.

To prove the first sentence, first observe that $z_n \rightarrow z$ and $z_n \geq 0$ for all n then $z \geq 0$. This is because if you take any number $a < 0$ then the interval $(3a/2, a/2)$ does not contain any non-negative number and so does not contain any z_n , so z_n can not converge to a . (The interval exhibited is nothing but $(a - \epsilon, a + \epsilon)$ where $\epsilon = -a/2 > 0$). So $z \geq 0$.

Finally to prove the first sentence, observe, $y_n - x_n \geq 0$ for all n and $y_n - x_n \rightarrow b - a$ so that $b - a \geq 0$, that is, $a \leq b$. ■

Theorem: $\lim(1 + \frac{1}{n})^n = e$

Recall $e = \lim x_n$ where $x_n = \sum_{k=0}^n \frac{1}{k!}$. Take $0! = 1$, such a notation helps writing x_n using Σ ; without such an agreement, you need to write longer expression.

Proof: Let $y_n = (1 + \frac{1}{n})^n$ Let $\lim y_n = f$. We already saw $y_n \leq x_n$. Indeed

$$\begin{aligned} y_n &= 1 + 1 + \frac{1}{2!}(1 - \frac{1}{n}) + \frac{1}{3!}(1 - \frac{1}{n})(1 - \frac{2}{n}) + \cdots \\ &\quad + \frac{1}{n!}(1 - \frac{1}{n})(1 - \frac{2}{n}) \cdots (1 - \frac{n-1}{n}) \\ &\leq 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!} = x_n \end{aligned}$$

(term by term inequality). Since both sequences converge we conclude $f = \lim y_n \leq \lim x_n = e$. We now show, $e \leq f$.

If this is done you can conclude $e = f$. This is one of the standard arguments we use without even mentioning. This is fine because, $a \leq b$ rules out the possibility of $a > b$ and $b \leq a$ rules out the possibility of $b > a$ and hence $a = b$ — remember exactly one holds: $a > b$, $b > a$, $a = b$.

Returning to the proof, we need to show $\lim x_n \leq f$. For this enough to show

that for each k , $x_k \leq f$. So fix a k . Having fixed k , take any $n > k$. Then

$$1 + 1 + \frac{1}{2!}\left(1 - \frac{1}{n}\right) + \frac{1}{3!}\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right) + \cdots + \frac{1}{k!}\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{n - (k - 1)}{n}\right) \leq y_n, \quad (\spadesuit)$$

because right side has all the terms as on the left side and some more non-negative terms, (see definition of y_n) till

$$\frac{1}{n!}\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{n - 1}{n}\right)$$

Now take limits on both sides of $(\spadesuit)$. On left side each factor $(1 - \frac{i}{n}) \rightarrow 1$ and products of such terms converge to 1; and sum (only $k + 1$ terms) converges to sum. Thus

$$1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{k!} \leq \lim y_n = f; \quad x_k \leq f$$

This completes the proof. ■

Theorem: $x_n \leq y_n \leq z_n$ and $\lim x_n = a$; $\lim y_n = b$; $\lim z_n = c$ then $a \leq b \leq c$. ■

This follows from earlier result. A special case arises when $a = c$. Then you need not assume that $\{y_n\}$ converges. You can show $\{y_n\}$ converges and converges to a . This is sometimes referred to as **sandwich principle** or **squeezing principle**.

Theorem: $x_n \leq y_n \leq z_n$ and $\lim x_n = a = \lim z_n$. Then $\lim y_n$ exists and equals a .

Proof is straight verification of the definition. So fix $\epsilon > 0$. By hypothesis there is N_1 such that $x_n \in (a - \epsilon, a + \epsilon)$ for all $n \geq N_1$. Also there is N_2 such that $z_n \in (a - \epsilon, a + \epsilon)$ for all $n \geq N_2$. If you take $n > N_1 + N_2$ then both these statements hold and so for $n > N_1 + N_2$ we have y_n which is between x_n, z_n is also in the interval. ■

Theorem: Fix number $0 < a < 1$. The sequence $\{a^n\}$ converges to zero. In fact $n^k a^n \rightarrow 0$ for any integer $k \geq 1$. In particular $n^{35} a^n, n^{2000} a^n$ converge to zero. In particular, $\lim n^{10000} / 2^n = 0$.

Proof: So I need to show that a^n is getting smaller and smaller. Let $1/a = 1 + h$ where $h > 0$. Remember, $(1 + h)^n \geq nh$, by binomial expansion, so that

$$a^n = \frac{1}{(1 + h)^n} \leq \frac{1}{n h}.$$

Given $\epsilon > 0$ we can choose N so that $1/n < \epsilon h$ for all $n \geq N$. For this N we see $a^n < \epsilon$ for all $n \geq N$.

Sometimes a careful observation of the proof gives better results. Use exactly same steps as above. But say $(1+h)^n \geq \binom{n}{2}h^2$ for $n \geq 2$. Thus for $n \geq 2$

$$na^n = n \frac{1}{(1+h)^n} \leq \frac{n}{\binom{n}{2}h^2} = \frac{1}{n-1} \frac{2}{h^2}$$

Given $\epsilon > 0$, choose $N \geq 2$ so that $1/(N-1) < \epsilon h^2/2$. this will do.

You can show now that $n^{1000}a^n \rightarrow 0$. ■

You may wonder why we took the above route. Firstly, you need to get an idea of how small/large is a^n to get the conclusion required. You can write $a = 1 - h$ because $0 < a < 1$. But unfortunately binomial expansion has negative and positive terms and estimation becomes messy. Instead of showing a^n is small, I decided to understand if $1/a^n$ is large. That is all.

Theorem: If $a > 0$, then the sequence $(\sqrt[n]{a})$ converges to 1.

If $a = 1$, there is nothing to prove. Let $a > 1$. Since $\sqrt[n]{a} > 1$, let $\sqrt[n]{a} = 1 + h_n$ where $h_n > 0$. Thus

$$a = (1 + h_n)^n \geq nh_n; \quad h_n \leq \frac{a}{n} \rightarrow 0.$$

Thus $\sqrt[n]{a} = 1 + h_n \rightarrow 1$. We have used the simple fact that $1 + h_n \rightarrow 1$ using that $h_n \rightarrow 0$.

If $0 < a < 1$, then use $(1/a) > 1$ and so $\sqrt[n]{1/a} \rightarrow 1$. Observe

$$(\sqrt[n]{a} \sqrt[n]{b})^n = (\sqrt[n]{a})^n (\sqrt[n]{b})^n = ab$$

First equality is by exponentiation 1 and second is by definition of n -th root. Thus, by definition of n -th root, $\sqrt[n]{ab} = \sqrt[n]{a} \sqrt[n]{b}$. In particular, taking $b = 1/a$,

$$\sqrt[n]{a} = \frac{1}{\sqrt[n]{1/a}} \rightarrow \frac{1}{1} = 1$$

We are now using rules and theorems without mentioning. Instead of reading blindly, convince yourself as to which theorem is applied at each equality. ■

Again you might wonder what made me write $\sqrt[n]{a} = 1 + h_n$. Let us think. I want to show $\sqrt[n]{a}$ converges to one. Just see the definition. I need to convince you that the difference $\sqrt[n]{a} - 1$ is getting close to zero, so I named it as h_n and want to estimate how small/large it is. You must remember n -th

root is existential statement, we have no formula; as long as you hang on to n -th root no further simplification is in sight. But n -th power is calculable. So the equation $\sqrt[n]{a} = 1 + h_n$ is written as $a = (1 + h_n)^n$ etc.

Theorem: $\sqrt[n]{n} \rightarrow 1$.

There is no new idea. This follows from careful observation of the earlier proof. Let, as above, $\sqrt[n]{n} = 1 + h_n$ where $h_n > 0$. Thus

$$n = (1 + h_n)^n \geq \binom{n}{2} h_n^2; \quad h_n^2 \leq \frac{n}{\binom{n}{2}} = \frac{2}{n-1} \rightarrow 0.$$

Thus $h_n^2 \rightarrow 0$. From this we can deduce that $h_n \rightarrow 0$ as follows. Let $\epsilon > 0$. Choose n_0 so that $h_n^2 \leq \epsilon^2$ for $n \geq n_0$. Clearly then, since we already know that $h_n > 0$, we conclude $0 < h_n < \epsilon$ for $n \geq n_0$. ■

Theorem: Given any two numbers $a < b$, there is a rational number x such that $a < x < b$. In fact there are infinitely many rational numbers.

Same statement holds with rational replaced by irrational.

Proof: Since $b - a > 0$ fix, by Archimedean property an integer $n \in \mathbb{N}$ such that $n > \frac{1}{b-a}$ or $\frac{1}{n} < b - a$. There is at least one integer k such that $k > na$ (apply Archimedean property with $x = na$, if $a < 0$, then $k = 0$ would do). Let us take least such integer m . Thus $m > na$ but $(m - 1) \leq na$. Then $m > na$ tells $(m/n) > a$. Further, $(m - 1) \leq na$ tells

$$\frac{m}{n} = \frac{m-1}{n} + \frac{1}{n} < \frac{na}{n} + (b-a) = b$$

This completes the proof.

To get infinitely many just consider the numbers

$$a, \frac{a+b}{2}, \frac{a+2b}{3}, \frac{a+3b}{4} \dots$$

These numbers are strictly increasing and each less than b . Pick one rational between each of them.

To get irrational number Apply the earlier fact to the numbers $a/\sqrt{2}$ and $b/\sqrt{2}$. Get rational x so that $a < x\sqrt{2} < b$. This $x\sqrt{2}$ is irrational, proving the result IF $x \neq 0$. Why? if it is rational, say y , then $\sqrt{2} = y/x$, a rational number too, which is not the case. But if $x = 0$, the proof is still incomplete because the number we exhibited is $x\sqrt{2} = 0$. But if $x = 0$, then instead of a, b , consider $0, b$ and get, using the earlier argument an irrational number x , such that $0 < x < b$ and this completes the proof. ■

Why is the above interesting? Because Every real number can be explained

using rationals, the high school fractions. Thus all unknown numbers can be explained using known numbers. In a sense, you know this already. Decimal expansion does precisely that.

Theorem: Given any number x , we can get a sequence of rationals $r_n \uparrow x$ and we can get a sequence of rationals $s_n \downarrow x$.

Proof: Consider the intervals

$$(x-1, x-\frac{1}{2}), (x-\frac{1}{2}, x-\frac{1}{3}), (x-\frac{1}{3}, x-\frac{1}{4}) \dots$$

Take r_k from k -th interval. They are increasing and $0 < x - r_k \leq 1/k$.
For decreasing sequence, consider

$$(x+\frac{1}{2}, x+1), (x+\frac{1}{3}, x+\frac{1}{2}), (x+\frac{1}{4}, x+\frac{1}{3}) \dots \blacksquare$$

Exponentiation 3: $x^{1/n}$ for $x > 0$, $n \in \mathbb{N}$

DEF: $x^{1/n}$ is the unique n -th root (positive) of x which we showed earlier.

1. $0 < x < 1$ implies $x^{1/n} < 1$ and $x > 1$ implies $x^{1/n} > 1$
2. $m < n$: If $0 < x < 1$ then $x^{1/m} < x^{1/n}$. If $x > 1$ then $x^{1/m} > x^{1/n}$.
3. $0 < x < y$ implies $x^{1/n} < y^{1/n}$
4. $(x^{1/m})^{1/n} = x^{1/mn}$
5. $(xy)^{1/n} = x^{1/n}y^{1/n}$
6. $(x^{1/m})^n = (x^n)^{1/m}$

We have actually proved some of them. But here again.

1. If $x^{1/n} = 1$ then $x = (x^{1/n})^n = 1$.
If $x^{1/n} > 1$ then $(x^{1/n})^n > 1$, this you can see by induction using axiom. (For example: $a > 1$ implies $(a-1) > 0$. If further $b > 0$ then $(a-1)b > 0$, that is $ab - b > 0$ so $ab > b$ but $b > 1$ so $ab > 1$. See how we used order $<$ is friendly with addition/multiplication and is transitive (?).
Similarly if $x^{1/n} < 1$ then $x < 1$.
It appears, superficially, that we promised ‘ A implies B ’ but we proved ‘ B implies A ’. Please convince yourself.

2. Let $0 < x < 1$. Want to show $x^{1/m}$ increases as m increases. Enough to show that for any m ,

$$x^{1/m} < x^{1/(m+1)} \quad \text{or} \quad [x^{1/m}]^{m+1} < [x^{1/(m+1)}]^{m+1} \quad (\clubsuit)$$

Then induction takes over to show for every $k \geq 1$; $x^{1/m} < x^{1/(m+k)}$ and you can take $k = n - m$.

$$[x^{1/m}]^{m+1} = [x^{1/m}]^m [x^{1/m}]^1 = x x^{1/m} < x = [x^{1/(m+1)}]^{m+1}$$

first equality is exponentiation 1; second is definition of m -th root; inequality is from noting $x^{1/m} < 1$. This proves (♣)

Do the case $x > 1$ in exactly same way.

3. If $x^{1/n} = y^{1/n}$ then $(x^{1/n})^n = (y^{1/n})^n$, that is $x = y$. If $x^{1/n} > y^{1/n}$ then $(x^{1/n})^n > (y^{1/n})^n$, that is $x > y$.

4. To show left side equals $(x^{1/mn})$ we show mn -th power of left side equals x . We use exponentiation 1. Let $a = x^{1/m}$

$$[(x^{1/m})^{1/n}]^{mn} = [a^{1/n}]^{nm} = ([a^{1/n}]^n)^m = a^m$$

5. We show right side raised to n -th power equals xy so that right side equals $(xy)^{1/n}$.

$$(x^{1/n}y^{1/n})^n = (x^{1/n})^n (y^{1/n})^n = xy$$

6. Shall show left side raised to m th power equals x^n , so that left side equals $(x^n)^{1/m}$. We use exponentiation 1.

$$[(x^{1/m})^n]^m = [(x^{1/m})^{nm}] = [(x^{1/m})^m]^n = [x]^n = x^n$$

Cauchy sequences:

How do we know if a given sequence (x_n) converges? One way is to take each real number x and see if $x_n \rightarrow x$. If the answer is yes for some number x , then stop and say that the sequence converges. If the answer is no for every x , say that the sequence does not converge. But it is difficult to test every x . Moreover, in this procedure we are looking outside the sequence.

Is there a way to tell whether the sequence converges, by just looking at the terms of the sequence. Yes. Here it is. If a sequence converges, then the terms are getting close to something. And hence, they must be getting close among themselves also. The interesting point is that if the terms of a sequence are getting closer among themselves, then the terms are actually getting close to something. We now show this.

Definition: A sequence (x_n) is Cauchy sequence if given any $\epsilon > 0$, there is N such that for all $m, n \geq N$ we have $|x_m - x_n| < \epsilon$.

$$(\forall \epsilon > 0) (\exists N) (\forall m, n \geq N) |x_m - x_n| < \epsilon$$

Thus any two terms of the sequence after N differ by at most ϵ .

Theorem: A sequence (x_n) converges iff it is Cauchy.

Proof: Let $x_n \rightarrow x$. Let $\epsilon > 0$. Get N so that $|x_n - x| < \epsilon/2$ for $n \geq N$. Clearly, if $m, n \geq N$ we have

$$|x_m - x_n| \leq |x_m - x| + |x - x_n| \leq \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Conversely, suppose that the sequence satisfies the given condition. We exhibit a number x and show that $x_n \rightarrow x$.

Step1: First we show that a Cauchy sequence is bounded. Pick N so that for $m, n \geq N$ we have $|x_m - x_n| < 1$. In particular for all $n \geq N$ we have $|x_n - x_N| < 1$ so that $|x_n| \leq |x_N| + 1$. Now take

$$C = \max\{|x_1|, |x_2|, \dots, |x_N|, |x_N| + 1\}$$

Easy to see that for ALL n we have $|x_n| \leq C$. Did you see there is no new idea here. It is the same idea used in proving that a convergence sequence is bounded.

Step 2: Put

$$a_n = \inf\{x_n, x_{n+1}, x_{n+2}, \dots\}; \quad b_n = \sup\{x_n, x_{n+1}, x_{n+2}, \dots\}$$

Thus $a_n \leq b_n$ and $a_n \uparrow$ and $b_n \downarrow$. We see

$$[a_1, b_1] \supset [a_2, b_2] \supset [a_3, b_3] \supset \dots$$

As seen once earlier (again no new idea, we have the same situation while proving $\mathbb{R}$ is not countable), $\{a_n\}$ is bounded by each b_k so that if $x = \sup a_n$ then $a_n \leq x \leq b_k$ for each n, k . In particular $x \in [a_n, b_n]$ for all n . We claim that there is no other such point. This is because given $\epsilon > 0$, there is N such that for $m, n \geq N$ we have $|x_n - x_m| \leq \epsilon/2$. In particular $x_N - \epsilon/2 \leq x_n \leq x_N + \epsilon/2$ for $n \geq N$. Equivalently, $x_N - \epsilon/2$ is a lower bound for $\{x_N, x_{N+1}, x_{N+2}, \dots\}$ and so $x_N - \epsilon/2 \leq a_N$. Similarly $x_N + \epsilon/2$ is an upper bound for $\{x_N, x_{N+1}, x_{N+2}, \dots\}$ so that $b_N \leq x_N + \epsilon/2$. Thus $x_N - \epsilon/2 \leq a_N \leq b_N \leq x_N + \epsilon/2$ giving $b_N - a_N < \epsilon$. Since the intervals are decreasing (?) we see $b_n - a_n < \epsilon$ for all $n \geq N$. This being true for every $\epsilon > 0$ we conclude that $b_n - a_n \rightarrow 0$. If now there are two different points $x < z$ common to all intervals, then we would have $b_n - a_n \geq z - x$ for all n which contradicts $b_n - a_n \rightarrow 0$.

Step 3: We now show that $x_n \rightarrow x$. Given $\epsilon > 0$, choose N so that $b_N - a_N < \epsilon$. Since $x \in [a_N, b_N]$ we conclude that $[a_N, b_N] \subset (x - \epsilon, x + \epsilon)$. You can see this as follows. $x \in [a_N, b_N]$, so $x \leq b_N$; hence $x - a_N \leq b_N - a_N < \epsilon$ or $x < a_N + \epsilon$ or $a_N > x - \epsilon$. And similarly $x \geq a_N$ or $-x \leq -a_N$; hence $b_N - x \leq b_N - a_N < \epsilon$ or $b_N < x + \epsilon$. By definition of a_N and b_N we see $\{x_n : n \geq N\} \subset [a_N, b_N]$. Thus $\{x_n : n \geq N\} \subset (x - \epsilon, x + \epsilon)$ as required. ■

Cantor intersection Theorem:

In showing that real numbers are uncountable and also in the arguments above we have come across a sequence of decreasing intervals of the type $[a, b]$ and got a point. This argument, due to Cantor, is powerful and appears often. So we need to separate it so that we do not repeat every time. A sequence of intervals $[a_n, b_n]$ is called nested if

$$[a_1, b_1] \supset [a_2, b_2] \supset [a_3, b_3] \supset \dots$$

Here a_i, b_i are all real numbers.

Cantor intersection theorem: Let $[a_n, b_n]$ be a nested sequence of intervals and $b_n - a_n \rightarrow 0$. Then there is exactly one point common to all these intervals. That is, there is exactly one number z such that $z \in [a_n, b_n]$ for all n .

Proof: Actually we have seen the proof earlier. But let us quickly recall. First note the following. If $[a, b] \supset [c, d]$, then both c and d are in the interval $[a, b]$ so that $a \leq c \leq d \leq b$. Thus we have

$$a_1 \leq a_2 \leq a_3 \leq \dots \leq b_3 \leq b_2 \leq b_1.$$

Thus any b_i is an upper bound for the increasing sequence (a_n) . Let s be the sup of $\{a_n\}$. In particular $a_n \leq s$ for each n . Moreover, each b_i being an upper bound for the sequence (a_n) we see that $s \leq b_n$ for each n . In other words $a_n \leq s \leq b_n$ for each n , showing that the point s is in all the intervals. Let x be any other point, say $s < x$. since $b_n - a_n \rightarrow 0$, fix k large so that $b_k - a_k < (x - s)/4$. Since $s \in [a_k, b_k]$ if x is also in this interval, then we would have $x - s \leq b_k - a_k$, contradiction. Thus there is only one point. ■

Now I want to return to the proof (completed) that Cauchy sequence converges. There we first step was to show that the sequence is bounded. Then some drama took place without using Cauchy-ness, At the end we returned to Cauchyness again to show there is exactly one point. Is there anything we

can learn from ‘that part of the’ drama? Yes. It has a beautiful and useful idea.

So let $\{x_n\}$ be a bounded sequence. I shall not repeat but we had nested sequence $[a_n, b_n]$ so that

$$(\bullet): b_n = \sup\{x_n, x_{n+1}, x_{n+2}, \dots\} \quad a_n = \inf\{x_n, x_{n+1}, x_{n+2}, \dots\}.$$

Thus $a_n \uparrow l$, say and $b_n \downarrow s$, say. When the sequence was Cauchy then there is only one point common and $l = s$ but we are NOT assuming that our sequence is Cauchy; Just that it is bounded.

For example: $0, 1, 0, 1, 0, 1, \dots$ Then $[a_n, b_n] = [0, 1] = [\alpha, \beta]$.

or the sequence: $\{0, 1, 2, 3, 0, 1, 2, 3, 0, 1, 2, 3, 0, 1, 2, 3, 0, 1, 2, 3, \dots\}$

then $[a_n, b_n] = [0, 3] = [a, b]$.

or $\{1, 1, 1/2, 3/2, 1/3, 4/3, 1/4, 5/4, 1/5, 6/5, \dots, 1/n, (n+1)/n, \dots\}$ Be careful, The terms $1/n, (n+1)/n$ are given for you to understand the sequence; definitely you should not think that $1/n$ is n -th term and $(n+1)/n$ is the $(n+1)$ -th term. Calculate and see.

or the countably many rationals $\{r : 4 < r < 5\}$ enumerated as a sequence $r_1, r_2, \dots$ Then $[a_n, b_n] = [4, 5]$ for all n .

Let us return to bounded sequence $\{x_n\}$. We claim the following:

(★) Given $\epsilon > 0$, there are infinitely many n such that $x_n \in (l - \epsilon, l + \epsilon)$. Be careful we are not saying $\lim x_n = l$. It is quite likely that infinitely many terms of our sequence may NOT be in this interval.

So fix $\epsilon > 0$. Give me any k . I show $n > k$ so that $l - \epsilon < x_n < l + \epsilon$. This will show there are infinitely n with this property. How come, showing one implies infinitely many? Because Take $k = 1$ and get $n_1 > 1$; take $k = n_1$ and get $n_2 > n_1$. Take $k = n_2$ and get $n_3 > n_2$. By induction, if you got $n_1, \dots, n_{100}$, take $k = n_{100}$ and get $n_{101} > n_{100}$.

To complete

(★) Given $\epsilon > 0$, there are infinitely many n such that $x_n \in (s - \epsilon, s + \epsilon)$.

Exactly the same proof above holds. Fix $\epsilon > 0$. Give me any k . I show $n > k$ so that $s - \epsilon < x_n < s + \epsilon$

This property (★) is an important property and has a name.

Let (x_n) be a sequence. A point x is a limit point of the sequence if for every $\epsilon > 0$, $x_n \in (x - \epsilon, x + \epsilon)$, for infinitely many n .

Of course for infinitely many n , x_n may be outside the interval too. Thus what we saw above is that α and β are limit points. We can say more.

($\bullet$): If $x < l$ then x is not a limit point of the sequence (x_n) .

Indeed if you take $x < l$ and take $\epsilon = (l - x)/4$, then $x + \epsilon < l$. Since $a_n \uparrow l$, there is a N such that $a_N > x + \epsilon$ which means for every $n > N$ we have $x_n > x + \epsilon$. Thus only finitely many terms of the sequence lie in the interval $(x - \epsilon, x + \epsilon)$.

($\bullet\bullet$): If $x > s$ then x is not a limit point of the sequence (x_n) .

Repeat 'same' argument: Take $x > s$ and take $\epsilon = (x - s)/4$, then $x - \epsilon > s$. Since $b_n \downarrow s$, there is a N such that $b_N < x - \epsilon$ which means for every $n > N$ we have $x_n < x - \epsilon$. Thus only finitely many terms of the sequence lie in the interval $(x - \epsilon, x + \epsilon)$.

These properties ($\bullet$) and ($\bullet\bullet$) deserve names.

The smallest limit point of a sequence is called $\liminf$, or limit inferior. Thus $l = \liminf x_n$. The largest limit point of a sequence is called $\limsup$ or limit superior of the sequence. Thus $s = \limsup x_n$.

Thus we have the following:

Theorem:

(i) Every bounded sequence (x_n) has at least one limit point.

This is called Bolzano or Bolzano-Weierstrass Theorem.

(ii) It has a smallest limit point, named, $\liminf x_n$ and it has a largest limit point, named $\limsup x_n$.

(iii) We always have $\liminf x_n \leq \limsup x_n$.

(iv) the sequence (x_n) converges iff $\liminf x_n = \limsup x_n$, equivalently, iff $\liminf x_n \geq \limsup x_n$.

(v) If z is a limit point of (x_n) we can get $n_1 < n_2 < n_3 < \dots$ so that the sequence $\{x_{n_1}, x_{n_2}, x_{n_3}, \dots\}$ converges to z . Conversely if we can get $n_1 < n_2 < n_3 < \dots$ so that the sequence $\{x_{n_1}, x_{n_2}, x_{n_3}, \dots\}$ converges to z , then z is a limit point of the sequence (x_n) .

(vi) If z_n is a limitpoint for each n and $z_n \rightarrow z$ then z is also a limit point. ■

In the above when we say the sequence $\{x_{n_1}, x_{n_2}, x_{n_3}, \dots\}$, we mean the sequence $(y_1, y_2, y_3, \dots)$ where $y_1 = x_{n_1}$, $y_2 = x_{n_2}$ and so on. This deserves a name.

Suppose (x_n) is a sequence. Suppose $1 \leq n_1 < n_2 < \dots$. Then the sequence $\{x_{n_1}, x_{n_2}, x_{n_3}, \dots\}$ is called a subsequence of (x_n) .

Proof: We have proved (i), (ii) and (iii). To prove (iv) observe that (x_n) converges iff it is Cauchy. If $l = s$, then $b_n - a_n \rightarrow 0$ and since $\{x_k : k \geq n\} \subset [a_n, b_n]$ we see the sequence is Cauchy. Conversely if the sequence is Cauchy, then $b_n - a_n \rightarrow 0$ so that $l = s$. Since always $l \leq s$, equality holds iff $l \geq s$. To prove (v), let z be a limit point. Remembering that each $(z - \epsilon, z + \epsilon)$ ($\epsilon > 0$) contains x_n for infinitely many values of n , first pick n_1 so that $x_{n_1} \in (z - 1, z + 1)$. Then pick $n_2 > n_1$ so that $x_{n_2} \in (z - 1/2, z + 1/2)$. If we picked $n_1 < n_2 < \dots < n_{k-1}$ so that $x_{n_j} \in (z - 1/j, z + 1/j)$ for $j = 1, 2, \dots, k-1$ we pick $n_k > n_{k-1}$ so that $x_{n_k} \in (z - 1/k, z + 1/k)$. This is possible because for infinitely many n we have x_n in this interval. Thus by induction we pick the integers. Clearly $\{x_{n_j}\}$ converges to z . Conversely if we have $n_1 < n_2 < n_3 < \dots$ so that $\{x_{n_k}\}$ converges to z , then given any interval $(z - \epsilon, z + \epsilon)$, for all large k we have $x_{n_k} \in (z - \epsilon, z + \epsilon)$ showing that for infinitely many n , $x_n \in (z - \epsilon, z + \epsilon)$. This completes the proof. To prove (vi), let $\epsilon > 0$. Using hypothesis get $z_k \in (z - \epsilon, z + \epsilon)$. Let $\eta = \min\{z_k - (z - \epsilon); (z + \epsilon) - z_k\} > 0$. Now z_k being a limit point of (x_n) , for infinitely many n we have $x_n \in (z_k - \eta, z_k + \eta) \subset (z - \epsilon, z + \epsilon)$, completing the proof. ■

Why did we say separately in (iv), the sequence converges iff $\liminf \geq \limsup$? Because, it is easier to show inequalities rather than exact equalities. In practice to show that a sequence converges we just show $l \geq s$. In fact we do not even exactly show this. We show $l \geq s - \epsilon$ for every $\epsilon > 0$. Then you must believe that $l \geq s$ and hence they are equal. Such an allowance with ϵ makes showing possible. When we use this I shall mention again.

on limits and convergence:

Limit of a sequence really depends on ‘tail’ of the sequence and not on initial terms. This is too loose a statement. Here are precise statements. Suppose $x_n \rightarrow z$.

You can delete first hundred terms. Suppose we define $y_1 = x_{101}, y_2 = x_{102}, \dots, y_n = x_{100+n}, \dots$. Then $y_n \rightarrow z$. Or

You can add hundred terms before x_1 . Suppose you define $u_1, u_2, \dots, u_{100}$ any numbers you like and put $u_{101} = x_1, u_{102} = x_2, \dots$ then $u_n \rightarrow z$.

Or You can change. Change first hundred terms to anything you like. Suppose you define v_1, v_2, v_{100} as you like and $v_{101} = x_{101}, v_{102} = x_{102}, \dots$. Then $v_n \rightarrow z$.

I have presented the concept of convergence as a way to discover new

numbers, like e . Actually, this concept is much more basic, much more fundamental than I made it out. *Everything* that we do (well, almost everything) depends on this limit concept — continuous functions, derivative, integral etc. Even to define functions rigorously we resort to infinite sums, essentially we consider polynomials of infinite degree. Limits arise in all disciplines.

If you are doing computations and providing an algorithm for finding root of an equation or for finding minimum of a cost function over a certain region, typically your algorithm would give an iterative procedure and runs like this: start with x_1 , do some thing to end up with x_2 , then keep on repeating the instructions to get $x_2, x_3, \dots$. Hope is that you will be heading to the quantity you are looking for. Thus your prescription of algorithm is not complete unless you show that the numbers you successively compute converge to the target.

You are having a course in physics and know how it appears. If you want to discuss about the air in this lecture hall, you will not write down equations of motion for each of the particles. Instead you will try to understand the question when there are only n particles and then take limits. This is because there are Huge number of particles. Similarly when you want to study the properties of a magnet, you first assume that a finite number n of atoms (usually arranged in a lattice) and understand the phenomenon and take limit as n becomes large.

Even in biological sciences it makes its appearance in an overwhelming manner. For example, you might be interested in: what happens to the species in the long run? (do they survive or become extinct) what happens to the mutant gene, in the long run, over generations (will it spread to the whole population?). Of course, most pertinent question is: what happens in the long run if we keep on using available natural resources at the current rate! Thus all these are questions about a limit.

The rigorous development of Calculus was initiated by Newton and Leibniz. But they were using infinitesimals, very small quantities etc. It was Cauchy and Bolzano who provided clear meaning of limits, though, of course, Euler, Laplace and several others before them already used infinite series and products. Did we not see that already Archimedes used the concept of limit, without saying so.

You should pay attention to this concept and think about it till you feel comfortable and till the concept appears *natural* to *you*.

The main difference between maths and reality is the following. In maths

we can think of the following sequence: First trillion terms are zero and all later terms are equal to one. Obviously, this sequence is converging to one, but looking at the first trillion terms gives no indication of this fact. However, such a thing does not occur in nature. That is, sequences that arise in most of the practical problems do give an indication of what is happening — with its first ten thousand or so terms. This can be considered as the gift of nature to us – naturally arising sequences give us an indication of what is going to happen in the long run.

This phenomenon helps us in simulations and see what the unknown is like. This also helps us in understanding numbers. More specifically, In theory, you say x_n converges to x . But in practice, its interpretation is this: If you are trying to calculate x_{10000} , do not bother, x_{10000} is like x — that is it is close to x . For example if you want to understand how the number $\sum_{k=0}^{10000} \frac{1}{k!}$ looks like, do not bother to calculate, it looks like e .

Tutorial: Thursday 28th Aug.

Quiz 1: Tuesday September 2, for first half hour of class. Based on the first three weeks of classes.

The concept of limit of a sequence is basic in all of science. This is used to define various concepts.

series:

By virtue of the rules regarding real numbers, we can add any two real numbers and hence, by induction, any finitely many real numbers. But many times, we do see that infinitely many real numbers can be ‘theoretically’ added, that is, we feel that the sum must be a particular number.

For example, let us consider something familiar from high school, adding

$$1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \frac{1}{2^5} + \frac{1}{2^6} + \cdots.$$

Of course, we do not have time to add all these infinitely many numbers one by one. But, if we do start adding, we get

$$1, \quad 2 - \frac{1}{2}, \quad 2 - \frac{1}{2^2}, \quad 2 - \frac{1}{2^3}, \quad 2 - \frac{1}{2^4}, \quad 2 - \frac{1}{2^5}, \quad 2 - \frac{1}{2^6}, \cdots.$$

Even though we are unable to add all the infinitely many numbers above, we see that the successive sums are nearing 2 and also they never go above 2. In other words whatever number strictly below 2 you give me, I can show you that the sums are above that sooner or later. Also no sum is above 2. Thus, it seems reasonable to *define* the infinite sum to be 2. This is what we are going to do now.

Definition: A series is simply an expression $x_1 + x_2 + x_3 + \cdots$ where each x_i is a real number. We also write series as $\sum x_n$ or $\sum_{n \geq 1} x_n$. The number x_n is called the n -th term of the series.

See the analogy: $x_1, x_2, x_3, \cdots$ was called a series, also written as $\{x_n\}$ or $\{x_n, n \geq 1\}$. Also x_n is called n -th term of the sequence.

Yes, as you said, this should actually be called ‘infinite series’. Even though we do talk about finite sums, we do not wish to say it is sum of a

finite series (though there is nothing wrong to say so). Thus for us series are infinite series. Remember sequences we talked about were also infinite sequences. In fact we do come across finite sequences also many-a-times.

There is no meaning yet for a series. Here are some examples of series:

$$1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \frac{1}{2^5} + \frac{1}{2^6} + \cdots .$$

$$1 + 1 + 1 + 1 + \cdots .$$

$$1 + (-1) + 1 + (-1) + 1 + (-1) + 1 + (-1) + \cdots .$$

This series is usually written as

$$1 - 1 + 1 - 1 + 1 - 1 + 1 - 1 + 1 \cdots .$$

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \cdots .$$

The first series appears meaningful and we should get the number 2, if we add *all* those terms. It is not obvious, but true and will be proved, that the last series also has a meaning.

However, the second series has the following sums if we keep on adding the terms: $1, 2, 3, 4, 5, 6, 7, \cdots$. The value of the sum, as we keep on adding, exceeds anything that you can think of — remember the Archimedean principle, namely, given any number x , after some stage all the integers are larger than x . Since there is a particular direction in which we are going — on the number line — if we keep on and on adding, it is tempting to say that the sum is infinity. Yes, it is convenient and we shall do that later. This needs introduction of funny animal $+\infty$, and for symmetry another animal $-\infty$. But remember, it is not just a matter of introducing these characters into the drama, we should also agree upon the rules as to what we can and cannot do with these new things. Otherwise, there may come a stage when we do not understand each other. We shall soon talk about $\pm\infty$.

The third series is interesting. If we keep on adding we get $1, 0, 1, 0, 1, 0, \cdots$. The addition of successive numbers is not leading us away, in a particular direction or to a particular number. The sums oscillate between zero and one. In any case, if we keep on adding the sums are not approaching any number.

Thus a series is simply a suggestion to add certain numbers. Just because we have used $+$, it does. not mean we are adding the numbers. Whether we

can really add or not is a different matter. Sometimes we have a feeling as to what we should get if we really add. We shall make this feeling precise now. Most mathematical definitions are just precise way of expressing some feeling we have.

Definition: Let $\sum x_n$ be a series. We define its partial sums to be the sequence of numbers,

$$s_1 = x_1; \quad s_n = x_1 + x_2 + \cdots + x_n \quad n \geq 1.$$

s_n is called the n -th partial sum. We say that the series $\sum x_n$ converges if the sequence (s_n) converges. If $s_n \rightarrow a$, we say that the value of the series is a . We express this by writing in any of the following ways.

$$\sum x_n = a; \quad \sum_{n \geq 1} x_n = a; \quad x_1 + x_2 + x_3 + \cdots = a.$$

You should keep in mind that we said the value of the series *is* a . We did not say the series approaches a or nearly a or close to a etc. It *is* exactly a .

You should note the distinction between series and sequence. Both have numbers in a particular order. The order is important in both of them. If you change the order, it becomes a different thing. In a sequence, we simply have numbers standing in a line or in a row. In a series, we have numbers again in a row with a suggestion to add them. This suggestion is shown by putting the plus sign between the terms. Of course, you should know, by now, that adding includes subtraction too, adding (-5) is same as subtracting 5. Whether we can execute the suggestion is a different matter and that is why we made the next definition regarding convergence or value of a series.

So, a sequence is not a series and a series is not a sequence. Of course, you may think that a series is just a sequence with plus signs between terms, instead of the commas. This is only a typographical or superficial recognition — though correct and helps you in remembering, it is not the spirit in which these concepts are to be understood.

We shall now discuss some properties and examples of convergence.

$$1: \quad 1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \frac{1}{2^5} + \frac{1}{2^6} + \cdots = 2.$$

We knew this already. We are lucky, because we could actually calculate all the partial sums and see what exactly is happening. In general it will be difficult to calculate all partial sums.

$1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \frac{1}{5!} + \frac{1}{6!} + \frac{1}{7!} + \cdots$ converges. We knew this. We cannot explicitly calculate partial sums here.

2. If $\sum x_n = a$ and $\sum y_n = b$ then $\sum z_n = a + b$ where $z_n = x_n + y_n$.

If $\sum x_n = a$ then $\sum (29x_n) = 29a$.

Indeed if $(s_n), (t_n), (w_n)$ are partial sums of $\sum x_n, \sum y_n, \sum z_n$ respectively, then $w_n = s_n + t_n$ and the result follows from properties of convergence of sequences. Similarly if (v_n) are partial sums of the series $\sum (29x_n)$, then $v_n = 29s_n \rightarrow 29a$.

You should be careful and maintain discipline. You can not simply say $\sum (x_n + y_n) = \sum x_n + \sum y_n$. The whole issue is whether you can distribute the summation sign to x_n and y_n separately. Yes, if the two series $\sum x_n$ and $\sum y_n$ converge, you can do. To understand subtle issues you should keep staring at examples. For example, when each $x_n = 1$ and each $y_n = -1$; the equality $\sum (1 - 1) = \sum 1 + \sum (-1)$ makes no sense. Notice left side summation is perfectly meaningful.

This also illustrates, you can not put brackets in a series as you like, The series $1 - 1 + 1 - 1 + 1 - 1 + \cdots$ is different from the series $(1 - 1) + (1 - 1) + (1 - 1) + \cdots$

3: $\sum x_n$ converges iff partial sum sequence $\{s_n, n \geq 1\}$ is Cauchy. Further $\sum x_n = a$ iff $s_n \rightarrow a$.

This is immediate from definition of convergence of series and properties of convergence of sequences.

4. $\sum x_n$ be a series where each $x_n \geq 0$. The series converges iff the sequence of partial sums is bounded.

Since the numbers x_n are positive (non-negative), the partial sums are increasing. We know that an increasing sequence converges iff it is bounded above. Also for an increasing sequence, bounded above is same as bounded.

5. If $\sum x_n$ converges then $x_n \rightarrow 0$ and the sequence $\{x_n\}$ is bounded.

To show $x_n \rightarrow 0$, let $\epsilon > 0$ be given. Fix N so that for $n, m \geq N$, $|s_m - s_n| < \epsilon$. In particular for $n \geq N + 1$, we have $|x_n| = |s_n - s_{n-1}| < \epsilon$. Convergent sequence is bounded and so $\{x_n\}$ is bounded.

6. Suppose $\sum x_n$ converges. If you add finitely many terms at the beginning, or delete finitely many terms at the beginning, or delete finitely many terms at the beginning, or alter finitely many terms that the beginning, then the resulting series converges.

Suppose we add k terms in the beginning. Thus the new series is $\sum y_n$

where $y_n, n \leq k$ are the given new numbers and $y_{k+i} = x_i$ for $i \geq 1$. If (s_n) and (t_n) are partial sums of the series $\sum x_n$ and $\sum y_n$, then clearly for $n, m \geq 1$ we have $|t_{n+k} - t_{m+k}| = |s_n - s_m|$ showing that if (s_n) is Cauchy then so is (t_n) .

Suppose we delete first k terms in the beginning. Thus the new series is $\sum y_n$ $y_n = x_{k+n}$ for $n \geq 1$. With same notation as above, we see $|t_n - t_m| = |s_{n+k} - s_{m+k}|$ to complete the proof.

The last statement is similarly proved.

7. if $\sum |x_n|$ converges then $\sum x_n$ converges.

If $(s_n), (t_n)$ are partial sums of $\sum |x_n|$ and $\sum x_n$, then for $n > m$ we see

$$|t_n - t_m| = |x_{m+1} + x_{m+2} + \cdots + x_n| \leq |x_{m+1}| + |x_{m+2}| + \cdots + |x_n| = |s_n - s_m|$$

and the proof is easily completed.

8. Ratio test: Let $\sum x_n$ be a series with $x_n \neq 0$ for all n . If there is a number $0 < c < 1$ such that $\frac{|x_{n+1}|}{|x_n|} \leq c$ after some stage then $\sum |x_n|$ converges and so $\sum x_n$ converges.

Let us say, $\frac{|x_{n+1}|}{|x_n|} \leq c$ holds for $n \geq k$. Thus

$$|x_{k+1}| \leq c|x_k|.$$

$$|x_{k+2}| \leq c|x_{k+1}| \leq c^2|x_k|.$$

If we showed $|x_{k+n}| \leq c^n|x_k|$ then

$$|x_{k+n+1}| \leq c|x_{k+n}| \leq c \cdot c^n|x_k| = c^{n+1}|x_k|$$

where we first used the given hypothesis and then induction hypothesis. Using sum of (finite) geometric series we see that partial sums of the series $|x_k| + c|x_k| + c^2|x_k| + c^3|x_k| + \cdots$ is bounded above. The above inequalities show that partial sums of the series $|x_k| + |x_{k+1}| + |x_{k+2}| + \cdots$ are also bounded and hence this series converges. since adding finitely many terms at the beginning will not alter the conclusion, we see $\sum |x_n|$ converges. From earlier result $\sum x_n$ also converges.

9. For any real number x , the following series converge.

$$E(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!} + \frac{x^7}{7!} + \cdots$$

$$S(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} \cdots$$

$$C(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \frac{x^{10}}{10!} + \cdots .$$

For First series, enough to show that

$$x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!} + \frac{x^7}{7!} + \cdots$$

converges. Again enough to consider $x \neq 0$. But then, observing that n -th term now is $x^n/n!$ we see the ratio

$$\frac{|x_{n+1}|}{|x_n|} = \frac{|x|^{n+1}}{(n+1)!} \frac{n!}{|x^n|} = \frac{|x|}{n+1}$$

which is smaller than $1/2$ after some stage. Apply earlier result.

The other two series are also similar.

We shall identify later: $E(x) = e^x$ and $S(x) = \sin x$ and $C(x) = \cos x$.

Digression:

Some of you were wondering why we are studying sequences and series. Yes, we should know. For sequences I have already explained. Motto in life: explain unknown things using known things, especially because we know very little and we know the existence of toooooo much that we do not know.

For series too, you are already familiar from high school about sum of an infinite geometric series with common ratio smaller than one. You arrived at that sum **not because you added all those infinitely many numbers**, but because **you did what you can** – namely add more and more terms – and saw the trend. But are you sure? Yes, for example if you consider $1 + (1/2) + (1/2^2) + (1/2^3) + \dots$, keep adding you get ‘as close to 2’ as anyone wants, but never exceed 2. That is the reason the sum is **defined** as 2. Again you see how sequences enter here.

There are other reasons too. One of the most important mathematical objects are functions: describe temperature as a function of time or distance travelled as a function of time or the air density at each point of this room etc. The simplest functions are again high school polynomials. The next simplest thing are ‘infinite degree’ polynomials; they are not called so because degree has to be finite according to our agreement; so they are called power series. Exponential, Cosine, Sine functions are of this kind. Instead of a superficial discussion, let us look at two concrete examples.

You learn in physics that most of the rules that describe phenomena lead to differential equations and you need to solve them. The solution is achieved, most of the time, NOT by any closed formula but by a series with ‘educated guess’. For example, you are looking for a function $y(x)$; ($x \in R$) such that $y(0) = 1$ and $y' = y$ (prime is derivative). Thus, for the function we are seeking, derivative at any point equals the value of the function itself. Since we need $y(0) = 1$, and constant functions are the simplest we know, the first guess $y_1(x) \equiv 1$. But this does not obey $y' = y$. Since $y' = y$ is same as $y(x) - y(0) = \int_0^x y(t)dt$ (you will officially learn this later, fundamental theorem of calculus) we need $y(x) = 1 + \int_0^x y(t)dt$. Let us put the first approximation on right side and integrate: you get $y_2(x) = 1 + x$. This also

does not satisfy our differential equation. So again use this approximation on right side and integrate to get $y_3(x) = 1 + x + (x^2/2!)$. You now guess whether the following ‘makes sense’ and solves our problem:

$$y(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$$

Yes, this does make sense and indeed satisfies our requirements.

You can say, this is just the function e^x we all know, why did you take such a long route and complicate my life? Firstly, you are repeating something which you were taught without thinking! Do you know e ? Can you raise it to the power x , say to the power $\sqrt[23]{88}$? Can you show derivative of e^x is e^x ? Think. Secondly, in a known territory (especially when you knew the answer already) this method may appear silly; but in an unknown territory this may be the only method!

Instead of powers of x and their sum, you can use trigonometric functions: $\sin x$, $\sin 2x$, $\sin 3x$ etc and their sums. Why do you want to do that? NO, intention is not to do something just because we can do. Instead of going too far and wasting your time let me say this. I have a rod of length 1, think of it as the interval $[0, 1]$. I know initially the temperature at each point x on the rod is $u(x)$. I maintain the end points of the rod at certain prescribed temperatures at each time t . Question: explain to me the temperature at time t at point x of the rod. How heat is conducted in the rod is given by a differential equation (now it is called partial differential equation) and you need to solve that. Historically, Fourier solved it for ‘certain initial’ conditions like end point at time t are maintained at temperature like $\sin t$, $\sin 2t$ etc. he could solve even when this data is linear combination of such functions. He felt: How nice if every initial data is only such, but possibly infinite, linear combination of trigonometric functions? Then I can solve my problem for any initial conditions.

When you read maths remember: To start with, we do not make arbitrary definitions and develop properties. We solve some interesting problems and in doing so come across certain ‘things’ again and again and again. Since this ‘thing’ keeps on coming up, we feel it is important. Then we give a name to this ‘thing’ and study its further properties and further uses. Once systematically developed these concepts find use in unexpected areas too. But to teach, we can not do this. If I discuss heat conduction, without telling you function, differentiation (which in turn needs convergence!) etc then we can not communicate with each other. Most mathematics, including set theory,

originated in understanding statements (some false) made by Fourier and making things clear and precise.

With this digression let us return to what we were doing. Usually Ratio test is stated (in an equivalent form) using $\limsup$ etc.

(8 Ratio test revisit) Let $\sum x_n$ be a series with $x_n \neq 0$ for all n . If $\limsup \frac{|x_{n+1}|}{|x_n|} < 1$ then $\sum |x_n|$ converges and so $\sum x_n$ converges. If $\liminf \frac{|x_{n+1}|}{|x_n|} > 1$ then $\sum x_n$ does not converge.

In this statement we are assuming that the ratios are bounded. It is unnecessary to assume so, but unfortunately we defined $\limsup$ and $\liminf$ only for bounded sequences.

In fact if $\limsup \frac{|x_{n+1}|}{|x_n|} = s < 1$, then with $c = (1+s)/2$ we see $0 < c < 1$ and after some stage the ratios are below c and (8) applies. On the other hand if $\liminf \frac{|x_{n+1}|}{|x_n|} = l > 1$, then with $c = (1+l)/2$ we have $c > 1$ and after some stage $|x_{n+1}| > c|x_n|$ so that $|x_{n+k}| > c^k|x_n|$. Since $x_n \neq 0$ and $c > 1$ we conclude that $\{x_n\}$ is not bounded and hence the series can not converge. ■

Can we have $\limsup(|x_{n+1}/x_n|) > 1$ and still the series $\sum x_n$ converge? Yes.

$$\frac{1}{3} + \frac{1}{2} + \frac{1}{3^2} + \frac{1}{2^2} + \frac{1}{3^3} + \frac{1}{2^3} + \frac{1}{3^4} + \frac{1}{2^4} + \cdots$$

Here $(1/2^n)/(1/3^n) = 3^n/2^n$ is unbounded and so $\limsup$ is not smaller than one (actually, with our present definitions, we can not talk $\limsup$ because this sequence is not bounded, later we make a better definition and see $\limsup$ is infinity). However the series converges because the partial sums are bounded by 2.

10. Root test. If $\limsup \sqrt[n]{|x_n|} < 1$, then the series $\sum |x_n|$ and hence $\sum x_n$ converges. If $\limsup \sqrt[n]{|x_n|} > 1$, then the series $\sum x_n$ does not converge. If $\limsup \sqrt[n]{|x_n|} = 1$, then the series $\sum x_n$ may converge in some cases and not converge in some other cases

Here we are assuming that the sequence $\{\sqrt[n]{|x_n|}\}$ is bounded. Again it is unnecessary to assume so, but at this stage we have defined $\limsup$ only for bounded sequences.

In the ratio test earlier the gap between convergence and nonconvergence criteria was too much; one depends on $\limsup$ and other on $\liminf$. But in the root test the gap is narrowed. Both on $\limsup$. But still there is a gap, we are unable to conclude in the 'boundary case' ($\limsup = 1$). Can we give a criterion which is decisive, without any such gaps? Not really, except re-

formulating the definitions (like Cauchyness of partial sums etc).

Proof of root test is simple, again a high school calculation. Suppose $\limsup \sqrt[n]{|x_n|} = s < 1$. let $c = (1 + s)/2$, then $s < c < 1$ and after some stage $\sqrt[n]{|x_n|} < c$, say for $n \geq N$. Then $|x_n| \leq c^n$, so partial sums of the series $|x_{N+1}| + |x_{N+2}| + \dots$ are bounded by $1/(1 - c)$ and hence the series converges. First few terms do not matter and so $\sum |x_n|$ converges, hence $\sum x_n$ converges too.

If $\limsup \sqrt[n]{|x_n|} = s > 1$, then taking $c = (1 + s)/2$, we see $1 < c < s$ so that for infinitely many n we have $\sqrt[n]{|x_n|} > c$ or $|x_n| > c^n$ for infinitely many n . In particular $x_n \not\rightarrow 0$ and so the series does not converge. ■

Unfortunately, none of these criteria decide convergence or nonconvergence of $\sum 1/n$ or $\sum 1/n^2$. Since $\lim |x_{n+1}|/|x_n| = \lim n/(n+1) = 1$ and $\lim n^2/(n+1)^2 = 1$ the ratio test fails. Root test also fails because $\sqrt[n]{1/n}$ and $\sqrt[n]{1/n^2}$ both have limit one.

11. Suppose $a_n \downarrow 0$. Then $\sum a_n$ converges iff the following series converges.

$$a_1 + 2a_2 + 2^2a_{2^2} + 2^3a_{2^3} + 2^4a_{2^4} + \dots$$

Proof is simple. Let the partial sums be denoted by

$$s_n = \sum_{i=1}^n a_i; \quad t_k = a_1 + 2a_2 + 2^2a_{2^2} + 2^3a_{2^3} + \dots + 2^k a_{2^k}$$

Since both the series consist of non-negative terms, their convergence is equivalent to boundedness of the partial sums. Thus the stated result follows from the following two claims.

- (i) For every n , there is a k such that $s_n \leq t_k$.
- (ii) For every k , there is an n such that $t_k \leq 2s_n$.

We prove (i) as follows. If $2^k \leq n < 2^{k+1}$, then

$$\begin{aligned} s_n &= a_1 + a_2 + a_3 + a_4 + \dots + a_n \leq \\ &a_1 + (a_2 + a_3) + (a_4 + \dots + a_7) + \dots + (a_{2^k} + \dots + a_{2^{k+1}-1}) \\ &\leq a_1 + 2a_2 + 2^2a_{2^2} + \dots + 2^k a_{2^k} = t_k. \end{aligned}$$

where, for the inequality we used that a_n are decreasing.

We prove (ii) as follows.

$$t_k = a_1 + 2a_2 + 2^2a_{2^2} + \dots + 2^k a_{2^k} \leq$$

$$\begin{aligned}
& 2\{a_1 + a_2 + 2a_4 + 2^2a_8 + \cdots + 2^{k-1}a_{2^k}\} \leq \\
& 2\{a_1 + a_2 + (a_3 + a_4) + (a_5 + \cdots + a_8) + \cdots + (a_{2^{k-1}+1} + \cdots + a_{2^k})\} \\
& = 2s_{2^k}. \quad \blacksquare
\end{aligned}$$

11. The series $\sum \frac{1}{n^p}$ converges iff $p > 1$.
 $a_n = n^{-p}$ and the terms are decreasing.

$$2^n a_{2^n} = 2^n (2^n)^{-p} = 2^{n(1-p)}$$

Thus $\sum 2^n a_{2^n}$ is a geometric series $\sum r^n$ where $r = 2^{1-p}$. ■

If the series $\sum |a_n|$ converges then we say that the series $\sum a_n$ converges absolutely. A series which is convergent but NOT absolutely convergent is called conditionally convergent.

Thus from earlier result, an absolutely convergent series is convergent. But as we see now a convergent series may not be absolutely convergent.

12. If $a_n \downarrow 0$, then the following series converges.

$$a_1 - a_2 + a_3 - a_4 + a_5 - a_6 + \cdots$$

Of course, the series is interesting only when all a_n are different from zero. Because, as soon as one a_n equals zero, the sum is actually finite sum and convergence issue is only theoretical.

In particular the series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \cdots$$

converges. Observe that this is not absolutely convergent.

Such series where the terms are alternatively positive and negative are called ‘alternating series’. Just to emphasize, you should not be deceived by appearance, the series above is alternating not because you see $\pm$ signs, you need to use that the numbers a_n are positive. A series $\sum b_n$ is alternating if $\text{sign}(b_n)$ and $\text{sign}(b_{n+1})$ are opposite for each n . For example even though you see signs changing in the following series, it is not alternating series.

$$1 - \frac{1}{2} - \frac{1}{2^2} + \frac{1}{2^3} - \frac{1}{2^4} - \frac{1}{2^5} + \cdots$$

Proof is simple. We show the following;

$$b_1 \geq b_2 \geq b_3 \geq \cdots \geq b_k \geq 0 \Rightarrow 0 \leq (b_1 - b_2 + b_3 - \cdots \pm b_k) \leq b_1.$$

Let us see what happens if this is done. Returning to our series, let s_n be its partial sums. If $m > n$, then the above conclusion tells us

$$|s_m - s_n| = |a_{n+1} - a_{n+2} + a_{n+3} - \cdots \pm a_m| \leq a_{n+1}.$$

Since $a_n \downarrow 0$, given $\epsilon > 0$, we can choose p such that $a_p < \epsilon$. The above inequality tells that after p -th stage any two partial sums differ by at most ϵ . In other words, partials sums form Cauchy sequence and hence the series converges.

We shall now prove the inequality claimed at the beginning of the proof. The left side equals $(b_1 - b_2) + (b_3 - b_4) + \cdots \geq 0$. In fact, each bracketed term is non-negative by decreasing nature of b_n ; if k is even there is nothing left over and if k is odd there is a last non-bracketed term which is positive. To see the other inequality, $b_1 - (b_2 - b_3) - (b_4 - b_5) - \cdots \leq b_1$. Each bracketed term is positive and is being subtracted from b_1 . If k is even, there is an unbracketed term b_k at the end which also appears with negative sign. This completes the proof.

13. Let $0 < r < 1$. Take any bounded sequence of numbers (α_n) . Then the following series converges.

$$\alpha_0 + \alpha_1 r + \alpha_2 r^2 + \alpha_3 r^3 + \alpha_4 r^4 + \alpha_5 r^5 + \alpha_6 r^6 + \cdots$$

This is immediate from the fact $|\alpha_n r^n| \leq C r^n$ where C is an upper bound for $\{|\alpha_n|, n \geq 0\}$ and the series $\sum C r^n$ converges.

Cauchy product of series:

We shall now define the concept of product of two series. For certain useful applications, we shall now consider series as $\sum_{n \geq 0} a_n$ rather than $\sum_{n \geq 1} a_n$.

In other words we consider series $a_0 + a_1 + a_2 + \cdots$ instead of, $a_1 + a_2 + a_3 + \cdots$ as has been done so far. You should not get confused. Either you can set your starting point a little back and think of zeroth term, first term etc (and zeroth partial sum, first partial sum etc). Or if you have trouble thinking like that, you can think that first term is a_0 , second term is a_1 and in general the n -th term is a_{n-1} . But it will help you if you get used to the first way of thinking. Thus we have partial sums $s_0, s_1, s_2 \cdots$.

So now let $\sum_{n \geq 0} a_n$ and $\sum_{n \geq 0} b_n$ be two series (of real numbers). We define

$$c_n = a_0 b_n + a_1 b_{n-1} + a_2 b_{n-2} + \cdots + a_n b_0; \quad n \geq 0.$$

Thus $c_0 = a_0b_0$; $c_1 = a_0b_1 + a_1b_0$. In general, c_n is the sum of all products a_ib_j where $i + j$ adds upto n . These are finite sums and you need not worry. The series $\sum_{n \geq 0} c_n$ is called the Cauchy product of the two series $\sum a_n$ and $\sum b_n$.

At first sight this looks complicated. But think of multiplying two polynomials

$$A(x) = \sum_0^p a_i t^i; \quad B(x) = \sum_0^q b_j t^j.$$

You know that product of two polynomials is again a polynomial. So let us say the product $A(t)B(t)$ is the polynomial $C(t) = \sum_0^{p+q} c_i t^i$. Then what are the coefficients? You can see $c_0 = a_0b_0$; $c_1 = a_0b_1 + a_1b_0$. In general, c_n is the sum of all products a_ib_j where $i + j$ adds upto n exactly as above. Of course, since a polynomial is a finite sum, when you reach $p + q$ there is only one term a_pb_q . In the infinite series case, there is always a_0, a_1 etc and finally a_n — all appearing in c_n . If you pretend that you have two infinite degree polynomials

$$A(t) = \sum_{n \geq 0} a_n t^n; \quad B(t) = \sum_{n \geq 0} b_n t^n.$$

and just like in the usual polynomial case, suppose you want to multiply these two infinite degree polynomials and write it again as a (infinite degree) polynomial by collecting powers of t , then you will exactly get $\sum c_n t^n$ where c_n are as defined above.

14. If $\sum_{n=0}^{\infty} a_n = A$ and $\sum_{n=0}^{\infty} b_n = B$ and if the series $\sum a_n$ is absolutely convergent then $\sum c_n = AB$ where $c_n = \sum_{k=0}^n a_k b_{n-k}$.

Proof: Let s_n, t_n and u_n be the partial sums of the series $\sum a_n, \sum b_n$ and $\sum c_n$ respectively. Known $s_n \rightarrow A$ and $t_n \rightarrow B$. Need to show that $u_n \rightarrow AB$. It is very difficult to handle AB because both A and B are values of infinite sums. So we shall reduce our job a little bit. We know $s_n B \rightarrow AB$. Thus if we can show that

$$u_n - s_n B \rightarrow 0 \quad (\spadesuit)$$

then $u_n = (u_n - s_n B) + s_n B \rightarrow 0 + AB$, as wanted.

$$u_n = a_0 b_0 + (a_0 b_1 + a_1 b_0) + \cdots + (a_0 b_n + a_1 b_{n-1} + \cdots + a_n b_0).$$

Collecting terms of a_0, a_1 etc together we get

$$u_n = a_0 t_n + a_1 t_{n-1} + a_2 t_{n-2} + \cdots + a_n t_0.$$

and

$$s_n B = a_0 B + a_1 B + a_2 B + \cdots + a_n B$$

Thus

$$\begin{aligned} u_n - s_n B &= a_0(t_n - B) + a_1(t_{n-1} - B) + a_2(t_{n-2} - B) + \cdots \\ &\quad + a_{n-1}(t_1 - B) + a_n(t_0 - B) \end{aligned}$$

Here is the feeling why this is small. If n is large, the first few terms on right are small because $(t_n - B)$ is small; remember that $t_n - B \rightarrow 0$. The last few terms are small because $a_n \rightarrow 0$, remember $\sum a_n$ converges. Not only the individual terms are each small but the entire sum is small, this is where absolute convergence is used. We execute the proof. But let us rewrite the above equation in a convenient way. If, (by chance) I take any integer $k < n$, then we can write the above as

$$\begin{aligned} u_n - s_n B &= a_0(t_n - B) + a_1(t_{n-1} - B) + a_2(t_{n-2} - B) + \cdots + a_{n-k}(t_k - B) \\ &\quad + a_{n-k+1}(t_{k-1} - B) + \cdots + a_n(t_0 - B) \quad (\spadesuit) \end{aligned}$$

since $t_n - B \rightarrow 0$, it is a bounded sequence. Also we know $\sum |a_n|$ is convergent. Thus pick a number M so that $|t_n - B| < M$ for all n and also $\sum |a_n| < M$.

Let $\epsilon > 0$ be given. We show that $|u_n - s_n B| < \epsilon$ for all large n .

First fix a K so that $|t_n - B| < \epsilon/(2M)$ for $n \geq K$. This is possible because $t_n - B \rightarrow 0$. Having fixed this K , now fix N so that $|a_n| < \epsilon/(2KM)$ for $n \geq N$. This is possible because $\sum a_n$ is convergent and hence $a_n \rightarrow 0$. Now let us take any $n > N + K$.

Recall $(\spadesuit)$ and use it with our chosen K , in place of k there. $n > N + K$ tells $n - K + 1 \geq N + 1$ so that $|a_j| < \epsilon/(2KM)$ for every a_j that appears in the second line there. Further each $|t_i - B|$ is at most M . Thus $|a_{n-i}(t_i - B)| \leq \epsilon/(2K)$. there are K terms in the second line and so second line is at most $\epsilon/2$ in modulus. $(\bullet)$

In the first line, each $|t_i - B| \leq \epsilon/(2M)$ because there $i \geq K$. Thus first row, in modulus is at most

$$\frac{\epsilon}{2M}(|a_{n-K}| + |a_{n-K-1}| + \cdots + |a_0|) \leq \frac{\epsilon}{2M} \sum |a_n| \leq \frac{\epsilon}{2M} M = \frac{\epsilon}{2}$$

Thus first line is at most $\epsilon/2$ in modulus. $(\bullet\bullet)$.

Thus $(\spadesuit), (\bullet), (\bullet\bullet)$ show $|u_n - s_n B| < \epsilon$ for all $n > N + K$. This completes the proof. $\blacksquare$

As you have seen in the proof, we used that $\sum a_n$ is absolutely convergent but did not use that $\sum b_n$ is absolutely convergent. Thus the theorem is true if one of the series is absolutely convergent. If none of them is absolutely convergent, then the Cauchy product may not converge. For example if we take both the series to be the alternating series

$$1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}} + - \dots$$

then the series converges. However their Cauchy product does not converge.

$$c_0 = 1; \quad c_1 = -\left(\frac{1}{\sqrt{1}}\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\frac{1}{\sqrt{1}}\right) \quad c_2 = \left(\frac{1}{\sqrt{1}}\frac{1}{\sqrt{3}} + \frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}}\frac{1}{\sqrt{1}}\right) \dots$$

$$c_n = (-1)^n \sum_{k=1}^{n+1} \frac{1}{\sqrt{k}\sqrt{n+2-k}} \quad n \geq 0$$

Using the AM-GM inequality you see $\sqrt{k}\sqrt{n+2-k} \leq \frac{n+2}{2}$. So

$$|c_n| \geq \sum_{k=1}^{n+1} \frac{2}{n+2} = 2 \frac{n+1}{n+2} \rightarrow 2,$$

Thus $\{c_n\}$ does not converge to zero, In particular $\sum c_n$ does not converge.

This result on Cauchy Product is very useful. For example let us take two numbers x and y and consider the series for $E(x)$ and $E(y)$. Since these are absolutely convergent, we conclude that their Cauchy product converges to $E(x)E(y)$. But computation shows that the Cauchy product is the series defining the number $E(x+y)$. Thus we conclude that

$$E(x+y) = E(x) \cdot E(y).$$

Since $E(0) = 1$, this says in particular

$$E(-x)E(x) = 1 \quad E(-x) = 1/E(x)$$

Since $E(1) = e$ by definition of the number e , we see that for every natural number $E(n) = e^n$. and $E(-n) = 1/E(n) = 1/e^n = e^{-n}$ we see that for every integer, positive or negative, $E(n) = e^n$. Thus

$$E(x) = e^x \quad x \in \mathbb{Z}$$

For an integer $q \geq 1$

$$E(1/q) \cdot E(1/q) \cdots E(1/q) \quad (q \text{ times}) = E(1) = e.$$

By definition of q -th root, it follows that $E(1/q)$ is q -th root of e . That is,

$$E(1/q) = e^{1/q}; \quad q \in N$$

It now follows (see exponentiation 4 coming soon) that for every rational number r , $E(r) = e^r$. Thus

$$E(x) = e^x, \quad x \in Q.$$

We shall show later, that the equality holds not only for rationals but for all real numbers x . Incidentally note that $E(x) > 0$ for all x . For $x = 0$ its value is 1, For $x > 0$ it is clearly positive and in fact larger than one. For $x < 0$, the equation $E(x) = 1/E(-x)$ shows it is positive and in fact smaller than one.

The sine and cosine functions are also defined by series and are not, at this moment, recognizable as the good old functions of high school. However, using Cauchy product, one can show

$$\sin(x+y) = \sin x \cos y + \cos x \sin y; \quad \cos(x+y) = \cos x \cos y - \sin x \sin y.$$

We are already using $\sin x$ and $\cos x$ for $S(x)$ and $C(x)$.

(♠) Exponentiation 4: x^r , $x > 0$ and $r \in Q$:

In Exp 1 we defined x^n for $x \neq 0$ and $n \in N$ and studied its properties. In Exp 2 we defined x^n for $x \neq 0$ and $n \in Z$ and studied its properties. In Exp 3 we defined $x^{1/n}$ for $x > 0$ and $n \in N$ and studied its properties. Note that n -th roots are so far defined and shown to exist for $x > 0$. Now we define x^r for $x > 0$ and r rational (positive, zero or negative).

Definition: If $r = m/n$ where $n \geq 1$ and both m, n integers, we define

$$x^r = (x^{1/n})^m = (x^m)^{1/n}$$

In Exp 3 we already showed that the second equality above is true. Here are the properties.

1. This is a good definition. Since a rational number can be written in several ways ($1/2$ or $3/6$) we need to show that we get the same answer, no matter how we express (denominator always positive).

If $r = \frac{m}{n} = \frac{p}{q}$ where $n, q \geq 1$, we need to show $(x^m)^{1/n} = (x^p)^{1/q}$. If you raise

both sides to (nq) power you get the equation $x^{mq} = x^{np}$. For example, using the previous rules

$$[(x^m)^{1/n}]^{nq} = \{[(x^m)^{1/n}]^n\}^q = (x^m)^q = x^{mq}$$

since $mq = np$, we conclude $x^{mq} = x^{np}$ is correct as required.

2. $x^{r+s} = x^r x^s$.

express r, s with same denominator and use earlier laws.

3. $x^{-r} = (1/x)^r$.

Use earlier laws, we already know $x^{-1/n} = (1/x)^{1/n}$.

4. $(xy)^r = x^r y^r$

Use earlier laws.

5. Let $r < s$. For $x > 1$ we have, $x^r < x^s$ and for $0 < x < 1$ we have $x^r > x^s$. Indeed if $x > 1$, showing $x^r < x^s$ reduces to, by earlier rules, to $x^{s-r} > 1$. But if $s - r = m/n$ both $m, n \geq 1$ then we knew by properties of n -th roots etc that $x^{1/n} > 1$ and $[x^{1/n}]^m > 1$.

6. If $r_n \rightarrow r$ then $x^{r_n} \rightarrow x^r$.

Enough to show: when $r_n \rightarrow 0$ then $x^{r_n} \rightarrow 1$. Then apply this result to $r_n - r$ to see $x^{r_n - r} \rightarrow 1$, so that $x^{r_n - r} \cdot x^r \rightarrow 1 \cdot x^r$ or $x^{r_n} \rightarrow x^r$ where we used earlier laws.

So let $\epsilon > 0$. Since $\sqrt[n]{x} \rightarrow 1$ and $\sqrt[n]{1/x} \rightarrow 1$ fix K so that

$$1 - \epsilon < x^{1/K} < 1 + \epsilon \qquad 1 - \epsilon < x^{-1/K} < 1 + \epsilon$$

Since $r_n \rightarrow 0$ there is N so that $|r_n| < 1/K$ for $n \geq N$. Now take any $n \geq N$. $(-1/K) < r_n < (1/K)$ so by monotonicity of x^r we see x^{r_n} is between $x^{-1/K}$ and $x^{1/K}$. But both these are in $(1 - \epsilon, 1 + \epsilon)$ and hence so is x^{r_n} .

One should be careful, we said ‘monotonicity’, but there are four cases corresponding to $x > 1$, $0 < x < 1$ and $r > 0$, $r < 0$ (we are writing r for r_n). But in all cases, ‘in between’ phrase used above is valid.

For example, say $x > 1$ and $r = r_n > 0$. Then $1 < x^r < x^{1/K} < 1 + \epsilon$ so that $1 - \epsilon < x^{r_n} < 1 + \epsilon$ as required. Suppose $0 < x < 1$ and $r = r_n > 0$. Then $1 > x^r > x^{1/K} > 1 - \epsilon$ so that $1 + \epsilon > x^{r_n} > 1 - \epsilon$ as required. The case $r < 0$ uses similar steps, uses inequalities for $x^{-1/K}$. ■

Later we shall have a final go to complete the full definition of exponentiation.

infinities: Confusing but Convenient symbols

It is time to introduce the objects $+\infty$ and $-\infty$. This is only for convenience. You should keep in mind that every statement made using $\pm\infty$ can also be made, conveying the same meaning, but without using these symbols – just that we need to make the statement long using if/then etc.. These

are NOT real numbers. These are not part of our family, they belong to extended family. The reason I postponed is this: handling them is simple if you are disciplined, otherwise leads to confusions.

We start with the picture first as to how these objects fit with our picture of the real number system. We put the object $+\infty$ at the right end of the real number line and the object $-\infty$ at the left end of the real number line. Shall discuss how to operate with these objects and the rules to which we agree upon now. First, we make a notational agreement. Just as we write 4 for $+4$ and if we want to say negative 4, we write -4 , now also we do the same. Instead of writing $+\infty$, we just write ∞ . So when I write ∞ , you do not ask me which infinity: plus or minus — Just as, when I write 4, you do not ask me whether I mean $+4$ or -4 .

Rule 1 (order): We agree (as the picture suggests) $x < \infty$ for all $x \in R$ and $-\infty < x$ for all $x \in R$. We agree to say, as is sensible now, $-\infty < +\infty$.

Does this mean every non-empty set now is bounded above by ∞ ? NO. We say a non-empty set S is bounded above if there is a real number a such that $x \leq a$ for all $x \in S$. This is same as earlier definition. Earlier I did not say real number because we had only real numbers and nothing else. Similarly we say that a non-empty set S is bounded below if there is a real number b such that $b \leq x$ for all $x \in S$. DO NOT think every set is now bounded because $-\infty$ and ∞ are staring at you.

Every set bounded above has a supremum. So far, a set which is not bounded above has no supremum. Now we make a definition. A set which is not bounded above also has a supremum and it is ∞ . Please have discipline, and observe the subtlety, we are not saying ∞ is an upper bound, we are saying it is supremum. The reason for such a convention is that we want a set of real numbers to be bounded if it is bounded in R as earlier.

Similarly, so far a set which not bounded below has no infimum. Now we agree to say that a set which is not bounded below also has a infimum and it is $-\infty$.

Let S be a non-empty set.

(i) For a non-empty set bounded above, its supremum is the least upper bound. In symbols,

$$a = \sup S \leftrightarrow [\forall x \in S)(x \leq a)] \& [\forall x \in S, x \leq b \rightarrow a \leq b].$$

We continue to have this. Remember if S is a bounded set of real numbers the above statement is correct by definition. The symbols express just this

fact, namely, supremum is an upper bound and any other upper bound is larger than this. Thus if you take the set S of all numbers which are larger than 5, we now have ∞ to be its supremum. The right side of the above statement is still correct. Unfortunately (to repeat what is said above), we do not say, in words, that ∞ is least upper bound of S , though you are invited to imagine so.

(ii) In a similar manner for sets bounded below, infimum is its greatest lower bound. In symbols,

$$l = \inf S \leftrightarrow [\forall x \in S)(l \leq x)] \& [\forall x \in S, b \leq x \rightarrow b \leq l].$$

Comments similar to above apply. For example, $-\infty$ is infimum of the set S of all numbers smaller than 5. Of course we do not say that it is the greatest lower bound.

Also the characterization of supremum and infimum remain correct provided, we formulate carefully. For a bounded set S , the characterization was the following:

$$s = \sup S \leftrightarrow [\forall x \in S, x \leq s] \& [\forall \epsilon > 0, \exists x \in S, x > s - \epsilon].$$

An equivalent formulation, not using ϵ , is the following

$$s = \sup S \leftrightarrow [\forall x \in S, x \leq s] \& [\forall b < s, \exists x \in S, x > b].$$

This formulation is still correct even if the supremum of the set is ∞ .

Similarly,

$$l = \inf S \leftrightarrow [\forall x \in S, l \leq x] \& [\forall b > l, \exists x \in S, x < b].$$

when we say sequence we still consider sequence of real numbers. The philosophy is this: we do not want to invite infinity on our own, but it may enter in our calculations on its own and we then should know how to handle it.

We say $x_n \rightarrow \infty$ if $\forall A \in R \exists N \forall n > N (x_n > A)$ Thus whatever A is given, after some stage ALL terms must be larger than A . Similarly, we say $x_n \rightarrow -\infty$ if $\forall A \in R \exists N \forall n > N (x_n < A)$ Thus whatever A is given, after some stage ALL terms must be smaller than A . We define $\limsup x_n$ as the limit of the tail sups. Thus if the sequence is not bounded above all the tail sups are ∞ and so $\limsup$ is ∞ . Thus you can

see that $\limsup$ is ∞ iff the set is not bounded above.

We define $\liminf x_n$ as the limit of the tail inf's. Thus if the sequence is not bounded below all the tail inf's are $-\infty$ and so $\liminf$ is $-\infty$. Clearly, $\liminf$ equals $-\infty$ iff the set is not bounded below.

For these statements it is important to keep in mind that we are considering sequences of real numbers. For example, we do NOT consider the sequence $\infty, \infty, 1, 1, 1, 1, 1, \dots$. Theoretically one can but we do not entertain.

Yes, $\limsup$ is still the largest limit point. But then you need a definition. We define ∞ as limit point of $\{x_n\}$ if given any A , there are infinitely many n such that $x_n > A$. Similarly, we define $-\infty$ as limit point of $\{x_n\}$ if given any A , there are infinitely many n such that $x_n < A$. When you learn topology later all these concepts fit in beautifully. We shall not bother now.

Rule 2 (addition): $x + \infty = \infty$ for all $x \in R$ as well as for $x = \infty$. $x - \infty = -\infty$ for all $x \in R$ as well as for $x = -\infty$. We do not talk about $\infty - \infty$. The reason is simple, certain things that we know are true still remain true even with this convention. No meaning of $\infty - \infty$ will validate certain existing statements. Also $-(\infty) = -\infty$.

For example A and B are (non-empty) sets of real numbers with supremums a and b , then the set $C = \{x + y : x \in A, y \in B\}$ has supremum to be $a + b$. We knew this if the numbers a and b are real. It remains true even if a and b are infinities. The sets being non-empty, the supremum can not be $-\infty$. For example, suppose $a \in R$ and $b = \infty$. This means that the set B is not bounded above. It is easy to see that C is not bounded above. similarly, if $a = \infty = b$, then C is not bounded above. similar remarks apply to infimums.

It is still true that if $x_n \rightarrow x$ and $y_n \rightarrow y$, then $x_n + y_n \rightarrow x + y$ provided $x + y$ makes sense, in other words, provided x, y are not infinities of opposite signs. This is very easy to prove.

Rule 3 (multiplication): Define $x \times \infty = +\infty$ if $x > 0$ or $x = \infty$; whereas it equals $-\infty$ if $x < 0$ or $x = -\infty$. Simialrly, define $x \times (-\infty) = -\infty$ if $x > 0$ or $x = \infty$; whereas it equals ∞ if $x < 0$ or $x = -\infty$. What we did not define is $\infty \times 0$ and $(-\infty) \times 0$. Actually there are reasons to define them to be zero, but right now we do not define them.

Theorem: The number e is not rational.

Suppose it is rational, say p/q where $p \geq 1, q \geq 1$ are integers. Observe that $q \neq 1$. This is because by looking the first two terms in the series you know $e > 2$. While showing that the partial sums are bounded we showed that $e < 3$. Thus $q \geq 2$. Let $s_n, n \geq 0$ be the partial sums of the series defining e . Since $s_n \uparrow e$, we see

$$\begin{aligned} 0 < e - s_q &= \frac{1}{(q+1)!} + \frac{1}{(q+2)!} + \frac{1}{(q+3)!} + \cdots \\ &= \frac{1}{(q+1)!} \left[1 + \frac{1}{(q+2)} + \frac{1}{(q+2)(q+3)} + \cdots \right] \\ &\leq \frac{1}{(q+1)!} \left[1 + \frac{1}{(q+1)} + \frac{1}{(q+1)^2} + \cdots \right] = \frac{1}{q! q} \end{aligned}$$

so that

$$0 < q! (e - s_q) \leq \frac{1}{q}$$

But by hypothesis $q!e$ is an integer and visibly $q!s_q$ is also an integer, so that the middle term above is an integer. This is a contradiction because $q \geq 2$. ■

Let us complete our discussion of exponentiation. Why are we wasting our time by doing in several slow steps? This is to make us familiar and comfortable with exponentiation. For example you should be as comfortable with the following as you are with $7^{2+35} = 7^2 \times 7^{35}$:

$$\left(\sqrt[21]{7} \right)^{\sqrt[5]{2} + \sqrt[9]{35}} = \left(\sqrt[21]{7} \right)^{\sqrt[5]{2}} \times \left(\sqrt[21]{7} \right)^{\sqrt[9]{35}}$$

Exponentiation is in a sense difficult and boring too. Need to do it slowly and see each step is a simple extension of the previous step.

Step 1: $x \neq 0; x^n, n \in \mathbb{N}$

Step 2: $x \neq 0; x^n, n \in \mathbb{Z}$

Step 3: $x > 0; x^{1/n}, n \in \mathbb{N}$

Step 4: $x > 0; x^r, r \in \mathbb{N}$

Step 5: $x \geq 1; x^a, a \in \mathbb{R}$

Step 6: $x > 0; x^a, a \in \mathbb{R}$

We have completed the first four steps. We shall now conclude by carrying out the final two steps.

♠ Step 5: x^a for $a \in R$ and $x \geq 1$

DEF: $x^a = \sup\{x^r : r \leq a\}$. This set is non-empty because x^r is in this set when $r \in (a-1, a)$. This is also bounded above by x^s where s is any rational in $(a, a+1)$. Note that we are using the monotonicity property of x^r . This was made possible because $x \geq 1$. usually in proving statements below we take $x > 1$, but you are advised to check that even when $x = 1$ the statement being proved is true. It is easy to see $1^a = 1$ for all a .

1. Agrees with earlier definition if a is rational. Also x^a increases with a . This is because, by monotonicity, $x^r < x^a$ if $r < a$ and also x^a itself is in the set above when a is rational. For the second statement, note that if $a < b$ then $\{x^r : r \leq a\} \subset \{x^r : r \leq b\}$.

2. Let r_n any rationals, $r_n \uparrow a$. Then $x^{r_n} \rightarrow x^a$. Note that this limit exists because x^{r_n} is increasing and bounded above. If a is rational, this is proved in step 4 already. So let a be not rational. Since $r_n \leq a$, by definition of x^a we see $x^{r_n} \leq x^a$ for each n and hence $\lim x^{r_n} \leq x^a$. On the other hand if x^r is in the set defining x^a , we see $r < a$ (a being not rational), thus there is an k such that $r < r_k$ ($r_n \uparrow a$ and $r < a$). Thus $x^r < x^{r_k} \leq \lim x^{r_n}$. This being true for every x^r that belongs to the set which defines x^a , we see $x^a \leq \lim x^{r_n}$.

This completes the proof.

We could have used this property to define x^a , namely, define x^a as the limit of x^{r_n} where r_n are rationals increasing to a . But then we need to show that this definition does not depend on the sequence (r_n) taken. This is easy too because if $r_n \uparrow a$ and $s_n \uparrow a$ (a not rational) then for every n there is k such that $r_n < s_k$ and for every k there is n such that $s_k < r_n$.

3. $x^{a+b} = x^a x^b$. In particular $x^{-a} = 1/x^a$. Take $r_n \uparrow a$ and $s_n \uparrow b$, then $r_n + s_n \uparrow a + b$. Using step 4,

$$x^{a+b} = \lim x^{r_n+s_n} = \lim x^{r_n} x^{s_n} = \lim x^{r_n} \lim x^{s_n} = x^a x^b$$

For the last statement, note $x^{-a+a} = x^0 = 1$

4. $(xy)^a = x^a y^a$. Proof is exactly as above.

5. if $a_n \rightarrow a$ then $x^{a_n} \rightarrow x^a$ for each fixed $x \geq 1$.

This proof is no different from the proof in previous step for rationals. By laws of indices shown above, enough to show: $b_n \rightarrow 0$ implies $x^{b_n} \rightarrow 1$. Let $\epsilon > 0$. Choose K so that $x^{1/K} < 1 + \epsilon$ and $x^{-1/K} > 1 - \epsilon$. Thus, noting that $x > 1$, we have $1 < x^{1/K} < 1 + \epsilon$ and $1 > x^{-1/K} > 1 - \epsilon$. Now take N so that for $n > N$ we have $|a_n| < 1/K$, that is $(-1/K) < a_n < (1/K)$. Then increasing nature of the map $a \mapsto x^a$ tells us

$$1 - \epsilon < x^{-1/K} < x^{a_n} < x^{1/K} < 1 + \epsilon$$

6. For each fixed $a \geq 0$, x^a increases with x . Remember $x \geq 1$.

This is because $1 \leq x < y$ implies $x^n < y^n$ by induction on n . Also $x^{1/m} < y^{1/m}$ (we proved it earlier: if you say $x^{1/m} > y^{1/m}$, raise to m -th powers to get contradiction). Putting these two together we see that x^r increases with x , as long as $r \geq 0$. Now take any $x < y$,

$$\begin{aligned} x^a &= \sup\{x^r : r \leq a\} = \sup\{x^r : 0 \leq r \leq a\} \\ &\leq \sup\{y^r : 0 \leq r \leq a\} = \sup\{y^r : r \leq a\} = y^a \end{aligned}$$

Here, first and last equalities are definitions of x^a and y^a respectively. The second equality is because for fixed $x \geq 1$ we know x^r increases with r and hence sup can be taken over non-negative r . Same argument applies for the penultimate equality. the middle inequality follows from what we have argued before the display.

8: if $x_n \rightarrow x$ then $x_n^a \rightarrow x^a$ for each fixed a .

Enough to prove this for $a > 0$ because $a = 0$ is easy and $a < 0$ follows from using (see 3.)

$$x_n^a = 1/(x_n)^{-a} \rightarrow 1/(x)^{-a} = x^a$$

Do for integers $a = k$, then $a = 1/k$ and $a \in Q$ and $a > 0$ (check.)

$$8 \ (x^a)^b = x^{ab}$$

If both a, b are rationals this is done earlier. Let a be rational and b any thing. Then take rationals $r_n \uparrow b$. Then both a, r_n being rational,

$$(x^a)^{r_n} = x^{ar_n}$$

Take limit and use 5 above. Finally a, b be anything. Take rationals $s_n \uparrow a$ then what we proved shows

$$(x^{s_n})^b = x^{s_n b}$$

From 5 and then 8, left side converges to $(x^a)^b$ and right side converges to x^{ab} . ■

♠ Step 6: x^a for $a \in R$ and $x > 0$

If $x > 1$ the above clause defines x^a . If $x = 1$, we put $x^a = 1$ whatever be a . If $0 < x < 1$, we put $x^a = (1/x)^{-a}$. This makes sense because, $1/x > 1$ and above clause applies.

If you have survived so far, you can prove the following properties. if you can not prove, return and start from step 1 and understand.

$$x^a \uparrow \text{ as } a \uparrow \text{ for } x > 1; \quad x^a \downarrow \text{ as } a \uparrow \text{ for } 0 < x < 1.$$

$$\begin{aligned} a_n \rightarrow a &\Rightarrow x^{a_n} \rightarrow x^a; & x_n \rightarrow x &\Rightarrow x_n^a \rightarrow x^a. \\ x^{a+b} &= x^a x^b; & (xy)^a &= x^a y^a. & (x^a)^b &= x^{ab} \end{aligned}$$

Power series:

A series of the form

$$P(x) = a_0 + a_1x + a_2x^2 + \cdots; \quad a_n \in R,$$

is called a power series. You can think of it as infinite degree polynomial in the variable x . But we do not say so because according to our agreement, degree of a polynomial must be finite. If you choose a real number α and replace x by α (and so x^n by α^n) the it becomes a series of real numbers. It may converge for some values of α and not converge for others. Can it converge when $\alpha = 1, 3$ and NOT converge when $\alpha = 2$? No

Theorem: Consider the power series $P(x) = \sum_0^\infty a_n x^n$.

- (i) If $\limsup \sqrt[n]{|a_n|} = 0$ then the series converges for every real number x . The series converges absolutely.
- (ii) If $\limsup \sqrt[n]{|a_n|} = \infty$ (this means $\{\sqrt[n]{|a_n|}\}$ is unbounded), then the series fails to converge for every $x \neq 0$.
- (iii) If $0 < \limsup \sqrt[n]{|a_n|} < \infty$ then the series converges for every number x with $|x| < \limsup \sqrt[n]{|a_n|}$ and does not converge for any number x with $|x| > \limsup \sqrt[n]{|a_n|}$.

The quantity

$$r = \frac{1}{\limsup \sqrt[n]{|a_n|}}$$

is called radius of convergence of the power series. The convention here is that when the denominator above is zero then we take $r = \infty$ and when the denominator equals infinity we take $r = 0$. thus what the theorem says is

that if $x \in (-r, r)$ the series $\sum a_n x^n$ converges absolutely, that is, $\sum |a_n x^n|$ converges. Also if x is a real number such that $|x| > r$ then the series $\sum a_n x^n$ does not converge. When $x = \pm r$ it may converge or may not. For example

$$P(x) = 1 + x + x^2 + x^3 + \cdots$$

converges for $|x| < 1$ and does not converge at both the points $x = \pm 1$. On the other hand

$$P(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^5}{5} + \cdots$$

has radius of convergence one; converges for $x = -1$ but does not converge for $x = 1$.

Proof of the theorem is simple and we have done it. Root test says the series converges if $\limsup \sqrt[n]{|a_n x^n|} < 1$, that is, $|x| \limsup \sqrt[n]{|a_n|} < 1$ that is $|x| < 1/\limsup \sqrt[n]{|a_n|}$. In that case it converges absolutely. On the other hand if $\limsup \sqrt[n]{|a_n x^n|} > 1$ it does not converge, that is when $|x| > 1/\limsup \sqrt[n]{|a_n|}$. ■

Thus for a power series there is an interval, symmetric, around zero where it converges. Thus we can define a real valued function f on the interval $(-r, r)$ by defining $f(x)$ for $x \in (-r, r)$ as the value of the sum of the series $\sum a_n x^n$. Instead of using a new symbol, we use the same symbol P or $P(x)$. Thus when you see $P(x)$ you should make out from the context whether we are referring to the symbol $\sum a_n x^n$ or the function $x \mapsto P(x)$ on the interval $(x : |x| < r)$.

Why is r called ‘radius’? Because, if you study complex numbers $a + ib$ instead of real numbers; study convergence of sequences and series of complex numbers and if you study power series $P(x)$ where x is allowed to be complex number, then the above theorem is true and has a nice geometric interpretation. The series converges if you substitute for x any complex number $a + ib$ with $a^2 + b^2 < r$; does not converge for any complex number $a + ib$ with $a^2 + b^2 > r$. Here then is the geometry. If you think of complex number $a + ib$ (a, b real) as the point (a, b) in the coordinate plane you are familiar from high school, then you realize that the set of convergence described above is simply the disc with radius r having centre zero. Even though there are no new ideas in proof of this statement, we shall not do it because it needs a digression to complex numbers. This is better done in your complex analysis course when the splendour of complex functions unfolds.

Returning to power series, the exponential series, sine series and cosine series have $r = \infty$ so that those are defined on all of the real line. Such functions $P(x)$ defined on the interval of convergence have an interesting and useful property.

Theorem: $P(x)$ be power series with radius of convergence r . If $x_i \rightarrow x_0$ and if $|x_i| < r$ for all i and $|x_0| < r$, then the $P(x_i) \rightarrow P(x_0)$.

Proof is simple. Let $\epsilon > 0$ is given; we show i_0 such that $|P(x_i) - P(x_0)| < \epsilon$ for all $i \geq i_0$.

Wishful thinking: How nice if P were a polynomial, with all coefficients $a_n = 0$ after some stage, say for $n \geq 100$. High School calculation does! Because of properties of convergence you know $x_i^k \rightarrow x_0^k$ for $k = 1, 2, \dots, 100$ so that

$$P(x_i) = \sum_{n=0}^{100} a_n x_i^n \rightarrow \sum_{n=0}^{100} a_n x_0^n = P(x_0)$$

But of course this wish may not be true. Maybe there is a number, say 100, so that for all $i = 0, 1, 2, \dots$ we have $|\sum_{n=100}^{\infty} a_n x_i^n|$ is small, Then the infinite sums are essentially finite sums with small error.

Proof execution: There are two indices i and n , Read carefully.

Since $-r < x_0 < r$, Take $\delta > 0$ so that $[x_0 - \delta, x_0 + \delta] \subset (-r, r)$. Let $c = \max(|x_0 - \delta|, |x_0 + \delta|)$. Then $0 < c < r$ and whenever $y \in [x_0 - \delta, x_0 + \delta]$ we have $|y| \leq c$. And because the series $P(x)$ is absolutely convergent in $(-r, r)$, the series $\sum |a_n| c^n$ is convergent. Fix k so that $\sum_k^{\infty} |a_n| c^n < \epsilon/4$. Since $x_i \rightarrow x_0$, fix i_1 so that $|x_i - x_0| < \delta$ for $i \geq i_1$. Since for $i = 0$ or $i \geq i_1$ we see $x_i \in [x_0 - \delta, x_0 + \delta]$ we conclude that

$$\left| \sum_k^{\infty} a_n x_i^n \right| \leq \sum_k^{\infty} |a_n| c^n < \epsilon/4; \quad i = 0, \quad i \geq i_1 \quad (\clubsuit)$$

Thus for $i \geq i_1$,

$$\begin{aligned} |P(x_i) - P(x_0)| &= \left| \sum_0^{k-1} a_n x_i^n + \sum_k^{\infty} a_n x_i^n - \sum_0^{k-1} a_n x_0^n - \sum_k^{\infty} a_n x_0^n \right| \\ &\leq \left| \sum_0^{k-1} a_n x_i^n - \sum_0^{k-1} a_n x_0^n \right| + \left| \sum_k^{\infty} a_n x_i^n \right| + \left| \sum_k^{\infty} a_n x_0^n \right| \\ &\leq \left| \sum_0^{k-1} a_n x_i^n - \sum_0^{k-1} a_n x_0^n \right| + \frac{\epsilon}{4} + \frac{\epsilon}{4} \end{aligned}$$

where we used ($\clubsuit$). Now choose i_0 so that for $i \geq i_0$ the first term above is smaller than $\epsilon/2$, possible because we have finite sums and the first sum converges to the second as $i \rightarrow \infty$. If necessary taking a larger value we can safely assume $i_0 > i_1$. Clearly now for $i \geq i_0$ we have

$$|P(x_i) - P(x_0)| < \frac{\epsilon}{2} + \frac{\epsilon}{4} + \frac{\epsilon}{4} = \epsilon$$

This completes the proof. ■

Continuous functions

There are several interesting things one could discuss about sequences and series. But we should stop our discussion on series to proceed to other stories. We shall now discuss continuity of functions. You are already familiar probably. We shall deal with functions defined on R or subsets of R and taking values in R .

We define, for real numbers $a < b$ the set $[a, b] = \{x : a \leq x \leq b\}$. This is called closed bounded interval. The set $(a, b) = \{x : a < x < b\}$ is called an open bounded interval. Thus in a closed bounded interval the end points are in the set. For open bounded interval the end points are NOT in the set. (Yes, later I tell you what is an interval, what is closed or open about it etc.) We can also talk about $(a, b] = \{x : a < x \leq b\}$ and $[a, b) = \{x : a \leq x < b\}$.

So let us start with a function f defined on a closed bounded interval $[a, b]$. When should we agree to say that it is continuous. Yes, it is continuous if it is continuous at each point z in this interval. So when shall we say that the function is continuous at a point z . The feeling was that you should be able to draw its graph ‘continuously’ without lifting your pen. That is, there should be no abrupt changes. Idea is the following. If you change z a little bit, then the value also should change only a little bit, not too much. Or equivalently, if x is close to z , then $f(x)$ should be close to $f(z)$. We have to make this precise. This is what we shall do now.

Theorem: Let $f : [a, b] \rightarrow R$ and $z \in [a, b]$. Then the following two statements are equivalent.

- (i) $x_n \rightarrow z$ implies $f(x_n) \rightarrow f(z)$ where all points x_n are in $[a, b]$.
- (ii) Given $\epsilon > 0$, we can find a $\delta > 0$ so that $|f(x) - f(z)| < \epsilon$ whenever $|x - z| < \delta$ and $z \in [a, b]$

Proof: Let (ii) hold. Let $x_n \rightarrow z$. Fix $\epsilon > 0$. We exhibit N so that $|f(x_n) - f(z)| < \epsilon$ for $n \geq N$. Go to the hypothesis (ii) and get $\delta > 0$ for

this ϵ . Choose N so that $|x_n - z| < \delta$ for $n \geq N$. But then, for these n , we have $|f(x_n) - f(z)| < \epsilon$.

Let (ii) fail. We show (i) fails by producing a sequence $x_n \rightarrow z$ but $f(x_n) \not\rightarrow f(a)$. Since (ii) is false, there is an $\epsilon > 0$ for which we cannot find $\delta > 0$ satisfying the condition of (ii). Fix such an $\epsilon > 0$. We will not change this ϵ . Using the fact that $\delta = 1/n$ does not fulfil condition of (ii), we can fix an $x_n \in [a, b]$ such that $|x_n - z| < 1/n$ and yet $|f(x_n) - f(z)| \geq \epsilon$. Clear that $x_n \rightarrow a$ but $f(x_n) \not\rightarrow f(a)$.

Definition: A function $f : [a, b] \rightarrow R$ is continuous at a point z if any one of the above conditions holds at the point z . We say that f is continuous if it is continuous at every point z .

The two conditions are interesting. Condition (i) allows us to verify discrete instances. That is, to verify this condition we fix any arbitrary sequence $x_n \rightarrow z$ and verify $f(x_n) \rightarrow f(z)$. Of course there are uncountably many sequences converging to z , but that is a different matter. Whenever you set to verify *one* instance, it is a discrete sequence. On the other hand condition (ii) is a non-discrete condition. Even for one instance, you fix $\epsilon > 0$ and produce a δ ; but this delta should verify something for every x with $z - \delta < x < a + \delta$, namely, for every such x we must verify $-\epsilon < f(x) - f(z) < \epsilon$. Remember this verification is to be done for every x in an interval. However, both are equivalent. Sometimes (i) and sometimes (ii) would be handy.

Many times we deal with functions f defined on (a, b) or on $(a, b]$ etc. In such a case we say that f is continuous at a point z in this interval if any of the following two equivalent conditions holds:

- (i) If x_n is in our interval for each n and $x_n \rightarrow z$, then $f(x_n) \rightarrow f(z)$.
- (ii) Given $\epsilon > 0$, we can find a $\delta > 0$ so that $|f(x) - f(z)| < \epsilon$ whenever $|x - z| < \delta$ and x is in our interval.

That these two are equivalent is seen with same argument as above. In other words the requirements should hold only for points in the interval. Later we shall discuss functions defined on subsets of R which are not necessarily intervals.

Sometimes the notion of continuity is defined after defining limits of functions. One defines righthand limit, lefthand limit at z and says that the function is continuous at z if both righthand limit and lefthand limit exist and equal $f(z)$. I find it difficult to begin with these concepts, which demand understanding continuous limits. It is not to say that these concepts are unimportant. They are simple and important. After you feel comfortable

with the existing concepts and assimilate them, you would have no problem discovering them yourself. We will do that sometime.

The collection of all real valued functions on a set S has a nice structure. If f and g are such functions, you can define the function $h(x) = f(x) + g(x)$ on the set S . This function is simply denoted by $f + g$. Similarly fg denotes the function whose value at $x \in S$ is $f(x)g(x)$. The function $55f$ is the function whose value at $x \in S$ is $55f(x)$. The function $\frac{f}{g}$ or f/g has value $\frac{f(x)}{g(x)}$ at $x \in S$; but now this function may no longer be defined on all of S . It is defined only for those points $x \in S$ such that $g(x) \neq 0$. The domain of this function may not be all of S .

Thus remember sum, product etc of functions is defined by us. On the other hand, the set of real numbers R is a set with operations of addition and multiplication defined on the set satisfying certain conditions. The set of all real valued functions defined on S did not come with such operations. We defined those operations.

Theorem: If $f : R \rightarrow R$ and $g : R \rightarrow R$ are continuous functions, then so are $f + g$, fg and $29f$. The function f/g is also continuous if $g(x) \neq 0$ for all $x \in R$.

Proof follows from properties of sequences that we know.

The above remains if we had functions defined only on an interval. We should say that $g(x) \neq 0$ for all x in the interval if we want to say about f/g . Fact: The following are continuous from R to R .

$$f(x) \equiv 49; \quad f(x) = x; \quad f(x) = x^{100}.$$

More generally, every polynomial is a continuous function.

Theorem: Let f be a real valued continuous function defined on a closed bounded interval $[a, b]$. Then f is bounded, that is there is a number C such that $|f(x)| \leq C$ for all x . In fact, there are points x_0 and x_1 in $[a, b]$ so that $f(x_0) \leq f(x) \leq f(x_1)$ for all x .

Thus the function is not only bounded but attains its minimum and maximum values at the points x_0 and x_1 respectively.

Proof: Let us start with an observation. If we proved that every continuous function is bounded, then the second part follows. Indeed, let $\alpha = \inf\{f(x) : a \leq x \leq b\}$. If there is no point where the value of f equals α , then $g = 1/(f - \alpha)$ is well defined because denominator is never zero

and also continuous. This is not bounded, because for any $n \geq 1$, there is a point x with $f(x) < \alpha + (1/n)$ so that $g(x) > n$. This contradicts the fact continuous functions are bounded. Thus there is a point x_0 so that $f(x_0) \leq f(x)$ for all x . Similarly, there is an x_1 so that $f(x) \leq f(x_1)$ for all x .

Let us now show that f is bounded. Let $I_1 = [a, b]$. If f is not bounded, then it is not bounded either on the left half $[a, (a+b)/2]$ or on the right half $[(a+b)/2, b]$ of $[a, b]$. Let it be I_1 . If f is unbounded on both halves, take the left half. Since f is unbounded on I_1 let I_2 be the left half of I_1 if f is unbounded there, otherwise right half. In this way [actually, I should use induction: if I have defined intervals upto $n-1$, nested and length of each is half of previous one and f unbounded on all these; then we produce n -th interval such that etc holds.] we get a sequence of closed bounded intervals with lengths decreasing to zero and hence by Cantor's theorem will have exactly one point, say α , in common. Clearly $\alpha \in [a, b]$. Since f is continuous on $[a, b]$ at the point α , get $\delta > 0$ so that $|f(x) - f(\alpha)| \leq 1$ for all $x \in [a, b]$ with $|x - \alpha| < \delta$. In particular, on the interval $(\alpha - \delta, \alpha + \delta)$ the function is bounded by $|f(\alpha)| + 1$. Pick k so that length (I_k) is smaller than $\delta/4$. since $\alpha \in I_k$ we see $I_k \subset (\alpha - \delta, \alpha + \delta)$. In other words f is bounded on I_k . But by choice f is unbounded on all the intervals I_n . This contradiction proves our result.

What if the function is not defined on a closed bounded interval? Clearly, you can not take your interval S to be unbounded. If S is unbounded, then $f(x) = x$ is a continuous function on S which is not bounded. Of course, if it is defined on an open interval (a, b) , then it need not be bounded. For instance, the function $f(x) = 1/(x-a)$ tells you this.

We shall now prove another fundamental theorem.

Theorem: Let f be continuous on a closed bounded interval $[a, b]$. Suppose $f(a) = \alpha < \beta = f(b)$. Let $\alpha < \gamma < \beta$. Then there is a point z in the interval such that $f(z) = \gamma$

This is called **Intermediate Value Theorem**. If the function assumes two values, it assumes all values in between. It is unimportant that we assumed $\alpha < \beta$; even if $\alpha > \beta$, same happens. Either use the similar proof or apply this to the function $-f$ and the point $-\gamma$.

Here is the proof: Let $S = \{x \in [a, b] : f(x) < \gamma\}$.

Is this non-empty? Yes $a \in S$. Is it bounded? Yes, bounded by b . Let $z = \sup S$.

Can $z = a$? No. If $\epsilon = (\gamma - \alpha)/4$, then, by continuity of f , there is a $\delta > 0$

such that for $x \in [a, a + \delta)$ we have $|f(x) - f(a)| < \epsilon$. Thus

$$x \in [a, a + \delta) \Rightarrow f(x) < f(a) + \epsilon = \alpha + \frac{\gamma - \alpha}{4} < \gamma.$$

Thus $[a, a + \delta) \subset S$. So $\sup S$ can not be a .

Can $z = b$? No. Now take $\epsilon = (\beta - \gamma)/4$ and get $\delta > 0$ so that

$$x \in (b - \delta, b] \Rightarrow f(x) > f(b) - \epsilon = \beta + \frac{\beta - \gamma}{4} > \gamma.$$

Thus $S \subset [a, b - \delta]$ and $\sup S$ can not be b . Thus $a < z < b$. From now on any δ that we get is assumed to be smaller than $\min(z - a, b - z)$; if necessary take a smaller value.

Is $f(z) < \gamma$? No. If $f(z) < \gamma$, taking $\epsilon = (\gamma - f(z))/4$, you can get $\delta > 0$ such that for points $x \in (z - \delta, z + \delta)$ we have $f(x) < f(z) + \epsilon < \gamma$. In other words $(z, z + \delta) \subset S$ and z could not have been $\sup S$.

Is $f(z) > \gamma$? No. If $f(z) > \gamma$, taking $\epsilon = (f(z) - \gamma)/4$, you can get $\delta > 0$ such that for points $x \in (z - \delta, z + \delta)$ we have $f(x) > f(z) - \epsilon > \gamma$. In other words $f(x) > \gamma$ on $(z - \delta, z]$. By saying $z = \sup S$ we accepted that S contains no point above z . The present inequality tells that points of $(z - \delta, z]$ are also not in S . Thus z could not have been $\sup S$.

Thus $f(z) = \gamma$ and the proof is complete.

About power series: Some of you were wondering if the theorem is true when the limit point is boundary point. That is, if $P(x)$ converges for $x = r$ and if $x_i \rightarrow r$ then whether $P(x_i) \rightarrow P(r)$. Yes it does, but needs different proof. In our proof we have crucially used all x_i and z are in $(-r, r)$ and absolute convergence of the power series. The hypothesis $P(x)$ converges at r does not tell us that the series $\sum a_n r^n$ converges absolutely. May not be true.

Some of you were wondering the $\epsilon/2, \epsilon/4$ etc business. Can we not just use ϵ all the time and finally end up with 4ϵ or 8ϵ , whatever, and say proof is complete? Yes, can be done. You may feel what is the difference any 4ϵ is also small. Yes true But however if you promised to show that some thing is smaller than ϵ then you better show that rather than something else. If you did something else you need to justify how that fulfils your promise – Matter of Discipline. Later when you become expert, then you have this freedom.

Theorem: f and g are two continuous functions defined on an interval I . If $f(r) = g(r)$ for all rationals in I , then $f(x) = g(x)$ for all $x \in I$. That is $f \equiv g$.

The main content of the theorem is that to show two functions are same you need not show values are same at each point; just show that values agree at rational points – provided your functions are both continuous.

Recall, by interval we mean $[a, b]$ or (a, b) or $[a, b)$ or $(a, b]$ with $a < b$. When $a = b$, the interval $[a, b]$ reduces to one point while the other intervals are empty. Unless stated explicitly, we do not consider this case when we say interval. Also when we say function, we mean real valued function unless we specifically say function into something else.

Proof is simple. Given x there are rationals $r_n \rightarrow x$ so that $f(x) = \lim f(r_n) = \lim g(r_n) = g(x)$ where first and last equalities are by continuity and the middle is hypothesis.

This theorem remains true even if we consider unbounded intervals like $(a, \infty) = \{x : x > a\}$ or $(-\infty, a) = \{x : x < a\}$ or $[a, \infty) = \{x : x \geq a\}$ or $(-\infty, a] = \{x : x \leq a\}$ or even $(-\infty, \infty) = \mathbb{R}$.

Theorem: Fix $x > 0$. Consider $f(a) = x^a$. Then $f : \mathbb{R} \rightarrow (0, \infty)$.

- (i) f is a continuous function satisfying two conditions: $f(a+b) = f(a)f(b)$.
- (ii) Suppose that f is any continuous function on $\mathbb{R}$ to $\mathbb{R}$ such that $f(a+b) = f(a)f(b)$ holds for all $a, b \in \mathbb{R}$. Then f necessarily takes values in $[0, \infty)$. Further, Either it is zero for all a or it is never zero. In the second case, it must be one of the functions $f(a)$ listed above, namely, $f(a) \equiv x^a$ with $x = f(1)$.
- (iii) If $x > 1$, then $f(a) \rightarrow \infty$ as $a \rightarrow \infty$; $f(a) \rightarrow 0$ as $a \rightarrow -\infty$. This means the following. Given any number c , there is A so that $f(a) > c$ for all $a \geq A$. Similarly, given any number $\epsilon > 0$ there is an A so that $f(a) < \epsilon$ for all $a \leq A$.

If $x < 1$, then $f(a) \rightarrow 0$ as $a \rightarrow \infty$; $f(a) \rightarrow \infty$ as $a \rightarrow -\infty$. This means the following. Given any number c , there is A so that $f(a) > c$ for all $a \leq A$. Similarly, given any number $\epsilon > 0$ there is an A so that $f(a) < \epsilon$ for all $a \geq A$.

Proof: (i) Continuity was already shown above. The equation is just law of indices.

- (ii) Since for any a , $f(a) = f(\frac{a}{2} + \frac{a}{2}) = [f(a/2)]^2$ we see that $f(a) \geq 0$

for every $a \in R$. Suppose that $f(b) = 0$ for some b . Then for any a ; $f(a) = f(b)f(a-b) = 0$. So either $f \equiv 0$ or f is never zero.

Let us now assume f is never zero. Also $f(0) = f(0+0) = f(0)f(0)$ Thus if $f(0) = c$ then $c = c^2$. Since $f(0) \neq 0$ we have $f(0) = 1$. Let $f(1) = x > 0$. $f(a+b) = f(a) + f(b)$ will now yield $f(n) = x^n$ for all $n \in N$ and also for all $n \in Z$ using $f(a)f(-a) = f(0) = 1$. Since $(1/2) + (1/2) = 1$ we see $[f(1/2)]^2 = x$ and since $f(1/2) > 0$ we conclude that $f(1/2)$ must be $\sqrt{x}$. You can now show that $[f(1/n)]^n = f(1)$ and since $f(1/n) > 0$ conclude that $f(1/n) = x^{1/n}$. Now it follows that $f(r) = x^r$ for every $r \in Q$. Since f is given to be continuous and $a \mapsto x^a$ is shown to be continuous function and since they agree at every $r \in Q$, we conclude that they agree at every $a \in R$. (iii) If $x > 1$, then we knew, $x^n \rightarrow \infty$ and hence by (monotonicity in a) we conclude $x^a \rightarrow \infty$ in the sense described above. The part $a \rightarrow -\infty$ follows from noting $x^a = (1/x)^{-a}$ and again monotonicity. The case $0 < x < 1$ is similar. ■

We can regard x^a as a function of a for each fixed $x > 0$ defined on all of R ; or as a function of x defined on $(0, \infty)$ for each fixed $a \in R$.

Theorem: Let $a \in R$. Define $g(x) = x^a$. Then $g : (0, \infty) \rightarrow (0, \infty)$ is continuous and satisfies $g(xy) = g(x)g(y)$.

The last part is law of indices. We take $a > 0$. Proof for $a < 0$ is similar; or you can observe that the map $x \mapsto x^a$ is composition of $x \mapsto (1/x)$ and $(1/x) \mapsto (1/x)^{-a}$. For $a = 0$, the map $x^0 \equiv 1$ constant map and so continuous.

Need to show $x_n \rightarrow x$ implies $x_n^a \rightarrow x^a$, that is $(x_n/x)^a \rightarrow 1$. (Subtle point used: $(x_n/x)^a x^a = [(x_n/x)x]^a = x_n^a$ so that $(x_n/x)^a = x_n^a/x^a$.) Since $(x_n/x) \rightarrow 1$; what we need to show finally amounts to the following: $y_n \rightarrow 1$ implies $y_n^a \rightarrow 1$.

Fix $\epsilon > 0$. we need to show N so that $1 - \epsilon < y_n^a < 1 + \epsilon$ for $n \geq N$. Fix integer $K > a$. Since $(1 - \epsilon)^{1/K} < 1 < (1 + \epsilon)^{1/K}$ and since $y_n \rightarrow 1$, choose N so that

$$n \geq N \Rightarrow (1 - \epsilon)^{1/K} < y_n < (1 + \epsilon)^{1/K}$$

Let $n > N$. In case $y_n \geq 1$ we have using $a < K$

$$1 \leq y_n^a \leq y_n^K < 1 + \epsilon$$

and in case $y_n \leq 1$ we have using $a < K$

$$1 \geq y_n^a \geq y_n^K > 1 - \epsilon$$

Thus in either case we have $1 - \epsilon < y_n^a < 1 + \epsilon$; completing the proof. ■

Suppose, instead of the general x^a ; you want only to prove that $x \mapsto \sqrt[k]{x}$ is continuous, on $(0, \infty)$ where $k \geq 2$ is an integer. Should we go through all the above. No need to do. You can argue as follows. Take $x_n \rightarrow x$. The sequence $(y_n) = (\sqrt[k]{x_n})$ is bounded, because (x_n) is bounded. So it has at least one limit point. Let z be a limit point. Take a subsequence, say $y_{n_1}, y_{n_2}, \dots$ converging to z . Their k -th powers must converge to z^k . Here we used $a \mapsto a^k$ is a continuous function. But their k -th powers form a subsequence of (x_n) and hence must converge to x . Thus $z^k = x$. In particular, $z > 0$, because $x > 0$. By definition of k -th root, we see that $z = x^{1/k}$. Thus any limit point of the bounded sequence (y_n) is $x^{1/k}$. So it has only one limit point and hence (remember bounded sequence) must converge to it, completing the proof.

Here is another way of proving the above result. This method will be useful in other situations. Let us prove the theorem first and apply to the current problem. We see other uses too.

Theorem: Let $f : (0, \infty) \rightarrow (0, \infty)$ be a continuous strictly increasing function which is onto. Then for every $y \in (0, \infty)$ there is a unique $x \in (0, \infty)$, to be denoted $g(y)$ such that $f(x) = y$. In other words g is inverse of f . Moreover g is a strictly increasing continuous function on $(0, \infty)$. The hypothesis ‘onto’ holds if for example; $f(1/n) \rightarrow 0$ and $f(n) \rightarrow \infty$.

Proof: The intermediate value theorem yields the last sentence. Since f is strictly increasing, it is one-to-one map. Thus there is an inverse map g as claimed. If $g(y_1) < g(y_2)$ then $y_1 = f(g(y_1)) < f(g(y_2)) = y_2$. So g is strictly increasing. Pause and think, we used proof by contradiction.

To see that g is continuous, let $b > 0$ and $\epsilon > 0$ be given. Let $g(b) = a$. Need to show $\delta > 0$ so that $g(y)$ is in the interval $(g(b) - \epsilon, g(b) + \epsilon) = (a - \epsilon, a + \epsilon)$ whenever y is in $(b - \delta, b + \delta)$. There is no loss to assume that $a - \epsilon > 0$ (take a smaller ϵ and remember $a = g(b) > 0$). Since $a - \epsilon < a$ and $a + \epsilon > a$, use f is strictly increasing, and $f(a) = b$ to get $\delta_1, \delta_2 > 0$ so that $f(a - \epsilon_1) = b - \delta_1$ and $f(a + \epsilon) = b + \delta_2$. Clearly the strict increasing nature of f tells you that $f(x) \in (b - \delta_1, b + \delta_2)$ for $x \in (a - \epsilon, a + \epsilon)$. Equivalently, when $y \in (b - \delta_1, b + \delta_2)$ we have $g(y) \in (a - \epsilon, a + \epsilon)$. Take $\delta = \min\{\delta_1, \delta_2\}$. This will do. ■

If you apply the theorem above to the function $f(x) = x^k$, you see that $g(y) = y^{1/k}$ is a continuous function. Thus existence of k -th roots and the fact that k -th root map on $(0, \infty)$ is a continuous map follow painlessly. But you do not want to wait til you develop continuous functions etc to say square

roots and cube roots exist.

In the theorem above, it is not necessary to have the interval $(0, \infty)$. Any two intervals will do. For example, $f : [a, b] \rightarrow [c, d]$ strictly increasing onto continuous map then there is an inverse $g : [c, d] \rightarrow [a, b]$ and it is continuous. Exactly the same proof works. Or $f : (a, b) \rightarrow (c, d)$ is strictly increasing onto continuous map, then similar conclusion holds. The essential part of the conclusion is continuity. Inverse is purely set theoretic because we are already assuming that the function is one-one and onto. The intervals could be unbounded too.

Theorem: Let f be continuous strictly increasing function on R onto $(0, \infty)$. Then it has an inverse; $g : (0, \infty) \rightarrow R$ and g is continuous.

Proof is as above. In particular applying the theorem to the function $f(x) = e^x$ we have:

Theorem: There is a continuous function $L : (0, \infty) \rightarrow R$ such that $L(e^x) = x$ for every $x \in R$ and $e^{L(x)} = x$ for every $x \in (0, \infty)$.

Remember $x \mapsto e^x$ is a continuous function; $e^x > 0$ for all x and $e^x \rightarrow 0$ as $x \rightarrow -\infty$ and $e^x \rightarrow \infty$ as $x \rightarrow \infty$ and so e^x assumes all values by intermediate value theorem. Also e^x is strictly increasing.

This inverse function is called logarithmic function, denoted $\log x$. observe we are not using different variables x and y ; we are using x for both exponential and log. When we say e^x then x could be any number, but when we say $\log x$ then x should be positive. The rules of exponential function are reflected in logarithmic function.

$$\log(xy) = \log x + \log y \text{ (for } x, y > 0\text{)}$$

If $\log x = a$ and $\log y = b$, then the above is just a restatement of the law we proved: $e^{a+b} = e^a e^b$.

$$\log(x^a) = a \log x$$

Indeed, if $\log x = b$ then $e^b = x$ so that $x^a = (e^b)^a = e^{ab}$ or $\log(x^a) = ab = a \log x$.

$\log$ is an increasing function on $(0, \infty)$;

and $\log x \rightarrow -\infty$ as $x \rightarrow 0$; and $\log x \rightarrow \infty$ as $x \rightarrow \infty$;

and $\log 1 = 0$; and $\log x < 0$ on $(0, 1)$ and $\log x > 0$ on $(1, \infty)$.

Remember the logarithmic function is NOT defined for $x \leq 0$.

Consider the symbol x^a . If we fix $x > 0$ then you can regard this as a

function of a on R , namely $f(a) = x^a$. we knew it is continuous on R , that is $a_n \rightarrow a$ implies $x^{a_n} \rightarrow x^a$. If we fix $a \in R$ then you can regard this as a function of x , now only on $(0, \infty)$. We knew this is continuous on $(0, \infty)$, that is $x_n \rightarrow x$ (all strictly positive) then $x_n^a \rightarrow x^a$. Interesting question: If $x_n \rightarrow x$ (all positive) and if $a_n \rightarrow a$ then can we say $x_n^{a_n} \rightarrow x^a$? You might think that the previous two observations (fixing a , fixing x etc) should give this. But it will not immediately give. However the statement is true. Simply because $\log x_n \rightarrow \log x$ by continuity of $\log$; since $a_n \rightarrow a$ we see $a_n \log x_n \rightarrow a \log x$; since e^y is continuous function of y we see $e^{a_n \log x_n} \rightarrow e^{a \log x}$; that is $x_n^{a_n} \rightarrow x^a$.

composition:

Suppose we have a continuous f on an interval I taking values in an interval J . Suppose we have a continuous function g on J . Then we can define the composition: $h(x) = g(f(x))$ for $x \in I$. then h is a continuous function. This is easy. If $x_n \rightarrow x$ in I , then $f_n(x) \rightarrow f(x)$ in J by continuity of f . And now $g(f_n(x)) \rightarrow g(f(x))$ by continuity of g . That is $h_n(x) \rightarrow h(x)$. For example the function $\cos(e^{(\sin x)^2})$ is a continuous function.

Uniform Continuity:

A function defined on an interval I is continuous if it is continuous at every point, that given any point $a \in I$, it is continuous at a ; that is given a and $\epsilon > 0$ there is a $\delta > 0$ such that $x \in I$ and $a - \delta < x < a + \delta$ implies $f(a) - \epsilon < f(x) < f(a) + \epsilon$. Clearly the δ depends not only on ϵ but also on the point a where you are verifying continuity.

Example: $I = [0, 55]$ and $f(x) = x^2$.

This is continuous, we know already. Given any two points $x, y \in I$

$$|f(x) - f(y)| = |(x - y)(x + y)| \leq 110|x - y|$$

Thus given any ϵ , if we choose $\delta = \epsilon/110$ then this choice will work for all points a . That is, If you now take any point $a \in I$ then $|f(x) - f(a)| < \epsilon$ when $|x - a| < \delta$ and $x \in I$.

Example: $I = (0, \infty)$ and $f(x) = x^2$.

This is continuous. Now you can not find a $\delta > 0$ that works for all points. I give you $\epsilon = 1$. Suppose you give me any $\delta > 0$. I shall show you a point a for which this does not work, this means there is x with $a - \delta < x < a + \delta$ but $|f(x) - f(a)| > 1$. Take $a = \frac{2}{\delta}$ and $x = \frac{2}{\delta} + \frac{\delta}{2}$. Then

$$|x - a| = \frac{\delta}{2} < \delta; \quad |f(x) - f(a)| = \left(\frac{2}{\delta} + \frac{\delta}{2}\right)^2 - \left(\frac{2}{\delta}\right)^2 > 2$$

You may think that the failure to get one δ that works for all x is due to the fact that the interval is unbounded. This is not true.

Example: $I = (0, 1)$ and $f(x) = 1/x$

If I give $\epsilon = 1$ again, there is no single δ that works for all points. This is immediate because the function is unbounded near one, so what ever $\delta > 0$ is given we can find x, a in $(0, \delta)$ with $|f(x) - f(a)| > 1$. We can even have examples where The function is bounded and the interval is also bounded.

Example: $I = (0, 1)$ and $f(x) = \sin(1/x)$

Here the graph oscillates between -1 and $+1$ in every interval $(0, \delta)$.

However if we have a continuous function on a closed bounded interval $[a, b]$ then this cannot happen. First a definition.

Definition: A function f defined on an interval I is said to be uniformly continuous on I if given $\epsilon > 0$ there is a $\delta > 0$ such that if $x, y \in I$ and $|x - y| < \delta$ then $|f(x) - f(y)| < \epsilon$.

Theorem: A continuous function f defined on a closed bounded interval $[a, b]$ is uniformly continuous.

Proof is simple. Let f be continuous on $[a, b]$. Let $\epsilon > 0$ be given. Suppose the conclusion is false. So you can not find $\delta > 0$ as required. Take any integer $n \geq 1$, then $\delta = 1/n$ does not work. So you can pick x_n, y_n in $[a, b]$ so that $|x_n - y_n| < 1/n$ but $|f(x_n) - f(y_n)| \geq \epsilon$

Now consider the sequence (x_n) . Wishful thinking: $x_n \rightarrow z$. Then $a \leq x_n \leq b$ tells you $a \leq z \leq b$ so that $z \in [a, b]$. Also $|y_n - z| \leq |y_n - x_n| + |x_n - z| \rightarrow 0$. In other words $y_n \rightarrow z$ too. By continuity $f(x_n) \rightarrow f(z)$ and $f(y_n) \rightarrow f(z)$ so that $f(x_n) - f(y_n) \rightarrow 0$. But this contradicts that $|f(x_n) - f(y_n)| \geq \epsilon$ for every n .

What do we do if our wish is not true? Since the sequence (x_n) is bounded it has a limit point. Take one, say, z . So we have a subsequence converging to z . Say $k_1 < k_2 < \dots < k_n \dots$ such that $x_{k_n} \rightarrow z$. Exactly the same argument above shows that $y_{k_n} \rightarrow z$ and $f(x_{k_n}) - f(y_{k_n}) \rightarrow 0$ where as $|f(x_{k_n}) - f(y_{k_n})| \geq \epsilon$ for every n .

Completes the proof ■

Convergence of sequences of functions:

Suppose that we have a sequence of continuous functions (f_n) defined on an interval I . Suppose that for every point $x \in I$, the sequence of numbers $\{f_n(x)\}$ converges to a number $\lim f_n(x)$. Let us denote this number by $f(x) = \lim f_n(x)$. Thus we have function f on I . Many times we will be

interested in knowing if this limit function f is continuous. That it is not always so is easy to see.

example: $I = [0, 1]$; $f_n(x) = x^n$. Take $f(x) = 0$ for $0 \leq x < 1$ while $f(1) = 1$. Then $f_n(x) \rightarrow f(x)$ for each x . But clearly f is not continuous.

Before we proceed, you should remember one reason to discuss convergence (anywhere) is to locate objects with required properties in a given set using known objects. For example if the set is R then points are real numbers and you want to locate a number whose cube equals 3. Or the set is collection of continuous functions on $[0, 1]$. And so here objects are continuous functions defined on $[0, 1]$. You may want to locate an object, that is a function f whose derivative is itself, that is, $f' = f$. We saw how we started with constant function one and made it 'better and better' and at the end (there is no end, what we mean is the limit) we did get f which satisfies the condition. This is a simple problem but most of the problems of practical importance are complicated and you will not have closed formulae and you need to show existence using convergence. You need to understand the properties of the object without having a formula!

$(f_n, n \geq 1)$ and f are functions (real valued) on a set S . We say $f_n \rightarrow f$ pointwise if for each $x \in S$ the sequence of numbers $\{f_n(x)\}$ converges to the number $f(x)$. We say $f_n \rightarrow f$ uniformly if $\sup\{|f_n(x) - f(x)| : x \in S\} \rightarrow 0$.

Thus uniform convergence means given $\epsilon > 0$ there is an N so that for every $n \geq N$ we have $\sup\{|f_n(x) - f(x)| : x \in S\} \leq \epsilon$. In other words for every $n \geq N$ and every $x \in S$ we should have $|f_n(x) - f(x)| \leq \epsilon$. In the earlier example of the sequence (x^n) this does not happen. Given any n we see that for $\sqrt[n]{\epsilon} < x < 1$ we have $x^n > \epsilon$. In symbols $f_n \rightarrow f$ pointwise means

$$\forall(x \in S) \forall(\epsilon > 0) \exists N \forall(n > N) |f_n(x) - f(x)| < \epsilon$$

Thus given x and $\epsilon > 0$ there is an N which can depend on the given x and ϵ such that something happens. $f_n \rightarrow f$ uniformly means

$$\forall(\epsilon > 0) \exists N \forall(n > N) \forall(x \in S) |f_n(x) - f(x)| < \epsilon$$

Here given $\epsilon > 0$ there is an N such that something happens for all x . Thus now ϵ does not depend on any x , it is something that works for all x .

Generally when we say $f_n \rightarrow f$ it means pointwise convergence. if we mean uniform convergence we must say $f_n \rightarrow f$ uniformly. Here is a condition when the limit function is continuous.

Theorem: If $f_n \rightarrow f$ uniformly on an interval I and if each f_n is continuous, then f is continuous.

Let $x \in I$ and $\epsilon > 0$. Choose N so that for all $x \in I$ and all $n \geq N$ (?) we have $|f_n(x) - f(x)| < \epsilon/3$ using uniform convergence. Now use continuity of f_N to get $\delta > 0$ so that $|f_N(y) - f_N(x)| < \epsilon/3$ whenever $y \in I$ and $|y - x| < \delta$. Now take any $y \in I$ and $|y - x| < \delta$. Then

$$|f(y) - f(x)| \leq |f(y) - f_N(y)| + |f_N(y) - f_N(x)| + |f_N(x) - f(x)|$$

The first and last terms are at most $\epsilon/3$ by choice of N and the middle term is also so by choice of $\delta > 0$. This completes the proof. ■

Just as we talked about sequences, we can talk about series of functions also. We say a series $\sum f_n$ converges pointwise to f if for each x the series of numbers $\sum f_n(x)$ converges to the number $f(x)$. This is same as saying that the sequence of partial sums converge pointwise. Here partial sums of the series $\sum f_n$, are $s_n = \sum_1^n f_i$ and the sequence $\{s_n\}$ is the partial sum sequence. these are functions.

We say $\sum f_n \rightarrow f$ uniformly if the partial sum sequence converges uniformly to f .

Theorem: Suppose $r > 0$ is radius of convergence of the power series $P(x) = \sum a_i x^i$ and $0 < c < r$. Then the series converges uniformly on $[-c, c]$.

Proof is simple. Remember, for points within the interval the series converges absolutely. Let $\epsilon > 0$. Since $0 < c < r$ and the series $\sum |a_i| c^i$ converges, we can get N such that $\sum_N^\infty |a_i c^i| < \epsilon/3$. Then for any $x \in [-c, c]$ and for any $n \geq N$ we have

$$\left| \sum_0^n a_i x^i - P(x) \right| = \left| \sum_{n+1}^\infty a_i x^i \right| \leq \sum_{n+1}^\infty |a_i c^i| \leq \epsilon/3$$

completing the proof. ■

In particular we have, a result which we already showed earlier.

Theorem: $P(x) = \sum a_i x^i$ defines a continuous function of x in the open interval $(-r, r)$ where r is its radius of convergence. We assume $r > 0$.

Indeed if $a \in (-r, r)$ then fix c so that $0 < |a| < c < r$ which is possible. Since P is continuous in $[-c, c]$ and $a \in (-c, c)$ we conclude that P is continuous at a . This is true for every $a \in (-r, r)$ and completes the proof. ■

You must remember that uniform convergence allows you to say that the limit is continuous. However if the limit function is continuous, then it is not

necessary that convergence be uniform.

Example: $f(x) = nx$ for $0 \leq x \leq 1/n$ and $f(x) = 1 - n(x - 1/n)$ for $1/n \leq x \leq 2/n$ and $f(x) = 0$ for $2/n \leq x \leq 1$ is a continuous function on $[0, 1]$. Graph of f_n is isosceles triangle base $[0, 2/n]$ and height one; flat after $2/n$. This sequence converges to the zero function pointwise and so limit function is continuous. However the convergence is not uniform, because each function takes the value one at some point.

You should not confuse between uniform continuity and uniform convergence. Whenever you can produce a number depending on a point to satisfy some requirement, we say it is uniform if the number works for all points. Of course this is too abstract to make sense. Here are two illustrations. Suppose $\epsilon > 0$ is given. Continuity of f means given a point x , you can get a $\delta > 0$ so that $|f(y) - f(x)| \leq \epsilon$ for $|y - x| < \delta$. This δ could depend on the given point x – as you have seen with x^2 or $1/x$. The function is uniformly continuous if you can produce $\delta > 0$ which works for all x . Similarly, suppose $f_n \rightarrow f$ pointwise. Given a point x you can get N such that $|f_n(x) - f(x)| \leq \epsilon$ for all $n \geq N$. However this N could depend on the point x . If you can get N which works for all points x , then the convergence is uniform. Once you understand the meaning of the adjective, its appearance would not confuse but allows you to see some common theme.

Discontinuous function:

Are functions which are not continuous unimportant? uninteresting? do such functions not arise in practice? NO, not at all. They are as interesting as continuous functions and many times we get functions which are not continuous but the discontinuity is not horrible. What does this mean? This is what we understand now. For instance if you consider share prices they vary nicely, but once in a while, there is a sudden jump in the price of shares. Thus share price jumps. Or if you keep pinching some one there is no reaction initially but as the force increases he may suddenly scream, so the output goes from zero to one. To put it in some technical language, initially the neurons do not ‘fire’ and once a threshold is crossed they ‘fire’. It is precisely this phenomenon that occurs in neural networks through what are called activation functionals. When you see a seismograph, the curve moves smoothly but once in a while you see abrupt jumps. You can think of several other instances from so called ‘phase transitions’.

Here are some examples of functions, you should draw their graphs and

see. (Seeing is better than reading!)

(1). $f(x) = x + 1$ for $x \geq 0$ and $f(x) = x - 1$ for $x < 0$.

This function is nice except at $x = 0$. Even here whenever $x_n \rightarrow 0$ and all $x_n > 0$ then $f(x_n)$ converges to a value and it equals $f(0)$. Further whenever $y_n \rightarrow 0$ and all $y_n < 0$ then also $f(y_n)$ converges to a value but it is different from $f(0)$. The function appears to be continuous at zero from the right side – very informal way.

(2). $f(x) = x + 1$ for $x > 0$ and $f(x) = x - 1$ for $x \leq 0$.

Again This function is nice except at $x = 0$. Even here whenever $x_n \rightarrow 0$ and all $x_n < 0$ then $f(x_n)$ converges to a value and it equals $f(0)$. But now whenever $y_n \rightarrow 0$ and all $y_n > 0$ then also $f(y_n)$ converges to a value but it is different from $f(0)$. The function appears to be continuous at zero from the left side, again very informal way.

(3). $f(x) = x + 1$ for $x > 0$ and $f(x) = x - 1$ for $x < 0$ and $f(0) = 0$.

Again zero is the only trouble some point. But again whenever $x_n \rightarrow 0$ and all $x_n < 0$ then $f(x_n)$ converges to a value and it is different from $f(0)$. Also whenever $y_n \rightarrow 0$ and all $y_n > 0$ then also $f(y_n)$ converges to a value but it is again different from $f(0)$. The function appears to be well behaved and reaches specific targets from right side as well as left side, but none of them equals $f(0)$ and hence fails to be continuous.

(4). $f(x) = \sin(1/x)$ for $x < 0$ and $f(0) = 0$ for $x \geq 0$.

This function is continuous from right side at zero; however when $x_n < 0$ and $x_n \rightarrow 0$ $f(x_n)$ is not converging to a specific value. It may so happen $f(x_n)$ may not converge to any number at all or may converge to -1 or $+1$ etc depending on the sequence.

By considering the $\sin 1/x$ curve on both sides of zero and so on you can manufacture several possibilities.

If the function is not continuous at a then several things can happen. The function may be well behaved getting close to something or it may behave in a wiggly manner without getting close to any particular value — as you come close to a .

Of course the behaviour mentioned above can happen to the right of the point a under consideration or to the left of the point or on both sides.

It may be well behaved from both sides getting close to particular values but they may be different from the value of the function at that point.

We now briefly study this phenomenon. In the following it is enough if the function is defined in a bounded open interval containing a .

Theorem: Let $f : R \rightarrow R$ be a function and $a \in R$. Let L be a real number. The following two statements are equivalent.

- (i) Whenever $x_n \rightarrow a$ and each $x_n < a$ we have $f(x_n) \rightarrow L$.
- (ii) Given $\epsilon > 0$, there is a $\delta > 0$ such that whenever $a - \delta < x < a$ we have $|f(x) - L| < \epsilon$.

Proof is simple and similar to the corresponding statements we made while introducing continuity. Let us go through it once again.

Suppose (ii) holds. We shall prove (i). So take a sequence (x_n) where $x_n < a$ for every n and $x_n \rightarrow a$. We show $f(x_n) \rightarrow L$. So fix $\epsilon > 0$. We show a natural number n_0 such that $|f(x_n) - L| < \epsilon$ whenever $n \geq n_0$. To do this, With the given $\epsilon > 0$ in hand, use (ii) and fix $\delta > 0$ as stated. Since $x_n \rightarrow a$, fix n_0 so that for $n \geq n_0$, we have $|x_n - a| < \delta$. Now if $n \geq n_0$, using the fact that $x_n < a$ we conclude that $a - \delta < x_n < a$ and hence choice of δ shows $|f(x_n) - L| < \epsilon$.

Suppose (ii) fails. We show that (i) fails. Fix one $\epsilon > 0$ for which we can not find $\delta > 0$ as stated. Thus taking $\delta = 1$ we get an x_1 such that $a - 1 < x_1 < a$ and $|f(x_1) - L| \geq \epsilon$. Taking $\delta = 1/2$ we get an x_2 such that $a - 1/2 < x_2 < a$ and $|f(x_2) - L| \geq \epsilon$. In general, taking $\delta = 1/n$ we get an x_n such that $a - 1/n < x_n < a$ and $|f(x_n) - L| \geq \epsilon$. Clearly $x_n \rightarrow a$; we have $|f(x_n) - L| \geq \epsilon$ for every n . This completes the proof. ■

Note that a number L as above may not exist at all, like in case of the examples involving $\sin(1/x)$. However if such a number exists then it is unique. There can not be two such numbers. Indeed, if L and L' are two such numbers, then $f(a - \frac{1}{n})$ converges to both L and L' . Since a sequence can not converge to two different points we conclude that $L = L'$.

If any one of the above two things happens we say that f has a left limit at the point a and the value of the left limit equals L . We express it as

$$\lim_{x < a; x \rightarrow a} f(x) = L; \quad \text{or} \quad f(a-) = L.$$

Note that we are not evaluating the function f at $a-$; there is no number called $a-$. It is only a notational convenience to express in that fashion.

Theorem: Let $f : R \rightarrow R$ be a function and $a \in R$. Let L be a real number. The following two statements are equivalent.

- (i) Whenever $x_n \rightarrow a$ and each $x_n > a$, we have $f(x_n) \rightarrow L$
- (ii) Given $\epsilon > 0$, there is a $\delta > 0$ such that whenever $a < x < a + \delta$ we have

$$|f(x) - L| < \epsilon. \quad \blacksquare$$

Proof is similar to above. As in the case of left limits, note that such an L , if exists, is unique. If any one of the above two things happens we say that f has a right limit at the point a and the value of the right limit equals L . We express it as

$$\lim_{x>a; x\rightarrow a} f(x) = L \quad \text{or} \quad f(a+) = L.$$

Note that we are not evaluating the function f at $a+$; there is no number called $a+$. It is only a notational convenience to express in that fashion.

The following is easy to prove:

Theorem: Let f be defined on an open interval including a point a . Then f is continuous at a iff $f(a+)$ and $f(a-)$ exist and both equal $f(a)$.

Definition: f defined on an open interval (a, b) and $a < x < b$. We define f to be right continuous at x if $f(x+)$ exists and $f(x+) = f(x)$. We define f to be left continuous at x if $f(x-)$ exists and $f(x-) = f(x)$.

Thus the previous theorem can be restated as follows:

f is continuous at x iff it is both right continuous and left continuous. $\blacksquare$

For example the function $f(x) = x$ for $x \geq 0$ and $f(x) = x - 1$ for $x < 0$ is right continuous but not left continuous. Same is true of the function $f(x) = x$ for $x \geq 0$ and $f(x) = \sin(1/x)$ for $x < 0$. To say that the function is right continuous, you only see if the right limit exists and equals $f(x)$. You are not concerned whether the left limit even exists or not. Similar comment applies for left continuity.

If the function f is defined on a closed interval $[a, b]$ then we can define the right limit and left limit at every x , $a < x < b$. But, we can not define left limit at a , however we can define the right limit at a . Similarly, we can not define the right limit at b but we can define the left limit at b . We can easily see that f defined on $[a, b]$ is continuous at b iff the $f(b-)$ exists and equals $f(b)$. Similarly, f is continuous at a iff $f(a+)$ exists and equals $f(a)$.

The Dirichlet function defined by: $f(x)$ equals one or zero according as x is rational or irrational ; has the property that it is discontinuous at every point. However monotone functions are not too bad, as the following theorem shows.

Theorem: Let f be a monotone function on R . Then at every point x , the left limit $f(x-)$ and right limit $f(x+)$ exist. If f is increasing, then $f(x-) \leq f(x) \leq f(x+)$ and when f is decreasing then $f(x-) \geq f(x) \geq f(x+)$. Further f is continuous except at countably many points.

Proof is simple. Let us assume that the function is increasing – does not mean strictly increasing; $x < y$ implies $f(x) \leq f(y)$.

Fix a point x and let $c = \sup\{f(y) : y < x\}$. This is well defined: the set is non-empty because $f(x-1)$ belongs; and the set is bounded by $f(x)$. We claim that $c = f(x-)$. Indeed take any sequence $y_n \rightarrow x$ and each $y_n < x$. Firstly, $f(y_n) \leq c$ for each n , by definition of c . Also given $\epsilon > 0$, using definition of supremum get $y < x$ so that $f(y) > c - \epsilon$. Since $y_n \rightarrow x$ after some stage $y_n > y$ and hence $f(y_n) \geq f(y) > c - \epsilon$. This shows that $\lim f(y_n) = c$. Similar argument shows that $f(x+) = \inf\{f(y) : y > x\}$. Also the proof already shows $f(x-) \leq f(x) \leq f(x+)$. In particular f is not continuous at x iff $f(x-) < f(x+)$.

Thus if x is a discontinuity point, then the interval $(f(x-), f(x+))$ is not empty. Suppose now that a, b are two discontinuity points of f say, $a < b$. Then $f(a-) \leq f(a) \leq f(a+) \leq f(b-) \leq f(b) \leq f(b+)$ showing that the intervals $(f(a-), f(a+))$ and $(f(b-), f(b+))$ are disjoint. For each discontinuity point a , if we select a rational number r_a in the interval $(f(a-), f(a+))$, then for $a \neq b$ we have $r_a \neq r_b$. Since the set of rational numbers is countable, the set $\{r_a : a \in D\}$ is a countable set. The observation that $r_a \neq r_b$ for $a \neq b$ shows that the set D is countable. Similar argument applies if the function is decreasing. ■

Suppose that the point a is a discontinuity point of the function f . We say that it is a simple discontinuity if both $f(a-)$ and $f(a+)$ exist. that is, the left and right limits exist at the point a . In other words simple discontinuity is first of all a discontinuity point, but the function has right and left limits at that point. Simple discontinuity is also called discontinuity of the first kind.

If f is defined on a closed bounded interval $[a, b]$, then the point b is a simple discontinuity if $f(b-)$ exists and is different from $f(b)$. Similarly, a is a simple discontinuity if $f(a+)$ exists and is different from $f(a)$.

Differentiation:

There are some important questions which all lead to the same concept.

Question 1: How to draw tangent to a curve at a point. Suppose the curve is given by $y = f(x)$ and at the point $(a, f(a))$ on the curve we are required to draw tangent. The first question is: what do you mean by tangent. well, one intuitive idea is that it is a straight line passing through the given point and meets the curve at that point only. This is not quite correct, you can try to draw tangent to the curve $y = \sin x$ at several points and see what happens. Of course, for a circle, tangents are precisely straight lines that meet the circle at exactly one point. (why?)

Another idea is the following. Take the point $(a, f(a))$ on the curve and a nearby point $(x, f(x))$ join these two points by a straight line. This is a 'chord'. A natural question is whether this chord has a limiting form as the point x approaches a . At first sight this explanation appears too complicated because we are talking about limiting form of straight lines. But it is not difficult to understand.

After all, a straight line is determined by its slope — apart from a point through which it passes. In our case the point is $(a, f(a))$. Thus chords we are talking about or the tangent we are looking for, all pass through this point. So we only need to specify the slope. Thus the question amounts to asking whether the slopes of the chords have a limiting value. of course, slope of the chord we have mentioned above is nothing but $[f(x) - f(a)]/[x - a]$. Thus we need to see if this has a limiting value as x approaches a . Thus if you wish to draw tangent at the point $(a, f(a))$ for the curve (graph of f)

the slope should be

$$\lim_{x \rightarrow a, x \neq a} \frac{f(x) - f(a)}{x - a} \quad (\spadesuit)$$

Question 2: To start an entirely different second line of thought, suppose a particle is starting at the origin at time zero and travelling on the real line.. How do we understand its velocity? If it is travelling so that at time t the particle is at $5t$, then matters are simple. At time instants $t_0 < t_1$ it is at $5t_0$ and $5t_1$ respectively, so that the distance travelled during this time duration $t_1 - t_0$ is $5t_1 - 5t_0$ and so the velocity, distance divided by time, equals 5 and does not depend on the two time points we have taken. But in practice particles accelerate and do not travel with ‘uniform speed’ as above.

Suppose that the particle is at point $f(t) = t^2$, again starting at zero. The distance travelled during the time period 1 to 2 is $4 - 1 = 3$ while the distance travelled during time period 10 to 11 equals $121 - 100 = 21$ and you can try to find out the distance travelled during time period 1000 to 1001. You see that the particle is going faster and faster as time elapses. Thus the concept of velocity does not make sense unless you specify a time point and ask: what is the velocity at this time. In other words one needs to talk about instantaneous velocity.

Suppose a time point $t = 5$ is given. What is the velocity at time instant 5? The blind interpretation, distance travelled at time instant t divided by etc makes no sense. Naturally it should depend on what is happening around this time instant and one idea is to take the distance travelled during the time period 5 to $5 + h$ and then take the ratio. In other words, to understand the velocity at time 5 we need to calculate $[f(5 + h) - f(5)]/h$ and then see if there is any particular value to which this ratio is getting close as h gets closer to zero, that is, whether the intuitive idea of distance travelled/time approaches any particular value as we consider durations nearer to the time point of interest. To change notation, if the time point under consideration is a , then we need to look at the ratio $[f(a) - f(x)]/[a - x]$ and see if it has a limiting value as the (time) point x approaches a . Thus we need to look at

$$\lim_{x \rightarrow a, x \neq a} \frac{f(x) - f(a)}{x - a} \quad (\clubsuit)$$

Question 3: To start a yet different, third line of thinking, let us understand complexity of functions. The simplest functions are constant functions. Suppose we are given a continuous function f on R and a point a , Which simplest function best approximates f near a . Obviously the constant function

$\varphi(x) \equiv f(a)$ is the best. When x is close to a , $\varphi(x)$ is close to $f(x)$, simply because $f(x)$ is close to the number $f(a)$ when x is close to a . In symbols, $f(x) - \varphi(x) \rightarrow 0$ as $x \rightarrow a$. This means, $f(x_n) - \varphi(x_n) \rightarrow 0$ whenever $x_n \rightarrow a$.

Suppose you are allowed to use a little more complicated functions than constant functions, namely, straight line functions. Can you do better? so what is meant by better? Earlier we simply said $f(x) - \varphi(x)$ should get close to zero as x gets close to a . We did not demand any quantitative measurement on how fast this quantity should approach zero. Now having allowed functions general than constant functions, we demand that not only this error gets close to zero, the ratio $[f(x) - \varphi(x)]/[x - a]$ should get closer to zero as x gets closer to a . That is change in values of f must be of ‘much smaller order’ than change in values of the argument x .

Note that this implies, in particular, that $f(x) - \varphi(x)$ gets closer to zero as x gets closer to a . But since both f and φ are continuous, this difference is getting closer to $f(a) - \varphi(a)$. In other words $\varphi(a) = f(a)$. Thus the straight line passes through $(a, f(a))$. If the straightline we are thinking is $\varphi(x) = mx + c$, then what we concluded just now amounts to $ma + c = f(a)$ or $c = f(a) - ma$. Thus the straight line is

$$\varphi(x) = mx + f(a) - ma = f(a) + (x - a)m.$$

Thus

$$\frac{f(x) - \varphi(x)}{x - a} = \frac{f(x) - f(a)}{x - a} - m.$$

Thus the demand that this ratio gets closer to zero as x gets closer to a , simply amounts to saying that the ratio $[f(x) - f(a)]/[x - a]$ should get closer to the number m , slope of the line we are looking for. Thus the slope of the straight line function is given by

$$\lim_{x \rightarrow a, x \neq a} \frac{f(x) - f(a)}{x - a} \quad (\diamond)$$

All these thought processes ($\spadesuit$), ($\clubsuit$), ($\diamond$) lead to one common conclusion: the rate of change of the function is an important quantity. This is what we study next.

Theorem Let $f : R \rightarrow R$ and $a \in R$. For a number $m \in R$, the following two statements are equivalent.

(i) If $x_n \rightarrow a$ and $x_n \neq a$ for each n ; then

$$\frac{f(x_n) - f(a)}{x_n - a} \rightarrow m.$$

(ii) Given $\epsilon > 0$, there is a $\delta > 0$ such that

$$0 < |x - a| < \delta \Rightarrow \left| \frac{f(x) - f(a)}{x - a} - m \right| < \epsilon.$$

Proof is simple and we have come across such a situation several times earlier — connection between discrete and continuous formulations of an idea.

Suppose (ii) holds. To prove (i), take $x_n \rightarrow a$ and $x_n \neq a$ for every n . Shall show n_0 such that

$$n \geq n_0 \Rightarrow \left| \frac{f(x_n) - f(a)}{x_n - a} - m \right| < \epsilon.$$

Use (ii) with the given ϵ to get a $\delta > 0$ and then use $x_n \rightarrow a$ to get n_0 so that $n \geq n_0 \Rightarrow |x_n - a| < \delta$. This will do.

Conversely, if (ii) is false, we exhibit a sequence for which (i) fails. Since (ii) is false, fix $\epsilon > 0$ for which we can not find $\delta > 0$ stisfying the stated condition. With $\delta = 1/n$ get x_n so that $0 < |x_n - a| < 1/n$ and yet

$$\left| \frac{f(x) - f(a)}{x - a} - m \right| \geq \epsilon.$$

This sequence shows failure of (i).

The above conditions are also stated, in an equivalent way, as follows:

(i) if $h_n \neq 0$ for all n and $h_n \rightarrow 0$ then

$$\frac{f(a + h_n) - f(a)}{h_n} \rightarrow m$$

and (ii) Given $\epsilon > 0$, there is a $\delta > 0$ such that

$$0 < |h| < \delta \Rightarrow \left| \frac{f(a + h) - f(a)}{h} - m \right| < \epsilon.$$

Definition: Let $f : R \rightarrow R$ and $a \in R$. We say that f is differentiable at the point a if there is a number m satisfying the above conditions. In such a case m is called derivative of f at a . If the function is differentiable at every point, then we say that f is differentiable.

There are several notations.

$$f'(a) = m; \quad \frac{df}{dx}(a) = m; \quad D_x f(a) = m \quad Df(a) = m$$

Another way of saying the same thing is the following: Define the function $\varphi(x) = [f(x) - f(a)]/[x - a]$. Of course, this is defined on R except at the point a . If the right limit and left limit of this function exist at a and equal, then we say that the function f is differentiable at the point a and this limit is the derivative of the function f at the point a .

Observe that the number m , when exists, is unique.

Theorem: Let $f : R \rightarrow R$. if f is differentiable at a , then f is continuous at a .

Proof: Let $x_n \rightarrow a$. We need to show $f(x_n) \rightarrow f(a)$. If all the x_n are different from a , then

$$f(x_n) - f(a) = \frac{f(x_n) - f(a)}{x_n - a} (x_n - a) \rightarrow f'(a) \cdot 0 = 0.$$

Suppose that there is an n_0 such that $x_n \neq a$ for all $n \geq n_0$. Then you can write the equation above for $n \geq n_0$ obtaining the result. if there is an n_0 such that $x_n = a$ for all $n \geq n_0$, then there is nothing to do.

Finally suppose that there are infinitely many n such that $x_n \neq a$ and infinitely many n such that $x_n = a$. Then you will get two subsequences corresponding to each and the above argument applies for the two corresponding subsequences of $f(x_n)$. Using an earlier observation, we conclude that $f(x_n) - f(a) \rightarrow 0$. ■

Here is another way to argue. As stated above, the function $\varphi(x) = (f(x) - f(a))/(x - a)$ for $x \neq a$ and $\varphi(a) = f'(a)$ is a continuous function at the point a . Note that $f(x) - f(a) = \varphi(x)(x - a)$ for all x . Thus $f(x) - f(a) \rightarrow 0$ as $x \rightarrow a$. It is instructive to argue using $\epsilon - \delta$. That is, fix $\epsilon > 0$ and show $\delta > 0$ such that $|f(x) - f(a)| < \epsilon$ whenever $|x - a| < \delta$. ■

Theorem: if $f : R \rightarrow R$ and $g : R \rightarrow R$ are differentiable at a then so are $f + g$ and $39f$ and fg . In fact

$$(f + g)'(a) = f'(a) + g'(a); \quad (39f)'(a) = 39f'(a);$$

$$(fg)'(a) = f(a)g'(a) + f'(a)g(a).$$

If $g(x) \neq 0$ for all x , then f/g is also differentiable at a and

$$\left(\frac{f}{g}\right)'(a) = \frac{f'(a)g(a) - f(a)g'(a)}{g^2(a)}.$$

Proof: if you take any sequence $x_n \rightarrow a$, $x_n \neq a$ for all n , then

$$\frac{(f + g)(x_n) - (f + g)(a)}{x_n - a} = \frac{f(x_n) - f(a)}{x_n - a} + \frac{g(x_n) - g(a)}{x_n - a}$$

$$\begin{aligned}
& \rightarrow f'(a) + g'(a). \\
\frac{(39f)(x_n) - (39f)(a)}{x_n - a} &= 39 \frac{f(x_n) - f(a)}{x_n - a} \rightarrow 39f'(a). \\
\frac{(fg)(x_n) - (fg)(a)}{x_n - a} &= \frac{f(x_n)g(x_n) - f(x_n)g(a) + f(x_n)g(a) - f(a)g(a)}{x_n - a} \\
&= f(x_n) \frac{g(x_n) - g(a)}{x_n - a} + g(a) \frac{f(x_n) - f(a)}{x_n - a} \\
&\rightarrow f(a)g'(a) - f'(a)g(a).
\end{aligned}$$

Here we used continuity of f at a .

We show $1/g$ is differentiable and calculate its derivative, then the above multiplication rule can be used for the product $f \cdot (1/g)$. Take a sequence $x_n \rightarrow a$, $x_n \neq a$ for all n . Then

$$\frac{(1/g)(x_n) - (1/g)(a)}{x_n - a} = \frac{g(a) - g(x_n)}{x_n - a} \frac{1}{g(x_n)g(a)} \rightarrow \frac{-g'(a)}{g^2(a)}.$$

here we have used continuity of g at a . ■

Theorem (Chain rule): Let $f : R \rightarrow R$ and $g : R \rightarrow R$ and $a \in R$. Assume that f is differentiable at a and g is differentiable at the point $f(a)$. Let $h(x) = g(f(x))$, the composition. then h is differentiable at a and

$$h'(a) = g'(f(a))f'(a).$$

Proof: Take $x_n \rightarrow a$, $x_n \neq a$ for all n .

Case 1: Suppose that $f(x_n) = f(a)$ for all n . Then note that

$$f'(a) = \lim \frac{f(x_n) - f(a)}{x_n - a} = 0$$

so that $g'(f(a))f'(a) = 0$. Of course, $h(x_n) = h(a)$ for all n proving existence of limit $\lim [h(x_n) - h(a)]/[x_n - a]$ as well as the stated equality $h'(a) = 0 = g'(f(a))f'(a)$; in the case under consideration. Note that the same proof holds to show the result if $f(x_n) = f(a)$ for all n after some stage.

Case 2: Assume that $f(x_n) \neq f(a)$ for all n . Then we have

$$\begin{aligned}
\frac{h(x_n) - h(a)}{x_n - a} &= \frac{g(f(x_n)) - g(f(a))}{f(x_n) - f(a)} \frac{f(x_n) - f(a)}{x_n - a} \\
&\rightarrow g'(f(a))f'(a).
\end{aligned}$$

Here we have used that f is continuous at a , thus $f(x_n) \rightarrow f(a)$. This proves existence of limit of $[h(x_n) - h(a)]/[x_n - a]$ and also the stated formula. Note that the same proof holds to show the result if $f(x_n) \neq f(a)$ for all n after some stage.

Case 3. If $f(x_n) = f(a)$ for infinitely many n and also $f(x_n) \neq f(a)$ for infinitely many n , then consider the two corresponding subsequences and conclude the result. Observe that both subsequences converge to the same value (zero). ■

So far we have not calculated derivative of any specific function.

if $f(x) = 55$ for all x then $f'(a) = 0$ for all a .

if $f(x) = x$ for all x , then $f'(a) = 1$ for every a .

if $f(x) = x^{39}$ for all x , then $f'(a) = 39a^{38}$ for every a .

Proof: First two statements are easy (please do and then say easy). Last statement follows from

$$\frac{x^{39} - a^{39}}{x - a} = x^{38} + x^{37}a + x^{36}a^2 + \cdots + a^{38} \rightarrow 39a^{38}.$$

Using the rules above, you can now show that polynomials are differentiable and be able to calculate their derivatives.

if you have understood the earlier calculations, then you can do some improvements. Why should we have functions defined on all of the real line? suppose that f is a function defined on an open interval $I = (\alpha, \beta)$ where $-\infty \leq \alpha < \beta \leq +\infty$. Let $a \in I$. We say that f is differentiable at a and $f'(a) = m$ if any one of the following two equivalent conditions hold; (that these two conditions are equivalent is to be proved, but the same earlier proof works, check!)

(i) $x_n \in I$ for all n , $x_n \neq a$ for all $n \Rightarrow \frac{f(x_n) - f(a)}{x_n - a} \rightarrow m$.

(ii) Given $\epsilon > 0$, there is $\delta > 0$ such that

$$0 < |x - a| < \delta; x \in I \Rightarrow \left| \frac{f(x) - f(a)}{x - a} - m \right| < \epsilon.$$

Of course the case $\alpha = -\infty$ and $\beta = +\infty$ corresponds to $I = R$, discussed above.

All the facts remain true. For sum, product and ratio of functions (defined on the interval I now) with exactly the same proofs. the composition rule also holds with exactly the same proof as follows: let f be real valued function defined on an interval $I = (\alpha, \beta)$. let g be a real valued function defined on another interval $J = (\gamma, \delta)$. let us assume that range of the function

f is contained in J . that is, $f(x) \in J$ for every $x \in I$. Then the composition makes sense: $h(x) = g(f(x))$ defined on I . let now $a \in I$. Assume that f is differentiable at a and g is differentiable at $f(a)$. The h is differentiable at a and $h'(a) = g'(f(a))f'(a)$.

what about functions defined on a closed interval. Suppose that f is defined on the interval $[23, 33]$. The above para defines the concept of differentiability at all points $a \in (23, 33)$. Can you define derivative at the points 23 and 33? Yes, at the point 23, you can define righthand derivative, namely, limit of $[f(x) - f(23)]/[x - 23]$ as $x \downarrow 23$

$$\lim_{x \rightarrow 23+} \frac{f(x) - f(23)}{x - 23}$$

as was done in discussing left and right limits of functions. of course, this limit may not exist. When it exists, we say that f is differentiable at the point 23 (more precisely f has right hand derivative at 23). Similarly, you can define left derivative at the point 33 as

$$\lim_{x \rightarrow 33-} \frac{f(x) - f(33)}{x - 33} = \lim_{x \rightarrow 33-} \frac{f(33) - f(x)}{33 - x},$$

provided, of course, this limit exists.

We shall now see how derivative of f helps us to understand the function f better. These properties can be used in curve tracing. remember, if you need to draw graph of a function f on R , in early high school, you plotted f values at some points and made a 'free-hand curve' joining them. Unfortunately this may not give you the actual picture. Curve tracing means you should show where graph of f hits x axis (that is where $f(x) = 0$) and where f is increasing and where it is decreasing, where it is maximum, where it is minimum and where it is constant etc. As you learn more and more this curve tracing becomes finer and finer.

Definition; Let f be a function defined on an interval I and $a \in I$. We say f has a local maximum at the point a , if there is an open interval (α, β) containing the point a such that $f(x) \leq f(a)$ whenever $x \in I$ and $\alpha < x < \beta$.

In other words, locally, that is, in the interval (α, β) the function f has maximum at the point a . Of course, there may be points x outside this interval which are in I and $f(x) > f(a)$. If $a \in I$ is such a point that $f(x) \leq f(a)$ for every x in I , then we say that a is a point of global maximum in the interval I . Of course It is local maximum, but more than that.

We have similar definition regarding minimum.

We say f has a local minimum at the point a , if there is an open interval (α, β) containing the point a such that $f(a) \leq f(x)$ whenever $x \in I$ and $\alpha < x < \beta$.

Thus locally, that is, in the interval (α, β) the function f has minimum at the point a . There may exist points $x \in S$ such that $x \leq \alpha$ or $x \geq \beta$ where $f(x) < f(a)$. If $a \in S$ is a such a point that $f(a) \leq f(x)$ for every x in S , then we say that a is a point of global minimum in the interval.

Def: A point a which is either a local maximum or local minimum is called a local extremum for that function.

Theorem: Let f be defined on an open interval (u, v) and $a \in (u, v)$ is a local maximum or local minimum. If f is differentiable at the point a then $f'(a) = 0$.

Proof is simple. suppose a is a local maximum. Let us take $a \in (\alpha, \beta) \subset (u, v)$ so that $f(x) \leq f(a)$ for all $x \in (\alpha, \beta)$. if you take a sequence of points $\{x_n\}$ in this interval so that each $x_n < a$ and $x_n \uparrow a$, then we see

$$f'(a) = \lim \frac{f(x_n) - f(a)}{x_n - a} \geq 0,$$

because both numerator and denominator above are ≤ 0 . similarly, if we take a sequence of points $y_n \downarrow a$ in this interval, each $y_n > a$, we see

$$f'(a) = \lim \frac{f(y_n) - f(a)}{y_n - a} \leq 0.$$

because now denominator is positive while numerator is still nonpositive, Hence $f'(a) = 0$. similar proof applies for local minimum.

The main point is the following. if you are looking for the (local) maximum or local minimum in an open interval, just search among points where the derivative vanishes (assuming that the function is differentiable). Of course you will ask, how do I know if we have max or min? The answer has to wait till we define second derivative.

Understand carefully what we said. a is a point of max or min implies $f'(a) = 0$. We did not say: $f'(a) = 0$ implies a is a point of max or min. if you consider $f(x) = x^3$, we see $f'(0) = 0$ but zero is neither max nor min.

We also said that the point of max or min must be inside the interval, not endpoint. For example $f(x) = x$ defined on $[3, 20]$ has minimum and maximum at end points and the derivative there exists and not zero. Of course, you should consider left derivative at the point 3 and left derivative at the point 20.

Suppose that f is a continuous function on a closed bounded interval $[u, v]$. we do know that f is bounded and attains the bounds. suppose a is a point where f assumes largest value. If we now assume that $u < a < v$ and f is differentiable at a we can conclude $f'(a) = 0$. Geometrically it says the following. The tangent at a is parallel to the x -axis, its slope is zero. We shall now show that there is a tangent parallel to any given chord of the graph of f .

Lagrange's Mean Value Theorem: Let $u, v \in R$ and $u < v$. Let f be a continuous function on an interval $[u, v]$ which is differentiable at every point $u < x < v$. Then there is a point $\theta \in (u, v)$ such that

$$f(v) - f(u) = (v - u)f'(\theta).$$

You can rewrite this as

$$f'(\theta) = \frac{f(v) - f(u)}{v - u}.$$

Observe that slope of the chord joining the two points $(u, f(u))$ and $(v, f(v))$ on the curve (graph of f) is precisely the right side above. The slope of the tangent (to the curve f) at the point $\theta \in (u, v)$ is precisely the left side above.

Proof is simple. Shall convert the problem to one we already solved. define

$$\varphi(x) = [f(v) - f(u)]x - [v - u]f(x).$$

then $\varphi(u) = uf(v) - vf(u) = \varphi(v)$ and φ is differentiable in (u, v) . Also $\varphi'(x) = [f(v) - f(u)] - [v - u]f'(x)$. Thus stated conclusion is same as saying there is a θ such that $\varphi'(\theta) = 0$. If φ is constant then any point $\theta \in (u, v)$ will do the job. Otherwise φ must take values either larger than the value at the end points or values smaller than at the end points. Thus either maximum or minimum must be attained in the open interval (u, v) . Such a point will do, by the previous fact.

This theorem is called *mean value theorem*. $[f(v) - f(u)]/[v - u]$ is the 'mean velocity' in this interval — ratio of distance travelled to the time duration. There is another interpretation. For ten numbers $\{a_i\}$ their mean

value or average value is $\sum a_i/10$. If you have a nice bounded function g on a finite interval, then its mean value or average value is $\int_a^b g/(b-a)$, integral over the interval (instead of sum) divided by length of the interval, thus the above ratio is nothing but the average value of f' . We shall postpone this interpretation to later when we do integration.

A special case of the above is the following:

Rolle's Theorem or Theorem of Bhaskara II: If f is a continuous function on $[a, b]$, differentiable on (a, b) with $f(a) = f(b)$, then there is $c \in (a, b)$ with $f'(c) = 0$. ■

Theorem: Let f be defined and continuous on $[u, v]$ differentiable at every point of (u, v) .

- (i) If $f'(x) \geq 0$ for all x then f is increasing function in this interval.
- (ii) If $f'(x) \leq 0$ for all x then f is decreasing function in this interval.
- (iii) If $f'(x) = 0$ for all x then f is a constant in this interval.

Proof is simple. (i) Let $x < y$, then for some point θ ,

$$f(y) - f(x) = (y - x)f'(\theta) \geq 0.$$

Note that even if $x = u$ or $y = v$ the proof works. Same applies to prove (ii). (iii) follows from (i) and (ii). ■

Power series:

Let

$$P(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n + a_{n+1}x^{n+1} + \cdots$$

be a power series with radius of convergence $r > 0$. Thus this defines a continuous function on the interval $(-r, r)$ and also in fact the series converges uniformly and absolutely in any interval $[-c, c]$ for $0 < c < r$. Can we differentiate this, just like a polynomial term by term? That is if

$$Q(x) = a_1 + 2a_2x + 3a_3x^2 + \cdots + (n+1)a_{n+1}x^n + \cdots$$

then does it converge in $(-r, r)$ and then is $P'(x) = Q(x)$? Yes

Theorem: If r is the radius of convergence of the power series P then radius of convergence of the power series Q also equals r and moreover, in the interval $(-r, r)$, the function P is differentiable and $P' = Q$.

The moral is term-by-term differentiation of the power series can be done, that is, what you did in high school with polynomials can be done for ‘infinite degree’ polynomials too.

Proof:

$$Q(x) = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + \dots$$

We need to show that Q also has radius of convergence r . Here the coefficient of x^n is $(n+1)a_{n+1}$ so for the radius of convergence of Q we would be involved in $\sqrt[n]{(n+1)|a_{n+1}|}$. On the other hand calculation of r involved $\sqrt[n]{|a_n|}$. We first sort out this mismatch between n and $n+1$. We claim that the series $Q(x)$ converges iff the following series converges.

$$xQ(x) = a_1x + 2a_2x^2 + 3a_3x^3 + 4a_4x^4 + \dots$$

In fact let x be any real number. if the series $Q(x)$ converges, then multiplying each term by x we get the second series and hence the second series also converges. conversely, suppose that the second series converges. If $x = 0$, then first series also converges (inspect what is the series when you put $x = 0$). So let $x \neq 0$. But then the first series is obtained from the second series by dividing each term by x .

As a consequence, the radius of convergence of the power series $Q(x)$ is given by

$$\frac{1}{\limsup \sqrt[n]{n|a_n|}}.$$

we only need to show that

$$\limsup \left(\sqrt[n]{n} \sqrt[n]{|a_n|} \right) = \limsup \sqrt[n]{|a_n|}.$$

But since $\sqrt[n]{n} \rightarrow 1$, this follows from an exercise you did.

Now we prove $P' = Q$. Fix $z \in (-r, r)$. Need to show that $[P(x) - P(z)]/[x - z]$ is close to $Q(z)$ when x is close to z . The idea is same as that used in proving continuity of P . Cut off at a finite stage and argue remainders are small.

We know that $[P_N(x) - P_N(z)]/[x - z]$ is close to $Q_N(z)$ when x is close to z where P_N and Q_N are the polynomials

$$P_N(x) = a_0 + a_1x + a_2x^2 + \dots + a_Nx^N.$$

$$Q_N(x) = a_1 + 2a_2x + 3a_3x^2 + \dots + Na_Nx^{N-1}.$$

If we can show that $[P(x) - P(z)]/[x - z]$ is close to $[P_N(x) - P_N(z)]/[x - z]$ and $Q(z)$ is close to $Q_N(z)$, we are done as earlier in showing continuity. this is what we execute now.

Let $\epsilon > 0$ be given. We shall exhibit $\delta > 0$ so that $(z - \delta, z + \delta) \subset (-r, r)$ and

$$0 < |x - z| < \delta \Rightarrow \left| \frac{P(x) - P(z)}{x - z} - Q(z) \right| < \epsilon.$$

This will show that P is differentiable and $P'(z) = Q(z)$.

First we fix $c > 0$ so that $-r < -c < z < c < r$. This is possible because $-r < z < r$. Since Q also has radius of convergence r and $0 < c < r$ we conclude that the series $Q(c)$ converges absolutely. That is, the series $\sum (i+1)|a_{i+1}|c^i$ is convergent. So its tail sums converge to zero. Fix (as in proof of continuity), N such that

$$\sum_{i \geq N} (i+1)|a_{i+1}|c^i < \frac{\epsilon}{3}.$$

This is our N to define

$$P_N(x) = a_0 + a_1x + a_2x^2 + \cdots + a_Nx^N.$$

$$Q_N(x) = a_1 + 2a_2x + 3a_3x^2 + \cdots + Na_Nx^{N-1}.$$

Let us also put

$$\widetilde{P}_N(x) = a_{N+1}x^{N+1} + a_{N+2}x^{N+2} + a_{N+3}x^{N+3} + \cdots.$$

$$\widetilde{Q}_N(x) = (N+1)a_{N+1}x^N + (N+2)a_{N+2}x^{N+1} + (N+3)a_{N+3}x^{N+2} + \cdots.$$

so that we always have

$$P(x) = P_N(x) + \widetilde{P}_N(x); \quad Q(x) = Q_N(x) + \widetilde{Q}_N(x); \quad (\spadesuit)$$

With this notation, choice of N tells us

$$|x| \leq c \Rightarrow |\widetilde{Q}_N(x)| \leq \epsilon/3. \quad (\spadesuit)$$

Note that by mean value theorem and $(\spadesuit)$, for $x \in [-c, c]$ we have the following.

$$\begin{aligned} \left| \frac{a_{N+1}x^{N+1} - a_{N+1}z^{N+1}}{x - z} \right| &\leq |(N+1)\alpha_{N+1}c^N| \leq \frac{\epsilon}{3}. \\ \left| \frac{a_{N+1}x^{N+1} + a_{N+2}x^{N+2} - a_{N+1}z^{N+1} - a_{N+2}z^{N+2}}{x - z} \right| \\ &\leq |(N+1)a_{N+1}c^N + (N+2)a_{N+2}c^{N+1}| \end{aligned}$$

$$\leq \frac{\epsilon}{3}.$$

More generally, for every $k \geq 1$, we have

$$\left| \frac{\sum_{i=N+1}^{N+k} a_i x^i - \sum_{i=N+1}^{N+k} a_i z^i}{x - z} \right| \leq \frac{\epsilon}{3}.$$

Since this is true for every $k \geq 1$ we see by taking limits (over k),

$$-c \leq x \leq c \Rightarrow \left| \frac{\widetilde{P}_N(x) - \widetilde{P}_N(z)}{x - z} \right| \leq \frac{\epsilon}{3}. \quad (\heartsuit)$$

As noted already and easy to see, derivative of the polynomial P_N is Q_N . Thus we can fix $\delta > 0$ so that $(z - \delta, z + \delta) \subset (-c, c)$ and

$$0 < |x - z| < \delta \Rightarrow \left| \frac{P_N(x) - P_N(z)}{x - z} - Q_N(z) \right| < \frac{\epsilon}{3} \quad (\clubsuit)$$

For $0 < |x - a| < \delta$ we have the following.

The first equality below uses $(\blacklozenge)$ and next inequality uses $|t + u + v| \leq |t| + |u| + |v|$ and final inequality uses $(\clubsuit), (\heartsuit), (\spadesuit)$.

$$\begin{aligned} & \left| \frac{P(x) - P(z)}{x - z} - Q(z) \right| \\ &= \left| \frac{P_N(x) - P_N(z)}{x - z} + \frac{\widetilde{P}_N(x) - \widetilde{P}_N(z)}{x - z} - Q_N(z) - \widetilde{Q}_N(z) \right| \\ &\leq \left| \frac{P_N(x) - P_N(z)}{x - z} - Q_N(z) \right| + \left| \frac{\widetilde{P}_N(x) - \widetilde{P}_N(z)}{x - z} \right| + |\widetilde{Q}_N(z)| \\ &\leq \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon. \end{aligned}$$

This completes the proof of the theorem. ■

This is an extremely useful result. For example if $f(x) = e^x$

$$f(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

then we see $f' = f$.

Note that $e^x : R \rightarrow (0, \infty)$ has its inverse function $\log : (0, \infty) \rightarrow R$. This is also differentiable. Indeed as $x \rightarrow a$, we know $\log x \rightarrow \log a$

$$\frac{\log x - \log a}{x - a} = \frac{\log x - \log a}{e^{\log x} - e^{\log a}} \rightarrow \frac{1}{e^{\log a}} = \frac{1}{a}$$

where we used derivative of e^x is e^x , never zero. Thus

$$D(\log x) = \frac{1}{x}$$

In fact the same argument shows the following.

$f : (a, b) \rightarrow (c, d)$ be increasing 1-1 onto function which is continuous. Then we know its inverse $g : (c, d) \rightarrow (a, b)$ is continuous. The above argument shows that if moreover f is differentiable and f' is never zero, then g is differentiable and $g'(z) = \frac{1}{f'(g(z))}$. This is nearly obvious because we are saying $f'(g(z))g'(z) = 1$ which follows from Chain rule after noting $f(g(z)) = z$. But the catch is unless you show g is also differentiable, you can not use chain rule! Indeed for any z using the fact $x \rightarrow z$ implies $g(x) \rightarrow g(z)$;

$$\frac{g(x) - g(z)}{x - z} = \frac{g(x) - g(z)}{f(g(x)) - f(g(z))} \rightarrow \frac{1}{f'(g(z))}$$

Note that $f' \neq 0$ is used above. Without that the statement is meaningless. The function $f(x) = x^3$ on R is 1-1 onto and has inverse $g(x) = \sqrt[3]{x}$. More precisely – since we defined roots only for non-negative numbers – $g(x) = \sqrt[3]{x}$ if $x \geq 0$ and $g(x) = -\sqrt[3]{-x}$ if $x \leq 0$. This g is not differentiable at zero even though f is differentiable.

We can now use the above with composition rule to find derivatives of more functions. For example if $f(x) = a^x$ where $a > 0$ is fixed, then $f(x) = e^{\log a^x} = e^{x \log a}$ so that

$$f'(x) = e^{x \log a} \log a = a^x \log a$$

If $\varphi(x) = x^a$ defined for $x \in (0, \infty)$, then $\varphi(x) = e^{a \log x}$ so that

$$\varphi'(x) = e^{a \log x} a \frac{1}{x} = x^a \frac{a}{x} = ax^{a-1}$$

(Confusion: For f , I said $a > 0$ and for φ , I said $x > 0$. You should not be confused. always ‘the base’ is positive. Since I wanted to use variable x for functions, this problem arises. If I used the same a^x ; first as function of x and then as function of a this issue does not arise. You should think which formulation you are happy with.)

If

$$s(x) = x - \frac{x^3}{3!} + \frac{x^7}{7!} - \dots$$

$$c(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$$

Then we see $s'(x) = c(x)$ and $c'(x) = -s(x)$. We can also painlessly see now that $c^2(x) + s^2(x) = 1$ for all x . Indeed this is a differentiable function. Further

$$[c^2(x) + s^2(x)]' = -2c(x)s(x) + 2s(x)c(x) = 0$$

Thus $c^2(x) + s^2(x)$ is constant and clearly has value one at $x = 0$ by direct calculation. This proves the result. ■ In particular you see the fact (not obvious from power series):

$$-1 \leq c(x) \leq 1 \qquad -1 \leq s(x) \leq 1$$

$$s(x) = 0 \Rightarrow c(x) = \pm 1; \qquad c(x) = 0 \Rightarrow s(x) = \pm 1.$$

From the earlier calculations and power series, you see

$$c'' + c = 0; \quad c(0) = 1 \quad c'(0) = 0$$

$$s'' + s = 0; \quad s(0) = 0 \quad s'(0) = 1$$

Thus both c and s are solutions of

$$\varphi'' + \varphi = 0 \qquad (\clubsuit)$$

Clearly, by linearity of derivative we see that for any two numbers p, q the function $pc(x) + qs(x)$ is a solution of $(\clubsuit)$. Indeed any solution of $(\clubsuit)$ is a linear combination of the above two functions.

To see this, first observe that if ψ is a solution of $(\clubsuit)$, then $\psi^2 + (\psi')^2$ is constant function. In fact the same argument as earlier tells

$$[\psi^2 + (\psi')^2]' = 2\psi\psi' + 2\psi'\psi'' \equiv 0$$

by $(\clubsuit)$. Now let f be any solution of $(\clubsuit)$. Then put $p = f(0)$ and $q = f'(0)$ and $\varphi(x) = pc(x) + qs(x)$. Then φ and hence $\psi = f - \varphi$ is a solution of $(\clubsuit)$. Thus $\psi^2 + (\psi')^2$ is constant function. By choice of p, q it is easy to see that $\psi(0) = 0$ and $\psi'(0) = 0$ and so this constant must be zero. In particular $\psi \equiv 0$. Thus $f = pc(x) + qs(x)$.

These should indicate you that c, s are your high school cosine and sine functions. Soon we shall identify to your complete satisfaction.

We shall complete our discussion of the sine and cosine functions. So far we have, by definition,

$$s(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \dots$$

$$c(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \dots$$

$$s' = c; \quad c' = -s$$

$$s'' + s \equiv 0; \quad s(0) = 0; \quad s'(0) = 1 \quad (\star)$$

$$c'' + c \equiv 0; \quad c(0) = 1; \quad c'(0) = 0 \quad (\star)$$

$$s^2 + c^2 \equiv 1; \quad -1 \leq s, c \leq 1$$

$$s(x) = 0 \Rightarrow c(x) = \pm 1; \quad c(x) = 0 \Rightarrow s(x) = \pm 1$$

$$\varphi'' + \varphi \equiv 0 \Rightarrow \varphi = pc + qs; \quad p = \varphi(0); \quad q = \varphi'(0) \quad (\spadesuit)$$

Thus s and c are the unique solutions of the differential equation in $(\spadesuit)$ with the initial conditions as stated in $(\star)$. Since you know from high school that sine and cosine functions (as defined in high school) satisfy these, we conclude that s is the sine function and c is the cosine function that you have learnt.

■ Digression:

One of you asked me: how is sine function defined in High school. I am not sure, you should know. Probably, In high school sine and cosine functions are defined for $0 < \theta < \pi/2$ by taking a right-angle triangle with one angle θ and putting $\sin \theta = \text{opposite side/hypotenuse}$ and $\cos \theta = \text{adjacent side/hypotenuse}$. To show that this definition does not depend on the triangle, you use property of similar triangles. The relation $c^2 + s^2 = 1$ is just Pythagoras theorem. You declare $\sin 0 = 0$ (the opposite side has length zero; do not ask me if the degenerate triangle is a triangle; but that was enough in high school But you can rigorously justify by taking triangles with angle $\theta > 0$ and decreasing to zero). We take $\sin(\pi/2) = 1$; opposite side and hypotenuse are parallel lines starting at same level (you can justify this by not taking angle exactly $\pi/2$;but $\theta < \pi/2$ and increasing to $\pi/2$). Then somehow you extend to all angles $[0, 2\pi]$ and extend periodically. You prove geometrically that $\sin x/x \rightarrow 1$ as $x \rightarrow 0$. I suppose, after proving formulae

for $\sin(\theta + h)$, $\cos(\theta + h)$ we calculate derivatives. ■

Incidentally, the series was also in the background already in high school. Recall DeMoivre formula ($\sqrt{-1} = i$):

$$e^{ix} = \cos x + i \sin x. \quad (\dagger)$$

But

$$\begin{aligned} e^{ix} &= 1 + ix + \frac{(ix)^2}{2!} + \frac{(ix)^3}{3!} + \frac{(ix)^4}{4!} + \frac{(ix)^5}{5!} + \frac{(ix)^6}{6!} + \frac{(ix)^7}{7!} + \cdots \\ &= \left[1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \cdots \right] + i \left[x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \cdots \right] \end{aligned}$$

comparing the above with $(\dagger)$ we get our definition. ■

Let us complete this circle of ideas by rigorously explaining π also. We know $s(0) = 0$. Is there a point $a \geq 0$ such that $c(a) = 0$? Yes. Argue as follows.

Since $c(0) = 1$ and c is a continuous function, get $z > 0$ so that $c > 0$ on the interval $[0, z]$. Thus $s'(x) > 0$ for $0 \leq x \leq z$. So s is strictly increasing in this interval and since $s(0) = 0$, we conclude that $s(z) = \eta > 0$.

We claim that $c(x) = 0$ for some $x \leq z + (2/\eta)$. If it were not so, then $c(x) > 0$ in the interval $z \leq x \leq z + (2/\eta)$ also. Since $s' = c$, this implies that s is strictly increasing in this interval as well and thus $s(x) > s(z) = \eta$, for $z < x \leq z + (2/\eta)$. Moreover by the mean value theorem,

$$|c(z + (2/\eta)) - c(z)| = s(\theta) \cdot (2/\eta)$$

for some number θ with $z < \theta < z + (2/\eta)$. But, no matter what this θ is, $s(\theta) > \eta$. Thus

$$|c(z + (2/\eta)) - c(z)| > 2$$

which is impossible because all values of c are between -1 and $+1$.

Thus c must assume the value zero. So both the functions c and s have zeros.

Let $\alpha > 0$ be the first zero of c . That is $\alpha = \inf\{u > 0 : c(u) = 0\}$. Is $c(\alpha) = 0$? Yes There is a sequence $\{u_n\}$ in this set converging to the infimum, and so $c(\alpha) = \lim c(u_n) = 0$. Can $\alpha = 0$? No, because in $[0, z]$ we know $c \neq 0$. Thus $c(x) \neq 0$ for $0 \leq x < \alpha$ and $c(\alpha) = 0$. What is $s(\alpha)$? Since $c(\alpha) = 0$ we know $s(\alpha) = \pm 1$. But s is strictly increasing in $[0, \alpha]$ and so $s(\alpha) = +1$.

Let us define the functions

$$\tilde{s}(x) = -c(\alpha + x); \quad \tilde{c}(x) = s(\alpha + x); \quad x \in R.$$

By composition rule $x \mapsto x + \alpha \mapsto s(\alpha + x)$ we see that $\tilde{s}(x)$ and $\tilde{c}(x)$ satisfy the same differential equation ($\spadesuit$). By comparing initial conditions

$$\tilde{s}(0) = -c(\alpha + 0) = 0; \quad \tilde{c}(0) = s(\alpha + 0) = 1$$

$$(\tilde{s})'(0) = s(\alpha) = 1; \quad (\tilde{c})'(0) = c(\alpha) = 0.$$

Thus these new functions, not only satisfy the same differential equations but also satisfy the same initial conditions. Thus by uniqueness of solutions we conclude that these functions are same as c and s . In other words, we have

$$s(\alpha + x) \equiv c(x); \quad c(\alpha + x) \equiv -s(x).$$

Hence

$$s(2\alpha + x) = s(\alpha + [\alpha + x]) = c(\alpha + x) = -s(x).$$

Similarly, $c(2\alpha + x) = -c(x)$. So one gets

$$s(4\alpha + x) = s(x); \quad c(4\alpha + x) = c(x).$$

a function φ on R to R is said to be periodic if there is a number $h > 0$ such that $\varphi(x + h) = \varphi(x)$ for every real number x . Smallest such number, if exists, is called the period of the function. It is not too difficult to show that if φ is continuous and periodic and is not constant function, then there is indeed such a smallest number h . However, we do not need this fact in our analysis.

Returning to our analysis s and c are periodic with a period 4α . Of course, one can also deduce, successively, $s(\alpha) = +1$, $s(2\alpha) = 0$, $s(3\alpha) = -1$ and $s(4\alpha) = s(0) = 0$. This also shows that the period is indeed 4α but you need not bother to understand this now.

You know from high school that $\alpha = \pi/2$. Actually one can define π as 2α . But from school you know what is π — it is the area of a disc of radius one in the plane. So let me quickly argue that $\alpha = \pi/2$ with your knowledge of π . We need integration (obviously, because I am depending on your high school knowledge of π as area and area is integral!). We use results about integration which you know, but will fully justify in our course later.

first observe that an integration by parts gives us

$$\int_0^1 \sqrt{1-u^2} \, du = \left[u\sqrt{1-u^2} \right]_0^1 - \int_0^1 \frac{-u^2}{\sqrt{1-u^2}} \, du$$

$$= \int_0^1 \frac{1}{\sqrt{1-u^2}} du - \int_0^1 \sqrt{1-u^2} du.$$

Thus

$$2 \int_0^1 \sqrt{1-u^2} du = \int_0^1 \frac{1}{\sqrt{1-u^2}} du.$$

If you recall the definition of area, the integral on the left side above is nothing but the area of a quarter of the unit disc. Thus

$$\frac{\pi}{2} = \int_0^1 \frac{1}{\sqrt{1-u^2}} du.$$

Since $c(0) = 1$ and $c(t) \geq 0$ on $[0, \alpha]$, we have $s'(x) = c(x) = +\sqrt{1-c^2(x)}$, we have $s'(x)/\sqrt{1-c^2(x)} = 1$ on $(0, \alpha)$. Hence

$$\alpha = \int_0^\alpha 1 dx = \int_0^\alpha \frac{s'(x)}{\sqrt{1-s^2(x)}} dx = \int_0^1 \frac{1}{\sqrt{1-u^2}} du,$$

where we substituted $u = s(x)$ so that $s'(x)dx = du$. And of course this last integral is seen above to be $\pi/2$. This completes our identification of α with $\pi/2$.

Rearrangements:

By a rearrangement of a series $\sum_{n \geq 1} a_n$ we mean a series $\sum b_n$ where $b_j = a_{\pi(j)}$ for some permutation π of N . A permutation of N means a bijection.

if a series is absolutely convergent, then any rearrangement also converges and converges to the same value.

Here is why. Let (s_n) be the sequence of partial sums of $\sum a_n$ and (t_n) be partial sums of $\sum b_n$, the rearrangement where $b_n = a_{\pi(n)}$. We know $s_n \rightarrow A$. If we show $t_n - s_n \rightarrow 0$ then, by adding these two, the proof will be complete. So let $\epsilon > 0$ be given. We propose to show N so that $|t_n - s_n| < \epsilon$ for all $n > N$. Let $\epsilon > 0$.

First observe that, since the series $\sum |a_n|$ is convergent, its partial sums are Cauchy. So we can choose K so that for $m > n \geq K$ we have

$$|a_{n+1}| + |a_{n+2}| + \cdots + |a_m| < \epsilon/2.$$

This being true for every $m > n \geq K$, taking $n = K$ we see $\sum_{i=K}^m |a_i| < \epsilon/2$ for every $m > K$. Hence

$$\sum_{i>K} |a_i| \leq \epsilon/2. \quad (\clubsuit)$$

If $\pi(i_1) = 1$, $\pi(i_2) = 2$ and so on, upto $\pi(i_K) = K$, then take $N = \max\{i_1, i_2, \dots, i_K\}$. Note that $t_N = \sum_{i=1}^N b_i$ includes a_1 because $i_1 < N$ and $b(i_1) = a_{\pi(i_1)} = a_1$. Similarly all $a_1, a_2, \dots, a_K$ appear in t_N and hence in t_n for any $n > N$. Now if we take any $n > N$ then in s_n also all terms $a_1, \dots, a_K$ appear. Also observe that $N \geq K$ (maximum of any K distinct positive integers is at least K). Thus for any $n > N$ if you consider $(s_n - t_n)$, then $a_1, \dots, a_K$ cancel out, possibly others too. In any case, what remain are terms $\pm a_j$ for $j > K$. (if the term that remains is from s_n then plus; if it is from t_n then minus). Thus, $|s_n - t_n| \leq \sum_{K+1}^{\infty} |a_i| < \epsilon$. This being true for all $n > N$, the proof is complete. ■

When you add finitely many numbers, you can change the order in which you add, and you still get the same answer. For infinite series this is no longer true, in general. This was Riemann's observation.

(Riemann): if a series is convergent but NOT absolutely convergent, then given any number α there is a rearrangement $\sum b_n$ so that $\sum b_n = \alpha$. More generally, given any two numbers $\alpha < \beta$ there is a rearrangement $\sum b_n$ so that its partial sum sequence has $\limsup$ equal to β and $\liminf$ equal to α .

We shall not use this theorem. Then why are we doing it? Firstly, If you are careless with infinite series, spectacular unwanted things will happen. You should know this; discipline. Secondly, you see how you need large vocabulary to implement a simple idea. In idea, you may say 'proceed like this, you will get permutation'. But saying is not enough. You must convince the listener that you **can** proceed forever as suggested. Who knows you may be able to do this till 1000 and not 1001! Also you need to convince that what you get is a permutation; you have to show one-one and onto. In the midst of all this when you want to proceed by induction, you must state 'beforehand' what you want to do. This is very important because at the n -th step, you say suppose we have done 'this' till $(n - 1)$, we shall now do for n . Right? What is 'this'? You can not cook up 'this' at n -th step which will enable you to proceed further; exercise in writing proofs.

If you understand this statement and be careful then it is sufficient, you can skip reading the proof.

Let $\sum a_n$ be conditionally convergent. Then the crucial observations are the following

(i) sum of all the positive terms of the sequence equals $+\infty$ and sum of all

negative terms equals $-\infty$.

(ii) $a_n \rightarrow 0$

Main idea is this: start using positive terms till you exceed β , which happens from (i). Just when you exceed, stop. Now add negative terms till you go below α , which happens due to (i) again. Just when you go below, stop. Start adding positive terms till you go above β and continue. Remember, at every step you use at least one term a_i . Thus you will be consuming all the terms a_n . But will your partial sum overshoot β by too much? No, if you keep doing, since you are consuming at least one term in each step, (ii) tells you overshoot is very small. Similarly your partial sum will be going only very little below α , not too much. This does.

The idea is beautiful and execution is neat. At the same time you realize that you need good vocabulary to communicate! Here is the execution.

Let us put $x_n = a_n$ if $a_n \geq 0$; otherwise put $x_n = 0$. Similarly, put $y_n = a_n$ if $a_n < 0$; otherwise put $y_n = 0$. Clearly we have

$$x_n + y_n = a_n; \quad x_n - y_n = |a_n|; \quad n \geq 1.$$

If you say that $\sum x_n$ is convergent, then because $\sum a_n$ is convergent then by properties of sums of series you see $\sum(a_n - x_n) = \sum y_n$ is convergent too. But then again from properties of series, $\sum |a_n| = \sum(x_n - y_n)$ is convergent too, contradicting the hypothesis that $\sum a_n$ is conditionally convergent. I keep saying ‘by properties of series’ because you are likely to be misled to think I am ‘putting brackets’ which of course is not allowed always. I am not putting brackets. I am using the theorem if $\sum a_n$ is convergent and if $\sum b_n$ is convergent then $\sum c_n$ is convergent where $c_n = a_n - b_n$, in other words $\sum(a_n - b_n)$ is convergent.

Note that $\sum x_n$ is a series of positive terms so that if it is not convergent then partial sums increase to ∞ . Similarly, we argue that $\sum y_n = -\infty$. What we mean is the following: the partial sums of $\sum x_n$ are not bounded above and partial sums of $\sum y_n$ are not bounded below. That is all.

Now we have to actually pick terms of our series and so need vocabulary to tell positive and negative terms. Let $f(1)$ be the first $i \geq 1$ such that $a_i \geq 0$. Having defined $f(1) < f(2) < \dots < f(n-1)$, we define $f(n)$ to be the least $i > f(n-1)$ such that $a_i \geq 0$. Similarly $g(1)$ is the first i such that $a_i < 0$. In general $g(n)$ is the least $i > g(n-1)$ such that $a_i < 0$. Since the positive terms and the negative terms of $\sum a_n$ add up to infinities, we deduce that $f(n)$ and $g(n)$ are defined for all $n \geq 1$.

Each integer $i \geq 1$ appears exactly once — either as an $f(n)$ or as a $g(n)$ but not as both. (♠)

Convergence of $\sum a_n$ combined with the facts $f(n) \uparrow \infty$, $g(n) \uparrow \infty$ gives us: $a_{f(n)} \rightarrow 0$; $a_{g(n)} \rightarrow 0$. (♣)

Let $s(n) = \sum_1^n a_{f(i)}$ and $t(n) = \sum_1^n a_{g(i)}$.

The fact observed above says $s_n \uparrow \infty$; $t_n \downarrow -\infty$. (♡)

We are given $\alpha < \beta$ and need to show a rearrangement of the series $\sum a_n$ so that lim sup of partial sums is β and lim inf is α . We assume that $\alpha \geq 0$, same proof works if $\alpha < 0$.

We pick two sequences of integers $n_1 < n_2 < \dots$, and $m_1 < m_2 < \dots$ such that the following hold.

(1) n_1 is the least integer with $s(n_1) > \beta$ and m_1 is the least integer such that $s(n_1) + t(m_1) < \alpha$.

(2) For $i \geq 2$: Having selected $n_1, m_1, n_2, m_2, \dots, n_{i-1}, m_{i-1}$ we select n_i to be the least integer $> n_{i-1}$ such that $s(n_i) + t(m_{i-1}) > \beta$ and m_i is the least integer $> m_{i-1}$ such that $s(n_i) + t(m_i) < \alpha$.

Existence of such sequences is established by induction. First choose n_1 as stated in (1) and then choose m_1 as stated in (2). Then choose n_2 as stated in (1) and then m_2 as stated in (2). That you can proceed for ever is a consequence of (♡).

Claim 1:

$$0 \leq s(n_1) - \beta \leq a_{f(n_1)}; \text{ for } k \geq 2, 0 \leq [s(n_k) + t(m_{k-1})] - \beta \leq a_{f(n_k)}.$$

Indeed if $n_1 = 1$, then $s(n_1) = a_{f(1)} > \beta \geq 0$ so that, $0 \leq s(n_1) - \beta \leq a_{f(n_1)}$ as stated. If $n_1 \geq 2$, then by choice of n_1 , $0 \leq s(n_1 - 1) \leq \beta < s(n_1)$ so that $0 \leq s(n_1) - \beta \leq s(n_1) - s(n_1 - 1) = a_{f(n_1)}$ as stated. Similarly, by choice of n_k , $s(n_k - 1) + t(m_{k-1}) \leq \beta < s(n_k) + t(m_{k-1})$, so that $0 \leq s(n_k) + t(m_{k-1}) - \beta \leq s(n_k) - s(n_k - 1) = a_{f(n_k)}$.

Claim 2: $0 \leq \alpha - [s(n_k) + t(m_k)] \leq |a_{g(m_k)}|$.

This is proved exactly as above.

Claim 3: $|s(n_k) + t(m_{k-1}) - \beta| \rightarrow 0$ and $|s(n_k) + t(m_k) - \alpha| \rightarrow 0$.

This follows from claims 1, 2 and (♣).

Here is the permutation (a one-to-one, onto map of natural numbers).

$$\begin{aligned}
\pi(i) &= f(i) && \text{for } i \leq n_1 \\
&= g(i - n_1) && \text{for } n_1 + 1 \leq i \leq n_1 + m_1 \\
&= f(i - n_1 - m_1) && \text{for } n_1 + m_1 + 1 \leq i \leq n_2 + m_1 \\
&= g(i - n_2 - m_1) && \text{for } n_2 + m_1 + 1 \leq i \leq n_2 + m_2 \\
&= f(i - n_2 - m_2) && \text{for } n_2 + m_2 + 1 \leq i \leq n_3 + m_2 \\
&= \dots
\end{aligned}$$

(♠) shows that π is indeed a permutation of natural numbers. Let $u(n)$ be the n -th partial sum of the rearranged series $\sum a_{\pi(n)}$, that is, $u(n) = \sum_{i=1}^n a_{\pi(i)}$.

Claim 4: α is liminf of partial sums of the rearranged series. In fact the liminf can not be above α follows that the partial sums ‘corresponding’ to $n_i + m_i$ are going below α . It can not be smaller than α because the partial sums do not go much below α . I am not using notation and just using words!

Claim 5: β is limsup of partial sums. This is also as above. The sums ‘corresponding’ to $n_i + m_{i-1}$ exceed β so limsup is at least β . But if you take any $\epsilon > 0$, then after some stage, no partial sum is above $\beta + \epsilon$.

This completes the proof. In particular, given any number, we can rearrange so that the rearranged series converges to that number.

Back to differentiation:

Example:

$$f_0(x) = \sin(1/x); \quad x \neq 0$$

Can we extend it as a continuous function to all of R , that is, define at zero so that it is continuous at zero? No. Convince yourself.

$$f_1(x) = x \sin(1/x); \quad f_1(0) = 0.$$

Yes convince yourself this is continuous function on R . is it differentiable? At points other than zero, yes, applying product rule

$$f_1'(x) = \sin(1/x) - x^{-1} \cos(1/x)$$

At $x = 0$, we have

$$\frac{f_1(0+h) - f_1(0)}{h} = \sin(1/h)$$

has no limit as $h \rightarrow 0$. So f_1 though continuous, is not differentiable at zero.

$$f_2(x) = x^2 \sin(1/x); \quad f_2(0) = 0.$$

(Many times we omit the second equation, value at zero here; whenever the formula – with blind substitution for value of x – becomes meaningless; you are supposed not to do that; instead take limit as x approaches that value. Of course when this limit does not exist you understand that the function is not defined there. If the limit exists then value of f at that point is this limit and not blind substitution in the formula. But in our course, I shall specify as above so that you need not worry.)

Now

$$f_2'(x) = 2x \sin(1/x) - \sin(1/x) \quad f_2'(0) = 0$$

because we now have

$$\frac{f_2(0+h) - f_2(0)}{h} = h \sin(1/h) \rightarrow 0$$

Thus f_2 is differentiable everywhere. Is the derivative continuous? No. You can see that as $x \rightarrow 0$, $f_2'(x)$ has no limit.

$$f_3(x) = x^3 \sin(1/x); \quad f_3(0) = 0.$$

Now you see f_3 is again differentiable everywhere; and now f_3' is continuous as well at zero.

$$f_3'(x) = 3x^2 \sin(1/x) - 2x \sin(1/x), \quad f_3'(0) = 0$$

is this differentiable? No, first term is differentiable (see f_2); but second term is not – at zero. Of course it is differentiable at all $x \neq 0$.

Hope you see the ‘smoothness’ or ‘how many times differentiable’ depends on the factor x^k .

If a function f , that is, $x \mapsto f(x)$ is differentiable at every point, then we have a function $x \mapsto f'(x)$, derivative of f . if this function $x \mapsto f'(x)$ is differentiable then we have a function $x \mapsto f''(x)$, the second derivative of f . Since we can not put too many ‘primes’ we use a notation. We denote $f^{(1)}$ the first derivative function of f , namely Df . In general we denote $f^{(n)}$ the n -th derivative function of f , namely $Df^{(n-1)}$, derivative of the $(n-1)$ -th derivative. Of course, provided these exist.

Thus in the above example we have example of a function f such that $f^{(1)}$ does not exist; example of a function f for which $f^{(1)}$ exists but not $f^{(2)}$ and so on. Are there functions f such that $f^{(n)}$ exists for every n ? Yes. Such functions are called ‘infinitely differentiable’. For example all polynomials are infinitely differentiable. All infinite degree polynomials are also so. More precisely, let $P(x) = \sum a_n x^n$ be a power series with radius of convergence r .

Then P defines an infinitely differentiable function in the interval $(-r, +r)$. This follows from our theorem on differentiation of power series. For example e^x and $\sin x$ and $\cos x$ are examples. There are others too.

Example: This example arises in several contexts. In our course we see this again during Taylor series later.

$$f(x) = e^{-1/x^2}; \quad x > 0; \quad f(x) = 0 \quad x \leq 0$$

To prove that this is infinitely differentiable, namely, For each n it is differentiable n times, we proceed as follows. First observe that if you make the statement

S_n : f is n times differentiable

then there is no hope of proving it by induction. To go from $(n-1)$ to n , it is NOT enough to know $(n-1)$ -th derivative exists; we should know something about it to be able to do calculations and say the next derivative exists.

$$S_n : \quad f^{(n)}(x) = P_n(1/x)e^{-1/x^2} \quad x > 0; \quad f^{(n)}(x) = 0 \quad x \leq 0. \quad n \geq 1.$$

where P_n is some polynomial in one variable. We are not saying that P_n is of degree n ; it is not.

We claim S_1 is true. Indeed if $x < 0$ then in the interval $(2x, x/2)$ the function is identically zero and so derivative exists and is zero and x is in this interval. If $x > 0$ then composition rule tells that f is differentiable and

$$f'(x) = e^{-1/x^2} [-1/x^2]' = 2 \frac{1}{x^3} e^{-1/x^2}$$

which is of the form with $P_1(z) = 2z^3$ so that $P_1(1/x) = 2(1/x)^3$. What about $x = 0$. In what follows we use without mention, $x^n e^{-x^2} \rightarrow 0$ as $x \rightarrow \infty$; whatever be n .

$$\lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^-} \frac{0}{h} = 0$$

$$\lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^+} \frac{e^{-1/h^2}}{h} = \lim_{x \rightarrow \infty} x e^{-x^2} = 0$$

Assuming it is true for $(n-1)$ then we show it is true for n . Again for $x < 0$ derivative of $f^{(n-1)}$ is zero because the function is zero. Also for $x > 0$ it is argued as follows. We use product rule and chain (composition) rule

$$[P_{n-1}(1/x)e^{-1/x^2}]' = [P_{n-1}(1/x)]' e^{-1/x^2} + P_{n-1}(1/x)[e^{-1/x^2}]'$$

$$= P'_{n-1}(1/x)[1/x]' + P_{n-1}(1/x)2\frac{1}{x^3} e^{-1/x^2} = P_n(1/x)e^{-1/x^2}$$

where $P_n(z) = -P'_{n-1}(z) z^2 + P_{n-1}(z)2z^3$. Since P_{n-1} being a polynomial, we know P' is a polynomial again and this show that the formula is valid for $x > 0$. How about zero? Exactly as earlier

$$\lim_{h \rightarrow 0^-} \frac{f^{(n-1)}(0+h) - f^{(n-1)}(0)}{h} = \lim_{h \rightarrow 0^-} \frac{0}{h} = 0$$

$$\lim_{h \rightarrow 0^+} \frac{f^{(n-1)}(0+h) - f^{(n-1)}(0)}{h} = \lim_{h \rightarrow 0^+} P_{n-1}(1/h) \frac{e^{-1/h^2}}{h} = \lim_{x \rightarrow \infty} x P_{n-1}(x) e^{-x^2} = 0$$

This completes the inductive step and thus f is infinitely differentiable.

Generalized Mean Value Theorem:

We shall now discuss two generalizations of the mean value theorem and use of these generalizations. First let us recall the mean value theorem. f is a continuous function on $[a, b]$ which is differentiable at every point of (a, b) . Rolle says that if $f(a) = f(b)$ then for some $\theta \in (a, b)$ we have $f'(\theta) = 0$. Lagrange says, without assuming that $f(a) = f(b)$, there is a number $\theta \in (a, b)$ so that

$$[f(b) - f(a)] = (b - a)f'(\theta).$$

We shall complicate it now. Let g be the function $g(x) = x$. Then note that $g' \equiv 1$. and $g(b) - g(a) = b - a$. Thus the above equation takes the form

$$[f(b) - f(a)]g'(\theta) = [g(b) - g(a)]f'(\theta).$$

This statement is true in general and is called the generalized mean value theorem. Here it is.

Cauchy MVT: Let f and g be two continuous functions on the interval $[a, b]$, both differentiable at every point of (a, b) . Then there is a number $\theta \in (a, b)$ such that

$$[f(b) - f(a)]g'(\theta) = [g(b) - g(a)]f'(\theta).$$

Proof is simple. As earlier define

$$\varphi(x) = [f(b) - f(a)]g(x) - [g(b) - g(a)]f(x).$$

Note φ is continuous on $[a, b]$ and differentiable at every point of (a, b) with

$$\varphi'(x) = [f(b) - f(a)]g'(x) - [g(b) - g(a)]f'(x).$$

Also $\varphi(b) = f(b)g(a) - f(a)g(b) = \varphi(a)$. Usual Mean value theorem applies to give a point θ where $\varphi'(\theta) = 0$. This is the required conclusion. ■

of course if you know g' is never zero, then you know by Rolle, $g(b) \neq g(a)$ and so you can say

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(\theta)}{g'(\theta)}$$

The beauty is that you have the same θ in numerator and denominator.

We use the generalized MVT to derive a nice theorem that goes by the name of L'Hopital's rule. This theorem helps to evaluate limits of ratios of functions. Suppose f and g are two functions and we want to evaluate $\lim_{x \rightarrow a} [f(x)/g(x)]$. If f and g converge to some finite non-zero numbers then this is easy, the ratio of the functions converges to the ratio of the two limits. However, both f and g converge to zero we would not be able to blindly apply the theorem on limits of ratios.

L'Hopital:

Theorem: Let (a, b) be a bounded interval; f and g be two differentiable functions on this interval. Assume the following.

$$\lim_{x \downarrow a} f(x) = 0; \quad \text{and} \quad \lim_{x \downarrow a} g(x) = 0 \quad \text{and} \quad g'(x) \neq 0 \text{ for all } x \in (a, b) \quad \text{and} \quad \lim_{x \downarrow a} \frac{f'(x)}{g'(x)} = \alpha \in R$$

$$\text{Then } \lim_{x \downarrow a} \frac{f(x)}{g(x)} = \alpha.$$

Hypothesis on f'/g' is meaningful because the denominator g' is never zero. Conclusion makes sense because if $g(x) = 0$ then there can be at most one such x_0 , otherwise Rolle tells you g' is zero somewhere. Since we are interested in the limit, we consider only $x < x_0$. In the proof we consider only x smaller than such x_0 , if any.

First two examples.

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{\cos x}{1} = 1$$

(used $\lim_{x \rightarrow 0}$, but to confirm to the theorem read as $\lim_{x \downarrow 0}$) This you can do using power series definition of $\sin x$ too.

$$\frac{\sin x}{x} = 1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \frac{x^6}{7!} + \frac{x^8}{9!} - \dots$$

This is a power series with radius of convergence ∞ and defines continuous function on R and its value at zero is visibly one.

For another example

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \lim_{x \rightarrow 0} \frac{\sin x}{2x} = \lim_{x \rightarrow 0} \frac{\cos x}{2} = \frac{1}{2}$$

Here we applied L'Hopital twice. This can again be done by using power series

$$1 - \cos x = 1 - \left[1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \dots \right]$$

again you will get a power series etc. Or you can also do using the earlier example and the fact $1 - \cos x = 2(\sin x/2)^2$.

Proof of the rule is simple. Let $\epsilon > 0$ be given. we need to show a $\delta > 0$ so that

$$a < x < a + \delta \Rightarrow \left| \frac{f(x)}{g(x)} - \alpha \right| < \epsilon.$$

Let us choose, using hypothesis, a number $\delta > 0$ so that

$$a < x < a + \delta \Rightarrow \left| \frac{f'(x)}{g'(x)} - \alpha \right| < \epsilon/2.$$

We show that the same δ serves our purpose, let us take $a < x < a + \delta$. Take also some y with $a < y < x$. Then by generalized MVT, the ratio $[f(x) - f(y)]/[g(x) - g(y)]$ equals $f'(\theta)/g'(\theta)$ where θ is in between x and y ; in particular $a < \theta < a + \delta$ so that we have

$$\left| \frac{f(x) - f(y)}{g(x) - g(y)} - \alpha \right| < \epsilon/2.$$

But this is true for every y with $a < y < x$. If we let $y \downarrow a$ we see $f(y) \rightarrow 0$ and $g(y) \rightarrow 0$. Thus taking limits (as $y \downarrow a$) in the above inequality we get

$$a < x < a + \delta \Rightarrow \left| \frac{f(x)}{g(x)} - \alpha \right| \leq \epsilon/2.$$

This completes the proof.

There is one subtle point. Can $g(y) = g(x)$? If so what we wrote above is meaningless. But no need to worry because then by Rolle, g' must be zero somewhere in between. But hypothesis rules it out. Can $g(x) = 0$? This we already ruled out by saying that there can be only one such x (again Rolle) and in talking about limit we consider only x smaller than such a point, if any.

If you are not comfortable with the phrase $y \downarrow a$, take a sequence of points $a < y_n < x$ such that $y_n \rightarrow a$. Write the inequalities only for these y_n and take limit as $n \rightarrow \infty$. ■

There are many questions: why did we take bounded interval? Why limit at a , why not at b . Why α is assumed finite? We shall sort out.

Why did we take α to be finite? It is not necessary, we can allow $\alpha = \infty$. Thus, suppose

$$\lim_{x \downarrow a} \frac{f'(x)}{g'(x)} = \infty$$

We show that

$$\lim_{x \downarrow a} \frac{f(x)}{g(x)} = \infty.$$

Same proof above works, just that we should not use $\infty + \epsilon$ etc. Actually there is no need if you understand what is to be proved. Let A be any given number. We shall show $\delta > 0$ so that

$$a < x < a + \delta \Rightarrow \frac{f(x)}{g(x)} > A.$$

Use hypothesis to get $\delta > 0$ so that

$$a < x < a + \delta \Rightarrow \frac{f'(x)}{g'(x)} > A.$$

As earlier this δ would do with exactly the same proof — take $a < x < a + \delta$, take any $a < y < x$, argue about the ratio $[f(x) - f(y)]/[g(x) - g(y)]$ and take $y \downarrow a$.

Can we take $\alpha = -\infty$. Yes, exactly the same argument as in the case of ∞ above would do.

Why did we take a finite? Can it be $-\infty$. Yes. But then we will be talking about $\lim$ as $x \downarrow -\infty$. So we should reformulate the argument and not use $-\infty + \delta$. For example let α be finite. Given $\epsilon > 0$ we need to show a number c so that $x < c$ implies the ratio $f(x)/g(x)$ is close to α upto ϵ . Proof is exactly the same. Same holds in case α is not finite.

Why did we take limit at a ? Yes we can take limit as $x \uparrow b$ and we can allow b to be ∞ too. How about taking a point $a < c < b$; assume that f and g converge to zero as $x \rightarrow c$ but the ratio $f'(x)/g'(x) \rightarrow \alpha$ as $x \rightarrow c$. Can we still say that $f(x)/g(x)$ converges to α as $x \rightarrow c$? Yes, argue in the intervals (a, c) and (c, b) .

I have stated simplest case, but all these embellishments are possible and useful too. There is one question, namely, why did you assume f and g are converging to zero? what if they converged to ∞ ? Instead of $0/0$ form we have ∞/∞ form. is the result true? Yes, it is true, needs different proof, shall do later.

L'Hopital continued:

We shall discuss two issues connected with L'Hopital rule. Recall, it tells us that if $f(x) \rightarrow 0$ and $g(x) \rightarrow 0$ as $x \rightarrow a$, and if $f'(x)/g'(x) \rightarrow \alpha$, then $f(x)/g(x) \rightarrow \alpha$. This is called the $0/0$ form. We shall now show that the same result holds even for ∞/∞ form.

L'Hopital's rule:

Let f and g be two differentiable functions on (a, b)

$\lim_{x \rightarrow 0^+} f(x) = \infty$ and $\lim_{x \rightarrow 0^+} g(x) = \infty$; $g'(x) \neq 0$ on (a, b) ; and

$\lim_{x \rightarrow 0^+} f'(x)/g'(x) \rightarrow \alpha$ as $x \rightarrow a$. Then $\lim_{x \rightarrow 0^+} f(x)/g(x) \rightarrow \alpha$ as $x \rightarrow a$.

The proof proceeds along the same lines as the $0/0$ case, but a little more involved.

We treat the case $\alpha \in \mathbb{R}$. Other cases, namely, $\alpha = \infty$ and $\alpha = -\infty$ are similar. So let $\epsilon > 0$ be given. Need to show a number $\delta > 0$ so that

$$x \in (a, a + \delta) \Rightarrow \left| \frac{f(x)}{g(x)} - \alpha \right| < \epsilon.$$

As earlier, using the hypothesis, fix $\delta_1 > 0$

$$x \in (a, a + \delta_1) \Rightarrow f(x) > 0; g(x) > 0; \alpha - \frac{\epsilon}{4} < \frac{f'(x)}{g'(x)} < \alpha + \frac{\epsilon}{4}.$$

Since both $f(0^+)$ and $g(0^+)$ are infinity, first two requirements can be achieved and because of hypothesis on f'/g' the third is possible. Let us fix a number y with $a < y < a + \delta_1$ (sort of a reference point and will not be changed in our calculations from now). The generalised MVT implies

$$a < x < y \Rightarrow \alpha - \frac{\epsilon}{4} < \frac{f(x) - f(y)}{g(x) - g(y)} < \alpha + \frac{\epsilon}{4}.$$

I need $f(x)/g(x)$ and so should get rid of the denominator $(g(x) - g(y))$ but should have $g(x)$ in the denominator. This is achieved if we multiply above inequality by $[g(x) - g(y)]/g(x)$. Unfortunately unless this quantity is positive, above inequalities get reversed. So, take δ_2 so that $0 < \delta_2 < \delta_1$ and for $x \in (a, a + \delta_2)$ we have $g(x) > g(y)$. This is possible because $g(x) \rightarrow \infty$

as $x \rightarrow a$.

Now let us take any $x \in (a, a + \delta_2)$ and multiply the above inequality by the positive number $[g(x) - g(y)]/g(x)$. Thus for $a < x < a + \delta_2$ we have

$$\left(\alpha - \frac{\epsilon}{4}\right) \frac{g(x) - g(y)}{g(x)} < \frac{f(x)}{g(x)} - \frac{f(y)}{g(x)} < \left(\alpha + \frac{\epsilon}{4}\right) \frac{g(x) - g(y)}{g(x)}.$$

equivalently,

$$\left(\alpha - \frac{\epsilon}{4}\right) \left[1 - \frac{g(y)}{g(x)}\right] < \frac{f(x)}{g(x)} - \frac{f(y)}{g(x)} < \left(\alpha + \frac{\epsilon}{4}\right) \left[1 - \frac{g(y)}{g(x)}\right].$$

Let us observe that $[g(y)]/g(x) \rightarrow 0$ as $x \rightarrow a$ because y is fixed and $g(x) \rightarrow \infty$. So left most term converges to $\left(\alpha - \frac{\epsilon}{4}\right) > \left(\alpha - \frac{\epsilon}{2}\right)$ as $x \rightarrow a$. Also right most term converges to $\left(\alpha + \frac{\epsilon}{4}\right) < \left(\alpha + \frac{\epsilon}{2}\right)$ as $x \rightarrow a$. So we can choose δ_3 so that $0 < \delta_3 < \delta_2$ and for $a < x < a + \delta_3$

$$\left(\alpha - \frac{\epsilon}{4}\right) \left[1 - \frac{g(y)}{g(x)}\right] > \alpha - \frac{\epsilon}{2}; \quad \left(\alpha + \frac{\epsilon}{4}\right) \left[1 - \frac{g(y)}{g(x)}\right] < \alpha + \frac{\epsilon}{2}.$$

Thus

$$a < x < a + \delta_3 \Rightarrow \left(\alpha - \frac{\epsilon}{2}\right) < \frac{f(x)}{g(x)} - \frac{f(y)}{g(x)} < \left(\alpha + \frac{\epsilon}{2}\right).$$

or

$$a < x < a + \delta_3 \Rightarrow \left(\alpha - \frac{\epsilon}{2}\right) + \frac{f(y)}{g(x)} < \frac{f(x)}{g(x)} < \left(\alpha + \frac{\epsilon}{2}\right) + \frac{f(y)}{g(x)}.$$

Since $f(y)/g(x) \rightarrow 0$ as $x \rightarrow a$, choose δ_4 so that $0 < \delta_4 < \delta_3$ and for $x \in (a, a + \delta_4)$ we have

$$-\epsilon/2 < \frac{f(y)}{g(x)} < +\epsilon/2.$$

Let now x be such that $a < x < a + \delta_4$ Then

$$\alpha - \epsilon = \alpha - \frac{\epsilon}{2} - \frac{\epsilon}{2} \leq \alpha - \frac{\epsilon}{2} + \frac{f(y)}{g(x)} < \frac{f(x)}{g(x)} <$$

$$\left(\alpha + \frac{\epsilon}{2}\right) + \frac{f(y)}{g(x)} \leq \left(\alpha + \frac{\epsilon}{2}\right) + \frac{\epsilon}{2} = \alpha + \epsilon$$

Thus

$$a < x < a + \delta_4 \Rightarrow \alpha - \epsilon < \frac{f(x)}{g(x)} < \alpha + \epsilon.$$

This completes proof of the rule. ■

Comments similar to the case of $0/0$ apply here too; for example α need not be finite (then you need to change wording, can not use $\alpha + \epsilon$ etc); nothing special about $x \downarrow a$ and can state with $x \uparrow b$; the interval may be infinite. Also you can state $-\infty/\infty$ form etc – trivial modification achieved changing f to $-f$ and so on.

For example evaluate $\lim x^x$ as $x \downarrow 0; x > 0$. We have $f(x) = e^{x \log x}$. If we calculate $\lim x \log x$ then by continuity of exponential function we can complete the calculation. There are several ways of doing this. One way is to use

$$x \log x = \frac{\log x}{1/x}; \quad \text{and} \quad \frac{x^{-1}}{-x^{-2}} = -x \rightarrow 0$$

So by L'Hopital $x^x \rightarrow e^0 = 1$. We used $-\infty/\infty$ form!

Here is another way, already familiar to us. Want $\lim x \log x$ as $x \downarrow 0$. Setting $\log x = -y$ we see $x = e^{-y}$. Thus we need $\lim ye^{-y}$ as $y \uparrow \infty$ which we saw, is zero.

Taylor:

We shall now generalize MVT in a different direction.

I have introduced earlier higher derivatives, without much ado. it is possible to motivate, at least the second derivative. Geometric picture is this. If the function is not a straight line, then it must be curving, deviating from being straight. What is the rate? How do you define and measure it?

Particle picture is this. If the velocity is not constant, then it is changing. What is its rate of change? Just as first derivative (velocity) is rate of change of distance travelled, second derivative is the rate of change of velocity. This is called acceleration.

Analytical picture is this. Given a function f and a point a we wanted first a constant function which approximates f near a . More precisely, wanted constant function $\varphi(x) \equiv c$ so that $f(x) - \varphi(x) \rightarrow 0$ as $x \rightarrow a$. The function is the constant function equal to the number $f(a)$. Then we allowed a little more general functions, namely straight line function φ but wanted better approximation. More precisely, wanted $\varphi(x) = \alpha + \beta x$ so that $[f(x) - \varphi(x)]/[x - a] \rightarrow 0$ as $x \rightarrow a$. of course this, in particular implies $f(x) - \varphi(x) \rightarrow 0$ as $x \rightarrow a$. We saw $\varphi(x) = f(a) + f'(a)(x - a)$.

We can allow quadratic functions but demand more better approximation, $[f(x) - \varphi(x)]/(x - a)^2 \rightarrow 0$. Remember, this implies, in particular, $f(x) - \varphi(x) \rightarrow 0$ and also $[f(x) - \varphi(x)]/(x - a) \rightarrow 0$. All these limits are as $x \rightarrow a$. We shall not pursue the details.

Returning to our discussion on higher derivatives, you may be able to talk of only third derivative but not fourth etc. it all depends on the function. Look at the example involving $x^8 \sin(1/x)$.

If our function f is given by a power series, then the fundamental theorem on power series tells us that f' is again given by a power series and hence f'' is again given by a power series. this goes on forever; repeated application of the fundamental theorem on power series.

Recall we use $f^{(k)}$ to denote the k -th derivative of f . Do not confuse it with k -th power of f , pay attention to the brackets. Thus for the first derivative we use $f^{(1)}$ or f' . Sometimes we use, for notational convenience, $f^{(0)}$ for f itself, zeroth derivative! We quickly went through the notation while discussing two examples last time.

Let us now restate MVT as follows. If f is a continuous function on $[a, b]$ which is differentiable at every point of (a, b) , then there is a number $\theta \in (a, b)$ such that

$$f(b) = f(a) + (b - a)f'(\theta).$$

The value of the function at b is explained in terms of $f(a)$ and derivative at some point.

Let us, from now on, suppose that f is defined on an open interval which includes $[a, b]$. suppose the function has second derivative. Can we explain $f(b)$ in terms of $f(a)$, $f'(a)$ and second derivative at some point, — probably a better explanation of $f(b)$? Yes. But before doing this, let us look at proof of the MVT. Let

$$\varphi(x) = f(x) - f(a) - C(x - a).$$

We see $\varphi(a) = 0$. the number C is so chosen that $\varphi(b) = 0$. In other words, $C(b - a) = f(b) - f(a)$. There is a number $\theta \in (a, b)$ such that $\varphi'(\theta) = 0$, a point where φ has max or min. In case φ is constant any point would do. Thus $\varphi'(\theta) = f'(\theta) - C = 0$ Or $f(b) - f(a) = C(b - a) = f'(\theta)(b - a)$. Thus

$$f(b) = f(a) + (b - a)f'(\theta).$$

Here is now the extension. let f be two times differentiable function on an open interval which includes two points a and b , say, $a < b$. Then there is a number $\theta \in (a, b)$ such that

$$f(b) = f(a) + (b - a)f'(a) + \frac{1}{2!}(b - a)^2 f''(\theta).$$

Proof is exactly as above. Put

$$\varphi(x) = f(x) - f(a) - (x - a)f'(a) - C\frac{1}{2!}(x - a)^2.$$

Note that $\varphi(a) = 0$. Choose C so that $\varphi(b) = 0$. that is

$$C \frac{1}{2!}(b-a)^2 = f(b) - f(a) - (b-a)f'(a).$$

We get $\xi \in (a, b)$ so that $\varphi'(\xi) = 0$. Note that

$$\varphi'(x) = f'(x) - f'(a) - C(x-a)$$

Observe that $\varphi'(a) = 0$ and MVT applied to φ' gives $\theta \in (a, \xi)$ such that $\varphi^{(2)}(\theta) = 0$. But $\varphi^{(2)} = f^{(2)} - C$. Thus $f^{(2)}(\theta) = C$. Hence,

$$\frac{1}{2!}(b-a)^2 f^{(2)}(\theta) = C \frac{1}{2!}(b-a)^2 = f(b) - f(a) - (b-a)f'(a).$$

That is

$$f(b) = f(a) + (b-a)f'(a) + \frac{1}{2!}(b-a)^2 f^{(2)}(\theta).$$

Suppose that f is three times differentiable on an interval which includes $a < b$. then there is a number $\theta \in (a, b)$ such that

$$f(b) = f(a) + (b-a)f'(a) + \frac{1}{2!}(b-a)^2 f^{(2)}(a) + \frac{1}{3!}(b-a)^3 f^{(3)}(\theta).$$

Proof is exactly as above. Define

$$\varphi(x) = f(x) - f(a) - (x-a)f'(a) - \frac{1}{2!}(x-a)^2 f^{(2)}(a) - C \frac{1}{3!}(x-a)^3.$$

$$\varphi'(x) = f'(x) - f'(a) - (x-a)f^{(2)}(a) - C \frac{1}{2!}(x-a)^2.$$

$$\varphi''(x) = f''(x) - f''(a) - C(x-a).$$

$$\varphi^{(3)}(x) = f^{(3)}(x) - C.$$

Note $\varphi(a) = \varphi'(a) = \varphi''(a) = 0$. Choose C so that $\varphi(b) = 0$. Apply MVT for φ to get $\xi \in (a, b)$ with $\varphi'(\xi) = 0$, then MVT for φ' to get $\eta \in (a, \xi)$ with $\varphi^{(2)}(\eta) = 0$, then MVT for φ'' to get $\theta \in (a, \eta)$ with $\varphi^{(3)}(\theta) = 0$. This will give the desired result.

If you have understood this, there is no difficulty in proving the following.

Theorem: If f is n times differentiable in an open interval which includes two points $a < b$, then there is a number $\theta \in (a, b)$ so that

$$f(b) = f(a) + (b-a)f'(a) + \frac{1}{2!}(b-a)^2 f^{(2)}(a) + \frac{1}{3!}(b-a)^3 f^{(3)}(a) + \dots$$

$$+ \frac{1}{(n-1)!} (b-a)^{(n-1)} f^{(n-1)}(a) + \frac{1}{n!} (b-a)^n f^{(n)}(\theta).$$

In compact notation

$$f(b) = \sum_0^{n-1} \frac{(b-a)^k}{k!} f^{(k)}(a) + \frac{1}{n!} (b-a)^n f^{(n)}(\theta). \quad \blacksquare$$

Recall that here we used $f^{(0)}$ for f . This is convenient notation to push it under the summation sign. Also we used $0! = 1$. Shall explain proof, just execution of what I said already. $\blacksquare$

Of course, there is nothing special in taking $a < b$, we have done it to fix ideas. We could have taken $a > b$. Exactly, the same formula is valid. Instead of saying $\theta \in (a, b)$ we say, θ is between a and b .

Let us look at a special case. suppose that f is a function defined on $(-r, r)$ where $r > 0$, differentiable n times. Let us take $a = 0$ and b any point of this interval. we get the following. There is a number θ between zero and b so that

$$f(b) = \sum_0^{n-1} \frac{b^k}{k!} f^{(k)}(0) + \frac{1}{n!} b^n f^{(n)}(\theta).$$

In other words, for every $x \in (-r, r)$, there is a number θ between zero and x ; (this θ depends on the point x) so that the following holds.

$$f(x) = \sum_0^{n-1} \frac{f^{(k)}(0)}{k!} x^k + \frac{f^{(n)}(\theta)}{n!} x^n.$$

In long hand,

$$f(x) = f(0) + \frac{f'(0)}{1!} x + \frac{f''(0)}{2!} x^2 + \cdots + \frac{f^{(n-1)}(0)}{(n-1)!} x^{n-1} + R_n(x)$$

where

$$R_n(x) = \frac{f^{(n)}(\theta)}{n!} x^n,$$

for some θ between zero and x . The number θ depends on the point x . and so the above expression is NOT a polynomial.

The above formula is called Taylor expansion of f around zero, with remainder R_n ; or simply, Taylor formula with remainder.

Assume, for a moment that our function f has derivatives of all orders, and they are bounded by a number M . That is, $f^{(n)}$ exists for every n and $|f^{(n)}(x)| \leq M$ for all x in the interval $(-r, r)$ and also for all integers $n \geq 1$. For example if we take the function e^x or $\sin x$ or $\cos x$ and $r = 1000$, this holds. These are not the only functions.

Then we can keep on writing the Taylor formula for every n . Interestingly,

$$|R_n(x)| \leq M \frac{x^n}{n!} \rightarrow 0,$$

because the series $\sum (x^n/n!)$ converges. Where does this lead us? For every $x \in (-r, r)$

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^n(0)}{n!}x^n + \cdots$$

simply because if you consider the n -th partial sum on the right side, it differs from left side by an amount that converges to zero.

What does all this mean? Under the conditions we assumed, the unknown function is actually a power series, no more complicated! Once you understand what is going on, you can do better. For example, you can assume that there is a number M such that the n -th derivative on the interval $(-r, r)$ is bounded by M^n instead of M . Exactly the same proof works.

you can even do better. Assume that for any given c , with $0 < c < r$; there is a number M (depending on c) so that $|f^{(n)}(x)| \leq M^n$ for every $x \in [-c, c]$ and every n . Of course these trivial statements look complicated at this stage and you need not bother. Just ignore this para, if you feel so.

Firstly, understand that there are functions which are differentiable exactly 30 times and no more. For such functions, you can not talk of Taylor series for $n > 30$. So let us consider only functions which are differentiable n times for every n . then the above thought process leads us to guess that probably, every such function is a power series.

There are two questions now. firstly, is the above guess true? Secondly, if the function is actually a power series, does Taylor series give us the same or some different series. Answer to the first question is in the negative as is clear from the example discussed last time: $\exp(-1/x^2)$ for $x > 0$ and zero for $x \leq 0$. This is because $f^{(n)}(0) = 0$ and thus Taylor series fails. There are functions which are infinitely differentiable, but do not come from a power

series.

Answer to the second question is: yes. This is simple to see. suppose that we do have a function f given by a power series.

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \cdots; \quad -r, x < r.$$

Clearly, $f(0) = a_0$. Using the fundamental theorem on power series, we see

$$f'(0) = a_1; \quad f''(0) = 2!a_2; \cdots, \quad f^{(k)}(0) = k!a_k.$$

Thus $f^{(k)}(0)/k! = a_k$ and Taylor series agrees with the given power series. Replace a_k by $f^{(k)}(0)/k!$ and read.

We shall prove Taylor formula now. Actually, I have done for once/twice/thrice differentiable cases so that you can prove the general case yourself. We need to prove:

If f is n times differentiable in an interval which includes two points $a < b$, then there is a number $\theta \in (a, b)$ so that

$$f(b) = \sum_0^{n-1} \frac{(b-a)^k}{k!} f^{(k)}(a) + \frac{1}{n!} (b-a)^n f^{(n)}(\theta). \quad \blacksquare$$

For the proof, you define

$$\varphi(x) = f(x) - \sum_0^{n-1} \frac{(x-a)^i}{i!} f^{(i)}(a) - \frac{1}{n!} (x-a)^n C.$$

Note again that $f^{(0)}(a) = f(a)$. Prove by induction that for $1 \leq k \leq n-1$,

$$\varphi^{(k)}(x) = f^{(k)}(x) - \sum_{i=k}^{n-1} \frac{(x-a)^{i-k}}{(i-k)!} f^{(i)}(a) + \frac{1}{(n-k)!} (x-a)^{n-k} C.$$

$$\varphi^{(n-1)}(x) = f^{(n-1)}(x) - (x-a)C; \quad \varphi^{(n)}(x) = f^{(n)}(x) - C.$$

Observe $\varphi^{(k)}(a) = 0$ for $k = 0, 1, 2, \dots, n-1$. Now let the constant C be so chosen that $\varphi(b) = 0$. That is

$$\frac{1}{n!} (b-a)^n C = f(b) - \sum_0^{n-1} \frac{(b-a)^k}{k!} f^{(k)}(a).$$

As earlier, applying MVT successively for $\varphi, \varphi', \varphi''$ etc you finally get $\theta \in (a, b)$ so that $\varphi^{(n)}(\theta) = 0$. In other words $f^{(n)}(\theta) = C$. Thus

$$\frac{1}{n!} (b-a)^n f^{(n)}(\theta) = f(b) - \sum_0^{n-1} \frac{(b-a)^k}{k!} f^{(k)}(a)$$

as was to be proved. $\blacksquare$

Before taking up integration, we shall collect a few results which are interesting and important.

Misc 1: Convex functions:

A real valued function φ defined on an interval I is said to be convex if $x, y \in I$ and $0 < \lambda < 1$ then

$$\varphi(\lambda x + (1 - \lambda)y) \leq \lambda\varphi(x) + (1 - \lambda)\varphi(y)$$

This is same as saying that the chord joining two points on the curve is above the curve (does not go below the curve). Indeed the points on the curve are $(x, \varphi(x))$ and $(y, \varphi(y))$ with $x < y$; then the equation of the chord is — using running coordinates (X, Y) (keep in mind x, y are some points we have fixed to start with)

$$Y - \varphi(x) = \frac{\varphi(y) - \varphi(x)}{y - x} (X - x)$$

Any point between x, y can be written as $\lambda x + (1 - \lambda)y$ for some $0 < \lambda < 1$. So taking $X = \lambda x + (1 - \lambda)y$ so that $(X - x) = (1 - \lambda)(y - x)$ we see

$$Y = \varphi(x) + [\varphi(y) - \varphi(x)](1 - \lambda) = \lambda\varphi(x) + (1 - \lambda)\varphi(y)$$

is the vertical coordinate of the chord at $\lambda x + (1 - \lambda)y$. showing that the geometric picture is same as the definition above.

Here is an example of a convex function: $I = [0, 1]$ and $\varphi(0) = \varphi(1) = 1$ and $\varphi(x) = 0$ for $0 < x < 1$. However convex functions defined on open intervals are nice.

From now on our convex functions φ are defined on an open interval (a, b) possibly an infinite interval.

(1) for $s < t < u$, the difference quotients are increasing (that is, non-decreasing). In other words,

$$\frac{\varphi(t) - \varphi(s)}{t - s} \leq \frac{\varphi(u) - \varphi(t)}{u - t}$$

Proof is simple algebra:

$$t = \frac{u - t}{u - s}s + \frac{t - s}{u - s}u$$

so that

$$\varphi(t) \leq \frac{u - t}{u - s}\varphi(s) + \frac{t - s}{u - s}\varphi(u)$$

or

$$(u - t)\varphi(t) + (t - s)\varphi(t) \leq (u - t)\varphi(s) + (t - s)\varphi(u)$$

Rearrange.

(2) for any four points $s < t < u < v$

$$\frac{\varphi(t) - \varphi(s)}{t - s} \leq \frac{\varphi(v) - \varphi(u)}{v - u}$$

Immediately follows from above.

(3) Let $[c, d] \subset (a, b)$ be a bounded interval. Then φ is continuous on (c, d) . Since $a < c$ and $d < b$ we can fix two points $a < w < z < c$ and $d < u < v < b$. Then we have for any $s < t$ in $[c, d]$

$$\frac{\varphi(z) - \varphi(w)}{z - w} \leq \frac{\varphi(t) - \varphi(s)}{t - s} \leq \frac{\varphi(v) - \varphi(u)}{v - u} \quad (\diamond)$$

So if $\frac{\varphi(t) - \varphi(s)}{t - s} \geq 0$ then so is the last term in $(\diamond)$ and thus

$$\left| \frac{\varphi(t) - \varphi(s)}{t - s} \right| \leq \left| \frac{\varphi(v) - \varphi(u)}{v - u} \right|$$

If $\frac{\varphi(t) - \varphi(s)}{t - s} \leq 0$ then so is the first term in $(\diamond)$ and thus

$$\left| \frac{\varphi(t) - \varphi(s)}{t - s} \right| \leq \left| \frac{\varphi(z) - \varphi(w)}{z - w} \right|$$

Thus, in any case, if you denote (Remember z, w, u, v are points fixed by us)

$$M = \max \left\{ \left| \frac{\varphi(z) - \varphi(w)}{z - w} \right|, \left| \frac{\varphi(v) - \varphi(u)}{v - u} \right| \right\}$$

then

$$|\varphi(t) - \varphi(s)| \leq M|t - s| \quad \forall s, t \in [c, d]$$

This shows continuity because whenever $t \in (c, d)$ and $s_n \rightarrow t$ then after some stage $s_n \in (c, d)$ and hence

$$|\varphi(t) - \varphi(s_n)| \leq M|t - s_n| \rightarrow 0$$

(4) for any $t \in (a, b)$ the left derivative $\varphi'_-(t)$ and right derivative $\varphi'_+(t)$ of φ exist, finite and $\varphi'_-(t) \leq \varphi'_+(t)$. Here is the definition of left and right derivatives.

$$\varphi'_-(t) = \lim_{s < t, s \uparrow t} \frac{\varphi(t) - \varphi(s)}{t - s}; \quad \varphi'_+(t) = \lim_{s > t, s \downarrow t} \frac{\varphi(t) - \varphi(s)}{t - s}$$

regarding left derivative, we show that if $s_1 < s_2 < t$ then

$$\frac{\varphi(t) - \varphi(s_1)}{t - s_1} \leq \frac{\varphi(t) - \varphi(s_2)}{t - s_2}$$

In other words the ‘difference quotients’ increase as s increases to t and hence have a limit, namely the supremum of the quotients. To prove the above is again routine. We already know

$$\begin{aligned}\frac{\varphi(s_2) - \varphi(s_1)}{s_2 - s_1} &\leq \frac{\varphi(t) - \varphi(s_2)}{t - s_2} \\ \frac{\varphi(t) - \varphi(s_1)}{s_2 - s_1} - \frac{\varphi(t) - \varphi(s_2)}{s_2 - s_1} &\leq \frac{\varphi(t) - \varphi(s_2)}{t - s_2} \\ \frac{\varphi(t) - \varphi(s_1)}{s_2 - s_1} &\leq \frac{\varphi(t) - \varphi(s_2)}{t - s_2} \frac{t - s_1}{s_2 - s_1}\end{aligned}$$

Also each left quotient is smaller than a right quotient, showing that the set of left quotients is bounded above. Thus it has a supremum. In particular whenever $s_n \uparrow t$ and $s_n < t$ for each n , then the quotients converge to the supremum, establishing existence of $\varphi'_-(t)$. The right derivative is similar.

(5) If $t < u$ then

$$\varphi'_-(t) \leq \varphi'_+(t) \leq \varphi'_-(u) \leq \varphi'_+(u)$$

In particular there are countably many points where φ is not differentiable.

The stated inequalities are already clear from the previous calculation. The last claim follows from the others because if φ is not differentiable at t then we have a non-degenerate interval $(\varphi'_-(t), \varphi'_+(t))$ and for different discontinuity points these intervals are disjoint – an argument we have seen earlier.

(6) suppose that φ is differentiable. φ is convex iff the first derivative is increasing: $s \leq t$ implies $\varphi'(s) \leq \varphi'(t)$.

convexity implies φ' is increasing from earlier result. Conversely, if φ' is increasing then for $s < t < u$ the following inequality holds by MVT:

$$\frac{\varphi(t) - \varphi(s)}{t - s} \leq \frac{\varphi(u) - \varphi(t)}{u - t}$$

This inequality is equivalent to convexity.

(7) Suppose φ is twice differentiable. Then φ is convex iff $\varphi'' \geq 0$. Immediate from the previous statement.

(8) Given a point of on the graph of φ we can draw a line passing through that point and lying below (not necessarily strictly below) the curve.

This line is called supporting line at that point for φ . You can think of it as tangent line at that point; but we did not say so because the function may not be differentiable at that point. Also note that the supporting line

may pass through several other points. For example, Consider the function on $[0, 1]$ whose graph consists of lines joining $(0, 1) \rightarrow (1/3, 0) \rightarrow (2/3, 0) \rightarrow (1, 1)$. You can consider the graph on $(0, 1)$ if you want domain to be open interval. This is clearly convex, and supporting line for any point on the curve strictly between $1/3$ and $2/3$ will obviously include the full line segment $(1/3, 0) \rightarrow (2/3, 0)$. Actually the same is true at points on the other two segments of the curve. However note that at the points $(1/3, 0)$ and $(2/3, 0)$ you can draw several supporting lines. Observe φ is not differentiable at these points $1/3$ and $2/3$.

Proof is simple. Let $(t, \varphi(t))$ be the given point. Set

$$\alpha = \sup_{s < t} \frac{\varphi(s) - \varphi(t)}{s - t}; \quad \beta = \inf_{t < u} \frac{\varphi(t) - \varphi(u)}{t - u}$$

Since each difference quotient that appears in making α is smaller than each of the difference quotients that make β we see that the first and second sets of difference quotients appearing in the display are bounded above and bounded below respectively. Thus α, β are finite and $\alpha \leq \beta$. Take any number m ; $\alpha \leq m \leq \beta$. Consider the line

$$y = mx + \varphi(t) - mt.$$

This is a straight line. It passes through $(t, \varphi(t))$. To show it lies below. Let $s < t$. We need to show $y(s) \leq \varphi(s)$; that is

$$ms + \varphi(t) - mt \leq \varphi(s) \quad \text{i.e.} \quad \varphi(t) - \varphi(s) \leq mt - ms$$

which is true by definition of m . Next let $u > t$. We need to show again $y(u) \leq \varphi(u)$; that is

$$mu + \varphi(t) - mu \leq \varphi(u) \quad \text{i.e.} \quad mu - mt \leq \varphi(t) - \varphi(u)$$

which is again true by definition of m .

(9) Here are some convex functions: $\varphi(x) = e^x$ on R ; $\varphi(x) = x^2$ on R ; $\varphi(x) = |x|$. You already saw others above.

Convex functions occur in several place in maths. In fact there is convex optimization which is very useful. Of course, you need to define convex functions of several variables; imagine their graphs etc. The supporting line above will then be called supporting hyperplane. (line/plane/hyperplane).

Misc 2: Newton's algorithm for zero

This is a computational application of the Taylor formula. It is to find a zero of a function. Shall present the most primitive version of an algorithm Newton discovered.

Suppose we have a function $f : R \rightarrow R$. Let z be a zero of f , that is, $f(z) = 0$. Find an algorithm to calculate z with required degree of accuracy, if we know roughly where it is located.

Start with a point z_0 . This is your initial or starting approximation to z (and hence, in general, you need to start close to z). Draw tangent to the curve, graph of f , at $(z_0, f(z_0))$. Suppose it cuts the x -axis at z_1 (so you need to assume that tangent is not parallel to the x -axis). This is your first approximation to z . Then draw tangent to the curve at $(z_1, f(z_1))$. Suppose that it cuts the x -axis at z_2 . This is your second approximation. Continue the process. The hope is you are heading towards to z , the actual zero.

What makes us hope so? Well, as is always the case, look at some examples. For instance consider the curve $f(x) = x^2 - 2$ and take $z_0 = 1$ and try. Take the curve $f(x) = x^3 - 2$ and try again. of course, the fact that it works in the examples we have seen is not good enough to believe that this is always true. In fact it is not always true that these numbers z_n so obtained converge to the actual zero. But, under fairly general conditions they do converge.

Let us see analytically what the algorithm says. The tangent at $(z_0, f(z_0))$ to the curve is given by

$$y - f(z_0) = f'(z_0)(x - z_0).$$

To find the point of intersection with the x -axis, set $y = 0$ to see

$$z_1 = z_0 - \frac{f(z_0)}{f'(z_0)}.$$

To get z_2 you repeat the same formula above. Here is the precise statement.

Theorem:

f is a twice differentiable function in an open interval (a, b) .

z is a point in (a, b) with $f(z) = 0$.

$f'(x)$ is never zero in this interval. $0 < \alpha < \infty$ where

$$\frac{\sup |f''(x)|}{\inf |f'(y)|} \leq \alpha \quad \text{i.e.} \quad \forall x \forall y \quad \frac{|f''(x)|}{|f'(y)|} \leq \alpha.$$

Choose $0 < \delta < 1/\alpha$ such that $(z - \delta, z + \delta) \subset (a, b)$.
start with $z_0 \in (z - \delta, z + \delta)$. Define recursively,

$$z_{n+1} = z_n - \frac{f(z_n)}{f'(z_n)}.$$

Then each of the points z_n is in the interval I and $z_n \rightarrow z$.

Here is the proof. Suppose $z_n \in I$. By Taylor expansion around z_n , there is a point θ between z_n and z such that

$$f(z) = f(z_n) + f'(z_n)(z - z_n) + \frac{1}{2}f''(\theta)(z - z_n)^2.$$

Since $f(z) = 0$ we get

$$f(z_n) = -f'(z_n)(z - z_n) - \frac{1}{2}f''(\theta)(z - z_n)^2.$$

so that by definition of z_{n+1}

$$z_{n+1} = z_n - \frac{f(z_n)}{f'(z_n)} = z_n + (z - z_n) + \frac{1}{2} \frac{f''(\theta)}{f'(z_n)}(z - z_n)^2.$$

$$|z_{n+1} - z| = \left| \frac{1}{2} \frac{f''(\theta)}{f'(z_n)}(z - z_n)^2 \right| \leq \frac{1}{2} \alpha |z_n - z|^2 \quad (\spadesuit)$$

Since $z_0 \in I$ we can apply $(\spadesuit)$ with $n = 0$ to see

$$|z_1 - z| \leq \frac{1}{2} \{ \alpha |z_0 - z| \} |z_0 - z| < \frac{1}{2} |z_0 - z|.$$

where we used $\alpha |z_0 - z| \leq \alpha \delta < 1$. This shows $z_1 \in (z - \delta, z + \delta)$ and in fact closer to z than z_0 . In particular $\alpha |z_1 - z| < 1$. Thus again $(\spadesuit)$ applies with $n = 1$ to give

$$|z_2 - z| \leq \frac{1}{2} \{ \alpha |z_1 - z| \} |z_1 - z| < \frac{1}{2} |z_1 - z| \leq \frac{1}{2^2} |z_0 - z|.$$

By induction, you can conclude the following for each n :

$$z_n \in I; \quad \alpha |z_{n-1} - z| < 1; \quad |z_n - z| \leq \left(\frac{1}{2} \right)^n |z_0 - z|.$$

This completes proof of the assertion made.

Misc 3. Euler Constant

This is called Euler-Mascheroni constant. We know that the Harmonic series $\sum(1/n)$ does not converge. How fast are the partial sums increasing? Answer: s_n is like $\log n$ upto an additive constant. Define

$$\gamma_n = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n} - \log n$$

We claim $\gamma_n \rightarrow \gamma$; $0 < \gamma < \infty$. This we show by proving (i) the sequence is decreasing and (ii) it is bounded below by a positive number. First as you suggested, by MVT, $\log(n+1) - \log n$ equals $1/x$ for some $n < x < n+1$. Thus

$$\frac{1}{n+1} < \log(n+1) - \log(n) < \frac{1}{n} \quad (\dagger)$$

Using this

$$\gamma_n - \gamma_{n+1} = -\frac{1}{n+1} + [\log(n+1) - \log n] > 0$$

This proves γ_n is decreasing. Again using $(\dagger)$

$$\gamma_n = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n} - \log n >$$

$$1 + [\log 3 - \log 2] + [\log 4 - \log 3] + \cdots [\log(n+1) - \log n] - \log n \\ 1 - \log 2 + [\log(n+1) - \log n] > 1 - \log 2$$

for all n . Note that $e > 2$ so that $\log 2 < 1$. This proves (ii). We still do not know whether γ is rational or not.

If you want to prove $(\dagger)$ without differentiation, here it is. We knew that $(1 + \frac{1}{n})^n \uparrow e$ and e is not rational. So $(1 + \frac{1}{n})^n < e$. That is $n[\log(n+1) - \log n] < \log e = 1$ or $[\log(n+1) - \log n] < 1/n$. Similar argument gives the other inequality if we show $a_n = (1 + \frac{1}{n})^{n+1} \downarrow e$. That it is converging to e is immediate because the extra factor $(1 + \frac{1}{n})$ is converging to one. We only need to show a_n is decreasing as n increases.

$$\frac{a_{n-1}}{a_n} = \left(\frac{n}{n-1}\right)^n \left(\frac{n}{n+1}\right)^{(n+1)} = \left(\frac{n^2}{n^2-1}\right)^n \left(\frac{n}{n+1}\right) \\ = \left(1 + \frac{1}{n^2-1}\right)^n \left(\frac{n}{n+1}\right) \geq \left(1 + \frac{n}{n^2-1}\right) \left(\frac{n}{n+1}\right)$$

where we used $(1+x)^n \geq 1+nx$

$$\geq \left(1 + \frac{n}{n^2}\right) \left(\frac{n}{n+1}\right) = 1$$

Returning to the alternating harmonic series, we can show that it adds to $\log 2$. Here is how.

$$\gamma_{2n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{2n} - \log(2n)$$

$$\gamma_n = 2\left[\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \cdots + \frac{1}{2n}\right] - \log(n)$$

so that

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{2n} - \log 2 = \gamma_{2n} - \gamma_n \rightarrow 0$$

That is

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{2n} \rightarrow \log 2$$

as stated. Note that Taylor, gave you a series for $\log(1+x)$ for $|x| < 1$, what the above says is that this Taylor series holds for $x = 1$ as well.

The above leads to the interesting formula

$$2 = \frac{e}{\sqrt[2]{e}} \frac{\sqrt[3]{e}}{\sqrt[4]{e}} \frac{\sqrt[5]{e}}{\sqrt[6]{e}} \frac{\sqrt[7]{e}}{\sqrt[8]{e}} \cdots$$

Where the infinite product means limit of finite partial products.

Misc 4. Circular and Hyperbolic functions:

We defined (and proved continuity, differentiability)

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots = \sum_0^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots = \sum_0^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

These are called **circular functions** because for each x , the point $(\cos x, \sin x)$ lies on circle of radius one centred at the origin. These are also called Trigonometric functions. Apparently these series representations were discovered by Madhava (of Kerala) three hundred year before Newton. Here are other trigonometric functions

$$\tan x = \frac{\cos x}{\sin x}; \quad \cot x = \frac{\sin x}{\cos x} = \frac{1}{\tan x}$$

$$\sec x = \frac{1}{\cos x} \quad \operatorname{cosec} x = \frac{1}{\sin x}$$

Though sine and cosine functions are defined for all $x \in \mathbb{R}$, these functions above are defined only when the denominator is nonzero. Thus their domain is union of a sequence of open intervals.

We have shown existence of zero for cosine function earlier, using derivative of cosine is sine. Here is another way.

$$\begin{aligned}\cos 2 &= 1 - \frac{2^2}{2!} + \frac{2^4}{4!} - \left(\frac{2^6}{6!} - \frac{2^8}{8!}\right) - (?-?) - \dots \\ &= -\frac{1}{3} - (\geq 0) - (\geq 0) - \dots \leq -\frac{1}{3}\end{aligned}$$

The inequality is because all odd partial sums after third are below $(-1/3)$ and hence so is the limit. (Even though we can justify the bracketing because the series is absolutely convergent; there is no need; the bracketing is only to say something about the odd **finite partial sums**).

If we have introduced complex exponential series then we see, equating real and imaginary parts, $e^{ix} = \cos x + i \sin x$ (De Moivre's formula of high school). This will help show, for example again by equating real and imaginary parts,

$$\sin(x+y) = \sin x \cos y + \cos x \sin y \quad \cos(x+y) = \cos x \cos y - \sin x \sin y$$

Since we have not discussed complex exponential, we do hands on calculation.

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \quad \cos y = 1 - \frac{y^2}{2!} + \frac{y^4}{4!} - \frac{y^6}{6!} + \dots$$

Use Cauchy product

$$\sin x \cos y = x - \left[x \frac{y^2}{2!} + 1 \cdot \frac{x^3}{3!} \right] + \left[x \frac{y^4}{4!} + \frac{x^3 y^2}{3! 2!} + \frac{x^5}{5!} \cdot 1 \right] - \dots \quad (*)$$

$$\cos x \sin y = y - \left[y \frac{x^2}{2!} + 1 \cdot \frac{y^3}{3!} \right] + \left[y \frac{x^4}{4!} + \frac{y^3 x^2}{3! 2!} + \frac{y^5}{5!} \cdot 1 \right] - \dots \quad (**)$$

Adding we get $\sin(x+y)$ formula. Note that invoking Cauchy product for the above two equations we used absolute convergence. However for adding them uses the simple rule if $\sum a_n = A$ and $\sum b_n = B$, then $\sum c_n = A + B$ where $c_n = a_n + b_n$.

Similarly You get $\cos(x+y)$. In fact you can try $\sin(x-y)$ and $\cos(x-y)$ for practice.

We define

$$\cosh x = \frac{e^x + e^{-x}}{2} \quad \sinh x = \frac{e^x - e^{-x}}{2}$$

These are called hyperbolic functions. For each x , the point $(\cosh x, \sinh x)$ lies on the hyperbola $X^2 - Y^2 = 1$. Just like circular functions, $\sinh x$ is an odd function and $\cosh x$ is an even function. $\sinh 0 = 0$ and $\cosh 0 = 1$. But $\cosh x$ never vanishes and $\sinh x$ vanishes only at zero. You can easily show the following. Now you can multiply the real exponentials you see without using Cauchy product.

$$\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y$$

$$\cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y$$

You can imitate Trigonometric functions and define $\tanh x = \sinh x / \cosh x$ and $\coth x = \cosh x / \sinh x$ and $\operatorname{sech} x = 1 / \cosh x$ And $\operatorname{cosech} x = 1 / \sinh x$. Hyperbolic functions have important use in Engineering and Physics.

Misc 5. Logarithm:

We defined the functions $f(x) = e^x$ and its inverse $g(x) = \log x$. Remember $f : R \rightarrow (0, \infty)$ whereas $g : (0, \infty) \rightarrow R$. This logarithm is called logarithm to the base e or natural logarithm. Actually it is denoted, not as $\log x$ that I am doing, but as $\ln x$. (Sorry, do not want to change notation now). Sometimes it is important to use other base. In fact we use logarithm to the base 10, called common logarithm. Here is the definition.

Let $a > 0, a \neq 1$. Logarithm to the base a is defined by $\log_a x = \frac{\log x}{\log a}$, defined for $x > 0$.

The log, without base, you see in the above definition are what we have already defined. So you see if you take $a = e$ you get $\log_e x = \log x$ because $\log e = 1$. Also observe

$$\log_a x = u \leftrightarrow \log x = u \log a \leftrightarrow \log x = \log a^u \leftrightarrow x = a^u$$

Thus $x \mapsto \log_a x$ is nothing but the inverse map of $u \mapsto a^u$ or the inverse map of $x \mapsto x^a$. Here are simple properties which are easy to verify (do it)

$$(i) \log_a(xy) = \log_a x + \log_a y$$

$$(ii) \log_a x = \frac{\log b}{\log a} \log_b x; \text{ called change of base formula. Here } b > 0, b \neq 1.$$

Misc 6. Topology of R

We have been using the word ‘open interval’ for (a, b) and closed interval for

$[a, b]$ without defining what is meant by open and what is meant by interval, etc separately. We shall do this now. Late when you learn Topology course, you learn more. When we say set below, it is a subset of R , thus does not include $\pm\infty$.

$A \subset R$ is called an interval if: $x, y \in A; x < z < y \Rightarrow z \in A$.

Thus interval means a set without any breaks; if two points are in the set, everything between is also in the set. You can verify that the following are intervals: Here $a, b \in R$ and $a \leq b$.

$$\begin{array}{ccccccc} R & (-\infty, b) & (-\infty, b] & (a, +\infty) & [a, +\infty) \\ & (a, b) & [a, b] & (a, b] & [a, b] \end{array}$$

Note that if $a = b$ then $[a, b] = \{a\}$ and $(a, b) = \emptyset$. Convince yourself that these are intervals.

In fact these nine types are the only intervals. Here is the proof. Let A be an interval. If $A = \emptyset$ then we already observed that it is listed above. So assume that $A \neq \emptyset$.

Case 1: A is neither bounded above nor bounded below. So given any z there is $y > z$ in A and $x < z$ in A and hence $z \in A$. This is true for all z . So $A = R$.

Case 2: A is bounded above but not below. Let $b = \sup A$. We claim $z < b$ implies $z \in A$. Indeed, not bounded below gives $x < z$ in A . Also $z < b$, supremum gives you $y > z$ in A . So $z \in A$. Thus, b being sup, $(-\infty, b) \subset A \subset (-\infty, b]$. In case $b \notin A$ then $A = (-\infty, b)$ and in case $b \in A$ then $A = (-\infty, b]$.

Case 3: A is bounded below but not above. Let $a = \inf A$. Convince yourself that A must be (a, ∞) or $[a, \infty)$.

Case 4: A is bounded above and below. $a = \inf A$ and $b = \sup A$. First observe, arguing as above that $(a, b) \subset A \subset [a, b]$. Either $a \in A$ or $a \notin A$. Also either $b \in A$ or $b \notin A$. These give rise to four cases. ■

$A \subset R$ is called an open set if $x \in A \Rightarrow (\exists \epsilon > 0)(x - \epsilon, x + \epsilon) \subset A$

$A \subset R$ is called a closed set if x is a limit point of A implies $x \in A$; equivalently, if there is a sequence $\{x_n\}$ of points in A such that $x_n \rightarrow x$ then $x \in A$.

Thus an open set must contain a small interval centered around each of its points. A closed set is 'closed' under limit points. Clearly (i) $\emptyset$ and R are open sets. (ii) Union of any number of open sets is open. In fact, a point in the union is in one of the sets and that set already contains a required interval

and so the union does. (iii) Intersection of finitely many open sets is open. In fact given a point in the intersection; if $\epsilon_i > 0$ satisfies the requirement for the i -th set then $\epsilon = \min \epsilon_i > 0$ satisfies for the intersection. Minimum of finitely many strictly positive numbers is strictly positive (why?).

Similarly (i) $\emptyset$ and R are closed sets. (ii) Union of any finite number of closed sets is closed. In fact, if a sequence in the union converges to a point, infinitely many terms of that sequence must be in one of the sets and so limit point must be there. (iii) Intersection of any number of closed sets is closed. In fact a sequence in the intersection is clearly a sequence in each of the sets and so its limit must be in each of the sets.

Using these you can cook up several examples of open sets and closed sets. $\cap(5 - 1/n, 5 + 1/n) = \{5\}$, so Intersection of infinitely many open sets need not be open.

Since we have list of all intervals above, it is easy to see that if a set is both an open set and an interval then it must be one of R , $(-\infty, b)$, (a, ∞) , (a, b) . This agrees with what we have been doing so far.

R and $\emptyset$ are both open and closed. Is there any other set which is both closed and open? NO. Here is the proof, you already know the idea, used in proving intermediate value theorem for continuous functions. Let A be both open and closed. If $A \neq \emptyset$ there must be a point in it; pick $x \in A$. If $A \neq R$ there must be a point not in it; pick $y \notin A$. We assume $x < y$; similar argument applies if $y < x$. Let

$$s = \sup\{z \in A : x \leq z \leq y\}$$

Firstly the sup makes sense because the set above is non-empty, it has x ; it also is bounded by y . Thus s is a real number. Since s sup of a set of points in A , there is a sequence in A converging to it and so $s \in A$, because A is closed. Can $s = y$? No because $y \notin A$. Thus $s < y$ and (s, y) is a non-empty interval. and $(s, y) \subset A^c$ by definition of s . But A being open, we must have $(s - \epsilon, s + \epsilon) \subset A$ which is not the case. This contradiction proves the result. ■

$A \subset R$ is called dense in R if every nonempty open set contains a point of A , equivalently, every nonempty open interval contains a point of A

For example set of rationals; set of irrationals; set of algebraic numbers are dense sets. Set of integers, finite sets, are not dense sets. If you take set of rational numbers in the interval $(0, 1)$ then it is not dense in R . However you can say (if you have made appropriate definition) that it is dense in $(0, 1)$. We shall not convert our course into a course in Topology.

Misc 7. Common continuity point:

We had an example of a function whose set of discontinuity points are exactly the set of rational numbers. We were curious if we can have a function whose set of discontinuity points is exactly the set of irrational numbers. I said ‘no’ but did not prove. Here is the proof. Let us use notation C_f for set of continuity points of f .

if f and g are functions defined on R and if both C_f and C_g are dense, then $C_f \cap C_g \neq \emptyset$

In other words there is a point where both f and g are continuous. Since set of rationals is dense and set of irrationals is also dense and they are disjoint, our question has a negative answer. Here is the proof of the above claim. You have seen the idea several times earlier (and that is why I am doing this): One best way to produce a number is to produce nested bounded closed intervals $[a_n, b_n]$ and consider their intersection (and thank Cantor). If the lengths of these intervals decrease to zero, they have exactly one point in common. Otherwise there may be more than one.

We start with an observation. Fix $\epsilon > 0$. Given $a < b$ we can get $[c, d] \subset (a, b)$ such that for $x, y \in [c, d]$ we have $|f(x) - f(y)| < \epsilon$ and also $|g(x) - g(y)| < \epsilon$. Proof is simple. Since C_f is dense get $z \in (a, b)$ such that $z \in C_f$. Choose $(z - \delta, z + \delta) \subset (a, b)$ such that for all x, y in this interval $|f(x) - f(y)| < \epsilon$. For this you only need to choose δ using definition of continuity of f at z so that $|f(x) - f(z)| < \epsilon/2$ for $|x - z| \leq \delta$. Cut down δ so that the stated interval fits in (a, b) . Now start with this interval $(z - \delta, z + \delta)$ instead of (a, b) and consider g instead of f to get $c < d$ so that $[c, d] \subset (z - \delta, z + \delta)$ so that $|g(x) - g(y)| < \epsilon$ for all $x, y \in [c, d]$. This does what we wanted for both f and g .

To complete the proof, Start with any $a < b$. Apply above with $\epsilon = 1/2$ to (a, b) to get $a_1 < b_1$ with $[a_1, b_1] \subset (a, b)$ as stated. Apply above with $\epsilon = 1/3$ to (a_1, b_1) to get $a_2 < b_2$ with $[a_2, b_2] \subset (a_1, b_1)$ as stated. Apply above with $\epsilon = 1/4$ and interval (a_2, b_2) and continue by induction. Take a point $u \in \cap [a_n, b_n]$. We claim both f and g are continuous at u . In fact to verify continuity of f at u , given $\epsilon > 0$ we take n so that $1/n < \epsilon$ and show the interval $[a_n, b_n]$. If you want explicit δ for f , then take $\min\{c - a_n, b_n - c\}$. Convince yourself this does. Similarly this works for g too. ■

As you realize I started with any $a < b$ and showed a point of common continuity. Thus indeed the set of common continuity points is a dense set again. You will learn a more general theorem in a Topology course later.

→ Quiz 2: November 4 Tuesday. ←

Misc 8. Abel's theorem:

Suppose we have a power series $P(x) = \sum a_n x^n$ with radius of convergence one. Then this defines a continuous function in the interval $(-1, 1)$. Suppose $\sum a_n$ converges. Then $P(x)$ is defined at $x = 1$ also. Thus P is defined on $(-1, 1]$. The question arose whether P is continuous on the interval $(-1, 1]$. We only need to discuss continuity at $x = 1$.

Abel's Theorem: If the power series $P(x) = \sum a_n x^n$ with radius of convergence $r = 1$ converges for $x = 1$, then it is continuous at $x = 1$ also. That is $P(1-) = P(1)$, that is, $\lim_{x \uparrow 1} P(x) = \sum a_n$.

Proof: Let $\sum a_n = A$. Let $\epsilon > 0$ be given. We show $\delta < 1$ so that if $\delta < x < 1$, then $|P(x) - A| < \epsilon$.

Let $s_0 = a_0$ and $s_n = \sum_{j=0}^n a_j$ be the partial sums. Thus $s_n \rightarrow A$. Also for $n \geq 1$ we have $a_n = s_n - s_{n-1}$. Hence

$$\begin{aligned} & a_0 + a_1 x + a_2 x^2 + \cdots + a_k x^k + a_{k+1} x^{k+1} \\ &= s_0 + (s_1 - s_0)x + (s_2 - s_1)x^2 + \cdots + (s_{k+1} - s_k)x^{k+1} \\ &= s_0(1-x) + s_1 x(1-x) + s_2 x^2(1-x) + \cdots + s_k x^k(1-x) + s_{k+1} x^{k+1} \\ &\rightarrow P(x) \end{aligned}$$

For $|x| < 1$ we have $x^{k+1} \rightarrow 0$ and also (s_n) is a bounded sequence (being convergent) and Hence $s_{k+1} x^{k+1} \rightarrow 0$. Thus we can say

$$(1-x)[s_0 + s_1 x + s_2 x^2 + s_3 x^3 + \cdots + s_k x^k] \rightarrow P(x)$$

That is,

$$P(x) = (1-x)[s_0 + s_1 x + s_2 x^2 + s_3 x^3 + \cdots]$$

Also from sum of geometric series we have for $|x| < 1$;

$$A = (1-x)[A + Ax + Ax^2 + Ax^3 + \cdots]$$

Thus

$$\begin{aligned} |P(x) - A| &= |(1-x)[(s_0 - A) + (s_1 - A)x + (s_2 - A)x^2 + (s_3 - A)x^3 + \cdots]| \\ &\leq (1-x) \sum_0^N |s_i - A| + (1-x) \sum_{N+1}^{\infty} |s_n - A| x^n \quad 0 < x < 1 \quad (\spadesuit) \end{aligned}$$

whatever be $N \geq 1$. Now choose N so that $n \geq N \rightarrow |s_n - A| < \epsilon/2$. Having fixed this N let us denote $1 + \sum_0^N |s_i - A| = C$. Choose $0 < \delta < 1$ so that $(1 - \delta)C < \epsilon/2$. That is, $\delta > 1 - (\epsilon/2C)$. Observe $\epsilon > 0, C > 0$ so $\delta < 1$. Make sure $\delta > 0$. Thus if $x \in (\delta, 1)$ then $(1 - x)C < (1 - \delta)C < \epsilon/2$. Thus if $x \in (\delta, 1)$, then from (♠)

$$|P(x) - A| \leq \frac{\epsilon}{2} + \frac{\epsilon}{2}(1 - x) \sum_{N+1}^{\infty} x^n < \epsilon$$

as required ■

Integration:

Here is the problem, familiar from high school. How do you calculate areas. if we have a rectangular region, we some how seem sure and agreed upon that its area is product of lengths of sides. Thus areas of rectangles are taken as known. From this we some how built up several other areas, for example area of a circle of radius r equals πr^2 . We also know how to calculate areas of triangles and some other figures like parallelogram, trapezium etc.

[Digression: What is area? We do not know; it is the amount of space in the plane region under consideration. But to make transactions – selling, buying, gifting etc – we need to know. People observed, in case of rectangles, how the amount of space changes. If you increase the length 3 times and breadth 5 times then the ‘amount of space’ gets multiplied by 15; they observed $[0, 3] \times [0, 5]$ is 15 rectangles similar to $[0, 1] \times [0, 1]$. Thus if you fix area of $[0, 1] \times [0, 1]$ as one then area of $[0, m] \times [0, n]$ should be mn .

In high school we had a graph sheet – a sheet of paper with closely and equally spaced vertical and horizontal lines dividing the paper into very very small squares. We used to draw circle of unit radius and count how many of these small squares are contained within the circle. Also count how many of these small squares intersect the inside of the circle, not necessary fully contained in the circle. These give lower and upper bound respectively for the area bounded by the circle. Unfortunately, such things are now considered waste of time and below standard! People forget that the main point here is not the act of drawing and counting, but **idea of how you arrive at a concept** is the main point.]

Let us consider a bounded function f defined on the interval $[0, 1]$. As a first step, let us assume that the function takes only non-negative values, so that the graph of the function is above the x -axis. We also assume that the function is bounded. Consider the region under the curve, bounded below

by the x -axis. On the sides it is bounded by the vertical lines at $x = 0$ and $x = 1$. Thus to the left, the region is bounded by the y -axis and to the right, it is bounded by the vertical line at $x = 1$.

Analytically, it is the region $\{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq f(x)\}$.

How nice if the function is a constant $f \equiv c > 0$. Then the graph of f is just a flat horizontal line at height c and thus the region is just a rectangle with sides of lengths one and c . Its area is c . Suppose the function is piece-wise constant, say, on $[0, 1/4]$ value of f is α ; on $(1/4, 1/3)$ value of f is β and finally on $[1/3, 1]$ value of f is γ . Then the region under consideration consists of three rectangles and so the ‘total’ area of the region equals $\alpha(1/4) + \beta(1/12) + \gamma(2/3)$.

If the function is not piece-wise constant, then the region is, of course, not made up of finitely many rectangles. Let us denote the area by A . The plan is to find a number which bounds A from below and to find a number which bounds A from above. For instance, suppose that there are reasons to believe that the unknown area A must at least be 3 and also can be at most 3. Then you must agree that the area must equal 3, no matter how you want to define area. Of course, if you can only conclude that the area must at least be 3 and can not exceed 4, then we are sure that it must be a number between 3 and 4; but not yet sure what exactly it should equal.

If you are trying to see a lower bound for the area, the best way is to fit in non-overlapping rectangles in our region and take their total area. this can be done in several ways. what is the most efficient way of doing it? here is what Riemann’s idea is. Break up the interval $[0, 1]$ into finitely many intervals, say $[0, a_1], [a_1, a_2] \cdots [a_{55}, 1]$ — 56 pieces. On each piece, let us calculate the infimum of the function. Let them be $m_0, m_1, \dots, m_{55}$. Consider the rectangles with base $[a_i, a_{i+1}]$ and height m_i for $i = 0, 1, 2, \dots, 55$. Clearly these rectangles are non-overlapping (have common sides) and are all within the region of our interest. Thus the area A must be at least sum of the areas of theses rectangles. Here we have used two intuitive feelings: (i) area adds up for disjoint regions and (ii) if a region S_1 is contained in another S_2 then area of S_1 can not exceed area of S_2 .

For every breaking up of the interval $[0, 1]$ we calculate a number, namely, sum of the areas of the rectangles constructed as above, by taking infimum of the function in each of the intervals. Let L be the set of numbers so obtained. Thus, we must agree to the following. Whatever be the area A , we

must have $\alpha \leq A$ for each $\alpha \in L$. In other words A is an upper bound for the set L . Let l be the least upper bound of the set L , that is, supremum of L . Thus we have $l \leq A$.

Let us use the same notation as in the earlier para, and now let us consider the supremum of the function in each of the intervals, say, $M_0, M_1, \dots, M_{55}$. Consider the rectangles with base $[a_i, a_{i+1}]$ and height M_i for $i = 0, 1, 2, \dots, 55$. Clearly these rectangles are non-overlapping (have common sides) and all the rectangles put together include the region of our interest. Thus the area A can at most be sum of the areas of these rectangles.

For every breaking up of the interval $[0, 1]$ we calculate a number, namely, sum of the areas of the rectangles constructed as above, by taking supremum of the function in each of the intervals. Let U be the set of numbers so obtained. Thus, we must agree to the following. Whatever be the area A , we must have $A \leq \alpha$ for each $\alpha \in U$. In other words A is a lower bound for the set U . Let u be the infimum or the glb of the set U . Thus we have $A \leq u$.

Thus we have calculated two numbers l and u and we are sure that $l \leq A \leq u$. In particular, if we are lucky and it so turns out that $l = u$ then it is sensible to define this number to be the area of the region and there is nothing for us to decide. What is amazing is that this equality holds in many cases. Thus this intuitive algorithm leads to an answer for the concept of area in many situations.

Of course, it does not lead to an answer in many other cases. For example, if our function f is given by: $f(x)$ equals one or zero according as x is rational or not. Then L consists of one number, namely 0; whereas U consists of one number 1. thus $l = 0$ and $u = 1$, unequal.

[Digression: I started with the problem of defining ‘amount of space’ – which we called area – in a given region in the plane. Soon I restricted to specific regions, area bounded by x -axis, graph of a positive function and two vertical lines. You can ask: What about other regions? From these you can build area for some more regions following the intuitive feeling that for disjoint regions the area should add up. We shall not continue this. Our idea is to get to integration.

Of course you may ask: are there ideas other than Riemann’s? Yes, you will learn in a later course another idea of Lebesgue, which actually is better. If that is better then why are we not doing that? it is a little abstract, but not difficult, needs maturity and in a sense essential for advanced analysis.]

Integration, Execution of idea:

We shall now start with definitions and execute the above idea. In what follows $[a, b]$ is a closed bounded interval. By a partition of this interval we mean a finite set of points $a = a_0 < a_1 < a_2 < \cdots < a_k = b$. We denote partition by P . Thus partition is just a finite subset of $[a, b]$ and $a \in P$ and $b \in P$. Of course a set consists of only points and there is nothing like first or second element in the set. When we think of a partition, we keep the order also in mind when we picturize it. A partition as above breaks the interval into finitely many subintervals, namely, $[a_0, a_1], [a_1, a_2], \cdots, [a_{k-1}, a_k]$. Sometimes we refer to these intervals as intervals of the partition. Definition of partition usually means cutting something into disjoint sets, but for these intervals we have ‘common boundary points’.

If P_1 and P_2 are two partitions, we say P_2 is finer than P_1 if every point of P_1 is also in P_2 , that is, $P_1 \subset P_2$. We write $P_1 \leq P_2$. Thus, for example if the interval is $[0, 1]$ then

$$\{0, 1/2, 1\}; \quad \{0, 1/4, 1/3, 1/2, 1\}; \quad \{0, 1/8, 3/8, 1\}$$

are all partitions. The second one is a refinement of the first one. The third one is not comparable to the other two. However we can think of a partition that refines both the second and third, namely

$$\{0, 1/8, 1/4, 1/3, 3/8, 1/2, 1\}.$$

In general refinement of two partitions P_1 and P_2 is just $P_1 \cup P_2$. Of course, you need to arrange the points of $P_1 \cup P_2$ in increasing order when you picturize. This is denoted by $P_1 \vee P_2$. Be clear, if a point x is in both partitions P_1 and P_2 then it does not appear twice in $P_1 \vee P_2$! Also it is not just all points of P_1, P_2 taken together in a bag, they have to be arranged in increasing order.

Now let f be a bounded real valued function on $[a, b]$. For a partition $P = \{a = a_0 < a_1 < a_2 < \cdots < a_k = b\}$, we define

$$U(P, f) = \sum_0^{k-1} M_i(a_{i+1} - a_i); \quad M_i = \sup\{f(x) : a_i \leq x \leq a_{i+1}\}.$$

$$L(P, f) = \sum_0^{k-1} m_i(a_{i+1} - a_i); \quad m_i = \inf\{f(x) : a_i \leq x \leq a_{i+1}\}.$$

$U(P, f)$ is the upper Riemann sum for the partition P and $L(P, f)$ is the lower Riemann sum for the partition P . We put

$$U(f) = \inf_P U(P, f); \quad L(f) = \sup_P L(P, f).$$

In the above, the inf and sup is over all the partitions of the interval $[a, b]$.

We say that f is *Riemann integrable* if $U(f) = L(f)$ and then define integral of f as this common value $U(f) = L(f)$. The integral is denoted $\int_a^b f$ or $\int_a^b f(x)dx$. Convince yourself that this is precisely what we thought at the beginning. The function f is called integrand. Since we do not handle any other integral in our course, we just say integrable to mean Riemann integrable.

[Digression: In $\int_a^b f(x)dx$, the variable x is ‘dummy’, in the sense, we could have written $\int_a^b f(z)dz$. It depends on what your variable is. I have not read history. This notation seems to have originated in trying to express what we did: we took value of the function “ $f(x)$ ” multiplied by length of a small interval at x , named “ dx ” and summed these – S for sum. Since this is limit of sums, not exactly sum, an elongated S .]

The main problem now is to understand which functions are integrable and how to calculate their integral. This is what try to understand.

1: For any partition P , $L(P, f) \leq U(P, f)$.

This follows from the fact $m_i \leq M_i$ for each i . You must also observe that the multiplier of m_i , namely, $(a_{i+1} - a_i)$ is nonnegative.

2. If $P_1 \leq P_2$, then $L(P_1, f) \leq L(P_2, f)$ and $U(P_1, f) \geq U(P_2, f)$.

First suppose that P_2 has only one extra point than P_1 ; say

$$P_1 = \{a = a_0 < a_1 < \cdots < a_k = b\}.$$

Suppose P_2 has one extra point α ; $a_j < \alpha < a_{j+1}$. Let

$$m' = \inf\{f(x) : a_j \leq x \leq \alpha\}; \quad m'' = \inf\{f(x) : \alpha \leq x \leq a_{j+1}\}.$$

Then m_j being $\inf\{f(x) : a_j \leq x \leq a_{j+1}\}$ we see $m_j \leq m'$ and $m_j \leq m''$; as the set gets larger, inf gets smaller. Thus

$$m_j(a_{j+1} - a_j) = m_j(\alpha - a_j) + m_j(a_{j+1} - \alpha)$$

$$\leq m'(\alpha - a_j) + m''(a_{j+1} - \alpha).$$

Observe that the only difference between $L(P_1, f)$ and $L(P_2, f)$ is the following. The term $m_j(a_{j+1} - a_j)$ appearing in $L(P_1, f)$ is replaced by the right side above in $L(P_2, f)$. Thus the above inequality shows that $L(P_1, f) \leq L(P_2, f)$. Similarly if

$$M' = \sup\{f(x) : a_j \leq x \leq \alpha\}; \quad M'' = \sup\{f(x) : \alpha \leq x \leq a_{j+1}\}.$$

then we see $M' \leq M_j$ and $M'' \leq M_j$ leading to the inequality $U(P_2, f) \leq U(P_1, f)$. Clearly, by induction, the inequalities follow if P_2 has k extra points than P_1 ; $k = 1, 2, \dots$. This completes the proof, since a partition has only finitely many points.

3. For any two partitions P_1 and P_2 , we have $L(P_1, f) \leq U(P_2, f)$.
In fact if $P = P_1 \vee P_2$, their refinement, we see using the two facts above

$$L(P_1, f) \leq L(P, f) \leq U(P, f) \leq U(P_2, f).$$

4. $L(f) \leq U(f)$.

The earlier fact says that every number $L(P, f)$ is a lower bound for the set $S = \{U(Q, f) : Q \text{ partition of } [a, b]\}$. Thus $L(f)$ is a lower bound for S . But $U(f)$ is the glb of S . Hence $L(f) \leq U(f)$.

5. If $\alpha \leq f(x) \leq \beta$ for all $x \in [a, b]$, then

$$\alpha(b - a) \leq L(f) \leq U(f) \leq \beta(b - a).$$

This follows from observing that for each partition P ,
 $\alpha(b - a) \leq L(P, f)$ and $U(P, f) \leq \beta(b - a)$.

6. f is integrable iff for every $\epsilon > 0$, there is a partition such that $U(P, f) - L(P, f) < \epsilon$

By properties of sup and inf, there are partitions P_1 and P_2 such that $L(P_1, f) > L(f) - \epsilon/2$ and $U(P_2, f) < U(f) + \epsilon/2$. If $P = P_1 \vee P_2$ then by property 2 above and using the fact that f is integrable, that is $L(f) = U(f)$;

$$U(P, f) - L(P, f) \leq U(P_2, f) - L(P_1, f) < \epsilon$$

7. Let f be a continuous function on $[a, b]$. Given $\epsilon > 0$, there is a $\delta > 0$ with the following property: whenever we take any partition with difference between successive points smaller than δ , then $U(P, f) - L(P, f) < \epsilon$.

In particular continuous functions are integrable.

Last sentence follows from noting that any continuous function on $[a, b]$ is bounded and property 6 above. To prove the first sentence, Use uniform continuity to get $\delta > 0$ so that

$$x, y \in [a, b]; |x - y| < \delta \Rightarrow |f(x) - f(y)| < \frac{\epsilon}{b - a}.$$

Now take any partition

$$P = \{a = a_0 < a_1 < \cdots < a_k = b\}$$

such that each interval of the partition has length smaller than δ . There are such partitions. For example, $a, a + (\delta/2), a + 2(\delta/2), \dots$. Of course, when $k(\delta/2)$ exceeds b , you stop and take the last point as b . If x and y are in $[a_j, a_{j+1}]$ we see $|f(x) - f(y)| < \epsilon/(b - a)$ and hence $M_j - m_j < \epsilon/(b - a)$. In fact, f being continuous, you can take x and y to be the points in the interval $[a_j, a_{j+1}]$ where the sup and inf are attained. Thus

$$U(P, f) - L(P, f) = \sum (M_j - m_j)(a_{j+1} - a_j) \leq \frac{\epsilon}{b - a} \sum (a_{j+1} - a_j) = \epsilon.$$

Thus we see that a large class of functions, namely continuous functions, are integrable.

Let us introduce a word that reduces our writing. Given a partition $P = \{a = a_0 < a_1 < a_2 < \cdots < a_k = b\}$, the maximum distance between successive points, $\max_i (a_{i+1} - a_i)$ is denoted by $\|P\|$, **norm** of P . The above proof shows the following.

9. Let f be a continuous function on $[a, b]$. Given $\epsilon > 0$, there is a $\delta > 0$ such that: whenever we take any partition $\|P\| \leq \delta$, then $U(P, f) - L(P, f) < \epsilon$., equivalently, $U(P, f) - \int_a^b f < \epsilon$ and $\int_a^b f - L(P, f) < \epsilon$.

We have been using lower sums and upper sums, which involve finding supremum and infimum of the function over intervals. We now describe Riemann sums that do not involve this step.

A **selection** for a partition is simply a selection of points from each partition interval; more precisely, a (finite) sequence of points $s = \{s_0 \leq s_1 \leq s_2 \leq \cdots s_{k-1}\}$ such that $s_i \in [a_i, a_{i+1}]$ for $i = 0, 1, 2, \dots, k - 1$. Given any interval, there are several possible partitions of the interval. Given one partition, there are several possible selections for the partition; we can pick any one point from each interval of the partition.

Given a partition P and a selection s for the partition, we define

$$R(P, f, s) = \sum_{i=0}^{k-1} f(s_i)[a_{i+1} - a_i].$$

We have denoted the selection by (s_i) . This is called the Riemann sum for the partition and selection. Recall that, instead of value of the function at the selected point, if we used the infimum and supremum in each partition interval we get the lower and upper Riemann sums.

10. Let f be a continuous function on $[a, b]$. Given $\epsilon > 0$, there is a $\delta > 0$ such that $|R(P, f, s) - \int f| < \epsilon$ for any partition P with $\|P\| < \delta$ and for any selection s for the partition.

We only need to observe that both $\int f$ and $R(P, f, s)$ are between $L(P, f)$ and $U(P, f)$. So the same δ as above would do.

11. Let $\{P_n\}$ be a sequence of partitions of $[a, b]$ with $\|P_n\| \rightarrow 0$ and for each n , let s_n be a selection for the partition P_n . Then for any continuous function f on $[a, b]$ we have:

$$U(P_n, f) \rightarrow \int f; \quad L(P_n, f) \rightarrow \int f; \quad R(P_n, f, s_n) \rightarrow \int f$$

This is clear from the previous statement. Thus even though the appearance of selection appears an extra complication, you should keep in mind that it is one more choice or freedom at our disposal and some times some one (like mean value theorem) may already make a choice for us. You will see this in the fundamental theorem of integral calculus.

Though we are concentrating on continuous functions now, one naturally wonders whether there are functions which are not continuous but integrable. the answer is yes. The evidence is also easy to get. Just consider the function $f(x) = 1$ for $0 < x < 1$; and $f(0) = 54, f(1) = 1/2$. This function is integrable. In fact lower sums and upper sums are easily calculable and they yield $U(f) = 1 = L(f)$.

But what is not easy to answer is the following question: what precisely are the functions which are integrable? The answer roughly is that f is integrable when and only when its set of discontinuities is 'small'. This is an important issue but shall not enter this discussion now. It is more important and basic to see how to calculate integrals and how to use integrals to our benefit. However, some of the observations we made above are true without assuming that we have a continuous function. Here is an example whose proof follows from what we have done above.

12. Let f be bounded function on $[a, b]$. Then f is integrable iff for every $\epsilon > 0$, there is a partition P such that $U(P, f) - L(P, f) < \epsilon$; or equivalently, for every $\epsilon > 0$ there is a partition P such that $|R(P, f, s) - R(P, f, s')| < \epsilon$ for any two selections s and s' .

For several of the statements below, continuity of the functions is not needed, integrability is enough; of course proofs have to be done with more care. But, as mentioned earlier, we shall not complicate life now.

13. if f and g are continuous, then

$$\int (f + g) = \int f + \int g; \quad \int (cf) = c \int f.$$

if $f \equiv 28$ on $[a, b]$, then $\int_a^b f = 28(b - a)$.

In other words integral is linear and for constant function it is the product of that constant and length of the interval.

This follows from noting $R(P, f+g, s) = R(P, f, s) + R(P, g, s)$ and $R(P, cf, s) = cR(P, f, s)$.

14. If f is continuous on $[a, b]$ and $a < c < b$, then

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

In other words integral adds over two disjoint intervals which have an end point in common.

Take a sequence of partitions P_n of $[a, b]$ such that $\|P_n\| \rightarrow 0$ and each P_n includes the point c . Then the set $Q_n = \{x \in P_n : x \leq c\}$ and the set $Q_n^* = \{x \in P_n : x \geq c\}$ will constitute partitions of $[a, c]$ and $[c, b]$ respectively. Clearly

$$\|Q_n\| \rightarrow 0; \quad \|Q_n^*\| \rightarrow 0; \quad U(P_n f) = U(Q_n, f) + U(Q_n^*, f)$$

and proof is completed by noting

$$U(P_n, f) \rightarrow \int_a^b f; \quad U(Q_n, f) \rightarrow \int_a^c f; \quad U(Q_n^*, f) \rightarrow \int_c^b f.$$

15. for any continuous function $|\int f| \leq \int |f|$.

In other words Analogous to sums: $|\sum a_n| \leq \sum |a_n|$

Note that $|f|$ is again a continuous function and hence its integral makes sense. proof is simple because for any partition

$$|U(P, f)| \leq U(P, |f|).$$

16. If f, g are continuous functions on $[a, b]$ and $f(x) \leq g(x)$ for every x then $\int f \leq \int g$.

In other words integral is monotone increasing.

This is immediate from $R(P, f, s) \leq R(P, g, s)$.

17. Fundamental Theorem of Integral Calculus:

let f be a continuous function on $[a, b]$.

(i) Define $F(a) = 0$ and $F(x) = \int_a^x f$ for $a < x \leq b$. Then F is continuous on $[a, b]$; it is differentiable; $F' = f$.

(ii) Let G be any continuous function on $[a, b]$ which is differentiable on (a, b) and $G'(x) = f(x)$ for $a < x < b$. Then $\int_a^b f = G(b) - G(a)$.

(iii) If G_1 and G_2 are two such functions as in (ii), then there is a number α such that $G_1(x) = \alpha + G_2(x)$ for every $x \in [a, b]$.

This is an extremely useful and powerful theorem. Firstly, it relates integration to derivatives. Secondly, it reduces our job of calculating integrals to finding functions whose derivative is the given function. This is easier than struggling with partitions, sups, infs, and limits.

Any function G as in (ii) is called a **primitive** for f . Part (i) says F is always a primitive, unfortunately, its definition depends on integration again. So it can not be used as a way to evaluate integrals. It is important and assures us of the existence of a primitive.

Sometimes only part (ii) is called fundamental theorem. Usually it is stated for integrable functions rather than only for continuous functions as we did above. For example if you take the function f which is one on $(0, 1)$ and our choice of values at $x = 0$ and $x = 1$; then f is not be continuous on $[0, 1]$ but the function $G(x) = x$ fits the bill.

However, one should take care in stating this theorem for noncontinuous Riemann integrable functions. So we shall not handle now. Under continuity we have $F'(x) = f(x)$ at **every** point. In fact it holds at a and b too provided we take the right derivative at a and left derivative at b .

Before proving this theorem, let us see two useful applications. The fundamental theorem makes it possible to translate theorems on derivatives to theorems on integrals, which help in calculating integrals. We 'translate' the

chain rule' and 'product rule' of differentiation.

18. Let φ be a strictly increasing continuously differentiable function on $[a, b]$ onto $[c, d]$. Let f be a continuous function on $[c, d]$. then

$$\int_a^b f(\varphi(x))\varphi'(x)dx = \int_c^d f(y)dy.$$

since φ' is assumed to be continuous, the integrand on the left side is continuous and hence integral makes sense. Also φ being strictly increasing and onto $[c, d]$ we have $\varphi(a) = c$ and $\varphi(b) = d$.

Thus integrating f on the interval $[c, d]$ is not simply integrating the composed function on $[a, b]$, but you need to multiply this composed function with φ' . the reason is that when you calculate Riemann sums on $[a, b]$ you multiply with length of partition interval. If $x_1 < x_2$ then length of the interval $[x_1, x_2]$ is $(x_2 - x_1)$. If $\varphi(x_1) = y_1$ and $\varphi(x_2) = y_2$ then length of the image interval $[y_1, y_2]$ is $\varphi(x_2) - \varphi(x_1)$ which is $\varphi'(\theta)[x_2 - x_1]$ for some number θ between x_1 and x_2 . If this interval is small, this is nearly $\varphi'(x_1)[x_2 - x_1]$ because φ' is continuous. Thus when an interval in domain is shifted to the range, the length gets magnified/reduced by a factor (since the factor could be less than one).

In practice, the method above, called **method of substitution**, is implemented as follows. You need to evaluate an integral which you recognize as the left side. you say put $y = \varphi(x)$ so that $dy = \varphi'(x)dx$ and the left side becomes the right side after noting that $y = c$ when $x = a$ and $y = d$ when $x = b$.

You can state a similar theorem when φ is decreasing. We need to multiply with $|\varphi'(x)| = -\varphi'(x)$ instead of simply $\varphi'(x)$.

Proof of the formula is simple. Let F be a primitive for f on $[c, d]$. Thus it is differentiable and $F'(y) = f(y)$ for $c < y < d$. Set $G(x) = F(\varphi(x))$ on $[a, b]$. Clearly, G is continuous on $[a, b]$, differentiable on (a, b) , and by chain rule,

$$G'(x) = F'(\varphi(x))\varphi'(x) = f(\varphi(x))\varphi'(x).$$

Thus by fundamental theorem

$$\begin{aligned} \int_a^b f(\varphi(x))\varphi'(x)dx &= G(b) - G(a) = F(\varphi(b)) - F(\varphi(a)) \\ &= F(d) - F(c) = \int_c^d f(y)dy. \end{aligned}$$

We have taken a simple route by using the extra concept of primitive assured by the fundamental theorem. You should remember that, such a method of taking easy way out will not help us when life does get complicated. But can we prove it starting from definition of integral? Yes. Here is another proof, you may ignore if you wish. This will be useful when you learn functions of several variable, that is, functions defined on R^2 , R^3 etc.

For any partition P of $[a, b]$, let $\varphi(P)$ denote the partition of $[c, d]$ obtained by taking images of points in P . This is indeed a partition of $[c, d]$. Further: If $\|P_n\| \rightarrow 0$, then $\|\varphi(P_n)\| \rightarrow 0$. This is a simple consequence of uniform continuity of φ . Also if s is a selection for P , then $t = \varphi(s)$ is a selection for the partition $\varphi(P)$. Here $\varphi(s) = \{\varphi(s_i) : 0 \leq i \leq k-1\}$ if $s = \{s_i : 0 \leq i \leq k-1\}$.

Let us temporarily name $g(x) = f(\varphi(x))\varphi'(x)$ defined on $[a, b]$. The second observation is the following. Given any partition P of $[a, b]$, there is a selection s for the partition P of $[a, b]$ so that

$$R(P, g, s) = R(\varphi(P), f, \varphi(s)).$$

To show the stated selector, if $[\alpha, \beta]$ is a partition interval or P , (that is, like $[a_i, a_{i+1}]$) in then by MVT, there is a $\gamma \in (\alpha, \beta)$ so that

$$[\varphi(\beta) - \varphi(\alpha)] = \varphi'(\gamma)[\beta - \alpha].$$

Let s be the selection for the partition P so that from any partition interval $[\alpha, \beta]$ it picks a point γ satisfying the above equation from this interval. MVT assures there is at least one such point. Then

$$g(\gamma)[\beta - \alpha] = f(\varphi(\gamma))\varphi'(\gamma)[\beta - \alpha] = f(\varphi(\gamma))[\varphi(\beta) - \varphi(\alpha)]$$

It is clear that this selector will do the job.

Now to complete the proof is simple. Take a sequence of partitions P_n of $[a, b]$ with $\|P_n\| \rightarrow 0$. Then $\|\varphi(P_n)\| \rightarrow 0$. For each n select s_n as described above for the partition P_n . Then

$$R(P_n, g, s_n) = R(\varphi(P_n), f, \varphi(s_n)).$$

Here the left side converges to $\int g$ and right side to $\int f$ over the appropriate intervals.

This completes the alternative proof. ■

Here is translation of product rule of differentiation.

19. Let F and G be two continuously differentiable functions on $[a, b]$ with derivatives f and g . Then

$$\int_a^b F(x)g(x)dx = \{F(b)G(b) - F(a)G(a)\} - \int_a^b f(x)G(x)dx.$$

In other words,

$$\int_a^b F(x)G'(x)dx = \{F(b)G(b) - F(a)G(a)\} - \int_a^b F'(x)G(x)dx.$$

This is called **integration by parts formula**.

Proof is trivial. The function FG is continuous and its derivative equals $Fg + fG$ and so is a primitive for the later. By Fundamental theorem

$$F(b)G(b) - F(a)G(a) = \int (Fg + fG) = \int Fg + \int fG.$$

The substitution rule and integration by parts formula, though very simple, are of utmost importance in most parts of higher studies, not only in maths but in physics and so on.

Let us now return to the proof of the fundamental theorem. Interestingly, the proof is simple.

(i) Shall show uniform continuity of F on $[a, b]$. Fix $\epsilon > 0$. Since f is continuous, it is bounded, say $|f(x)| < M$ for all x . Take $\delta = \epsilon/M$. Now take x, y with $|x - y| < \delta$. No loss to assume $x < y$. If $x = a$ then $F(a) = 0$ and so

$$|F(y) - F(a)| = |F(y)| = \left| \int_a^y f \right| \leq \int_a^y |f| \leq M(y - a) < \epsilon.$$

Observe, we are using properties of integrals proved above, for the two inequalities in the string here.

If $a < x < y$ our earlier observation tells

$$\int_a^y f = \int_a^x f + \int_x^y f; \quad i.e., \quad F(y) = F(x) + \int_x^y f;$$

so that

$$|F(y) - F(x)| = \left| \int_x^y f \right| \leq \int_x^y |f| \leq M(y - x) \leq \epsilon.$$

We show that F is differentiable and $F'(x) = f(x)$. Fix a point, say, x_0 and let us denote $f(x_0) = \alpha$. All the points appearing below are in the interval $[a, b]$. Let $\epsilon > 0$. We exhibit $\delta > 0$ so that

$$0 < y - x_0 < \delta \Rightarrow \left| \frac{F(y) - F(x_0)}{y - x_0} - \alpha \right| < \epsilon.$$

and

$$0 < x_0 - y < \delta \Rightarrow \left| \frac{F(y) - F(x_0)}{y - x_0} - \alpha \right| < \epsilon.$$

This will prove the stated result. Do not worry, if your point $x_0 = b$ then first implication can not arise and when $x_0 = a$ the second can not. Using continuity of f at x_0 , We fix $\delta > 0$ so that

$$y \in [a, b]; |y - x_0| < \delta \Rightarrow |f(y) - \alpha| < \epsilon.$$

For any $y > x_0$, we have (use splitting of integral over intervals; linearity of integral)

$$F(y) - F(x_0) = \int_{x_0}^y f(t)dt; \quad \int_{x_0}^y \alpha dt = \alpha(y - x_0).$$

$$\left| \frac{F(y) - F(x_0)}{y - x_0} - \alpha \right| = \left| \int_{x_0}^y \frac{1}{y - x_0} [f(t) - \alpha] dt \right| \leq \frac{1}{y - x_0} \int_{x_0}^y |f(t) - \alpha| dt.$$

If $y - x_0 < \delta$ then for every $t \in [x_0, y]$ we have $|t - x_0| < \delta$ so that the integrand above is at most ϵ and so the integral is at most $\epsilon(y - x_0)$ showing what we wanted.

Similar computation yields the result for $0 < x_0 - y < \delta$.

This completes proof of (i).

Proof of (ii). Let $G' = f$. First we observe the following. Given any partition P , there is a selection s so that $G(b) - G(a) = R(P, f, s)$. To see this let $P = \{a = a_0 < a_1 < a_2 < \cdots < a_k = b\}$. For each i let $s_i \in (a_i, a_{i+1})$ be given by the MVT to satisfy

$$G(a_{i+1}) - G(a_i) = f(s_i)(a_{i+1} - a_i).$$

This is the selection s . then

$$G(b) - G(a) = \sum_{i=0}^{k-1} [G(a_{i+1}) - G(a_i)] = \sum f(s_i)(a_{i+1} - a_i) = R(P, f, s).$$

The proof is completed as follows. Take a sequence of partitions P_n with $\|P_n\| \rightarrow 0$. For each n , get selector s_n for P_n as above. Then note

$$R(P_n, f, s_n) \rightarrow \int f; \quad R(P_n, f, s_n) = G(b) - G(a) \quad \text{for all } n.$$

This completes proof of (ii)

Proof of (iii): suppose that there are two functions G_1 and G_2 having the

same derivative f . Then $G_1 - G_2$ has zero derivative and hence is a constant in the interval (a, b) . Since $G_1 - G_2$ is continuous on the interval $[a, b]$ and is a constant in the interval (a, b) it must be constant in the interval $[a, b]$.

This completes proof of the fundamental theorem. ■

As suggested by one of you, you can also prove (iii) after proving (i) and then use it to show (ii) in a painless way.

We have excellent tools before us to evaluate integrals and also put them to use. sometimes we use a notation as follows: $\int f = F$ without mentioning any interval $[a, b]$. We use the same variable x for both f and F . This is to be interpreted as saying that F is a primitive for f . It simply means that over a 'meaningful' interval $F' = f$. Thus if you have two numbers $a < b$ in this interval then $\int_a^b f(x)dx = F(b) - F(a)$.

I assume that you have come across the following in high school and so go over them fast to reach new and interesting things. If you did not go through them in high school or if you do not remember, then you should convince yourself about their truth. Do not take anything for granted.

$$\int x^n = \frac{x^{n+1}}{n+1}. \quad n \neq -1$$

You should interpret this as follows. If you take $n \geq 0$ then for any bounded interval $[a, b]$ the integral $\int_a^b$ of left side equals the difference in values of the right side. take $x^0 = 1$. However when $n \leq -2$, you should consider intervals $[a, b] \subset (-\infty, 0)$ or $[a, b] \subset (0, \infty)$. You should not blindly take, for example $\int_a^b x^{-2}$ with $a = -2$ and $b = 3$; integrand is not defined at zero. It would be a nice exercise to calculate $\int x^n$ for positive integers n , following the same idea used by Archimedes.

$$\int x^\alpha dx = \frac{x^{\alpha+1}}{\alpha+1} \quad \alpha \neq -1.$$

You should interpret this to say for any $[a, b] \subset (0, \infty)$. For example when $\alpha = 1/2$ you should not take the interval $[-2, -1]$ because the integrand does not make sense to us.

Here are some more you should verify.

$$\int e^x dx = e^x, \quad x \in R. \quad \int \frac{1}{x} dx = \log x, \quad x > 0.$$

$$\int \cos x dx = \sin x. \quad \int \sin x dx = -\cos x.$$

$$\int \frac{1}{\sqrt{1+x^2}} dx = \log(x + \sqrt{1+x^2}).$$

$$\int \frac{1}{\sqrt{x^2-1}} dx = \log(x + \sqrt{x^2-1}) \quad |x| > 1.$$

You should interpret this for intervals $[a, b] \subset (-\infty, 0)$ or $[a, b] \subset (0, \infty)$.

$$\int \frac{1}{1-x^2} dx = \frac{1}{2} \log \frac{1+x}{1-x} \quad |x| < 1.$$

$$\int \frac{1}{1-x^2} dx = \frac{1}{2} \log \frac{x+1}{x-1} \quad |x| > 1.$$

Walli's product for π :

Using the above formulae you can calculate several integrals. I show you the power of the simple things you learnt now.

$$\int_0^{\pi/2} \sin^0 x dx = \frac{\pi}{2}; \quad \int_0^{\pi/2} \sin x dx = \cos 0 - \cos(\pi/2) = 1.$$

If $m > 1$, then integration by parts gives

$$\begin{aligned} \int_0^{\pi/2} \sin^m x dx &= \int_0^{\pi/2} \sin^{m-1} x (-\cos x)' dx \\ &= \int_0^{\pi/2} \cos x (m-1) \sin^{m-2} x \cos x dx \\ &= (m-1) \int_0^{\pi/2} \sin^{m-2} x dx - (m-1) \int_0^{\pi/2} \sin^m x dx \end{aligned}$$

so that

$$\int_0^{\pi/2} \sin^m x dx = \frac{m-1}{m} \int_0^{\pi/2} \sin^{m-2} x dx.$$

Thus

$$\begin{aligned} \int_0^{\pi/2} \sin^{2m} x dx &= \frac{2m-1}{2m} \frac{2m-3}{2m-2} \frac{2m-5}{2m-4} \cdots \frac{3}{4} \frac{1}{2} \frac{\pi}{2}. \\ \int_0^{\pi/2} \sin^{2m+1} x dx &= \frac{2m}{2m+1} \frac{2m-2}{2m-1} \frac{2m-4}{2m-3} \cdots \frac{4}{5} \frac{2}{3} 1. \end{aligned}$$

dividing the first equation by the second

$$\frac{\int_0^{\pi/2} \sin^{2m} x dx}{\int_0^{\pi/2} \sin^{2m+1} x dx} = \frac{(2m-1)(2m+1)}{(2m)^2} \frac{(2m-3)(2m-1)}{(2m-2)^2} \cdots$$

$$\begin{aligned} & \dots \frac{3 \times 5}{4^2} \frac{1 \times 3}{2^2} \frac{\pi}{2}. \\ \frac{\pi}{2} = & \frac{2^2}{1 \cdot 3} \frac{4^2}{3 \cdot 5} \frac{6^2}{5 \cdot 7} \dots \frac{(2m-2)^2}{(2m-3)(2m-1)} \frac{(2m)^2}{(2m-1)(2m+1)} \\ & \times \frac{\int_0^{\pi/2} \sin^{2m} x dx}{\int_0^{\pi/2} \sin^{2m+1} x dx}. \end{aligned}$$

We shall now show that as $m \rightarrow \infty$;

$$\frac{\int_0^{\pi/2} \sin^{2m} x dx}{\int_0^{\pi/2} \sin^{2m+1} x dx} \rightarrow 1. \quad (\spadesuit)$$

It will then follow that

$$\frac{\pi}{2} = \lim_{m \rightarrow \infty} \frac{2^2}{1 \cdot 3} \frac{4^2}{3 \cdot 5} \frac{6^2}{5 \cdot 7} \dots \frac{(2m-2)^2}{(2m-3)(2m-1)} \frac{(2m)^2}{(2m-1)(2m+1)}.$$

This is called Walli's product. We write it in a simpler way now.

$$\frac{\pi}{2} = \lim_{m \rightarrow \infty} \frac{2^{2m}(m!)^2}{3^2 \cdot 5^2 \dots (2m-1)^2(2m+1)} = \lim_{m \rightarrow \infty} \frac{2^{4m}(m!)^4}{[(2m)!]^2(2m+1)}.$$

Or

$$\sqrt{\frac{\pi}{2}} = \lim_{m \rightarrow \infty} \frac{2^{2m}(m!)^2}{(2m)! \sqrt{(2m+1)}}$$

Or

$$\sqrt{\pi} = \lim_{m \rightarrow \infty} \frac{2^{2m}(m!)^2}{(2m)! \sqrt{(m+1/2)}}$$

Since $\sqrt{m}/\sqrt{m+1/2} \rightarrow 1$. we get

$$\sqrt{\pi} = \lim_{m \rightarrow \infty} \frac{2^{2m}(m!)^2}{(2m)! \sqrt{m}} = \lim_{m \rightarrow \infty} 2^{2m} \frac{1}{\binom{2m}{m}} \frac{1}{\sqrt{m}}$$

This is called Walli's formula for $\sqrt{\pi}$.

You can rewrite above as

$$\lim_{m \rightarrow \infty} \binom{2m}{m} 2^{-2m} \sqrt{m} = 1/\sqrt{\pi}$$

To complete our proof of Walli, we should show

$$\frac{\int_0^{\pi/2} \sin^{2m} x dx}{\int_0^{\pi/2} \sin^{2m+1} x dx} \rightarrow 1. \quad (\spadesuit)$$

This is simple. Since $0 \leq \sin x \leq 1$ for $0 \leq x \leq \pi/2$, we see

$$\sin^{2m+1} x \leq \sin^{2m} x \leq \sin^{2m-1} x; \quad 0 \leq x \leq \pi/2$$

Hence

$$\int_0^{\pi/2} \sin^{2m+1} x \leq \int_0^{\pi/2} \sin^{2m} x \leq \int_0^{\pi/2} \sin^{2m-1} x$$

or, noting that all integrals are strictly positive,

$$1 \leq \frac{\int_0^{\pi/2} \sin^{2m} x}{\int_0^{\pi/2} \sin^{2m+1} x} \leq \frac{\int_0^{\pi/2} \sin^{2m-1} x}{\int_0^{\pi/2} \sin^{2m+1} x}$$

Or using the recurrence relation for the integral

$$1 \leq \frac{\int_0^{\pi/2} \sin^{2m} x}{\int_0^{\pi/2} \sin^{2m+1} x} \leq \frac{2m+1}{2m}$$

taking limits, since both the ends converge to one, we have our result. $\blacksquare$

This is a neat formula for π in terms of integers. You will see its use soon.

Stirling:

You know that as n becomes large $n!$ becomes very very large. How large? Can we explain? Yes it says that $n!$ is like $\sqrt{2\pi}e^{-n}n^{n+1/2}$. This is to be interpreted in the following sense. Their ratio converges to one.

$$\frac{n!}{\sqrt{2\pi} e^{-n} n^{n+1/2}} \rightarrow 1 \quad n \rightarrow \infty$$

When we say that a sequence (a_n) is like another sequence (b_n) (both are sequences of strictly positive numbers) there are two ways of understanding. Either $a_n - b_n \rightarrow 0$ (additive closeness) or $a_n/b_n \rightarrow 1$ (multiplicative closeness) Of course when the first happens, then the second also happens. However the other way is not in general true.

For example (n) is like $(n + 1/n)$ in the first sense and hence also in the second sense. The sequence (n^2) is like $(n^2 + n)$ in the second sense but not in the first sense; In fact the difference between the two sequences is n which becomes larger and larger. But then in what sense are they like each other? well, Both numbers are becoming large, when you replace one by the other, the relative error is going to zero.

If you are measuring length of this room, if you are off by a hundred meters then the error is indeed very huge. On the other hand if you are measuring distance of earth to Sun, if you are off by a mile or even hundred miles, the error is very very small. So the absolute error is many times unimportant and it is the relative error (relative to the magnitude of the quantity you are measuring) that matters.

You might wonder if this approximation is worth anything at all! After all, you can explain to any school student $100!$; just multiply 1,2,3, etc upto hundred. On the other hand to explain the approximation we stated, you need to explain e , explain π , etc. Have we complicated a simple quantity? No. This is useful in many calculations. In fact this is behind the notion of Entropy too.

If you see the denominator, apart from $\sqrt{2\pi}$ it has $e^{-n} n^{n+1/2}$. To understand this take logarithm: $(n + 1/2) \log n - n$. Now if you, for a moment, ignore $1/2$, this reminds you $n \log n - n$ or $x \log x - x$. So you should guess this has something to do with $\int \log x$ or area under the log function. Yes, We shall try to approximate the area of the function $f(x) = \log x$ from $x = 1$ to $x = n$.

Let us first make two observations. Consider the function $f(x) = \log x$ for $1 \leq x < \infty$. Then $f''(x) = -1/x^2 < 0$. The function is concave and chord joining two points on the curve lies below the curve. Also the tangent at any point is above the curve. We see this as follows. The equation of the tangent at $(u, f(u))$ is

$$y = f(u) + f'(u)(x - u).$$

Let us take any other point w in $[1, \infty)$. We need to show

$$f(w) \leq f(u) + f'(u)(w - u).$$

Case 1: $u < w$. Need to show

$$\frac{f(w) - f(u)}{w - u} \leq f'(u).$$

But the left side is $f'(\theta)$ for some $u < \theta < w$ and since f' is decreasing, the inequality is verified.

Case 2: $u > w$. Need to show

$$\frac{f(w) - f(u)}{w - u} \geq f'(u).$$

Again f' is decreasing and MVT does the job.

Thus, for $k \geq 1$, (area means area above x -axis)
the area under the chord joining $(k, \log k)$, $(k+1, \log(k+1))$ $\leq$
the area under the curve $y = \log x$ between k to $k+1$ $\leq$
the area under the tangent at $x + 1/2$ between k and $k+1$.
Thus

$$\frac{1}{2} \log(k+1) + \frac{1}{2} \log k \leq \int_k^{k+1} \log x \, dx \leq \log(k+1/2).$$

Adding these for $k = 1, 2, \dots, n-1$ and remembering that $x \log x - x$ is a primitive for $\log x$ we get

$$\log(n!) - \frac{1}{2} \log n \leq n \log n - n + 1 \leq \sum_1^{n-1} \log(k+1/2).$$

Let

$$a_n = n \log n - n + 1 - [\log(n!) - \frac{1}{2} \log n] = \log \left\{ \frac{e^{-n} n^{n+1/2}}{n!} \right\} + 1.$$

Then a_n is the area between the curve $y = \log x$ and the chords explained above, from $x = 1$ to $x = n$. Thus we see

$$a_n \geq 0; \quad a_n \uparrow. \quad (\spadesuit)$$

Also

$$\begin{aligned} a_n &\leq \sum_1^{n-1} \left\{ \log(k+1/2) - \frac{1}{2} \log(k+1) - \frac{1}{2} \log k \right\} \\ &= \frac{1}{2} \sum_1^{n-1} \left\{ \log \frac{(k+1/2)}{k} - \log \frac{(k+1)}{k+1/2} \right\} \\ &\leq \frac{1}{2} \sum_1^{n-1} \left\{ \log \left(1 + \frac{1}{2k} \right) - \log \left(1 + \frac{1}{2(k+1/2)} \right) \right\} \\ &\leq \frac{1}{2} \sum_1^{n-1} \left\{ \log \left(1 + \frac{1}{2k} \right) - \log \left(1 + \frac{1}{2(k+1)} \right) \right\} \\ &= \frac{1}{2} \log \frac{3}{2} - \frac{1}{2} \log \left(1 + \frac{1}{2n} \right) \leq \frac{1}{2} \log \frac{3}{2} \end{aligned}$$

As a consequence a_n is bounded above . So $(\spadesuit)$ implies that a_n converges.
Say $a_n \uparrow c$

$$\log \left\{ \frac{e^{-n} n^{n+1/2}}{n!} \right\} = a_n - 1 \uparrow c - 1.$$

Or

$$\log \left\{ \frac{n!}{e^{-n} n^{n+1/2}} \right\} \rightarrow c^*; \quad \frac{n!}{e^{-n} n^{n+1/2}} \rightarrow e^{c^*} = k > 0 \quad \text{say}$$

for some constant k . Or

$$\frac{n!}{k e^{-n} n^{n+1/2}} \rightarrow 1. \quad (\clubsuit)$$

We shall now evaluate the constant k by using the above limit in a known case, namely, Walli's product. We know

$$\frac{2^{2n} (n!)^2}{(2n)! \sqrt{n}} \rightarrow \sqrt{\pi}. \quad (\bullet)$$

From $(\clubsuit)$ we see

$$\frac{(n!)^2}{k^2 e^{-n} e^{-n} n^{n+1/2} n^{n+1/2}} \rightarrow 1$$

Taking the even subsequence in $(\clubsuit)$ we see

$$\frac{(2n)!}{k e^{-2n} (2n)^{2n+1/2}} \rightarrow 1$$

From these two we see

$$\frac{(n!)^2}{(2n)!} \frac{k e^{-2n} 2^{2n+1/2} n^{2n+1/2}}{k^2 e^{-2n} n^{2n+1}} \rightarrow 1$$

That is

$$\frac{(n!)^2}{(2n)!} \frac{2^{2n+1/2}}{k \sqrt{n}} \rightarrow 1; \quad \text{or} \quad \frac{(n!)^2}{(2n)!} \frac{2^{2n}}{\sqrt{n}} \frac{\sqrt{2}}{k} \rightarrow 1$$

Thus

$$\frac{(n!)^2}{(2n)!} \frac{2^{2n}}{\sqrt{n}} \rightarrow \frac{k}{\sqrt{2}}$$

comparing with $(\bullet)$ we get

$$\frac{k}{\sqrt{2}} = \sqrt{\pi} \quad \text{or} \quad k = \sqrt{2\pi}$$

Using in $(\clubsuit)$ gives

$$\frac{n!}{\sqrt{2\pi} e^{-n} n^{n+1/2}} \rightarrow 1.$$

There are several other things you can get with integration. But first we go back to integration and complete the process.

Fine tuning of integral:

For the few of you who are going a little slow: You can go through the definitions; you can ignore the prose that shows all properties remain valid; you can just accept them on faith if you wish.

After defining upper sums; lower sums; integrability and after showing that very continuous function on a closed bounded interval is integrable, we have been specializing to continuous functions. However, one does come across functions which are not continuous or functions which are bounded continuous but defined only on an open interval, may be unbounded interval; and so on.

(A) Regarding non-continuous functions we have already seen the following. Take a continuous f on $[0, 1]$ and change it at finitely many points by assigning any values of your choice. Then it is integrable; in fact, its integral is exactly same as $\int f$. Of course if you change at countably many points, then it may no longer be integrable. Take constant function one and change it to zero at rationals. Then it is no longer integrable.

(B) Now let us consider bounded function but the interval where it is defined is an open interval (a, b) . For example the function $\sin(1/x)$ is defined on the open interval $(0, 1)$, continuous and bounded. We, at this moment, are unable to talk about integrability because we did everything on a closed bounded interval. Discussing open intervals poses no serious problems. Exactly the same analysis goes through. This we do first. (Question: if the analysis is same why did we take closed interval earlier? This was only done to fix ideas and have concrete picture in mind.)

So let us take a bounded open interval (a, b) and a bounded function f on this interval. As earlier, partition is a finite sequence of points

$$P = \{a = a_0 < a_1 < a_2 < \cdots < a_k = b\}.$$

Given a partition P we define $U(P, f)$ and $L(P, f)$ the upper and lower sums as earlier; just that for the first and last interval we take supremum and infimum only over $(a, a_1]$ and $[a_{k-1}, b)$ — obviously because function is not defined at a, b . As earlier, we say that f is Riemann integrable if supremum of all lower sums equals infimum of all upper sums and in that case, the

common value is called integral of f and is denoted

$$\int_a^b f; \quad \int_a^b f(x)dx.$$

The fact that every lower sum is smaller than every upper sum is obvious and also the fact that upper sums decrease whereas lower sums increase as the partition becomes finer. It is also easy to show that f is integrable iff for any given $\epsilon > 0$, we can get a partition P so that $U(P, f) - L(P, f) < \epsilon$.

We can show that a bounded continuous function is integrable. Earlier we used uniform continuity, but now this is no longer possible because continuous function on an open interval need not be uniformly continuous; for example, $\sin 1/x$ function on $(0, 1)$. However, we can take advantage of the fact that the function is bounded.

Let $|f(x)| \leq M$ for all $x \in (a, b)$. Let $\epsilon > 0$. Let us choose $\delta > 0$ so that $2M\delta < \epsilon/4$. Note that on any subinterval, sup minus inf of f is at most $2M$. Thus in particular, if you consider the interval $(a, a + \delta]$ or the interval $(b - \delta, b]$ you see that sup minus inf over that interval times delta is smaller than $\epsilon/4$.

The function being uniformly continuous on $[a + \delta, b - \delta]$, get a partition P_1 of this $[a + \delta, b - \delta]$ so that $U(P_1, f) - L(P_1, f) < \epsilon/4$. The partition P for (a, b) is simply the points in P_1 along with a at the beginning and b at the end. Thus the first interval of this partition is $(a, a + \delta]$. and the last interval of this partition is $(b - \delta, b)$. It is easy to see that

$$U(P, f) - L(P, f) < \frac{\epsilon}{4} + \frac{\epsilon}{4} + \frac{\epsilon}{4} < \epsilon.$$

The argument also shows that if $\|P_n\| \rightarrow 0$ then the lower or upper sums converge to the integral.

it is interesting to see that in the notation for integral, $\int_a^b$ there is nothing to show whether we have open interval or closed interval. Since area of a line is zero, it makes no difference whether you include the lines at the end points or not, in calculating the area under the curve.

More precisely, suppose f were actually continuous on $[a, b]$ and you ignore it and consider the function only on (a, b) and **calculate the integral on this open interval as defined above**. You get the same answer as you would get when you calculated for the closed interval, using earlier prescription. The proof is very simple, for each partition whether you calculate the sums over (a, b) or $[a, b]$ you get the same answer. After all, the only difference is the end intervals and there continuity of the function tells you whether you include

end point or not, you get the same value for the sup or inf. Thus we need not write

$$\int_{[a, b]} f \quad \text{or} \quad \int_{(a, b)} f$$

just use $\int_a^b f$.

You can also show that selections (given a partition, select points from each partition interval) and Riemann sums also lead to the value of the integral, even on an open interval. In fact, whenever $\|P_n\| \rightarrow 0$ the sums $R(P_n, f, s_n)$ converge to the integral whatever be the selection s_n for P_n – as long as f is bounded continuous function. One now understands that the selection does not allow a from first interval and b from last interval. In fact even in the partition, the first interval is $(a, a_1]$ and last interval is $[a_{k-1}, b)$. The main observation is that as earlier, $L(P, f) \leq R(P, f, s) \leq U(P, f)$.

The idea in the above argument has something more to offer. suppose that f is a bounded function on $[a, b]$ but only known to be continuous on (a, b) . Is it integrable? Yes, the argument above show that it is integrable and equals the integral of f over the open interval (a, b) .

The arguments used so far above also show you the following. Suppose f is a bounded continuous function on (a, b) . For each n so that $\frac{2}{n} < b - a$, denote integral of f over the interval $[a + 1/n, b - 1/n]$ by α_n . then $\alpha_n \rightarrow \int_a^b f$. In

Other words

$$\int_{[a+1/n, b-1/n]} f \rightarrow \int_{(a, b)} f$$

In fact if you take any two sequences, $a < a_n < b_n < b$ with $a_n \downarrow a$ and $b_n \uparrow b$ then

$$\int_{a_n}^{b_n} f \rightarrow \int_a^b f \quad (\clubsuit)$$

All the earlier results hold good with same proofs using $R(P_n, f, s_n)$ If f, g are bounded Riemann integrable on (a, b) , then so is $f + g$ and

$$\int_a^b (f + g) = \int_a^b f + \int_a^b g \text{ and } \int_a^b 39f = 39 \int_a^b f \text{ and } \int_a^b f = \int_a^c f + \int_c^b f \text{ for } a < c < b$$

and if $f \equiv 55$ then $\int_a^b f = 55(b - a)$;

and if $\varphi : (a, b) \rightarrow (c, d)$ strictly increasing continuously differentiable onto function; then for bounded continuous f on (c, d) we have

$$\int_a^b f(\varphi(x))\varphi'(x)dx = \int_c^d f(y)dy$$

To prove this, you only need to take $[c_n, d_n] \uparrow (c, d)$ and consider a_n, b_n so that $\varphi(a_n) = c_n$ and $\varphi(b_n) = d_n$ and apply earlier results to see

$$\int_{a_n}^{b_n} f(\varphi(x))\varphi'(x)dx = \int_{c_n}^{d_n} f(y)dy$$

and take limits, use ($\clubsuit$).

One can state fundamental theorem of integration: If f continuous bounded on (a, b) and $F' = f$ then $\int_a^b f = \lim[F(b_n) - F(a_n)]$ where $a_n \downarrow a$ and $b_n \uparrow b$ are points in (a, b) . Immediate from ($\clubsuit$) again. You can also state integration by parts formula in a similar way.

You can also consider $[a, b)$ and $(a, b]$. (Though looks unnatural, whatever be the interval, you just ignore end points and just consider the open interval and do).

We have discussed integrals of bounded functions over bounded intervals, both open as well as closed. Integrals where either the function is unbounded or the interval is unbounded are called **improper integrals**. There is nothing improper about them, just that you can not use Riemann sums blindly. Just as infinite sums are defined as limits of finite (partial) sums, so are these integrals. We shall first discuss unbounded functions over bounded interval.

Of course there are several possibilities. For example, you can consider the function

$$f(x) = \frac{1}{\sqrt{x}} + \frac{1}{\sqrt{x-1}} + \frac{1}{\sqrt{x-2}}; \quad x \in (0, 3), x \neq 1, 2; \quad f(1) = f(2) = 0.$$

Then f is unbounded near three points, namely, zero, one and two. We shall not consider such functions. We discuss simple situations. Though most of the discussion can be made general, we discuss continuous functions in what follows.

(C) Bounded interval, function unbounded: Let us consider function f on an open interval (a, b) which is unbounded only near a , that is for any $\delta > 0$, it is bounded on $(a + \delta, b)$. Then for every $\delta > 0$, the integral $\int_{a+\delta}^b f$ makes sense and is defined by earlier considerations. We say that $\int_a^b f$ exists if

$$\lim_{\delta \downarrow 0} \int_{a+\delta}^b f$$

exists and then we define this limit as $\int_a^b f$. You must recall the meaning of limit $\delta \downarrow 0$. Thus $\int_a^b f$ exists if there is a number α such that for every sequence $a_n \downarrow a$, $a_n > a$ for all n , we have

$$\int_{a_n}^b f \rightarrow \alpha.$$

In that case this number α is defined to be the value of the integral of f over the interval (a, b) .

For example consider

$$\int_0^1 \frac{1}{x} dx.$$

This does not exist because for integral over $(\delta, 1)$, you get $\log \delta$ and does not converge to a finite limit as $\delta \downarrow 0$. Actually the same situation occurs and the integral

$$\int_0^1 \frac{1}{x^\alpha} dx$$

does not exist for $\alpha \geq 1$. On the other hand the integral exists for $\alpha < 1$. Of course if $\alpha \leq 0$, then the function is bounded and actually continuous on the closed interval $[0, 1]$ and we need not discuss. So let $0 < \alpha < 1$.

$$\int_\delta^1 \frac{1}{x^\alpha} dx = \frac{1}{1-\alpha} (1 - \delta^{-\alpha+1}) \rightarrow \frac{1}{1-\alpha},$$

as $\delta \downarrow 0$.

In a similar way, if the function f defined on an open interval (a, b) is unbounded only near b , that is for any $\delta > 0$, it is bounded on $(a, b - \delta)$ (whenever we make such statements it is assumed $a < b - \delta$ etc) for every $\delta > 0$, the integral $\int_a^{b-\delta} f$ makes sense and is defined by earlier considerations.

We say that $\int_a^b f$ exists if

$$\lim_{\delta \downarrow 0} \int_a^{b-\delta} f$$

exists and then we define this limit as $\int_a^b f$.

We can, in a fashion analogous to the above, show that the integral

$$\int_a^b \frac{1}{(b-x)^\alpha} dx$$

exists if $\alpha < 1$ and does not exist if $\alpha \geq 1$.

The substitution rule and integration by parts hold good for these integrals. For example suppose φ is a continuously differentiable strictly increasing function on a bounded interval (a, b) onto the bounded interval (c, d) . Suppose that f is a function integrable on (a, b) , and f possibly unbounded. Then integral of $f(\varphi(x))\varphi'(x)$ exists over (a, b) and we have

$$\int_a^b f(\varphi(x))\varphi'(x)dx = \int_c^d f(y)dy.$$

Suppose that f is unbounded only near a , then verify that the equality holds when you consider $a + \delta$ to b on left side and from $\varphi(a + \delta)$ to d on the right side and then take limits. Similar remark applies if the infinity occurs only near b .

Actually there is another possibility: f may indeed be nice but left side may not be nice because φ' is unbounded as happens when you take $\varphi(x) = \sqrt{x}$ on $(0, 1)$ onto $(0, 1)$. But once you understand simple things and ideas involved, you can sort out other things — no new ideas are needed.

Suppose that the function is unbounded at both end points, like

$$f(x) = \frac{1}{\sqrt{x(1-x)}} \quad 0 < x < 1$$

Then how do we define? Say f is continuous function defined on (a, b) but is unbounded in each interval $(a, a + \epsilon)$ as well as unbounded in each interval $(b - \eta, b)$. Clearly, on each $[a^*, b^*]$ it is bounded when $a < a^* < b^* < b$. Then we take $a < c < b$ and define

$$\int_a^b f = \int_a^c f + \int_c^b f$$

provided both integrals on the right exist. Will it depend on the point c we have chosen? No, it is easy to show that this does not depend on the point.

Thus we have given meaning to $\int_a^b f$ over a bounded interval when f may be unbounded at one or both end points (remember, throughout we are considering only continuous functions).

Consistency:

The first question that arises is whether for bounded functions on bounded intervals this recipe agrees with the earlier definition. This we have already seen earlier. If you have a bounded function on a bounded interval (a, b) , you can follow two procedures and they both lead to the same answer.

The first method is to take partitions of the open interval (a, b) , calculate the lower and upper Riemann sums; say that the function is integrable if sup of lower sums agrees with inf of upper sums and declare the common value as integral.

Second method (as defined just now) is to take integral over $[a + \epsilon, b)$ or over $(a, b - \eta]$ and take the limit as $\epsilon \downarrow 0$ (or $\eta \downarrow 0$). Both lead to the same answer.

usual rules:

If $\int_a^b f$ and $\int_a^b g$ exist then $\int_a^b (f + g)$ exists and

$$\int_a^b (f + g) = \int_a^b f + \int_a^b g; \quad \int_a^b 39f = 39 \int_a^b f$$

This is simply by using properties of limits. There is one subtle point. It is quite possible that f is unbounded at an end point but g is a bounded continuous function on the open interval (a, b) . Then you might argue that we defined $\int g$ through Riemann sums and not through limits. So how do you justify? But as mentioned above, consistency, in such a case

$$\int_a^b g = \lim_{a^* \downarrow a} \int_{a^*}^b g \quad (\bullet)$$

So the above argument is perfectly fine. There are subtle points in writing proof. For example f may be unbounded at one end point and g at another. Or it may so happen that both f and g are unbounded (at one or both end points) but $f + g$ may be bounded. So $(\bullet)$ saves to unify the arguments.

The substitution rule also holds: φ continuously differentiable strictly increasing function on (a, b) onto (c, d) and f defined on (c, d) and integrable. Then

$$\int_c^d f(y)dy = \int_a^b f(\varphi(x))\varphi'(x)dx$$

That is, right side exists and equals left side. This is simply because if you take $a^* > a$ and denote $c^* = \varphi(a^*)$ then from what we know

$$\int_{c^*}^d f(y)dy = \int_{a^*}^b f(\varphi(x))\varphi'(x)dx$$

Also as $a^* \downarrow a$ then $c^* = \varphi(a^*) \downarrow c$ and since left side limit exists (by hypothesis) we conclude right side limit also exists and they are equal. If

f is unbounded at both end points then you can modify this argument and show the result.

(D) Let us now consider the case of unbounded interval. Suppose that f is a function defined on the interval $(0, \infty)$ which is bounded on every interval $(0, a)$. We say that the integral

$$\int_0^\infty f$$

exists if the limit

$$\lim_{a \rightarrow \infty} \int_0^a f$$

exists and then we define this limit as the value of the integral. In other words, there is a number α such that $\lim_{a_n \rightarrow \infty} \int_0^{a_n} f$ exists and equals α whenever $a_n \rightarrow \infty$. It is not enough to have this limit for one sequence of numbers $a_n \rightarrow \infty$.

for example

$$\int_0^\infty e^{-x} dx$$

exists and equals one.

$$\int_0^\infty x^n e^{-x} dx$$

exists and equals $(n-1)!$ if n is a positive integer. you prove it by induction on n and integration by parts over $(0, a)$. You need to use $a^k e^{-a} \rightarrow 0$ as $a \rightarrow \infty$.

Similarly you define integral of a continuous function over all of R . Take a number c and define

$$\int_{-\infty}^\infty f = \int_{-\infty}^c f + \int_c^\infty f$$

provided the two integrals on the right exist. This does not depend on the point c .

There are other possibilities: $\int_0^\infty x^{-1/2} e^{-x} dx$. Here the interval is unbounded and also the function is unbounded near zero. We shall see soon, you combine both recipes.

usual rules:

One question that needs to be looked into is whether the usual simple rules —for sum, constant multiple, etc — apply. Yes, it is just a matter of

using them for the case we know and applying limits. We have already seen for unbounded function on a bounded interval. Same applies in other cases too: interval of integration is infinite or/and function is unbounded.

If $a < c < b$, then

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

This means if both integrals on the right side exist then so does the integral on the left side; if the integral on the left side exists then so do both integrals on the right side; and then equality holds. Remember that we are assuming f is unbounded only near the end points, either both end points or only one. In other words, if we take any $a < \alpha < \beta < b$, then f is bounded on $[\alpha, \beta]$. You can prove again by taking $a < \alpha < c < \beta < b$ and using known result over the interval $[\alpha, \beta]$ and then taking limits.

Same argument applies over infinite intervals too.

integration by parts formula also holds. You apply for bounded subinterval and take limits.

substitution:

Recall that the method of substitution says the following.

$$\varphi : (a, b) \rightarrow (c, d)$$

strictly increasing C^1 function with bounded derivative and f is a continuous function on (c, d) . Then

$$\int_a^b f(\varphi(x))\varphi'(x) = \int_c^d f(y).$$

The same result holds for infinite intervals etc.

Just as we use the term converges for series, for integral also we use the term **integral converges** instead of it exists; in case of improper integrals, not for bounded functions on bounded intervals (just as you do not say converges for finite sums).

beta integral:

The integral

$$\int_0^1 x^{a-1}(1-x)^{b-1}dx$$

is called beta integral. The integral converges for $a > 0$ and $b > 0$. It does not converge if either $a \leq 0$ or $b \leq 0$. The value of the integral is denoted by $\beta(a, b)$. Thus this is defined only when both a and b are strictly positive. Draw/Sketch the curve for various values of a, b .

When $a = 1, b = 1$, then the integrand is one and hence so is the value of the integral. For $a \geq 1, b \geq 1$ the integrand is a nice continuous function on the closed interval $[0, 1]$ and the integral is the usual one and is not improper. You may say it is a proper integral.

If $b \geq 1$ but $a < 1$ then the integrand is unbounded at zero, and the integral is improper. Let now $0 < a < 1$. Then for any $0 < \alpha < 1$ we have

$$\int_{\alpha}^1 x^{a-1}(1-x)^{b-1} \leq \int_{\alpha}^1 x^{a-1} \leq \frac{1}{a} - \frac{\alpha^a}{a} \leq \frac{1}{a}.$$

and hence the integral converges. of course, this only shows that these integrals are bounded above by $1/a$, but do they converge? Yes, the integrand is positive and the integrals are increasing and hence converge. In fact the limit is supremum of the set

$$\left\{ \int_{\alpha}^1 x^{a-1}(1-x)^{b-1} : 0 < \alpha < 1 \right\}$$

Let us continue assuming that $b \geq 1$ but now $a \leq 0$. Then

$$\begin{aligned} \int_{\alpha}^1 x^{a-1}(1-x)^{b-1} &\geq \int_{\alpha}^{1/2} x^{a-1}(1-x)^{b-1} \geq (1/2)^{b-1} \int_{\alpha}^1 x^{a-1} \\ &= (1/2)^{b-1} \left[\frac{1}{a} - \frac{\alpha^a}{a} \right]. \end{aligned}$$

Since $a < 0$ we see that this last quantity increases to infinity as $\alpha \rightarrow 0$. Thus the integral does not converge.

Exactly similar arguments apply when $a \geq 1$ and $0 < b < 1$.

When $0 < a < 1$ and $0 < b < 1$, the integrand is unbounded at both end points of the interval $(0, 1)$. We argue that integral over $(0, 1/2)$ converges and integral over $(1/2, 1)$ converges and argue as earlier that the integral over $(0, 1)$ converges and actually it converges to the sum of the above two, namely, integral over $(0, 1/2)$ and integral over $(1/2, 1)$.

Finally when both $a < 0$ and $b < 0$, the integral does not converge. Ideas needed are already present in the above argument. the value of the integral has a notation:

$$\beta(a, b) = \int_0^1 x^{a-1}(1-x)^{b-1} dx; \quad a > 0, b > 0$$

gamma integral:

The integral

$$\int_0^{\infty} e^{-x} x^{a-1} dx$$

is called Gamma integral. It converges for $a > 0$ and does not converge for $a \leq 0$. For $a > 0$, the value of the integral is denoted by $\Gamma(a)$. We have already seen that for $a = 1$, the integrand is just e^{-x} and the integral converges and has value one. It will be good exercise for you to draw shape of the function for several values of a . First note that the integrand is positive.

Case $a \geq 1$:

the integrand is bounded at zero. It is improper only because range of integration is infinite interval. We show that

$$\lim_{A \rightarrow \infty} \int_0^A e^{-x} x^{a-1}$$

exists. Denote this integral by I_A . Since integrand is positive, we conclude that I_A increases with A . It suffices to show that $\{I_A : A > 0\}$ is bounded.

If this is done, then I_A converges to the supremum. Indeed let

$$c = \sup\{I_n : n = 1, 2, 3, \dots\}.$$

Then $I_n \uparrow c$. We argue that $I_A \uparrow c$ as follows. Let $\epsilon > 0$ be given. Choose m so that $I_m > c - \epsilon$. Then for any $A > m$, we have $I_A \geq I_m > c - \epsilon$, more precisely, $c - \epsilon \leq I_A \leq c$ for $A > m$.

Let $k > a - 1$ be any integer. We know that $e^{-x/2} x^k \rightarrow 0$ as $x \rightarrow \infty$ and hence

$$e^{-x/2} x^{a-1} \rightarrow 0; \quad \text{as } x \rightarrow \infty.$$

Say, it is smaller than one for $x \geq \alpha$. On the interval $[0, \alpha]$ the function $e^{-x/2} x^{a-1}$ is continuous and hence bounded; remember that $a \geq 1$. Thus there is a number M so that

$$e^{-x/2} x^{a-1} \leq M; \quad \forall x \geq 0.$$

Thus

$$e^{-x} x^{a-1} = e^{-x/2} e^{-x/2} x^{a-1} \leq e^{-x/2} M; \quad \forall x \geq 0.$$

So for any $A > 0$

$$\int_0^A e^{-x} x^{a-1} \leq \int_0^A M e^{-x/2} = M 2[1 - e^{-A/2}] \leq 2M.$$

showing that the set $\{I_A : A > 0\}$ is bounded.

Case $0 < a < 1$:

In this case the integrand is unbounded at zero and also the range of integration is unbounded. For any $0 < \alpha < 1$, we have

$$\int_\alpha^1 e^{-x} x^{a-1} \leq \int_\alpha^1 x^{a-1} = \frac{1}{a} - \frac{\alpha^a}{a} \leq \frac{1}{a}.$$

Thus the integral converges over the interval $(0, 1)$. More precisely, there is a number c_1 such that

$$\int_{\alpha}^1 e^{-x} x^{a-1} \rightarrow c_1; \quad \text{as } \alpha \rightarrow 0.$$

Also for any $\alpha > 1$, noting that $a - 1 < 0$ and hence $x^{a-1} < 1$ for $x > 1$, we have

$$\int_1^{\alpha} e^{-x} x^{a-1} \leq \int_1^{\alpha} e^{-x} = e^{-1} - e^{-\alpha} \leq 1.$$

Thus the integral converges over the interval $(1, \infty)$. More precisely, there is a number c_2 such that

$$\int_1^{\alpha} e^{-x} x^{a-1} \rightarrow c_2; \quad \text{as } \alpha \rightarrow \infty.$$

Thus the integral converges over $(0, \infty)$. In fact, if we take $\alpha_n \rightarrow 0$ and $\beta_n \rightarrow \infty$, then after some stage $\alpha_n < 1 < \beta_n$. Thus after this stage,

$$\int_{\alpha_n}^{\beta_n} e^{-x} x^{a-1} = \int_{\alpha_n}^1 e^{-x} x^{a-1} + \int_1^{\beta_n} e^{-x} x^{a-1} \rightarrow c_1 + c_2.$$

An integration by parts (and induction) shows that for integers $n = 1, 2, \dots$

$$\Gamma(n) = (n-1)!$$

More generally, integration by parts shows that for $a > 0$,

$$\Gamma(a+1) = a\Gamma(a).$$

This simple equation has interesting consequences. For example, let us take $a = -1/2$. — **which we are not supposed to take**. Use the equation above, $\Gamma(-\frac{1}{2} + 1) = (-1/2)\Gamma(-1/2)$. It is reasonable to define

$$\Gamma(-1/2) = -2\Gamma(1/2).$$

Now we can define $\Gamma(-3/2)$ etc. In other words the equation above which is a theorem for $a > 0$ can be used for defining $\Gamma(a)$ for negative values of a . Thus one can define the Gamma function for all real values *except* for $a = 0, -1, -2, \dots$, that is, except for non-positive integers. In fact it can be extended for complex numbers a also. Such an extension is not only fun, but also has interesting consequences. But you must keep in mind that $\Gamma(-1/2)$ is NOT defined via integral formula but by a different formula.

There is a close relation between the beta and gamma integrals. These are

two important integrals that arise in practice. These are what we discuss next.

[Question: Darboux integral we find confusing. Yes, I also do. When you try a new problem, situation is confusing, you do not know how to proceed, try various things, finally something works out. Then you ignore all other ideas and proceed with this which succeeded. Exactly the same happens with concepts too. **No one know what is integral, but everyone has a rough idea of what it should be.** So we formalize, several trials. Darboux, Henstock and several others arose. Finally Once something that is useful is arrived we continue with that. So we continued with Riemann integral. I do not always remember the various definitions, do not remember history! It is quite likely that the upper and lower sums are Darboux whereas sums using selections are due to Riemann. It is also possible that, when $U(f) > L(f)$, you still have an urge to define integral and so name $U(f)$ as upper integral and $L(f)$ as lower integral. In fact one can have a full course on the history of integration itself.

However, as far as our course is concerned, we shall NOT define upper/lower integral etc. In fact even our integral has some weak points; later you will learn Lebesgue integral – the ultimate! That is what one uses in most situations.]

Sometimes an improper integral can be converted to a regular integral.

$$\int_1^\infty \frac{x}{1+x^3} = \int_0^1 \frac{1}{1+x^3}$$

Firstly the integral on left is improper because it is over an infinite interval. It exists because integrand is non-negative and smaller than $x/x^3 = 1/x^2$ which is integrable. If we put $x = 1/y$ so that $dx = (-1/y^2)dy$ and as x ranges from 1 to infinity, y ranges from 1 to zero. The integral changes as stated.

series and integrals:

Our knowledge about integrals can be used to discuss convergence of series.

Let f be a decreasing function on the interval $[1, \infty)$ which is strictly positive.

Let $\sum_1^\infty a_n$ be a series of non-negative numbers..

If $a_n \leq f(n)$ for all n and $\int_1^\infty f$ converges then so does the series $\sum a_n$.

If $f(n) \leq a_n$ for all n and $\sum_1^\infty a_n$ converges then so does $\int_0^\infty f$

To prove the first statement, for each $k \geq 1$, by hypothesis $a_{k+1} \leq f(k+1) \leq f(x)$ for all $x \in [k, k+1]$; the last inequality from the fact that f is decreasing. Thus $a_{k+1} \leq \int_k^{k+1} f$ and hence $\sum_2^n a_k \leq \int_1^n f$. Thus partial sums of the series $\sum a_n$ are bounded above by $a_1 + \int_1^\infty f$. So the series converges.

The second part is similar, bound $\int_1^n f$ by $\sum a_k$. ■

you can use this to show the series $\sum n^{-p}$ converges for $p > 1$ and $\sum (1/n)$ does not. You only need to observe that $\int_1^\infty x^{-p}$ converges for $p > 1$; while for $p = 1$ it does not. ■

We now discuss two integrals; very useful; that appear in probability, Differential equations (Heat equation), physics and harmonic analysis and functional analysis and so on.

Normal integral:

$$\int_{-\infty}^\infty \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx = 1.$$

You will learn later double integrals, use them to prove this equality, in a painless and smart way, The point you should appreciate is that high school

knowledge and our definition of integral is enough to evaluate. Because of the symmetry, by changing variable $x = -y$, you see that integral over the two halves $[0, \infty)$ and $(-\infty, 0]$ are equal. Thus we only need to show

$$\int_0^\infty e^{-x^2/2} dx = \sqrt{\pi/2}$$

First of all does this integral converge? Yes. $\int_0^1$ is usual integral of continuous function on a closed bounded interval. But on $[1, \infty)$ you see $x^2 \geq x$ so that integrand is smaller than $e^{-x/2}$ which we know is integrable on $[1, \infty)$. Denote

$$a_n = \int_0^\infty x^n e^{-x^2/2} dx; \quad n = 0, 1, 2, \dots$$

$$a_0 = ?; \quad a_1 = 1.$$

Integration by parts gives $a_k = (k-1)a_{k-2}$ for $k > 1$ leading to

$$a_{2n} = (2n-1)(2n-3) \cdots (3)(1)a_0 = \frac{(2n)!}{n! 2^n} a_0$$

$$a_{2n+1} = (2n)(2n-2) \cdots (2)1 = n! 2^n$$

Fix $k \geq 1$. Note that for any $\lambda \in R$ we have

$$\int_0^\infty x^{k-1} (\lambda x - 1)^2 e^{-x^2/2} dx > 0$$

because integrand is non-negative, continuous and is not the zero function. Thus

$$\lambda^2 a_{k+1} - 2\lambda a_k + a_{k-1} > 0; \quad \forall \lambda \in R$$

hence high school tells you,

$$a_k^2 \leq a_{k+1} a_{k-1}; \quad \text{or} \quad a_k \leq \sqrt{a_{k+1} a_{k-1}}$$

Now $a_{2n} \leq \sqrt{a_{2n+1} a_{2n-1}}$ gives you

$$\frac{(2n)!}{n! 2^n} a_0 \leq \sqrt{n! (n-1)! 2^n 2^{n-1}} = n! 2^n \frac{1}{\sqrt{2n}}$$

Or

$$a_0 \leq \frac{n! n! 2^n 2^n}{(2n)! \sqrt{2n}} \quad (\spadesuit)$$

Now use $a_{2n+1} \leq \sqrt{a_{2n+2} a_{2n}}$ to see

$$n! 2^n \leq \sqrt{\frac{(2n)!}{n! 2^n} a_0 \frac{(2n+2)!}{(n+1)! 2^{n+1}} a_0} = a_0 \frac{(2n)!}{n! 2^n} \sqrt{(2n+1)}$$

Or

$$a_0 \geq \frac{n! \, n! \, 2^n \, 2^n}{(2n)!} \frac{1}{\sqrt{2n+1}} \quad (\clubsuit)$$

Thus from $(\spadesuit)$ and $(\clubsuit)$

$$\frac{n! \, n! \, 2^n \, 2^n}{(2n)!} \frac{1}{\sqrt{2n+1}} \leq a_0 \leq \frac{n! \, n! \, 2^n \, 2^n}{(2n)! \sqrt{2n}}$$

Walli's product completes the calculation because quantities on both ends converge to $\sqrt{\pi/2}$.

A Trigonometric Integral:

$$\int_0^\infty \frac{\sin x}{x} = \frac{\pi}{2}$$

We first show that the integral is convergent. That is,

$$I_A = \int_0^A \frac{\sin x}{x}$$

has a finite limit as $A \rightarrow \infty$. Note that the integral is improper only at infinity, the integrand is bounded near 0.

We first show that given $\epsilon > 0$, there is a number A_0 such that $\int_A^B \frac{\sin x}{x} < \epsilon$ for $B > A > A_0$. Indeed take $A_0 > 4/\epsilon$. Let $B > A > A_0$. Then integration by parts gives

$$\int_A^B \frac{\sin x}{x} = -\frac{\cos B}{B} + \frac{\cos A}{A} + \int_A^B \frac{\cos x}{x^2}$$

so that

$$\left| \int_A^B \frac{\sin x}{x} \right| \leq \frac{2}{A} + \int_A^B \frac{1}{x^2} \leq \frac{3}{A} < \epsilon.$$

Just as Cauchy sequences have limits, such functions $I(A) = \int_0^A (\sin x/x)$ have limit (as $A \rightarrow \infty$) and is argued as follows. Take any sequence converging to ∞ , say $\{A_n\}$. Clearly, the above inequality shows that $\{I_{A_n}\}$ is a Cauchy sequence and hence has a finite limit, say c . Thus $I_A \rightarrow c$. (You may still have a doubt: if I take two sequences (A_n) and (B_n) increasing to infinity, they may have different limits. If that is so, then it is easy to see that the Sequence C_n defined by $C_{2n} = A_n$ and $C_{2n+1} = B_n$ is also a sequence converging to infinity and I_{C_n} does not converge.)

We now need to show that this limit equals $\pi/2$. Now that we know limit exists, we can take our convenient sequence and show limit is $\pi/2$. We take $A_n = (n + 1/2)\pi$ and show $\int_0^{A_n} \frac{\sin x}{x} \rightarrow \pi/2$. That is, we shall show

$$\int_0^{(n+1/2)\pi} \frac{\sin x}{x} \rightarrow \pi/2$$

By substitution, this is same as showing

$$\int_0^\pi \frac{\sin(n + 1/2)x}{x} \rightarrow \pi/2 \quad (\spadesuit)$$

Why did we settle on this sequence. Because we know

$$\int_0^\pi \frac{\sin(n + \frac{1}{2})x}{2 \sin \frac{x}{2}} = \frac{\pi}{2}. \quad (\clubsuit)$$

How do we know this? High School Trigonometry. The sine and cosine formulae give

$$2 \sin \frac{x}{2} \cos kx = \sin(k + \frac{1}{2})x - \sin(k - \frac{1}{2})x.$$

Adding from $k = 1$ to $k = n$, we see

$$2 \sin \frac{x}{2} \sum_1^n \cos kx = \sin(n + \frac{1}{2})x - \sin \frac{x}{2}.$$

or

$$\frac{\sin(n + \frac{1}{2})x}{2 \sin \frac{x}{2}} = \sum_1^n \cos kx + \frac{1}{2}$$

(This equality holds at $x = 0$ by direct verification) integrating and noting that cosine integrals vanish, we get

$$\int_0^\pi \frac{\sin(n + \frac{1}{2})x}{2 \sin \frac{x}{2}} = \frac{\pi}{2}.$$

proving $(\clubsuit)$. Thus to prove $(\spadesuit)$ it suffices to show

$$\int_0^\pi \left(\frac{1}{x} - \frac{1}{2 \sin \frac{x}{2}} \right) \sin(n + \frac{1}{2})x \rightarrow 0$$

This is an immediate consequence of the following two claims: Let $[a, b]$ be a closed bounded interval.

$$\lim_{\lambda \rightarrow \infty} \int_a^b \varphi(x) \sin \lambda x dx = 0 \quad \text{for any } C^1 \text{ function } \varphi. \quad (*)$$

$$\left(\frac{1}{x} - \frac{1}{2 \sin \frac{x}{2}}\right) \text{ is a } C^1 \text{ function on } [0, \pi]. \quad (**)$$

Proof of (*) is again integration by parts as earlier.

$$\int_a^b \varphi(x) \sin \lambda x = \frac{\varphi(a) \cos \lambda a - \varphi(b) \cos \lambda b}{\lambda} + \int_a^b \varphi' \frac{\cos \lambda x}{\lambda}$$

So, setting a bound M for the two continuous functions $|\varphi|$ and $|\varphi'|$ on $[a, b]$

$$\left| \int_a^b \varphi(x) \sin \lambda x \right| \leq \frac{2M}{\lambda} + \frac{M(b-a)}{\lambda} \rightarrow 0.$$

as $\lambda \rightarrow \infty$.

Proof of (**) is L'Hospital's rule. Of course as it stands value of the function at zero is not defined but one shows that the function converges to zero as $x \rightarrow 0$ and hence by defining the value at zero to be zero we see it is a continuous function on $[0, \pi]$. To see that the limit at zero is zero, differentiate numerator and denominator of

$$\frac{2 \sin \frac{x}{2} - x}{2x \sin \frac{x}{2}}$$

twice and see. Or smart way is the following

$$\frac{2 \sin \frac{x}{2} - x}{2x \sin \frac{x}{2}} = \frac{2 \left[(x/2) - (x/2)^3(3!)^{-1} + (x/2)^5(5!)^{-1} - \dots \right] - x}{2x \left[(x/2) - (x/2)^3(3!)^{-1} + (x/2)^5(5!)^{-1} - \dots \right]}$$

You see that in numerator x term gets cancelled and the leading term is x^3 and in the denominator leading term is x^2 . Thus for $x \neq 0$, you cancel x^2 from numerator and denominator and take limit as $x \rightarrow 0$ or put $x = 0$ because our power series in numerator and denominator are continuous.

Advantage is the following. You need to calculate derivative at zero. Since $\varphi(0) = 0$ (φ is the function under consideration on $[0, \pi]$) derivative is limit of $\varphi(x)/x$. If you did this then leading terms in both numerator and denominator are both x^3 and you can easily calculate $\varphi'(0)$. Of course you still need to calculate $\varphi'(x)$ for $x > 0$ and show $\lim \varphi'(x)$ is that value. This completes the proof. ■

We have Mean Value Theorem and generalized Mean value Theorem for integrals just like for derivatives.

MVT for integrals:

f continuous on a closed bounded interval $[a, b]$. Then there is a number $\theta \in [a, b]$ such that

$$\int_a^b f = f(\theta)(b - a)$$

Proof is simple. If m and M are the minimum and maximum of f over $[a, b]$ then $m \leq f \leq M$, so that

$$\int_a^b m \leq \int_a^b f \leq \int_a^b M; \quad m(b - a) \leq \int_a^b f \leq M(b - a)$$

So there is a number k such that $m \leq k \leq M$ and $\int_a^b f = k(b - a)$. By the intermediate value property for continuous functions we get θ such that $k = f(\theta)$.

Actually, this is just MVT we learnt for derivatives. If you denote F as primitive for f ; then above equation is nothing but $F(b) - F(a) = F'(\theta)(b - a)$.

Generalized MVT for integrals:

f continuous on a closed bounded interval $[a, b]$ and g continuous non-negative on $[a, b]$ Then there is a number $\theta \in [a, b]$ such that

$$\int_a^b fg = f(\theta) \int_a^b g$$

Exactly the same proof. With above notation, g being non-negative we have $mg \leq fg \leq Mg$ etc.

Taylor revisited with Cauchy form of remainder:

f is $(n + 1)$ times continuously differentiable on $[a, b]$. Then

$$\begin{aligned} f(b) = f(a) + \frac{(b-a)}{1!} f^{(1)}(a) + \frac{(b-a)^2}{2!} f^{(2)}(a) + \frac{(b-a)^3}{3!} f^{(3)}(a) + \\ \dots + \frac{(b-a)^n}{n!} f^{(n)}(a) + \frac{1}{n!} \int_a^b f^{(n+1)}(t)(b-t)^n dt \quad \blacksquare \end{aligned}$$

Thus

$$f(b) = f(a) + \int_a^b f^{(1)}(t) dt$$

$$f(b) = f(a) + \frac{b-a}{1!} f^{(1)}(a) + \frac{1}{1!} \int_a^b f^{(2)}(t)(b-t) dt$$

$$f(b) = f(a) + \frac{b-a}{1!} f^{(1)}(a) + \frac{(b-a)^2}{2!} f^{(2)}(a) + \frac{1}{2!} \int_a^b f^{(3)}(t)(b-t)^2 dt$$

Proof is immediate, induction, integrate by parts the last integral. ■

Binomial series:

Let $\alpha \neq 0$ be any number. Then

$$(1+x)^\alpha = 1 + \binom{\alpha}{1}x + \binom{\alpha}{2}x^2 + \cdots; \quad -1 < x < 1$$

where

$$\binom{\alpha}{k} = \frac{\alpha(\alpha-1)(\alpha-2)\cdots(\alpha-k+1)}{1 \times 2 \times 3 \cdots k}$$

Did you realize that $\binom{\alpha}{k} = \frac{\alpha!}{k!(\alpha-k)!}$ did not make sense and so we looked at another equivalent expression for $\binom{n}{k}$ and took it as definition.

To prove this We use Taylor with Cauchy remainder and show remainder term goes to zero. Let $f(x) = (1+x)^\alpha$. We consider $0 < x < 1$.

$$f(0) = 1; \quad f^{(1)}(x) = \alpha(1+x)^{\alpha-1}; \quad f^{(2)}(x) = \alpha(\alpha-1)(1+x)^{\alpha-2}$$

$$f^{(n)}(x) = \alpha(\alpha-1)\cdots(\alpha-n+1)(1+x)^{\alpha-n}$$

$$f^{(n+1)}(x) = \alpha(\alpha-1)\cdots(\alpha-n+1)(\alpha-n)(1+x)^{\alpha-n-1}$$

consider the interval $(-1, 1)$ Taylor around zero ($a = 0, b = x$)

$$\begin{aligned} f(x) &= f(0) + \frac{x}{1!}f^{(1)}(0) + \frac{x^2}{2!}f^{(2)}(0) + \frac{x^3}{3!}f^{(3)}(0) + \\ &\quad \cdots + \frac{x^n}{n!}f^{(n)}(0) + \frac{1}{n!} \int_0^x f^{(n+1)}(t)(x-t)^n dt \\ &= 1 + \binom{\alpha}{1}x + \binom{\alpha}{2}x^2 + \cdots + \binom{\alpha}{n}x^n + R_n \\ R_n &= \frac{1}{n!} \int_0^x f^{(n+1)}(t)(x-t)^n dt \end{aligned}$$

Proof is completed by showing that $R_n \rightarrow 0$. Keep in mind $0 < x < 1$. First by MVT there is a z between zero and x such that

$$R_n = \frac{1}{n!} f^{(n+1)}(z)(x-z)^n x$$

As you pointed out we could say this with earlier version of Taylor, not necessarily going through the recent version.

$$R_n = \frac{1}{n!} \alpha(\alpha-1)\cdots(\alpha-n+1)(\alpha-n)(1+z)^{\alpha-n-1} (x-z)^n$$

$$= \left(1 - \frac{\alpha}{1}\right) \left(1 - \frac{\alpha}{2}\right) \cdots \left(1 - \frac{\alpha}{n}\right) \left(\frac{x-z}{1+z}\right)^n (-1)^n \alpha (1+z)^{\alpha-1}$$

Denote $(x-z)/(1+z) = c$. Since $0 \leq z \leq x < 1$ we see $0 < c < 1$. Since $(-1)^n = \pm 1$ and the last term in the display above does not involve n we only need to show,

$$\left(1 - \frac{\alpha}{1}\right) \left(1 - \frac{\alpha}{2}\right) \cdots \left(1 - \frac{\alpha}{n}\right) c^n \rightarrow 0 \quad (\star)$$

Since α may be positive or negative and could be large in magnitude, you should not jump to the conclusion that $(1 - \frac{\alpha}{2}) < 1$.

$\alpha > 0$: Since $\alpha/n \downarrow 0$, There is an K such that $0 < (1 - \frac{\alpha}{K}) < 1$ and hence for all $n > K$ we have $0 < (1 - \frac{\alpha}{n}) < 1$. Thus the quantity in $(\star)$ is at most

$$\left(1 - \frac{\alpha}{1}\right) \left(1 - \frac{\alpha}{2}\right) \cdots \left(1 - \frac{\alpha}{K}\right) c^n \quad n > K$$

and hence converges to zero, remember $0 < c < 1$.

$\alpha < 0$: Since $0 < c < 1$ fix d so that $0 < c < d < 1$. Since $\alpha/n \uparrow 0$, we observe that $(1 - \frac{\alpha}{n})c \downarrow c$. So There is an K such that $0 < (1 - \frac{\alpha}{K})c < d$ and hence for all $n > K$ we have $0 < (1 - \frac{\alpha}{n})c < d$. Thus the quantity in $(\star)$ is at most

$$\left(1 - \frac{\alpha}{1}\right) \left(1 - \frac{\alpha}{2}\right) \cdots \left(1 - \frac{\alpha}{K}\right) c^K d^{n-K} \quad n > K$$

and hence converges to zero, remember $0 < d < 1$.

The case $-1 < x < 0$ is similarly handled. This completes the proof of the binomial expansion. ■

There is another way of showing this. Define on $(-1, 1)$

$$f(x) = (1+x)^\alpha$$

$$g(x) = 1 + \binom{\alpha}{1}x + \binom{\alpha}{2}x^2 + \cdots$$

This makes sense, by ratio test the power series converges for $|x| < 1$. Both solve the following:

$$(1+x)\varphi'(x) = \alpha\varphi(x); \quad \varphi(0) = 1 \quad -1 < x < 1$$

For f this is direct verification; for g this is term by term differentiation and the fact that the series is absolutely convergent so that you can combine

terms. But there is only one solution for the above problem and hence $f = g$. Done.

How do you show there is only one solution? if there are two solutions then by taking difference you have ζ on $(-1, 1)$ so that

$$\zeta'(x) = \frac{\alpha}{1+x}\zeta(x); \quad \zeta(0) = 0$$

To show $\zeta \equiv 0$ fix any $0 < q < 1$. We show $\zeta(x) = 0$ for $-q < x < q$.

Case: $0 \leq x \leq q$
we have

$$|\zeta(x)| = \left| \int_0^x \zeta'(t) dt \right| \leq \int_0^x |\zeta'(t)| dt \leq \int_0^x \left| \frac{\alpha}{1+t} \right| |\zeta(t)| dt$$

Let $\varphi(x) = |\zeta(x)|$. denoting $c = |\alpha|$ and noting $\left| \frac{\alpha}{1+t} \right| \leq c$ the above inequality says

$$\varphi(x) \leq \int_0^x c \varphi(t) dt; \quad 0 \leq x \leq q \quad (\bullet)$$

You can use this and get self-improvement. φ being continuous and positive, say $\varphi(x) \leq M$ for $0 \leq x \leq q$. $(\bullet)$ says that $\varphi(x) \leq M c x$ for all $x \in [0, q]$. Before you get confused, I restate as: $\varphi(t) \leq M c t$ for all $t \in [0, q]$. Use $(\bullet)$ to see

$$\varphi(x) \leq \int_0^x c M c t dt \leq M \frac{(cx)^2}{2!}$$

Use $(\bullet)$ again

$$\varphi(x) \leq \int_0^x c M c^2 \frac{t^2}{2!} dt \leq M \frac{(cx)^3}{3!}$$

Thus you get $\varphi(x) \leq M (cx)^n / n!$ which converges to zero.

Case: $-q \leq x \leq 0$. Similar proof. This gives an alternative proof of the binomial expansion. ■

Note that when $\alpha > 1$ is an integer, then after some stage $\binom{\alpha}{n}$ equals zero and so the series is finite and is the usual binomial expansion. Being finite series it makes sense for all x and equality is true for all x . This is done by induction.

some terms that I did not introduce but often used in literature:

These terms will not be used in exam! I am explaining so that you need not panic if you come across when you read books.

Among numbers, there are simple numbers like positive integers, integers. Then rational numbers m/n where $n \geq 1$ and m are integers. Then we have

algebraic numbers, they are roots of polynomials with integer coefficients. Then we have transcendental numbers.

Similarly classes of Functions $f : R \rightarrow R$ also have names. A **monomial** is a function $f(x) = ax^m$ where $a \in R$ and $m \geq 0$ integer. A **Polynomial** is a sum of monomials. Thus a polynomial is a function $f(x) = a_0 + a_1x + a_2x^2 \cdots + a_mx^m$ for some integer $m \geq 0$. Here a_i are real numbers. A **rational function** is ratio of polynomials, where the denominator is not the identically zero polynomial. Thus it is of the form $f(x) = P(x)/Q(x)$ where P, Q are polynomials and $Q \not\equiv 0$. Of course Q may still have finitely many points where it is zero; and f is not defined at those points. Sometimes it may still be possible to define at such points if it is a common root of P and Q . Thus, for example $f(x) = (x^2 - 9)/(x - 3)$ is not defined at $x = 3$. But you can define it in a meaningful way!.

Then we have **algebraic functions**. $f(x) = \sqrt{x}$ for $[0, \infty)$. This is solution to $f^2 - x = 0$, $f \geq 0$ and $x \geq 0$. Similarly semicircle in upper half plane over $[-1, 1]$ is solution of $f^2 + x^2 - 1 = 0$ and $f \geq 0$ and $x \in [-1, +1]$. You can verify $f(x) = \sqrt{x} + \sqrt{x+1}$ is also algebraic. Thus a function f is algebraic on an interval I , if there is a polynomial in two variables $P(f, x)$ and an interval J such that for each $x \in I$, there is a unique $y \in J$ with $P(y, x) = 0$ and $f(x)$ is this y . When you prescribe algebraic function, you should tell I and J . You may have same I but you can change J as in the case of semicircle – we could have chosen lower half. Then there are **transcendental functions** – functions which are not algebraic.

Sometimes functions $f(x) = a^x$ where $a > 0$ is a fixed number are called **exponential functions**. This is because it is a function of the ‘exponent’, the argument is in the exponent. We called $f(x) = e^x$ as exponential function. More generally $f(x) = e^{cx}$ is called exponential function. There is no confusion because this is nothing but a^x where $a = e^c$.

On the other hand $f(x) = x^a$ is called **power function**. Domain depends on a , for example, if $a = 1/2$, then this is not defined for $x < 0$ (when you learn complex numbers you can do better).

sine, cosine, tangent, cosecant, secant, cotangent are called **Trigonometric** or **circular functions**. sinh, cosh, tanh etc are called **hyperbolic functions** for reasons discussed earlier.

Suppose f is a real valued function and you say f **vanishes** at a point a in its domain. This does not mean that the function disappears at a , but just that $f(a) = 0$. ■

Convergence of sequence or series is a **Tail phenomenon**. This means it

depends only on tail of the sequence or series. More precisely, suppose (a_n) and (b_n) are sequences which are equal after some stage. That is, say for n larger than a billion, we have $a_n = b_n$. Then if one of them converges, so does the other. Similarly in such a case $\sum a_n$ converges iff $\sum b_n$ converges.

Even if you interpret tail equivalence in a wider sense the statement remains true: there is an N and K such that $a_n = b_{n+K}$ for all $n > N$. Then one sequence (or series) converges iff the other does. ■

Continuity and differentiability is a **local phenomenon**. More precisely suppose that f and g are two functions defined on an open interval I and $a \in I$. Suppose they equal in a small interval around a — for example $f(x) = g(x)$ for all $x \in (a - 2^{10}, a + 2^{20})$. Then f is continuous at a iff g is. f is differentiable at a iff g is. ■

Final as scheduled.

Time: For two hours,

syllabus: whatever we did.

Marks: 50.

Misc 9: continuous but nowhere differentiable function:

This is not in the syllabus and you can safely ignore..

We can easily construct continuous functions which are not differentiable at finitely many points and with some work make continuous functions which are not differentiable at a sequence of points. Interestingly there are continuous functions which are not differentiable at all points. Here is one due to Van der Waerden. Basic idea is the following. The tent function, namely, $2x$ on $[0, 1/2]$ and $(2 - 2x)$ on $[1/2, 1]$, is not differentiable at $1/2$. You can make two tents, one in $[0, 1/2]$ and another in $[1/2, 1]$ and add to the above. continue adding and hopefully the resulting function is not differentiable at all points. But we should be careful because, in the process several tents may come close and smoothen out. Here is the execution.

Let $\langle x \rangle$ denote distance of x to the nearest integer. The graph of the function $x \mapsto \langle x \rangle$ is a series of tents with height zero at integers and reaching height $1/2$ at half-integers and then falls down. This is defined on all of $\mathbb{R}$. It is continuous. Clearly,

(★). If k is an integer, then $\langle x + k \rangle = \langle x \rangle$.

Let k be an integer. Note that if x, y are both in $[k, k + 1/2]$ then

$$\langle x \rangle - \langle y \rangle = (x - k) - (y - k) = x - y = \pm \langle x - y \rangle.$$

the sign depends on whether $x > y$ or not. And if both x, y are in the right half $[k + 1/2, k + 1]$ then

$$\langle x \rangle - \langle y \rangle = (k + 1 - x) - (k + 1 - y) = y - x = \pm \langle x - y \rangle.$$

Thus

(♠) x, y are in the same half between two integers $\Rightarrow \langle x \rangle - \langle y \rangle = \pm \langle x - y \rangle$.

Consider the function:

$$f(x) = \sum_{n=0}^{\infty} \frac{\langle 10^n x \rangle}{10^n}$$

Since $0 \leq \langle z \rangle \leq 1/2$ we see that the above series of continuous function is uniformly convergent and thus f is a continuous function. We now show

that f is nowhere differentiable. Because of periodicity of the function, it is enough to consider $0 \leq x \leq 1$.

So fix x . Let its decimal expansion be $x = 0.a_1a_2a_3 \dots$. When there are two expansions, we take the terminating one. We show a sequence $h_m \rightarrow 0$ but $\frac{f(x+h_m)-f(x)}{h_m}$ does not converge.

$$h_m = -10^{-m} \quad \text{when} \quad a_m = 4, 9; \quad h_m = +10^{-m} \quad \text{otherwise.}$$

what is special about 4, 9? Observe $x + h_m$ is just subtracting/adding one to the m -th decimal digit. Thus we see

(♣). x and $x + h_m$ are in the same half.

We now understand the difference quotient.

$$\begin{aligned} \frac{f(x+h_m)-f(x)}{h_m} &= \pm 10^m \sum_{n=0}^{\infty} \frac{\langle 10^n(x \pm 10^{-m}) \rangle - \langle 10^n x \rangle}{10^n} \\ &= \pm \sum_{n=0}^{\infty} \frac{\langle 10^n x \pm 10^{n-m} \rangle - \langle 10^n x \rangle}{10^{n-m}} \end{aligned}$$

If $n \geq m$ then 10^{n-m} is an integer, so by (★) numerator is zero.

If $n < m$, then $10^n x + h_m$ and $10^n x$ are in the same half by (♣). Indeed

$$10^n x = (\text{integer}L) + \cdot a_{n+1}a_{n+2} \dots a_m \dots;$$

$$10^n x + h_m = (\text{integer}L) + \cdot a_{n+1}a_{n+2} \dots (a_m \pm 1) \dots$$

Thus if $n < m-1$ the first decimal digits in both are same and so they are in the same half; if $n = m-1$, then choice of h_m as explained in (♣) shows the claim. Thus by (♠), the numerator equals 10^{n-m} . $\langle 10^n x \pm 10^{n-m} \rangle - \langle 10^n x \rangle = \pm 10^{n-m}$. Thus the ratio in the sum above is ± 1 . Thus

$$\frac{f(x+h_m)-f(x)}{h_m} = \pm \sum_{n=0}^{m-1} \pm 1$$

Be careful with the two $\pm$ signs; one may be plus and other minus. Thus as m changes, the difference between successive terms is ± 1 and so the sequence of difference quotients is not Cauchy and can not converge as claimed.

Misc 10: Weierstrass Polynomial approximation:

In high school you learnt polynomials. We now have continuous functions. Interestingly these are not too far from polynomials.

Given any continuous function f on $[0, 1]$ and given any $\epsilon > 0$ we can get a polynomial P (in one variable) such that $|f(x) - P(x)| < \epsilon$ for all $x \in [0, 1]$.

Thus you can uniformly approximate any continuous function on a closed bounded interval by a polynomial. Or geometrically, if you draw the graph

f and draw ϵ band around it — that is graphs of $f - \epsilon$ and $f + \epsilon$ — then graph of P lies within this band.

You might think that the Taylor expansion would help. Unfortunately it does not always help. Consider the infinitely differentiable function f which is zero for $x \leq 0$ and equals $\exp(-1/x^2)$ for $x > 0$. Then on the interval $[-1, 1]$, if you wish to approximate by a polynomial using Taylor expansion at zero, then it will not help.

We shall not prove this theorem now.

Final as scheduled November 26 Wednesday.

Time: two hours: 9:30-11:30,

syllabus: whatever we did before this class.

Marks: 50.